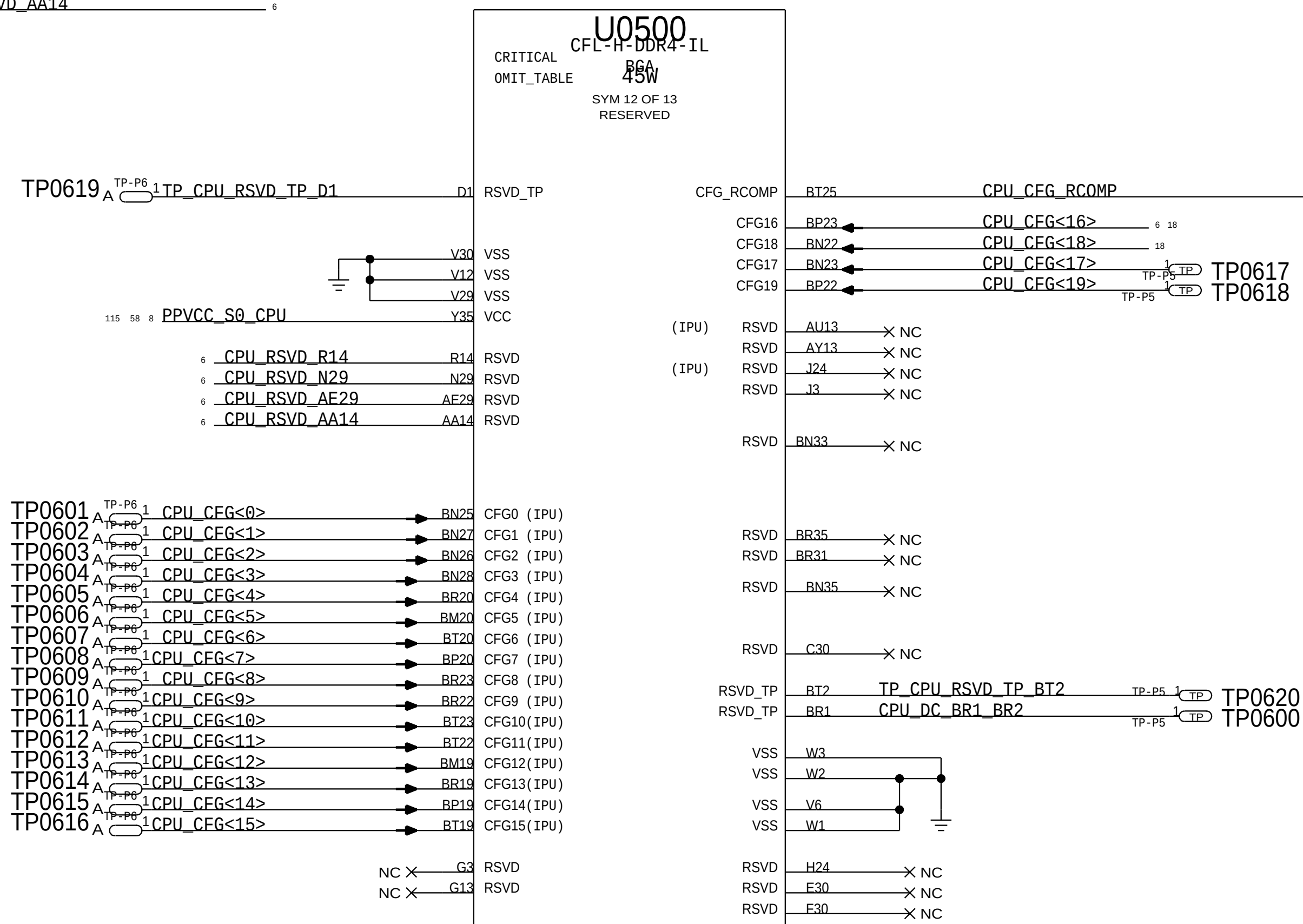
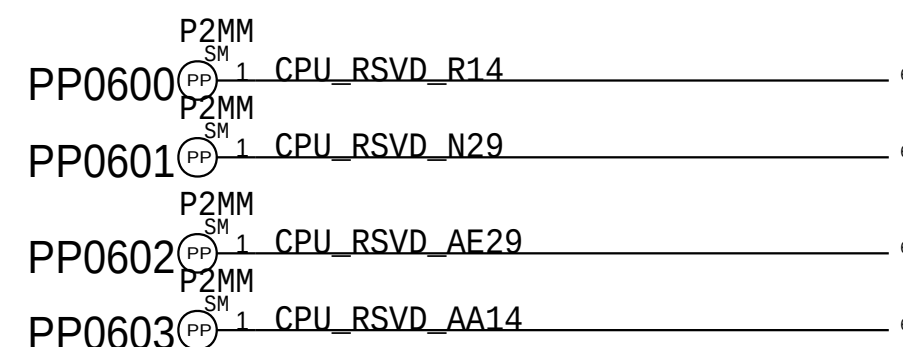
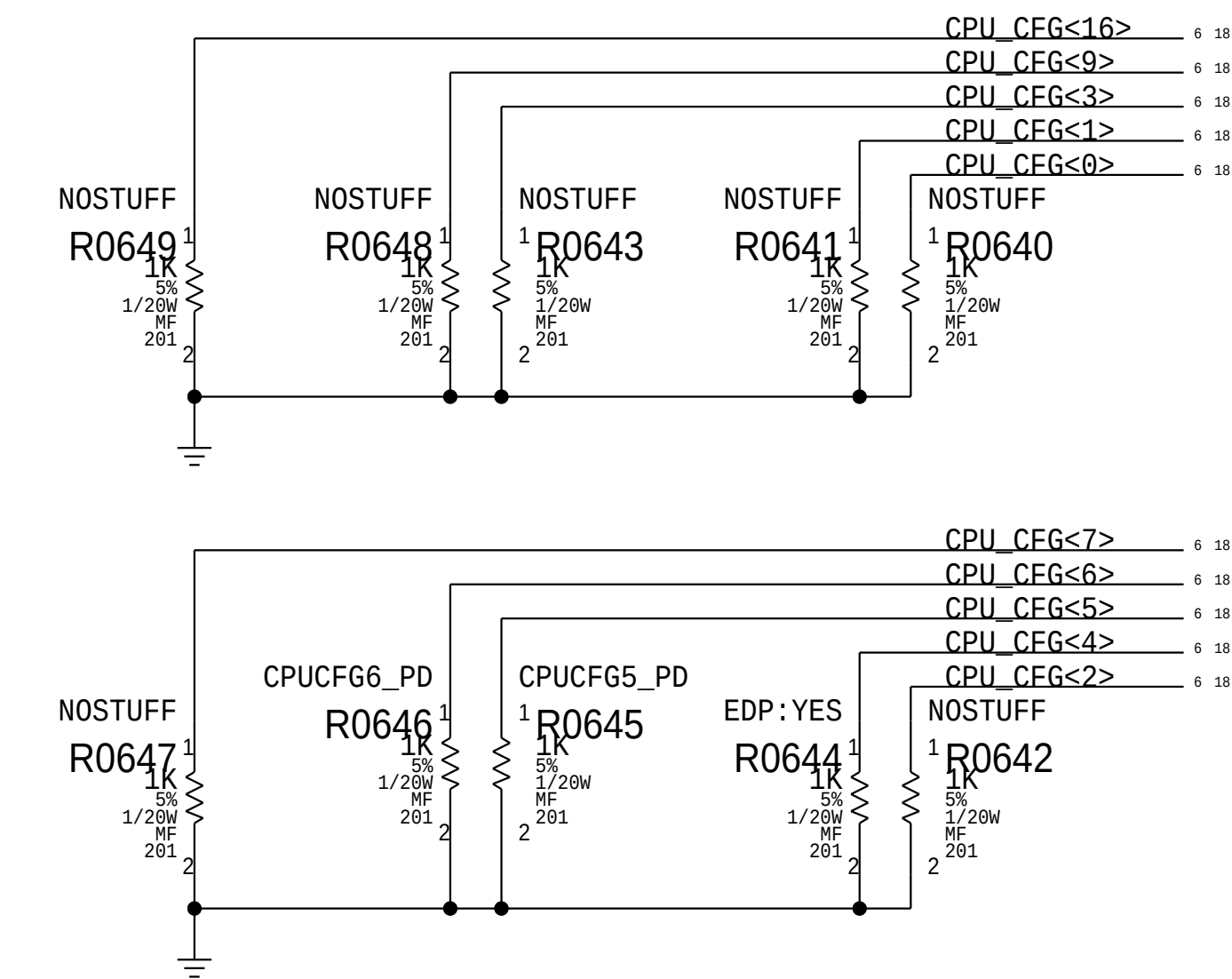
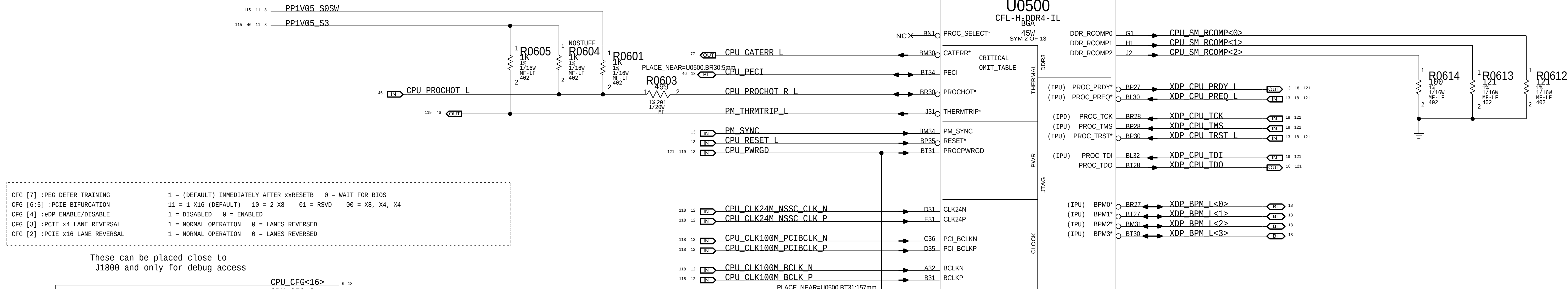


	8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																				
D	GPU Baffin <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>337S00493</td><td>1</td><td>GPU, AMD, BAFFIN, ULX, A3, 0.83, QS, BGA769</td><td>UA000</td><td>CRITICAL</td><td>BAFFIN_ULX</td></tr><tr><td>337S00494</td><td>1</td><td>GPU, AMD, BAFFIN, PROX, A3, 0.83, QS, BGA769</td><td>UA000</td><td>CRITICAL</td><td>BAFFIN_PROX</td></tr><tr><td>998-04866</td><td>1</td><td>INTERPOSER, AMD, C989, BGA769, VDDCI_MVDD</td><td>UA000</td><td>CRITICAL</td><td>STARDUST:VDDCI_MVDD</td></tr><tr><td>998-04867</td><td>1</td><td>INTERPOSER, AMD, C988, BGA769, VDDC</td><td>UA000</td><td>CRITICAL</td><td>STARDUST:VDDC</td></tr></table> FB VDRAM Parts <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>333S00074</td><td>4</td><td>IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, 8, 170 BGA</td><td>UA400, UA450, UA500, UA550</td><td>CRITICAL</td><td>FB_4GB_SAMSUNG</td></tr><tr><td>333S00075</td><td>4</td><td>IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, A, 170 BGA</td><td>UA400, UA450, UA500, UA550</td><td>CRITICAL</td><td>FB_4GB_MICRON</td></tr><tr><td>333S00100</td><td>4</td><td>IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, A, 170 BGA</td><td>UA400, UA450, UA500, UA550</td><td>CRITICAL</td><td>FB_4GB_HYNIX</td></tr><tr><td>333S00173</td><td>4</td><td>IC, GDDR5, 8Gb, 7Gbps, 1.5V, 12NM, A, 170 BGA</td><td>UA400, UA450, UA500, UA550</td><td>CRITICAL</td><td>FB_4GB_MICRON_1Z</td></tr><tr><td>333S00175</td><td>4</td><td>IC, GDDR5, 8Gb, 7Gbps, 1.5V, 12NM, A, 170 BGA</td><td>UA400, UA450, UA500, UA550</td><td>CRITICAL</td><td>FB_4GB_HYNIX_1Z</td></tr></table> CPU Parts CFL (Needs to be updated) <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>337S00509</td><td>1</td><td>CPU, CFLH, QP87, QS, U0, 2.2, 45W, 1.1, LGA1440</td><td>U0500</td><td>CRITICAL</td><td>CPU_CFL:2.2</td></tr><tr><td>337S00510</td><td>1</td><td>CPU, CFL-H, QP86, QS, U0, 2.6, 6C, 45W, 1.15, BGA1440</td><td>U0500</td><td>CRITICAL</td><td>CPU_CFL:2.6</td></tr><tr><td>337S00513</td><td>1</td><td>CPU, CFL-H, QP06, QS, U0, 2.9, 6C, 45W, 1.2, BGA1440</td><td>U0500</td><td>CRITICAL</td><td>CPU_CFL:2.9</td></tr><tr><td>998-12472</td><td>1</td><td>INTERPOSER, CFH-H, BGA1440</td><td>U0500</td><td>CRITICAL</td><td>CPU_SKL:SOCKET</td></tr></table> Main DRAM Parts <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>333S00131</td><td>16</td><td>IC, SDRAM, DDR4-2400, 16GBIT, 20NM, BGA78</td><td>U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900</td><td>CRITICAL</td><td>32G_MICRON_2400</td></tr><tr><td>333S00134</td><td>16</td><td>IC, SDRAM, DDR4-2400, 16GBIT, 20NM, BGA78</td><td>U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, 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U9900</td><td>CRITICAL</td><td>16G_HYNIX_2400</td></tr><tr><td>333S00163</td><td>16</td><td>IC, SDRAM, DDR4-2400, 8GBIT, 20NM, BGA78</td><td>U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900</td><td>CRITICAL</td><td>16G_MICRON_2400</td></tr></table> PCH CNL-H (Needs to be updated) <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>337S00523</td><td>1</td><td>IC, CNL, PCH-H, USFF, QNYP, QS, B0, BGA499</td><td>U1200</td><td>CRITICAL</td><td></td></tr></table> ACE & Ridges (Needs to Be Updated) <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>353S01442</td><td>4</td><td>IC, C03215, ACE, C0, USB PMR SW, BLNK, BGA96</td><td>U3100, U3200, UB300, UB400</td><td>CRITICAL</td><td></td></tr><tr><td>338S00408</td><td>2</td><td>IC, TBT, TITAN RIDGE DP, QUJK, QS, C1, CSP337</td><td>U2800, UB000</td><td>CRITICAL</td><td></td></tr></table> Hall Effect AMR <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>677-10608</td><td>2</td><td>SUBASSY T800 PCB, AMR, INTERPOSER, X1190</td><td>J4800, J4801</td><td>CRITICAL</td><td></td></tr></table> Power Controllers <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>353S01525</td><td>1</td><td>IC, ISL9240HB1Z, PMU, SOUWA, WSCP40, 2.1X3.3MM</td><td>U7000</td><td>CRITICAL</td><td></td></tr><tr><td>338S00267</td><td>1</td><td>IC, PMU, P650839, 7X7MM, BGA168</td><td>U7800</td><td>CRITICAL</td><td></td></tr><tr><td>353S01229</td><td>1</td><td>IC, SUPPLY, INTERSL, ISL6277AHRZ, SV12.0, QFN48</td><td>UA600</td><td>CRITICAL</td><td></td></tr><tr><td>353S00928</td><td>1</td><td>IC, ISL9582BA, IMPV8 CPU REG, QFN48, 6X8MM</td><td>U7100</td><td>CRITICAL</td><td></td></tr></table> Harpoon <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>339S00458</td><td>1</td><td>IC, MODULE, MFI/BT, MURATA, HARPOON, M, EST.7, LGA385</td><td>U3730</td><td>CRITICAL</td><td></td></tr></table> <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>339S00428</td><td>339S00458</td><td></td><td>ALL</td><td>IC_MODULE_MFI2_BT_HARPOON_EST_EST.7_LGA385</td></tr></table> USB-C Connector <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>998-12556</td><td>2</td><td>CONN, RCPT, 56+4, P+0.35, H+0.7, SHLD, THOR</td><td>J3300, JB500</td><td>CRITICAL</td><td></td></tr></table> Ocarina <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>338S00410</td><td>1</td><td>IC, PMU, OCARINA, D2499A0, OTP-AG, ML CSP56</td><td>U9000</td><td>CRITICAL</td><td></td></tr><tr><td>338S00410</td><td>1</td><td>IC, PMU, OCARINA, D2499A0, OTP-AG, ML CSP56</td><td>U9500</td><td>CRITICAL</td><td>OCARINA_2</td></tr></table>									PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	337S00493	1	GPU, AMD, BAFFIN, ULX, A3, 0.83, QS, BGA769	UA000	CRITICAL	BAFFIN_ULX	337S00494	1	GPU, AMD, BAFFIN, PROX, A3, 0.83, QS, BGA769	UA000	CRITICAL	BAFFIN_PROX	998-04866	1	INTERPOSER, AMD, C989, BGA769, VDDCI_MVDD	UA000	CRITICAL	STARDUST:VDDCI_MVDD	998-04867	1	INTERPOSER, AMD, C988, BGA769, VDDC	UA000	CRITICAL	STARDUST:VDDC	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	333S00074	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, 8, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_SAMSUNG	333S00075	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, A, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_MICRON	333S00100	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, A, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_HYNIX	333S00173	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 12NM, A, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_MICRON_1Z	333S00175	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 12NM, A, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_HYNIX_1Z	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	337S00509	1	CPU, CFLH, QP87, QS, U0, 2.2, 45W, 1.1, LGA1440	U0500	CRITICAL	CPU_CFL:2.2	337S00510	1	CPU, CFL-H, QP86, QS, U0, 2.6, 6C, 45W, 1.15, BGA1440	U0500	CRITICAL	CPU_CFL:2.6	337S00513	1	CPU, CFL-H, QP06, QS, U0, 2.9, 6C, 45W, 1.2, BGA1440	U0500	CRITICAL	CPU_CFL:2.9	998-12472	1	INTERPOSER, CFH-H, BGA1440	U0500	CRITICAL	CPU_SKL:SOCKET	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	333S00131	16	IC, SDRAM, DDR4-2400, 16GBIT, 20NM, BGA78	U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900	CRITICAL	32G_MICRON_2400	333S00134	16	IC, SDRAM, DDR4-2400, 16GBIT, 20NM, BGA78	U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900	CRITICAL	32G_HYNIX_2400	333S00147	16	IC, SDRAM, DDR4-2400, 8GBIT, 20NM, BGA78	U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900	CRITICAL	16G_HYNIX_2400	333S00163	16	IC, SDRAM, DDR4-2400, 8GBIT, 20NM, BGA78	U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900	CRITICAL	16G_MICRON_2400	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	337S00523	1	IC, CNL, PCH-H, USFF, QNYP, QS, B0, BGA499	U1200	CRITICAL		PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	353S01442	4	IC, C03215, ACE, C0, USB PMR SW, BLNK, BGA96	U3100, U3200, UB300, UB400	CRITICAL		338S00408	2	IC, TBT, TITAN RIDGE DP, QUJK, QS, C1, CSP337	U2800, UB000	CRITICAL		PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	677-10608	2	SUBASSY T800 PCB, AMR, INTERPOSER, X1190	J4800, J4801	CRITICAL		PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	353S01525	1	IC, ISL9240HB1Z, PMU, SOUWA, WSCP40, 2.1X3.3MM	U7000	CRITICAL		338S00267	1	IC, PMU, P650839, 7X7MM, BGA168	U7800	CRITICAL		353S01229	1	IC, SUPPLY, INTERSL, ISL6277AHRZ, SV12.0, QFN48	UA600	CRITICAL		353S00928	1	IC, ISL9582BA, IMPV8 CPU REG, QFN48, 6X8MM	U7100	CRITICAL		PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	339S00458	1	IC, MODULE, MFI/BT, MURATA, HARPOON, M, EST.7, LGA385	U3730	CRITICAL		PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	339S00428	339S00458		ALL	IC_MODULE_MFI2_BT_HARPOON_EST_EST.7_LGA385	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	998-12556	2	CONN, RCPT, 56+4, P+0.35, H+0.7, SHLD, THOR	J3300, JB500	CRITICAL		PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	338S00410	1	IC, PMU, OCARINA, D2499A0, OTP-AG, ML CSP56	U9000	CRITICAL		338S00410	1	IC, PMU, OCARINA, D2499A0, OTP-AG, ML CSP56	U9500	CRITICAL	OCARINA_2									
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
337S00493	1	GPU, AMD, BAFFIN, ULX, A3, 0.83, QS, BGA769	UA000	CRITICAL	BAFFIN_ULX																																																																																																																																																																																																																																																																							
337S00494	1	GPU, AMD, BAFFIN, PROX, A3, 0.83, QS, BGA769	UA000	CRITICAL	BAFFIN_PROX																																																																																																																																																																																																																																																																							
998-04866	1	INTERPOSER, AMD, C989, BGA769, VDDCI_MVDD	UA000	CRITICAL	STARDUST:VDDCI_MVDD																																																																																																																																																																																																																																																																							
998-04867	1	INTERPOSER, AMD, C988, BGA769, VDDC	UA000	CRITICAL	STARDUST:VDDC																																																																																																																																																																																																																																																																							
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
333S00074	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, 8, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_SAMSUNG																																																																																																																																																																																																																																																																							
333S00075	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, A, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_MICRON																																																																																																																																																																																																																																																																							
333S00100	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 25NM, A, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_HYNIX																																																																																																																																																																																																																																																																							
333S00173	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 12NM, A, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_MICRON_1Z																																																																																																																																																																																																																																																																							
333S00175	4	IC, GDDR5, 8Gb, 7Gbps, 1.5V, 12NM, A, 170 BGA	UA400, UA450, UA500, UA550	CRITICAL	FB_4GB_HYNIX_1Z																																																																																																																																																																																																																																																																							
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
337S00509	1	CPU, CFLH, QP87, QS, U0, 2.2, 45W, 1.1, LGA1440	U0500	CRITICAL	CPU_CFL:2.2																																																																																																																																																																																																																																																																							
337S00510	1	CPU, CFL-H, QP86, QS, U0, 2.6, 6C, 45W, 1.15, BGA1440	U0500	CRITICAL	CPU_CFL:2.6																																																																																																																																																																																																																																																																							
337S00513	1	CPU, CFL-H, QP06, QS, U0, 2.9, 6C, 45W, 1.2, BGA1440	U0500	CRITICAL	CPU_CFL:2.9																																																																																																																																																																																																																																																																							
998-12472	1	INTERPOSER, CFH-H, BGA1440	U0500	CRITICAL	CPU_SKL:SOCKET																																																																																																																																																																																																																																																																							
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
333S00131	16	IC, SDRAM, DDR4-2400, 16GBIT, 20NM, BGA78	U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900	CRITICAL	32G_MICRON_2400																																																																																																																																																																																																																																																																							
333S00134	16	IC, SDRAM, DDR4-2400, 16GBIT, 20NM, BGA78	U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900	CRITICAL	32G_HYNIX_2400																																																																																																																																																																																																																																																																							
333S00147	16	IC, SDRAM, DDR4-2400, 8GBIT, 20NM, BGA78	U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900	CRITICAL	16G_HYNIX_2400																																																																																																																																																																																																																																																																							
333S00163	16	IC, SDRAM, DDR4-2400, 8GBIT, 20NM, BGA78	U0500, U0600, U0700, U0800, U0900, U1000, U1100, U1200, U1300, U1400, U1500, U1600, U1700, U1800, U1900, U2000, U2100, U2200, U2300, U2400, U2500, U2600, U2700, U2800, U2900, U3000, U3100, U3200, U3300, U3400, U3500, U3600, U3700, U3800, U3900, U4000, U4100, U4200, U4300, U4400, U4500, U4600, U4700, U4800, U4900, U5000, U5100, U5200, U5300, U5400, U5500, U5600, U5700, U5800, U5900, U6000, U6100, U6200, U6300, U6400, U6500, U6600, U6700, U6800, U6900, U7000, U7100, U7200, U7300, U7400, U7500, U7600, U7700, U7800, U7900, U8000, U8100, U8200, U8300, U8400, U8500, U8600, U8700, U8800, U8900, U9000, U9100, U9200, U9300, U9400, U9500, U9600, U9700, U9800, U9900	CRITICAL	16G_MICRON_2400																																																																																																																																																																																																																																																																							
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
337S00523	1	IC, CNL, PCH-H, USFF, QNYP, QS, B0, BGA499	U1200	CRITICAL																																																																																																																																																																																																																																																																								
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
353S01442	4	IC, C03215, ACE, C0, USB PMR SW, BLNK, BGA96	U3100, U3200, UB300, UB400	CRITICAL																																																																																																																																																																																																																																																																								
338S00408	2	IC, TBT, TITAN RIDGE DP, QUJK, QS, C1, CSP337	U2800, UB000	CRITICAL																																																																																																																																																																																																																																																																								
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
677-10608	2	SUBASSY T800 PCB, AMR, INTERPOSER, X1190	J4800, J4801	CRITICAL																																																																																																																																																																																																																																																																								
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
353S01525	1	IC, ISL9240HB1Z, PMU, SOUWA, WSCP40, 2.1X3.3MM	U7000	CRITICAL																																																																																																																																																																																																																																																																								
338S00267	1	IC, PMU, P650839, 7X7MM, BGA168	U7800	CRITICAL																																																																																																																																																																																																																																																																								
353S01229	1	IC, SUPPLY, INTERSL, ISL6277AHRZ, SV12.0, QFN48	UA600	CRITICAL																																																																																																																																																																																																																																																																								
353S00928	1	IC, ISL9582BA, IMPV8 CPU REG, QFN48, 6X8MM	U7100	CRITICAL																																																																																																																																																																																																																																																																								
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
339S00458	1	IC, MODULE, MFI/BT, MURATA, HARPOON, M, EST.7, LGA385	U3730	CRITICAL																																																																																																																																																																																																																																																																								
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:																																																																																																																																																																																																																																																																								
339S00428	339S00458		ALL	IC_MODULE_MFI2_BT_HARPOON_EST_EST.7_LGA385																																																																																																																																																																																																																																																																								
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
998-12556	2	CONN, RCPT, 56+4, P+0.35, H+0.7, SHLD, THOR	J3300, JB500	CRITICAL																																																																																																																																																																																																																																																																								
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																							
338S00410	1	IC, PMU, OCARINA, D2499A0, OTP-AG, ML CSP56	U9000	CRITICAL																																																																																																																																																																																																																																																																								
338S00410	1	IC, PMU, OCARINA, D2499A0, OTP-AG, ML CSP56	U9500	CRITICAL	OCARINA_2																																																																																																																																																																																																																																																																							
	8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																				
	Programmable Parts <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00199</td><td>1</td><td>IC, 1MbIT SERIAL FLASH 2X3X0.6MM UFDFPN8 PKG</td><td>UA701</td><td>CRITICAL</td><td>GPUROM:BLANK</td></tr><tr><td>341S01017</td><td>1</td><td>T29, TR1 (V2.1), NEW-PROTO-2, X1190</td><td>U2890</td><td>CRITICAL</td><td>UPCROM_PROG:P1</td></tr><tr><td>341S01018</td><td>1</td><td>T29, TR2 (V2.1), NEW-PROTO-2, X1190</td><td>UB090</td><td>CRITICAL</td><td>UPCROM_PROG:P1</td></tr><tr><td>341S01025</td><td>1</td><td>IC, BT ROM, EVT,VXX, X1190</td><td>U3750</td><td>CRITICAL</td><td>BT_ROM:P2</td></tr><tr><td>341S00725</td><td>1</td><td>IC, WIFI ROM, PROTO0, X1190</td><td>U3710</td><td>CRITICAL</td><td>WIFI_ROM:P0</td></tr></table> H9M <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>339S00373</td><td>1</td><td>POP, GIBRALTAR+2GB 21NM, H, B0, SCK, CSP1406</td><td>U3900</td><td>CRITICAL</td><td>SOC:2GB</td></tr></table> <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>339S00373</td><td>339S00372</td><td></td><td>ALL</td><td>2GB, 20NM, M, B0, SCK, CSP1406</td></tr><tr><td>339S00378</td><td>339S00372</td><td></td><td>ALL</td><td>2GB, 21NM, H, B0, ATK, CSP1406</td></tr><tr><td>339S00377</td><td>339S00372</td><td></td><td>ALL</td><td>2GB, 20NM, M, B0, ATK, CSP1406</td></tr></table> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>339S00376</td><td>1</td><td>POP, GIBRALTAR+1GB 21NM, H, B0, ATK, CSP1406</td><td>U3900</td><td>CRITICAL</td><td>SOC:1GB</td></tr></table> <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>339S00375</td><td>339S00376</td><td></td><td>ALL</td><td>1GB, 20NM, M, B0, ATK, CSP1406</td></tr><tr><td>339S00371</td><td>339S00376</td><td></td><td>ALL</td><td>1GB, 21NM, H, B0, SCK, CSP1406</td></tr><tr><td>339S00370</td><td>339S00376</td><td></td><td>ALL</td><td>1GB, 21NM, M, B0, SCK, CSP1406</td></tr></table> S4E Parts 256GB <table><tr><td>335S00324</td><td>4</td><td>NAND, 3DV3, 856BP, XXX, S4E, 170G, SD, ULGA110</td><td>U8600, U8700, U8800, U8900</td><td>CRITICAL</td><td>S4E_256_WD</td></tr><tr><td>998-12418</td><td>4</td><td>NAND, 3DV3, 856BP, S4E, 170G, T, SUBX, ULGA110</td><td>U8600, U8700, U8800, U8900</td><td>CRITICAL</td><td>S4E_256_TB</td></tr></table> <table><tr><td>998-12419</td><td>998-12418</td><td></td><td>ALL</td><td>SUBW</td></tr></table> 512GB <table><tr><td>335S00325</td><td>4</td><td>NAND, 3DV3, 128GBP, XXX, S4E, 170G, SD, ULGA110</td><td>U8600, U8700, U8800, U8900</td><td>CRITICAL</td><td>S4E_512_WD</td></tr><tr><td>998-12420</td><td>4</td><td>NAND, 3DV3, 128GBP, S4E, 170G, T, SUBX, ULGA110</td><td>U8600, U8700, U8800, U8900</td><td>CRITICAL</td><td>S4E_512_TB</td></tr></table> <table><tr><td>998-12421</td><td>998-12420</td><td></td><td>ALL</td><td>SUBW</td></tr></table> 1TB <table><tr><td>335S00326</td><td>6</td><td>NAND, 3DV3, 170GBP, XXX, S4E, 170G, SD, ULGA110</td><td>U8600, U8700, U8800, U8900, U9100, U9200</td><td>CRITICAL</td><td>S4E_1TB_WD</td></tr><tr><td>998-12422</td><td>6</td><td>NAND, 3DV3, 170GBP, S4E, 170G, T, SUBX, ULGA110</td><td>U8600, U8700, U8800, U8900, U9100, U9200</td><td>CRITICAL</td><td>S4E_1TB_TB</td></tr></table> <table><tr><td>998-12423</td><td>998-12422</td><td></td><td>ALL</td><td>SUBW</td></tr></table> 2TB <table><tr><td>335S00327</td><td>8</td><td>NAND, 3DV3, 256GBP, XXX, S4E, 170G, SD, ULGA110</td><td>U8600, U8700, U8800, U8900, U9100, U9200</td><td>CRITICAL</td><td>S4E_2TB_WD</td></tr><tr><td>335S00321</td><td>8</td><td>NAND, 3DV4, 256GBP, XXX, S4E, 256G, SS, ULGA110</td><td>U8600, U8700, U8800, U8900, U9100, U9200, U9400</td><td>CRITICAL</td><td>S4E_2TB_SM</td></tr><tr><td>998-12424</td><td>8</td><td>NAND, 3DV3, 256GBP, S4E, 170G, T, SUBX, ULGA110</td><td>U8600, U8700, U8800, U8900, U9100, U9200, U9400</td><td>CRITICAL</td><td>S4E_2TB_TB</td></tr></table> <table><tr><td>998-12426</td><td>998-12424</td><td></td><td>ALL</td><td>SUBW</td></tr></table> 4TB <table><tr><td>335S00322</td><td>8</td><td>NAND, 3DV4, 512GBP, S4E, 256G, SS, ULGA110</td><td>U8600, U8700, U8800, U8900, U9100, U9200, U9400, U9600</td><td>CRITICAL</td><td>S4E_4TB_SM</td></tr></table> 2TB TLC <table><tr><td>335S00228</td><td>8</td><td>NAND, 3DV3, 256GBT, S4E, 256G, T, SLGA110</td><td>U8600, U8700, U8800, U8900, U9100, U9200, U9400, U9600</td><td>CRITICAL</td><td>TLC_2TB_TB</td></tr><tr><td>335S00247</td><td>8</td><td>NAND, 3DV3, 256GBT, S4E, 256G, SD, SLGA110</td><td>U8600, U8700, U8800, U8900, U9100, U9200, U9400, U9600</td><td>CRITICAL</td><td>TLC_2TB_WD</td></tr></table>									PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	335S00199	1	IC, 1MbIT SERIAL FLASH 2X3X0.6MM UFDFPN8 PKG	UA701	CRITICAL	GPUROM:BLANK	341S01017	1	T29, TR1 (V2.1), NEW-PROTO-2, X1190	U2890	CRITICAL	UPCROM_PROG:P1	341S01018	1	T29, TR2 (V2.1), NEW-PROTO-2, X1190	UB090	CRITICAL	UPCROM_PROG:P1	341S01025	1	IC, BT ROM, EVT,VXX, X1190	U3750	CRITICAL	BT_ROM:P2	341S00725	1	IC, WIFI ROM, PROTO0, X1190	U3710	CRITICAL	WIFI_ROM:P0	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	339S00373	1	POP, GIBRALTAR+2GB 21NM, H, B0, SCK, CSP1406	U3900	CRITICAL	SOC:2GB	PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	339S00373	339S00372		ALL	2GB, 20NM, M, B0, SCK, CSP1406	339S00378	339S00372		ALL	2GB, 21NM, H, B0, ATK, CSP1406	339S00377	339S00372		ALL	2GB, 20NM, M, B0, ATK, CSP1406	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	339S00376	1	POP, GIBRALTAR+1GB 21NM, H, B0, ATK, CSP1406	U3900	CRITICAL	SOC:1GB	PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	339S00375	339S00376		ALL	1GB, 20NM, M, B0, ATK, CSP1406	339S00371	339S00376		ALL	1GB, 21NM, H, B0, SCK, CSP1406	339S00370	339S00376		ALL	1GB, 21NM, M, B0, SCK, CSP1406	335S00324	4	NAND, 3DV3, 856BP, XXX, S4E, 170G, SD, ULGA110	U8600, U8700, U8800, U8900	CRITICAL	S4E_256_WD	998-12418	4	NAND, 3DV3, 856BP, S4E, 170G, T, SUBX, ULGA110	U8600, U8700, U8800, U8900	CRITICAL	S4E_256_TB	998-12419	998-12418		ALL	SUBW	335S00325	4	NAND, 3DV3, 128GBP, XXX, S4E, 170G, SD, ULGA110	U8600, U8700, U8800, U8900	CRITICAL	S4E_512_WD	998-12420	4	NAND, 3DV3, 128GBP, S4E, 170G, T, SUBX, ULGA110	U8600, U8700, U8800, U8900	CRITICAL	S4E_512_TB	998-12421	998-12420		ALL	SUBW	335S00326	6	NAND, 3DV3, 170GBP, XXX, S4E, 170G, SD, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200	CRITICAL	S4E_1TB_WD	998-12422	6	NAND, 3DV3, 170GBP, S4E, 170G, T, SUBX, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200	CRITICAL	S4E_1TB_TB	998-12423	998-12422		ALL	SUBW	335S00327	8	NAND, 3DV3, 256GBP, XXX, S4E, 170G, SD, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200	CRITICAL	S4E_2TB_WD	335S00321	8	NAND, 3DV4, 256GBP, XXX, S4E, 256G, SS, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200, U9400	CRITICAL	S4E_2TB_SM	998-12424	8	NAND, 3DV3, 256GBP, S4E, 170G, T, SUBX, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200, U9400	CRITICAL	S4E_2TB_TB	998-12426	998-12424		ALL	SUBW	335S00322	8	NAND, 3DV4, 512GBP, S4E, 256G, SS, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200, U9400, U9600	CRITICAL	S4E_4TB_SM	335S00228	8	NAND, 3DV3, 256GBT, S4E, 256G, T, SLGA110	U8600, U8700, U8800, U8900, U9100, U9200, U9400, U9600	CRITICAL	TLC_2TB_TB	335S00247	8	NAND, 3DV3, 256GBT, S4E, 256G, SD, SLGA110	U8600, U8700, U8800, U8900, U9100, U9200, U9400, U9600	CRITICAL	TLC_2TB_WD																																																																			
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341S01017	1	T29, TR1 (V2.1), NEW-PROTO-2, X1190	U2890	CRITICAL	UPCROM_PROG:P1																																																																																																																																																																																																																																																																							
341S01018	1	T29, TR2 (V2.1), NEW-PROTO-2, X1190	UB090	CRITICAL	UPCROM_PROG:P1																																																																																																																																																																																																																																																																							
341S01025	1	IC, BT ROM, EVT,VXX, X1190	U3750	CRITICAL	BT_ROM:P2																																																																																																																																																																																																																																																																							
341S00725	1	IC, WIFI ROM, PROTO0, X1190	U3710	CRITICAL	WIFI_ROM:P0																																																																																																																																																																																																																																																																							
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339S00375	339S00376		ALL	1GB, 20NM, M, B0, ATK, CSP1406																																																																																																																																																																																																																																																																								
339S00371	339S00376		ALL	1GB, 21NM, H, B0, SCK, CSP1406																																																																																																																																																																																																																																																																								
339S00370	339S00376		ALL	1GB, 21NM, M, B0, SCK, CSP1406																																																																																																																																																																																																																																																																								
335S00324	4	NAND, 3DV3, 856BP, XXX, S4E, 170G, SD, ULGA110	U8600, U8700, U8800, U8900	CRITICAL	S4E_256_WD																																																																																																																																																																																																																																																																							
998-12418	4	NAND, 3DV3, 856BP, S4E, 170G, T, SUBX, ULGA110	U8600, U8700, U8800, U8900	CRITICAL	S4E_256_TB																																																																																																																																																																																																																																																																							
998-12419	998-12418		ALL	SUBW																																																																																																																																																																																																																																																																								
335S00325	4	NAND, 3DV3, 128GBP, XXX, S4E, 170G, SD, ULGA110	U8600, U8700, U8800, U8900	CRITICAL	S4E_512_WD																																																																																																																																																																																																																																																																							
998-12420	4	NAND, 3DV3, 128GBP, S4E, 170G, T, SUBX, ULGA110	U8600, U8700, U8800, U8900	CRITICAL	S4E_512_TB																																																																																																																																																																																																																																																																							
998-12421	998-12420		ALL	SUBW																																																																																																																																																																																																																																																																								
335S00326	6	NAND, 3DV3, 170GBP, XXX, S4E, 170G, SD, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200	CRITICAL	S4E_1TB_WD																																																																																																																																																																																																																																																																							
998-12422	6	NAND, 3DV3, 170GBP, S4E, 170G, T, SUBX, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200	CRITICAL	S4E_1TB_TB																																																																																																																																																																																																																																																																							
998-12423	998-12422		ALL	SUBW																																																																																																																																																																																																																																																																								
335S00327	8	NAND, 3DV3, 256GBP, XXX, S4E, 170G, SD, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200	CRITICAL	S4E_2TB_WD																																																																																																																																																																																																																																																																							
335S00321	8	NAND, 3DV4, 256GBP, XXX, S4E, 256G, SS, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200, U9400	CRITICAL	S4E_2TB_SM																																																																																																																																																																																																																																																																							
998-12424	8	NAND, 3DV3, 256GBP, S4E, 170G, T, SUBX, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200, U9400	CRITICAL	S4E_2TB_TB																																																																																																																																																																																																																																																																							
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335S00322	8	NAND, 3DV4, 512GBP, S4E, 256G, SS, ULGA110	U8600, U8700, U8800, U8900, U9100, U9200, U9400, U9600	CRITICAL	S4E_4TB_SM																																																																																																																																																																																																																																																																							
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A	BOM Configuration 2 <table><tr><td rowspan="2"> Apple Inc.</td><td>DRAWING NUMBER</td><td>SIZE</td></tr><tr><td>051-02643</td><td>D</td></tr><tr><td rowspan="3">NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED</td><td>REVISION</td><td></td></tr><tr><td>4.0.0</td><td></td></tr><tr><td>BRANCH</td><td>evt-0</td></tr><tr><td rowspan="2"></td><td>PAGE</td><td>3 OF 200</td></tr><tr><td>SHEET</td><td>3 OF 131</td></tr></table>									 Apple Inc.	DRAWING NUMBER	SIZE	051-02643	D	NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION		4.0.0		BRANCH	evt-0		PAGE	3 OF 200	SHEET	3 OF 131																																																																																																																																																																																																																																																		
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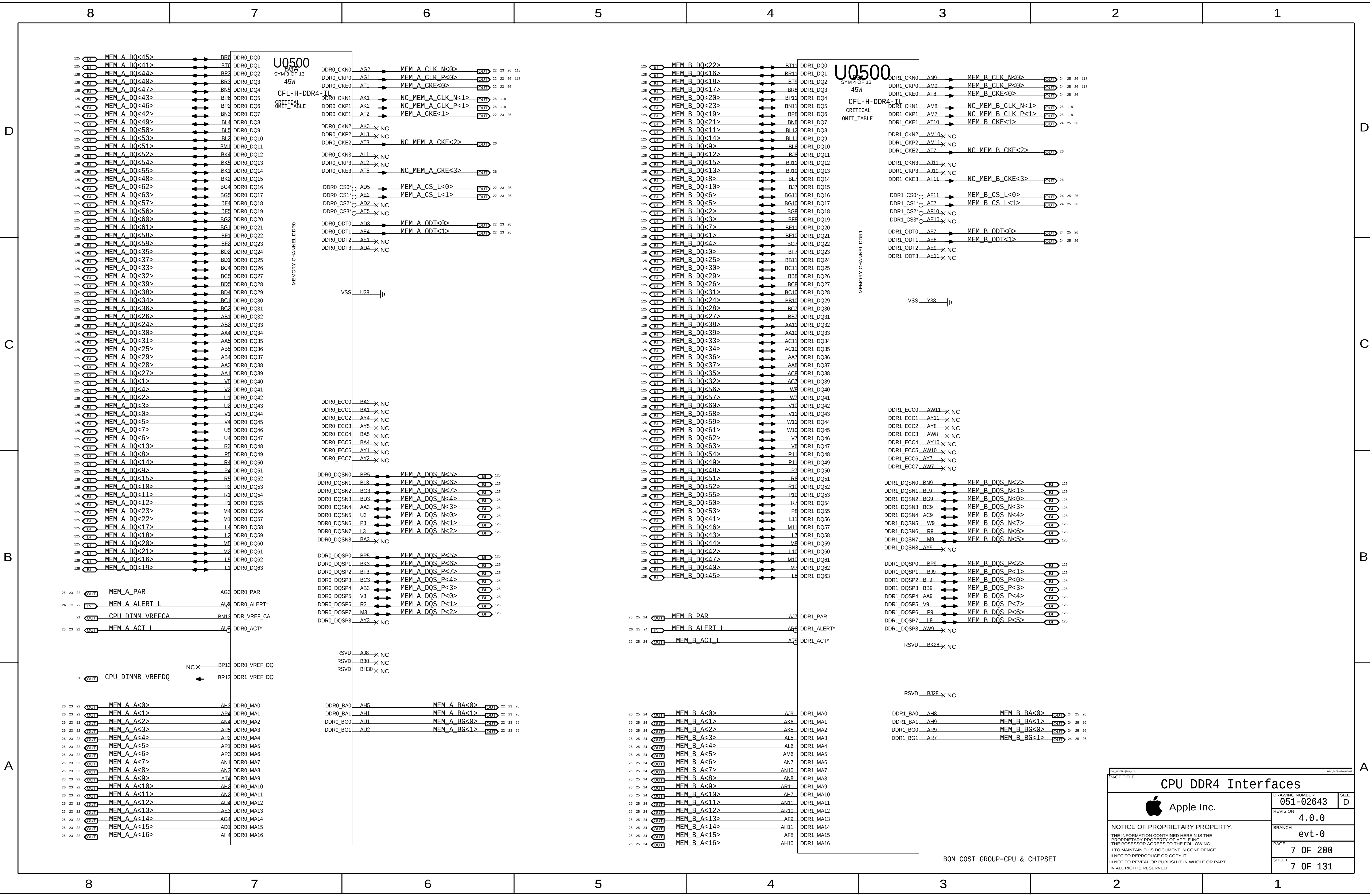
BOM GROUP	BOM OPTIONS
CPUPEG:X8X8	CPUCFG5_PD
CPUPEG:X8X4X4	CPUCFG6_PD, CPUCFG5_PD

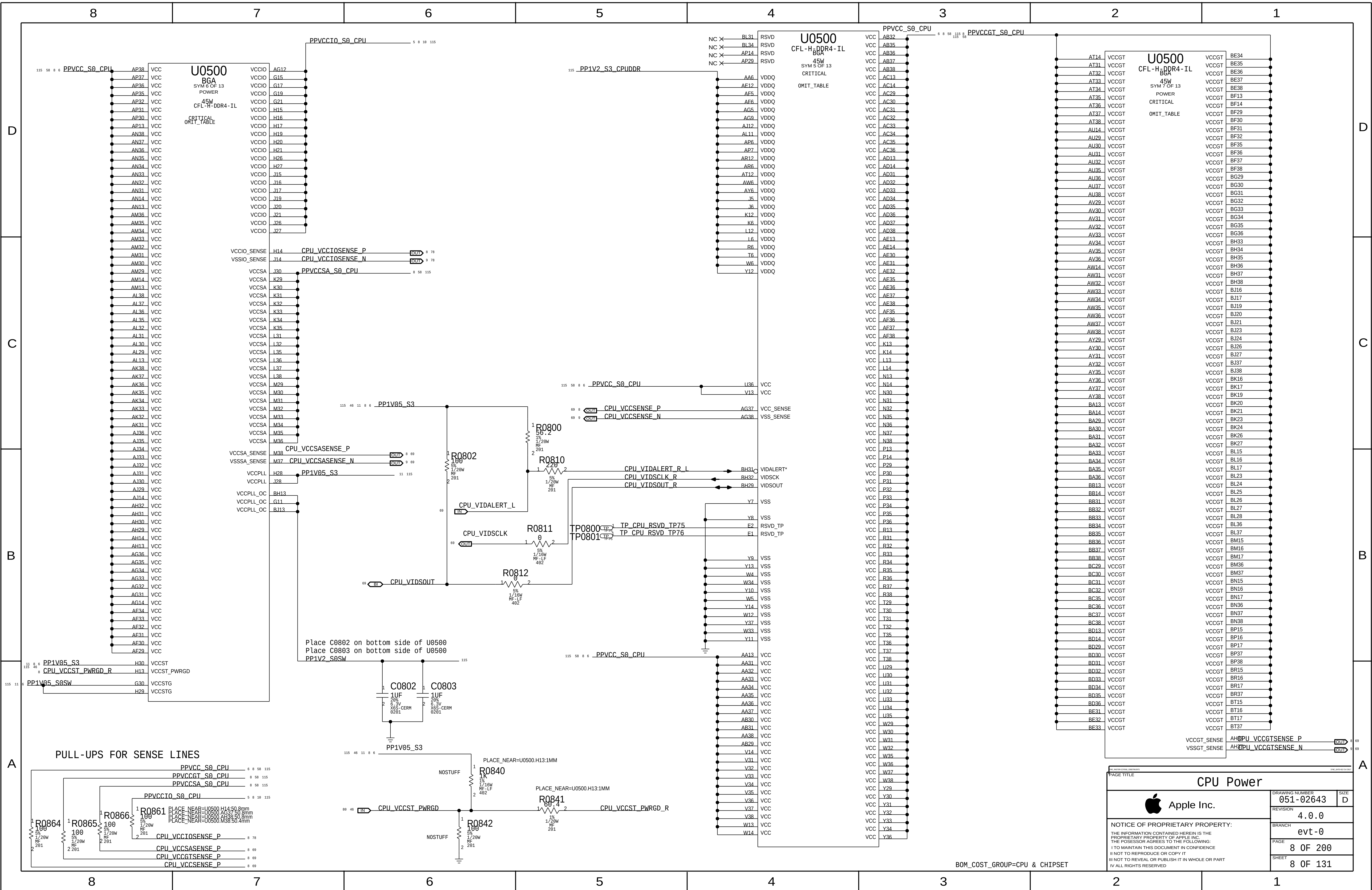
To use PEG X16 configuration, simply remove CPUPEG:X8X8 and CPUPEG:X8X4X4 from BOMs.

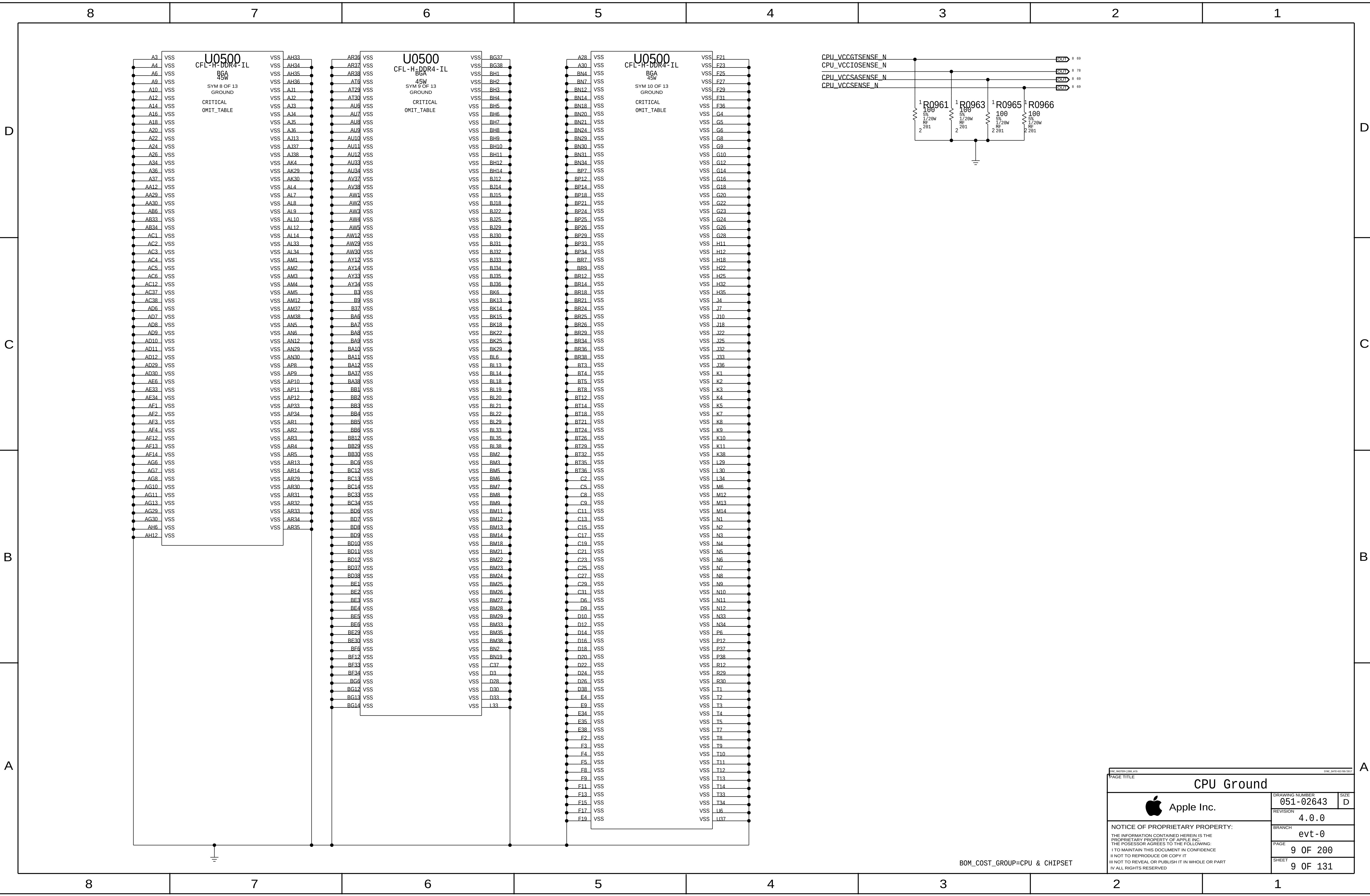
CFL H		
Name	Ball#	Signal Name
BN25	CFG[0]	NOA_N_0
BN27	CFG[1]	NOA_N_1
BN26	CFG[2]	NOA_N_2
BN28	CFG[3]	NOA_N_3
BR20	CFG[4]	NOA_N_4
BM20	CFG[5]	NOA_N_5
BT20	CFG[6]	NOA_N_6
BP20	CFG[7]	NOA_N_7
BR23	CFG[8]	NOA_N_8
BR22	CFG[9]	NOA_N_9
BT23	CFG[10]	NOA_N_10
BT22	CFG[11]	NOA_N_11
BM19	CFG[12]	NOA_N_12
BR19	CFG[13]	NOA_N_13
BP19	CFG[14]	NOA_N_14
BT19	CFG[15]	NOA_N_15
BN23	CFG[17]	NOA_STB_P_0
BP22	CFG[19]	NOA_STB_P_1
D1	RSVD_TP	PEG_VIEW0
E1	RSVD_TP	PEG_VIEW1
E2	RSVD_TP	PEG_VIEW3
E3	IST_TRIG	PEG_VIEW2
BT2	RSVD_TP	DDR_VIEW1
BR1	RSVD_TP	DDR_VIEW0

ENC. MASTER/388, A10	ENC. SAT/315/30, 316/30						
PAGE TITLE							
CPU Clock/Misc/JTAG/CFG							
 Apple Inc.	<table border="1"> <tr> <td data-bbox="2874 1729 3011 1741">DRAWING NUMBER</td> <td data-bbox="3011 1729 3033 1741">SIZE</td> </tr> <tr> <td data-bbox="2874 1741 3011 1751">051-02643</td> <td data-bbox="3011 1741 3033 1751">D</td> </tr> </table>	DRAWING NUMBER	SIZE	051-02643	D		
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PAGE	6 OF 200						
SHEET	6 OF 131						

BOM_COST_GROUP=CPU & CHIPSET







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CPU Ground

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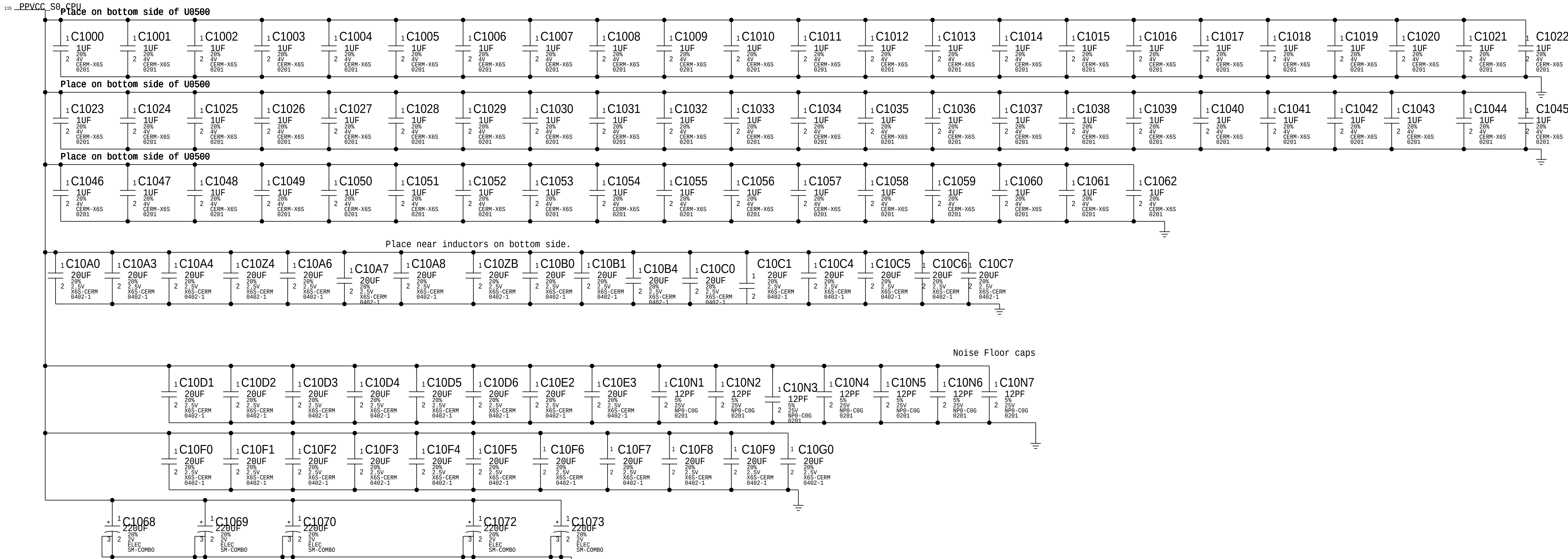
BOM_COST_GROUP=CPU & CHIPSET

CPU VCORE Decoupling

Intel recommendation: 5x 220uF ESR 5m ohms ESL 1.9nH each, 4x 47uF 0805 8x22uF 0603, 28x 10uF 0402, 3x 10uF 0402, 69x 1uF 0201
Apple Implementation:

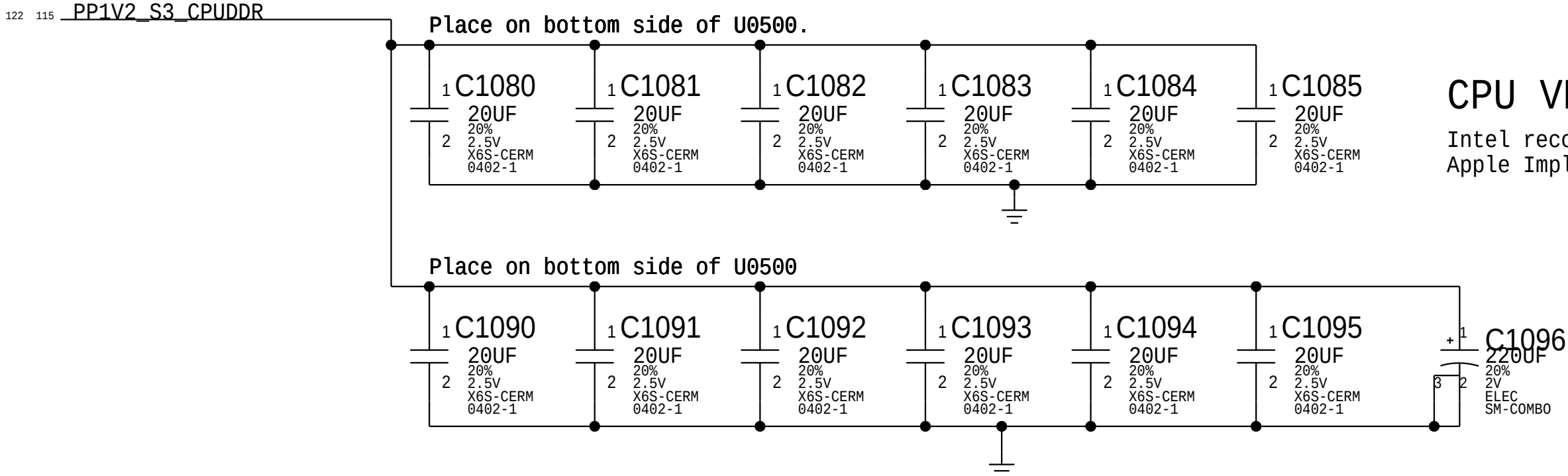
Vcc CPU Core Decoupling from 20140905 BOM

Board Edge: 2x 220uF, 4x 47uF rest on the back side



CPU VDDQ Decoupling

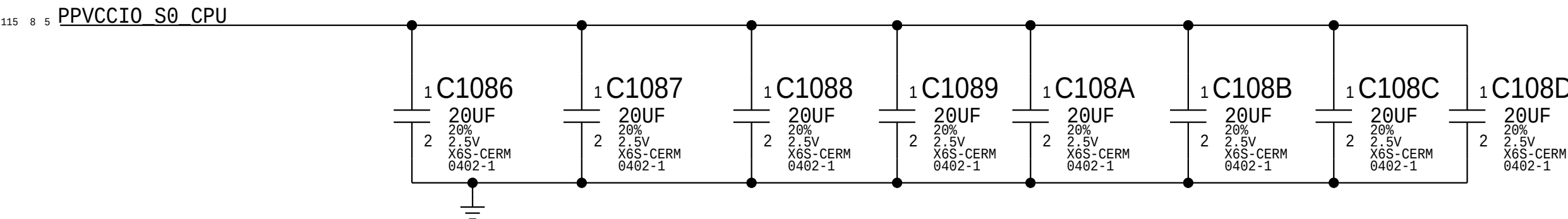
Intel recommendation: 10x 10uF 0402, 4x 22uF 0602
Apple Implementation:



CPU VCCIO Decoupling

Intel recommendation: 3x 10uF 0402 (opposite CPU)
Apple Implementation:

Place near U0500 on bottom side



NOTE: Intel decoupling recommendations from CBR schematics for Skylake H doc#557227 and PDG section 48.1 (document# 546884)

BOM_COST_GROUP=CPU & CHIPSET

CPU Decoupling 1		
	DRAWING NUMBER	051-02643
	REVISION	4.0.0
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	PAGE	10 OF 200
	SHEET	10 OF 131

CPU VGTSlice Decoupling

Intel recommendation: 7x 220uF, 6x 47uF 0805, 6x 22uF 0603, 35x 10uF 0402, 68 1uF 0201
Apple Implementation:

Vcc GT Slice Core Decoupling from 20140905 BOM

Board Edge: 4x220uF, 7x 47uF rest on back side

D

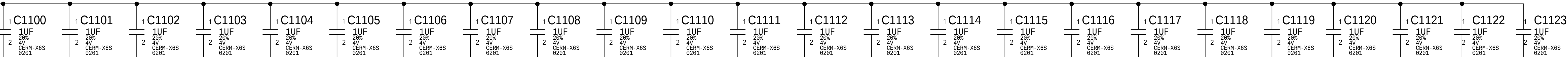
C

B

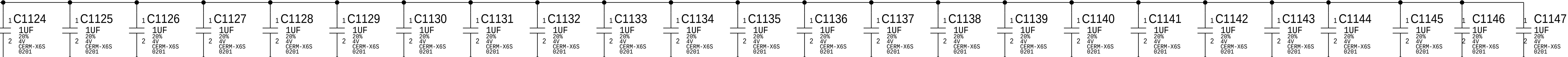
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122 115 PPVCCGT_S0_CPU

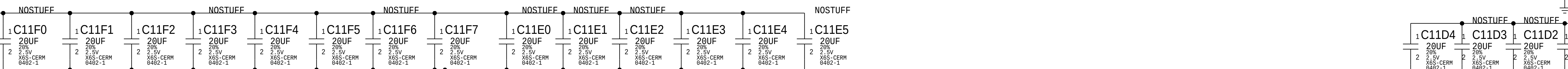
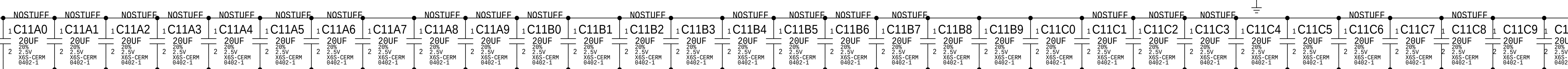
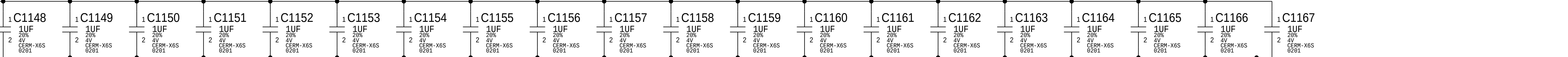
Place on bottom side of U0500



Place on bottom side of U0500

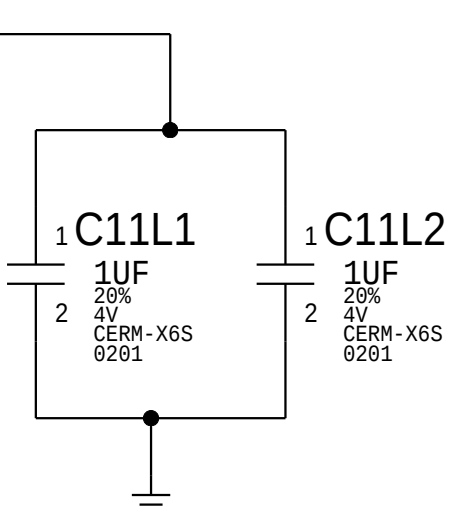


Place on bottom side of U0500

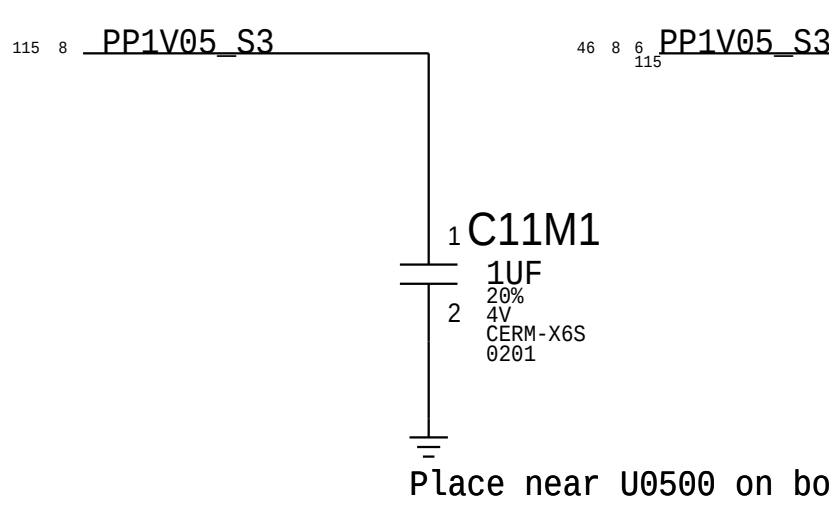


CPU VCCSTG Decoupling

115 8 6 PP1V05_S0SW
Place near U0500 on bottom side



CPU VCCPLL and VCCST Decoupling



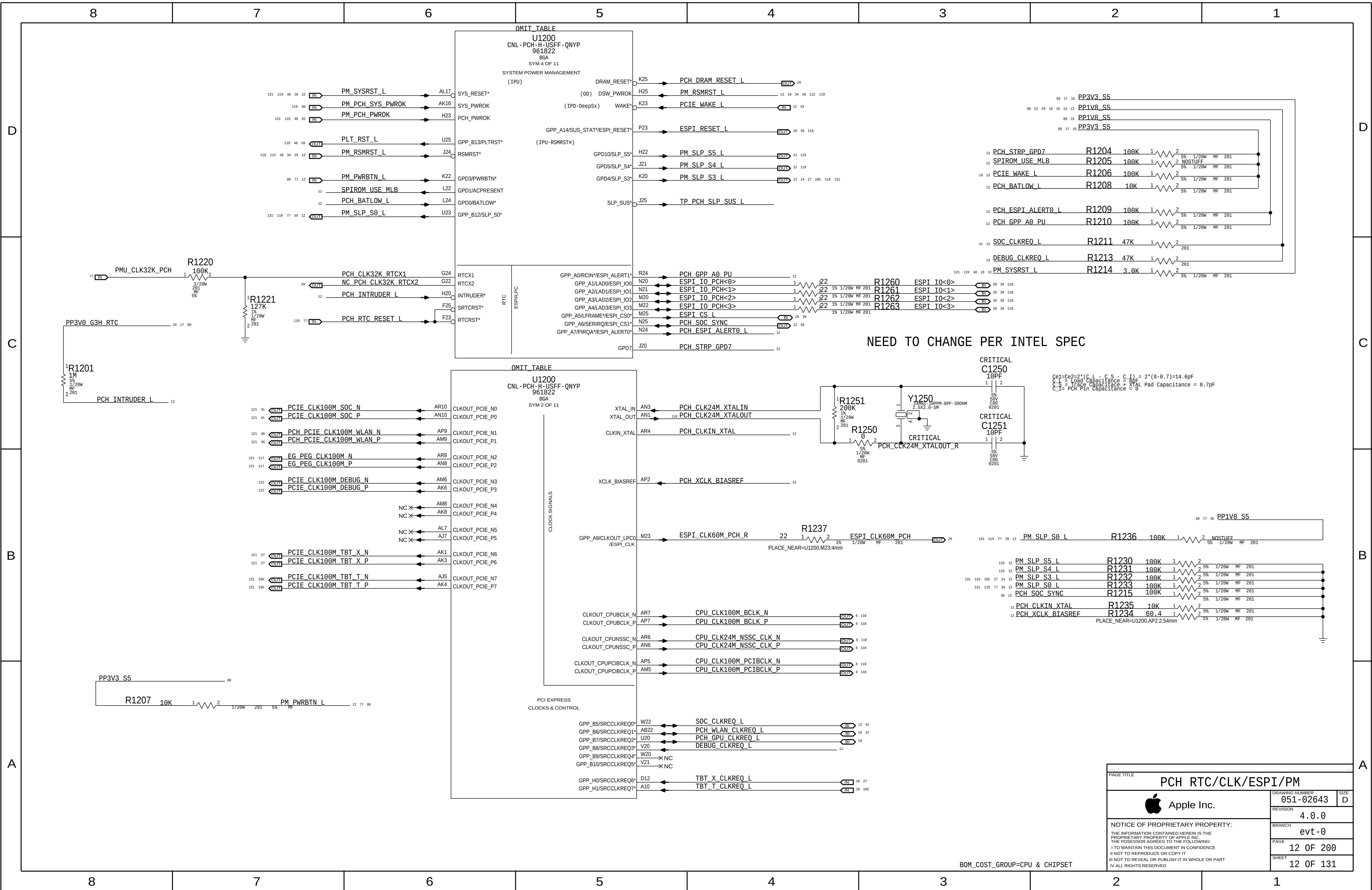
CPU VCCSA Decoupling

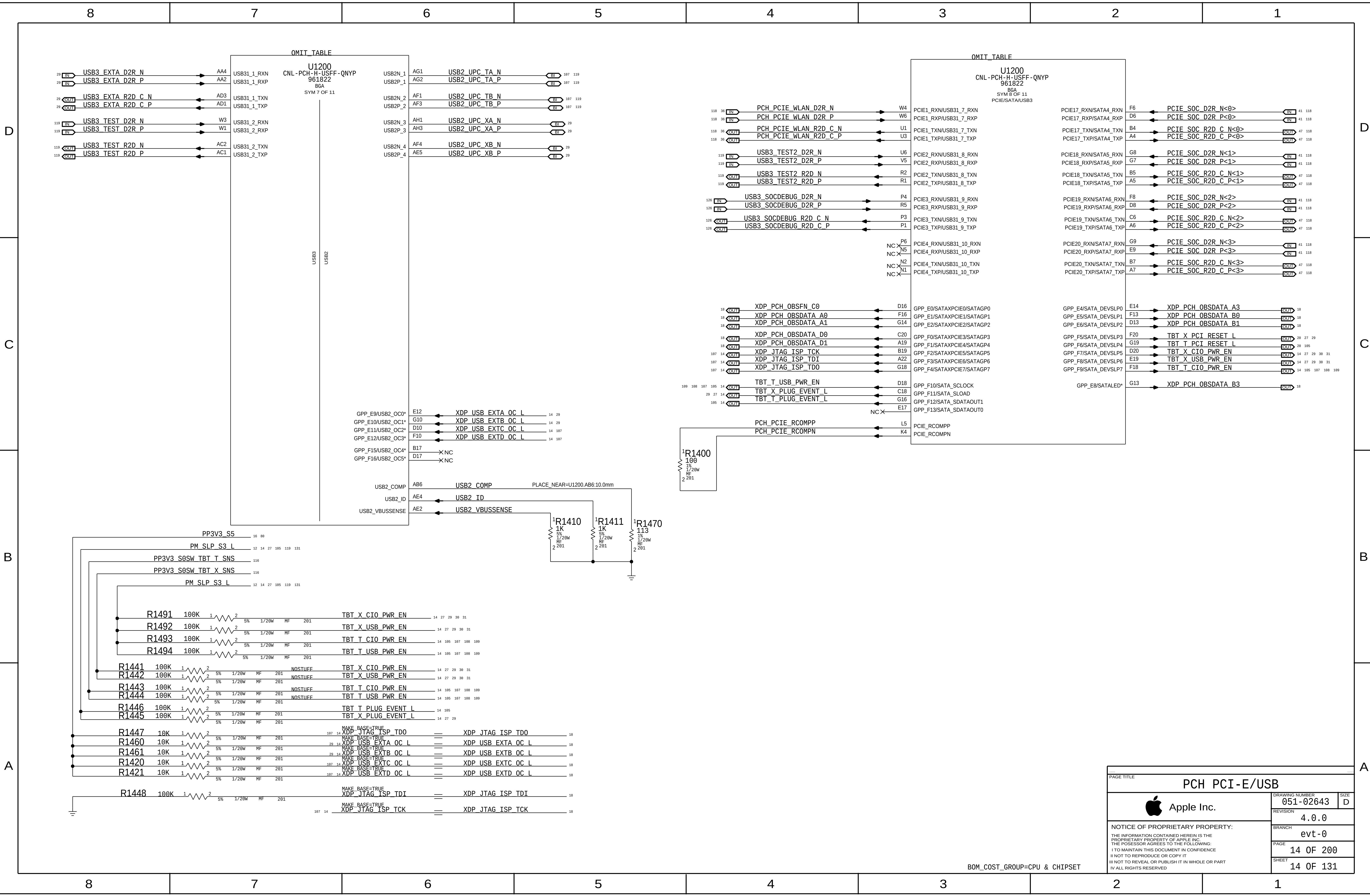
Intel recommendation: 2x 220uF, 1x 47uF 0805, 1x 22uF, 7x 10uF 0402, 3x 1uF 0201
Apple Implementation: 2x 220uF, 1x 22uF on board edge, everything else on back side

NOTE: Intel decoupling recommendations from CBR schematics for Skylake H doc#557227 and PDG section 48.1 (document# 546884)

BOM_COST_GROUP=CPU & CHIPSET

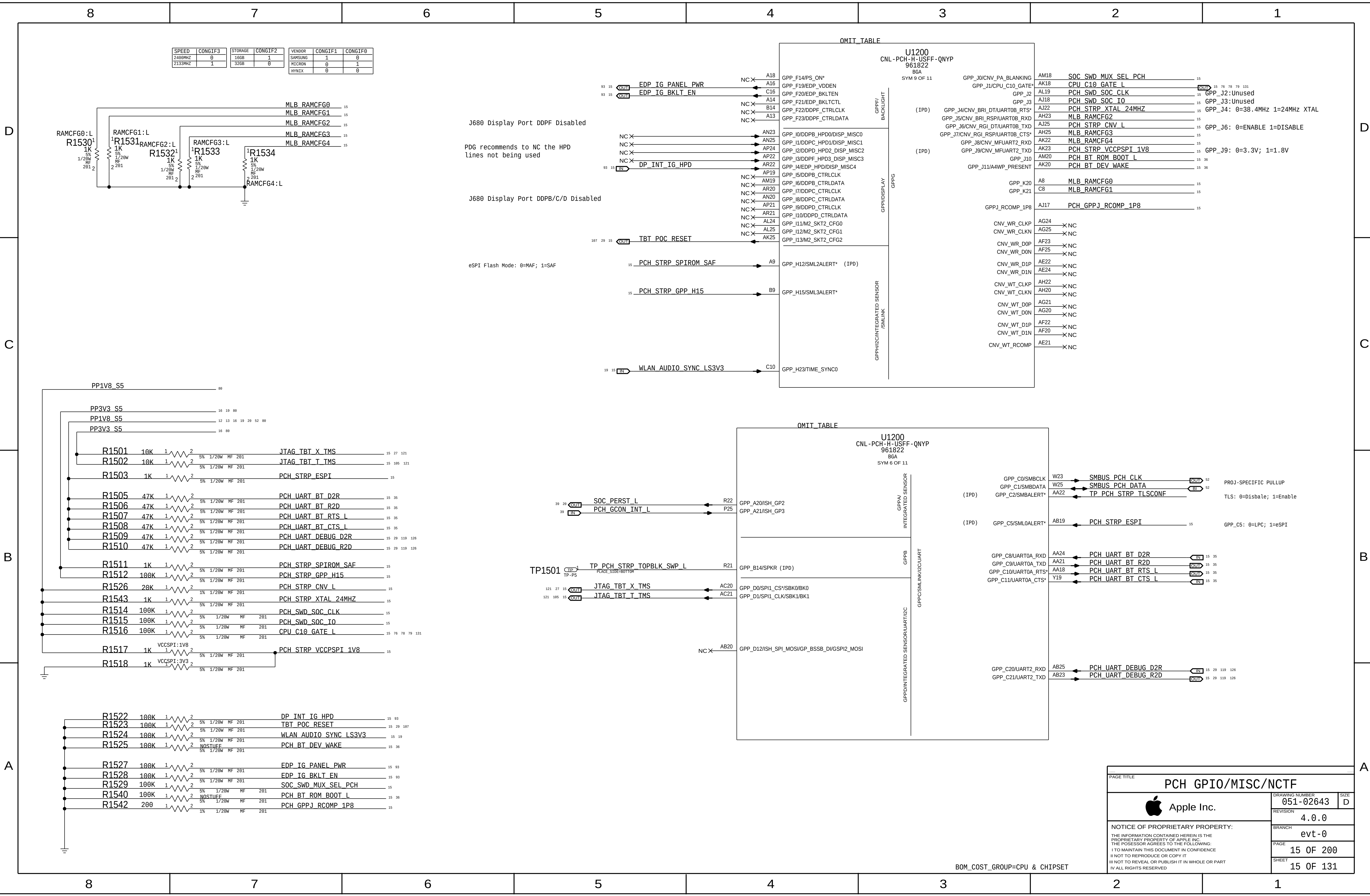
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	SHEET	11 OF 131

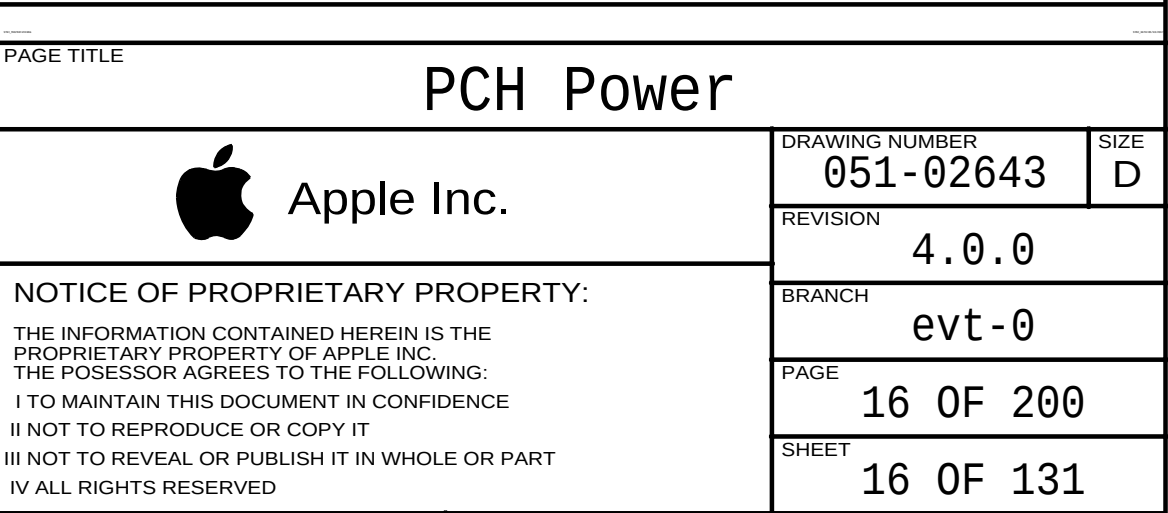


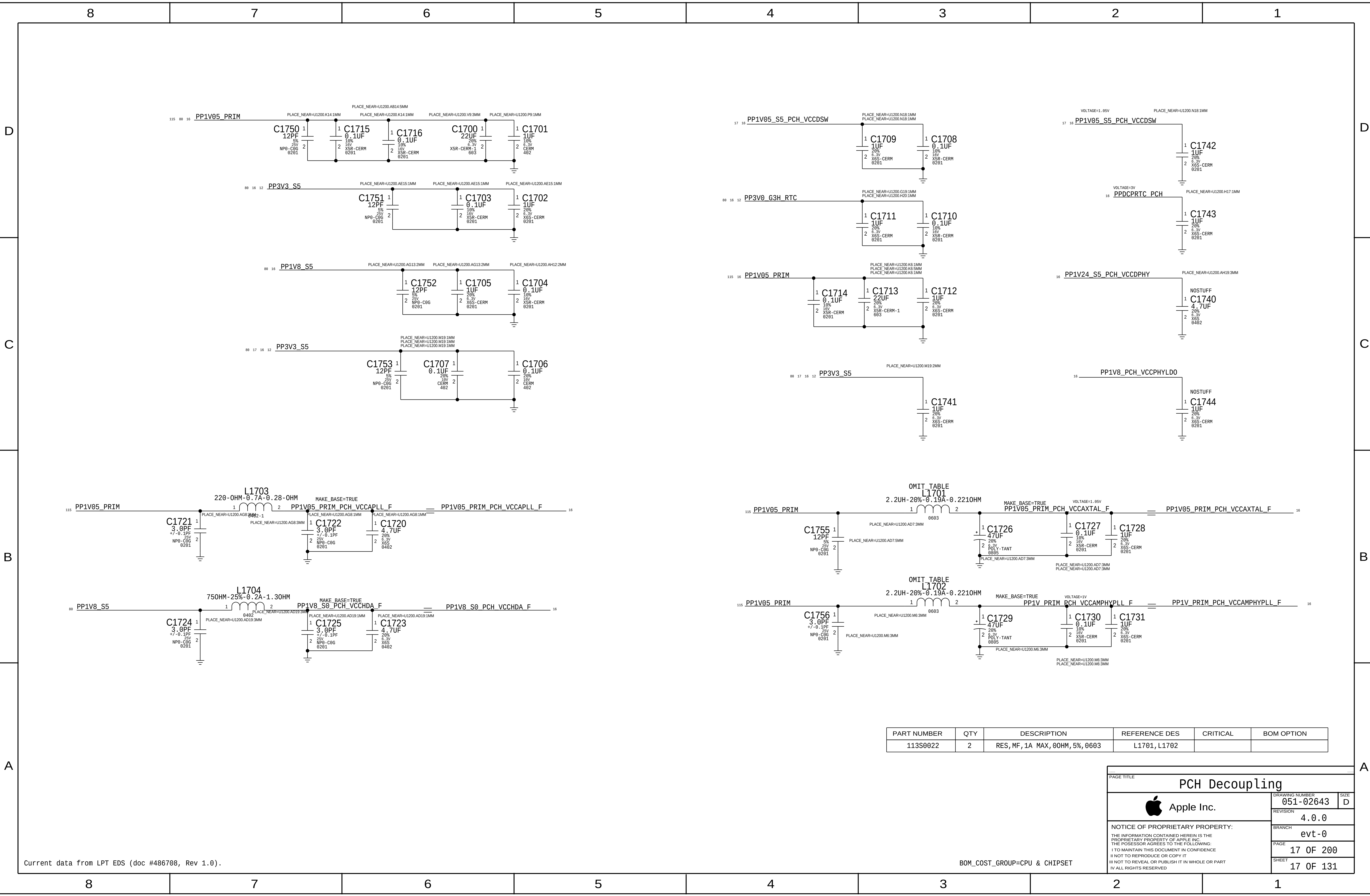


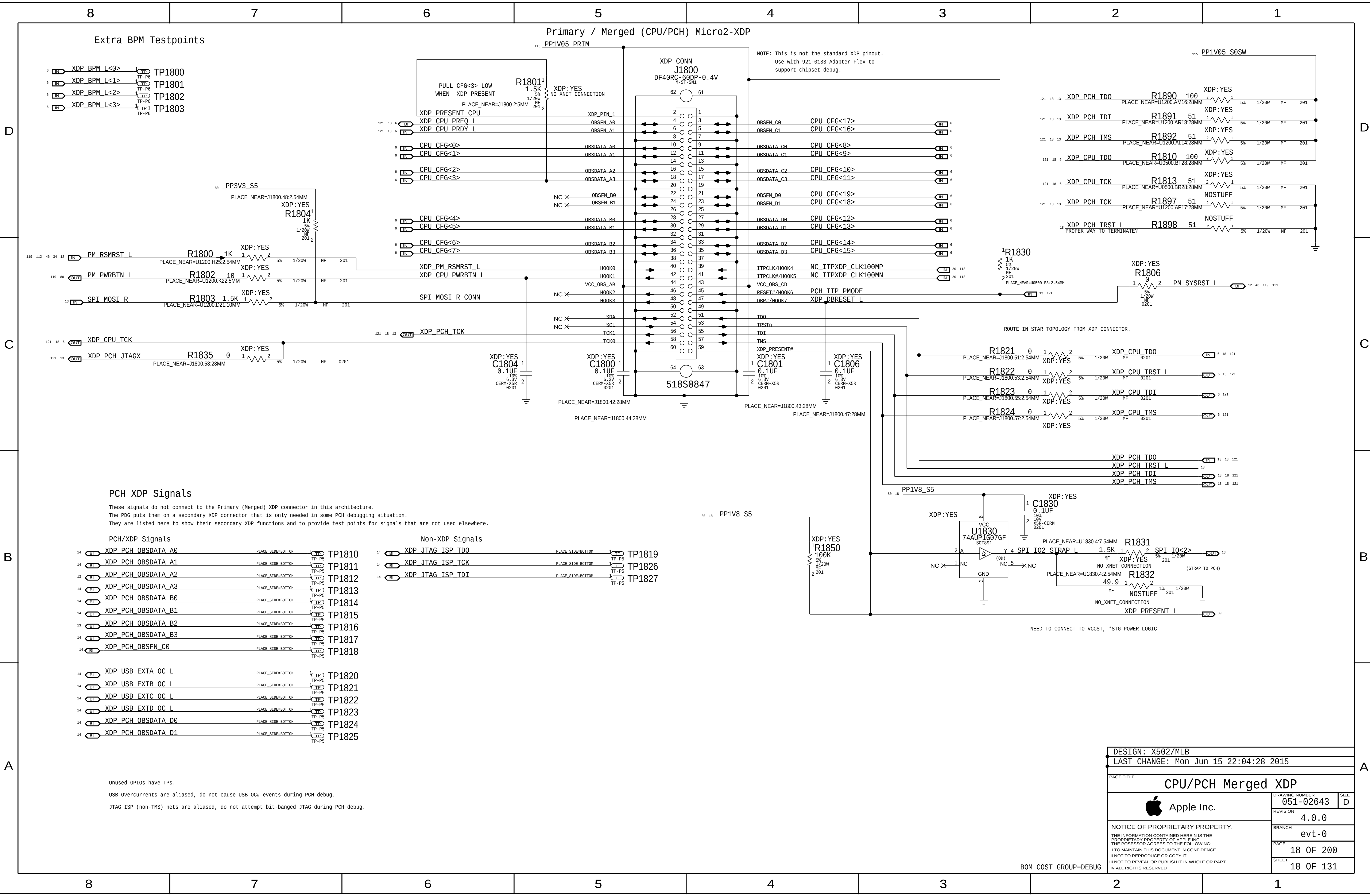
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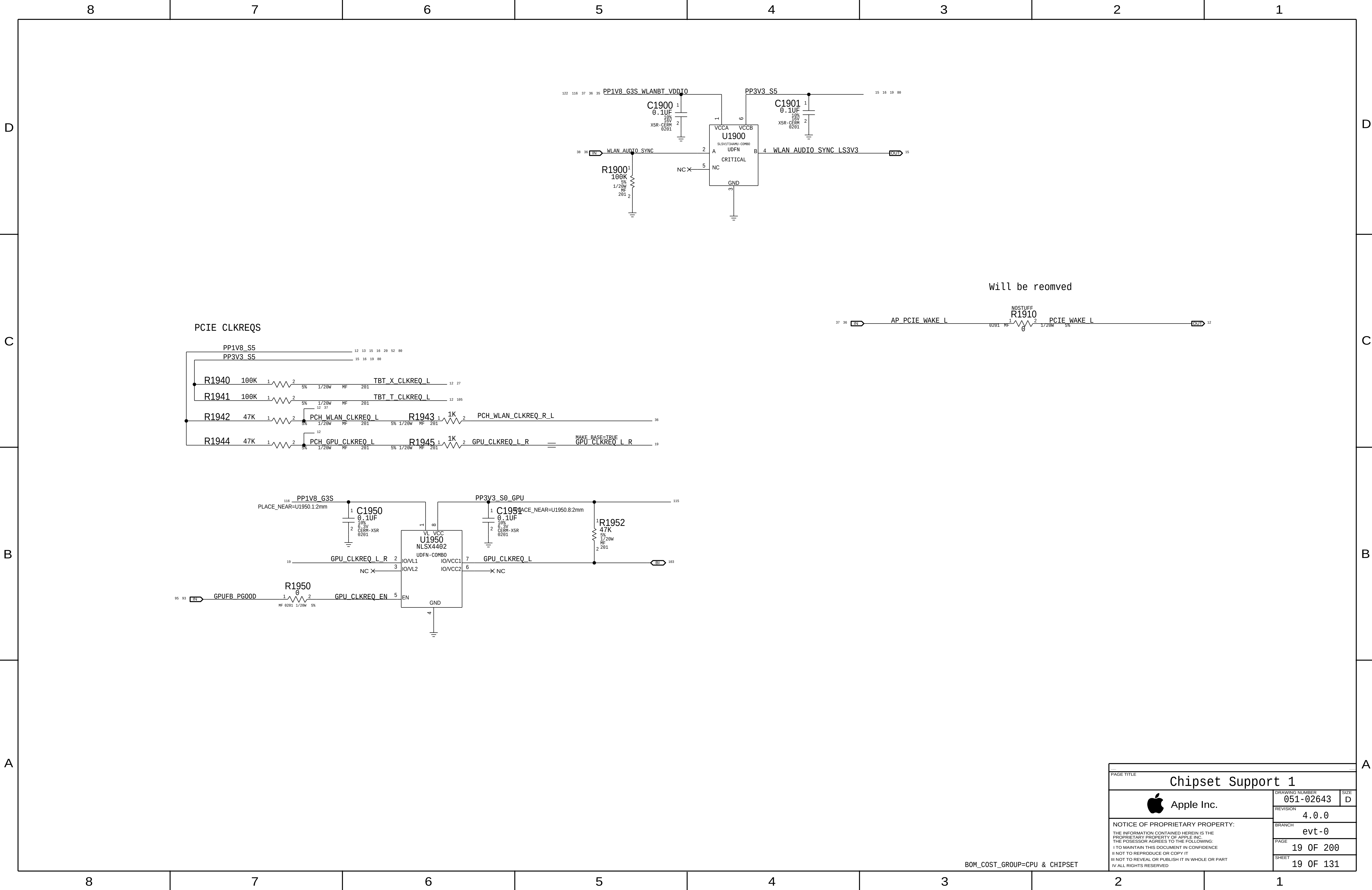
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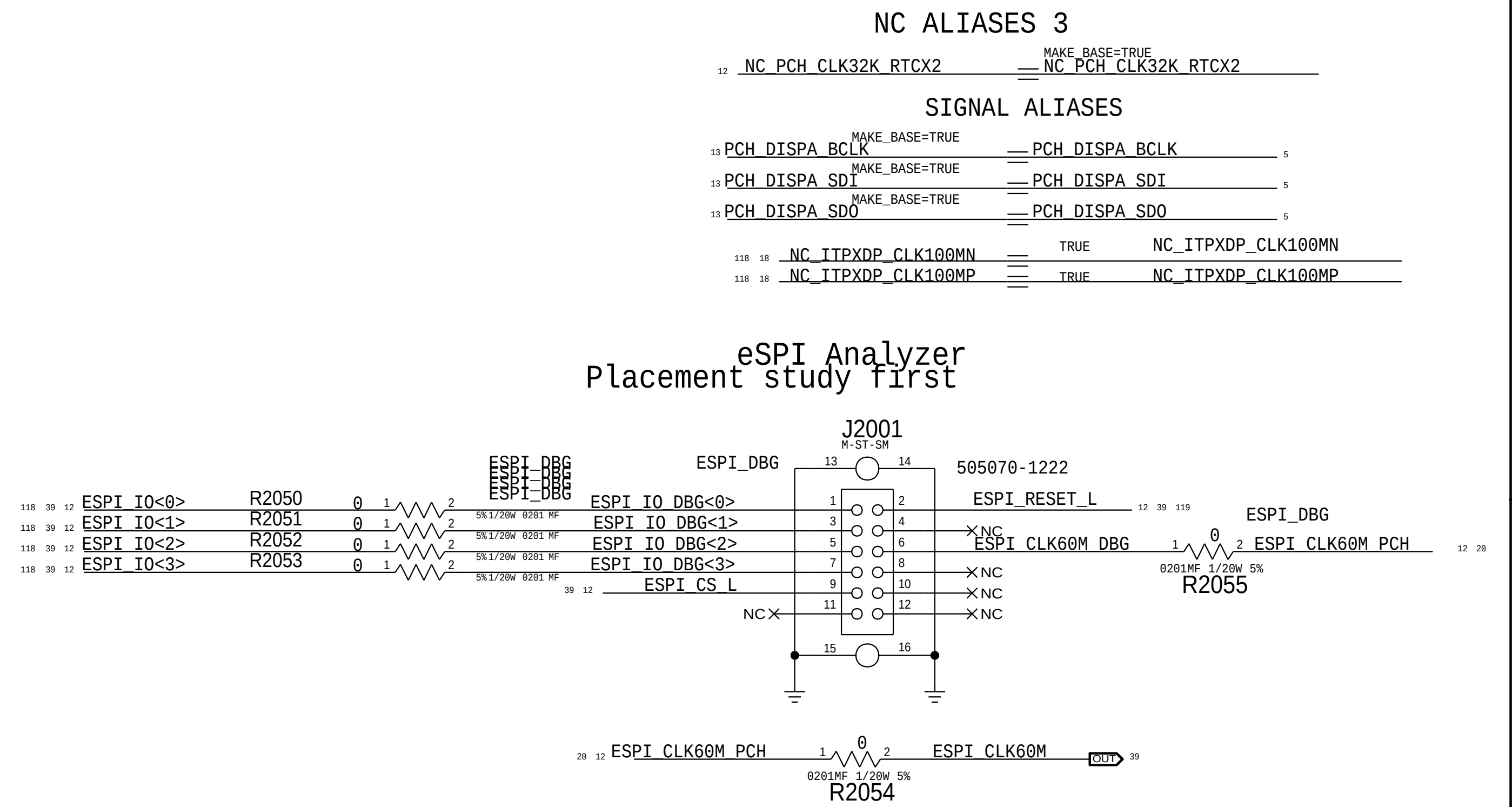
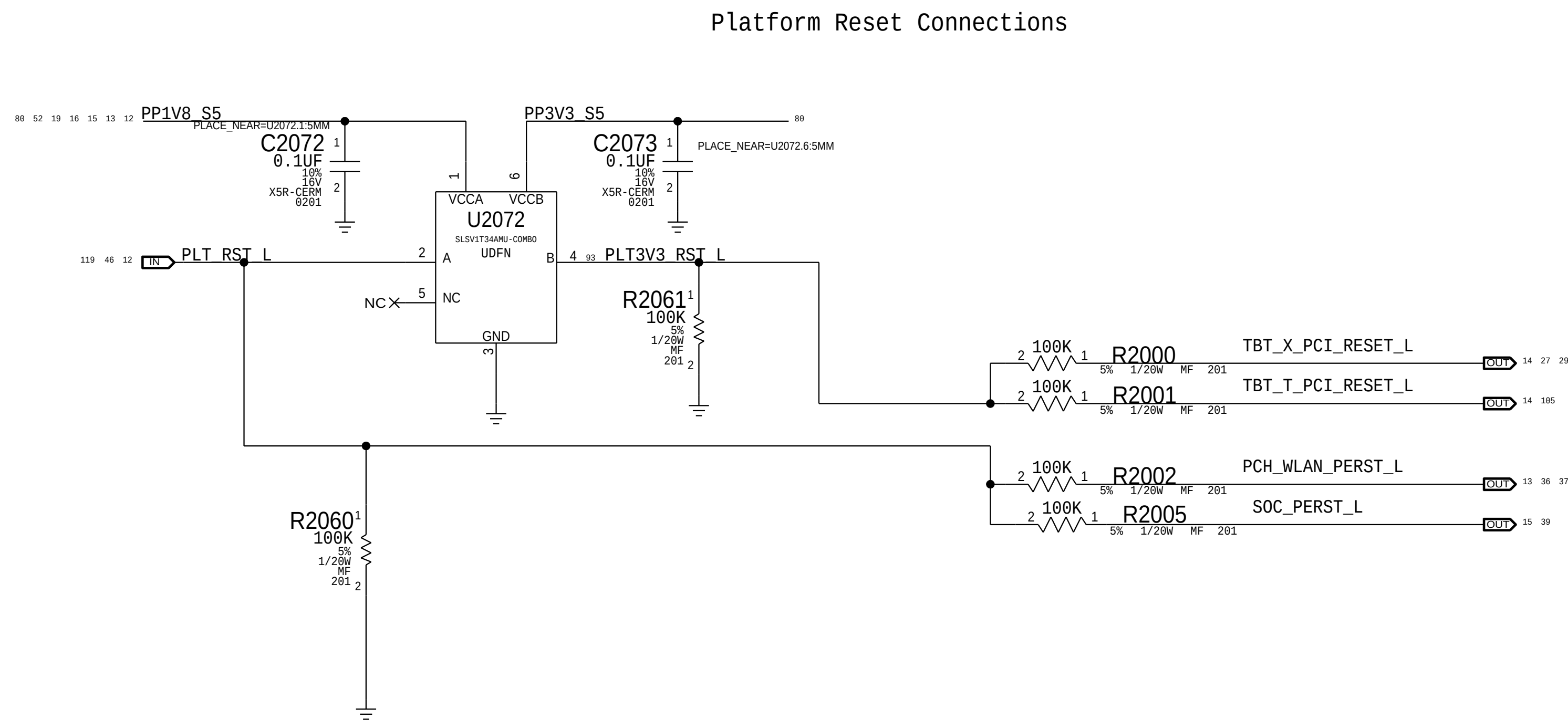


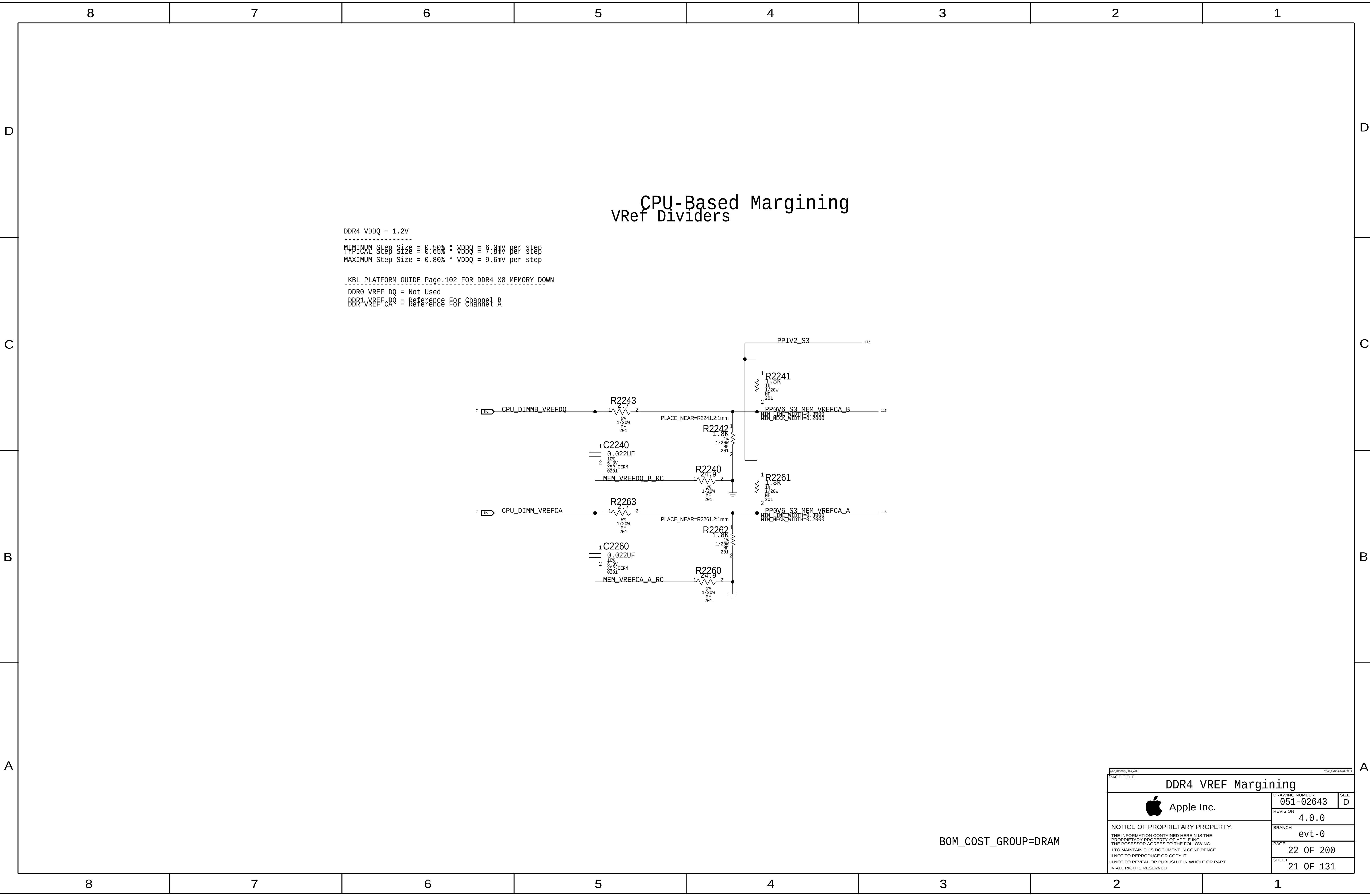


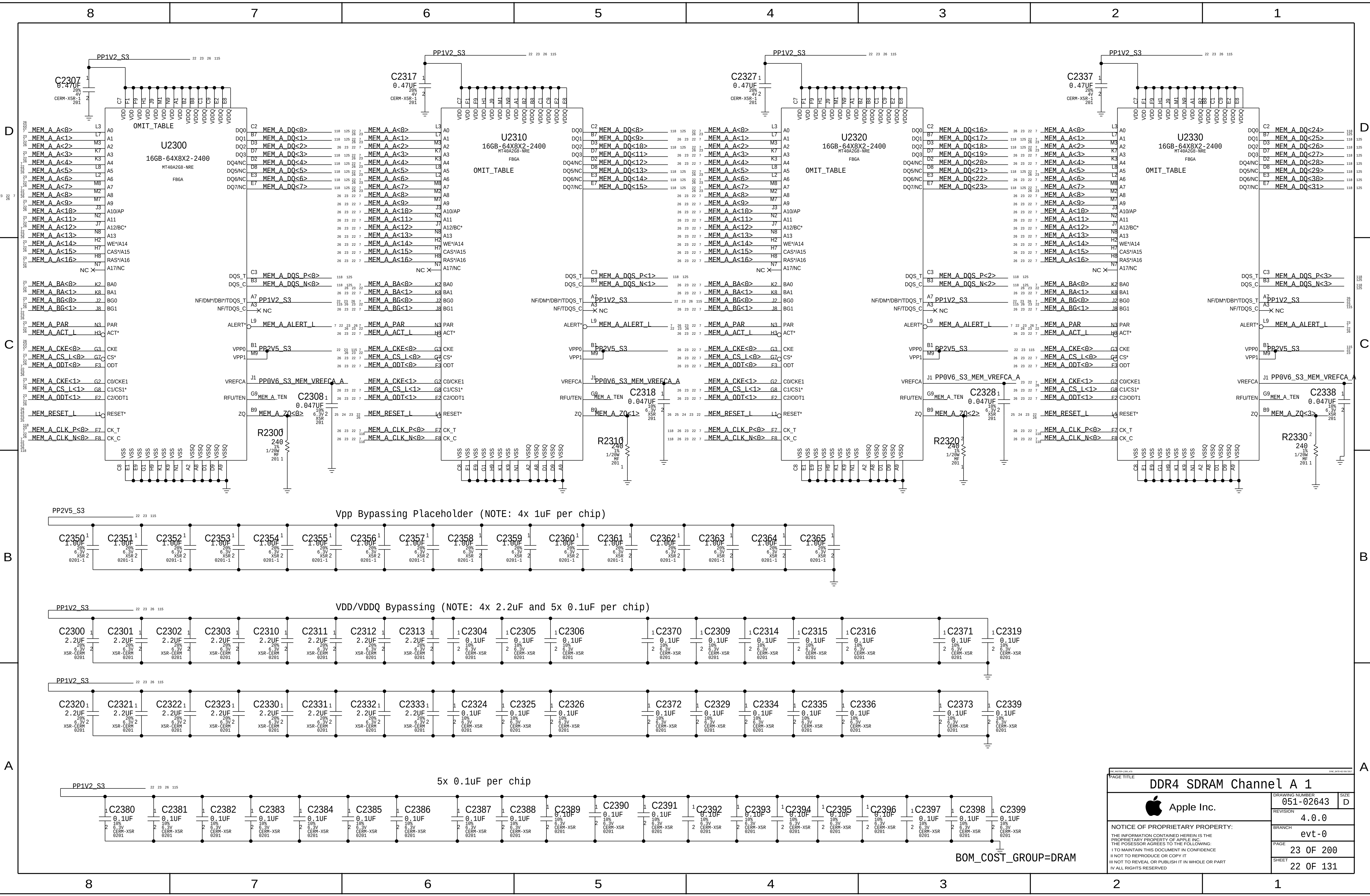




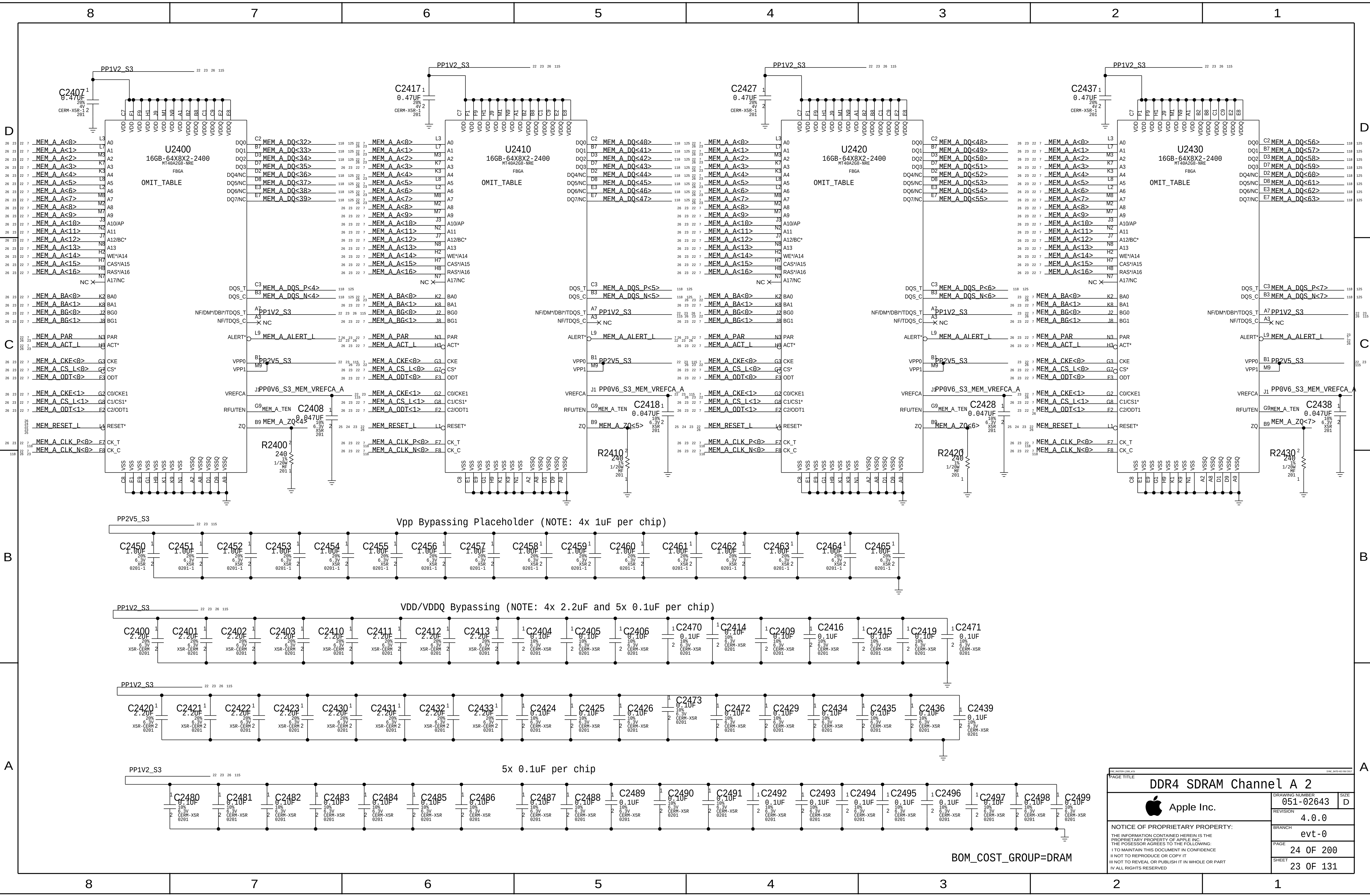
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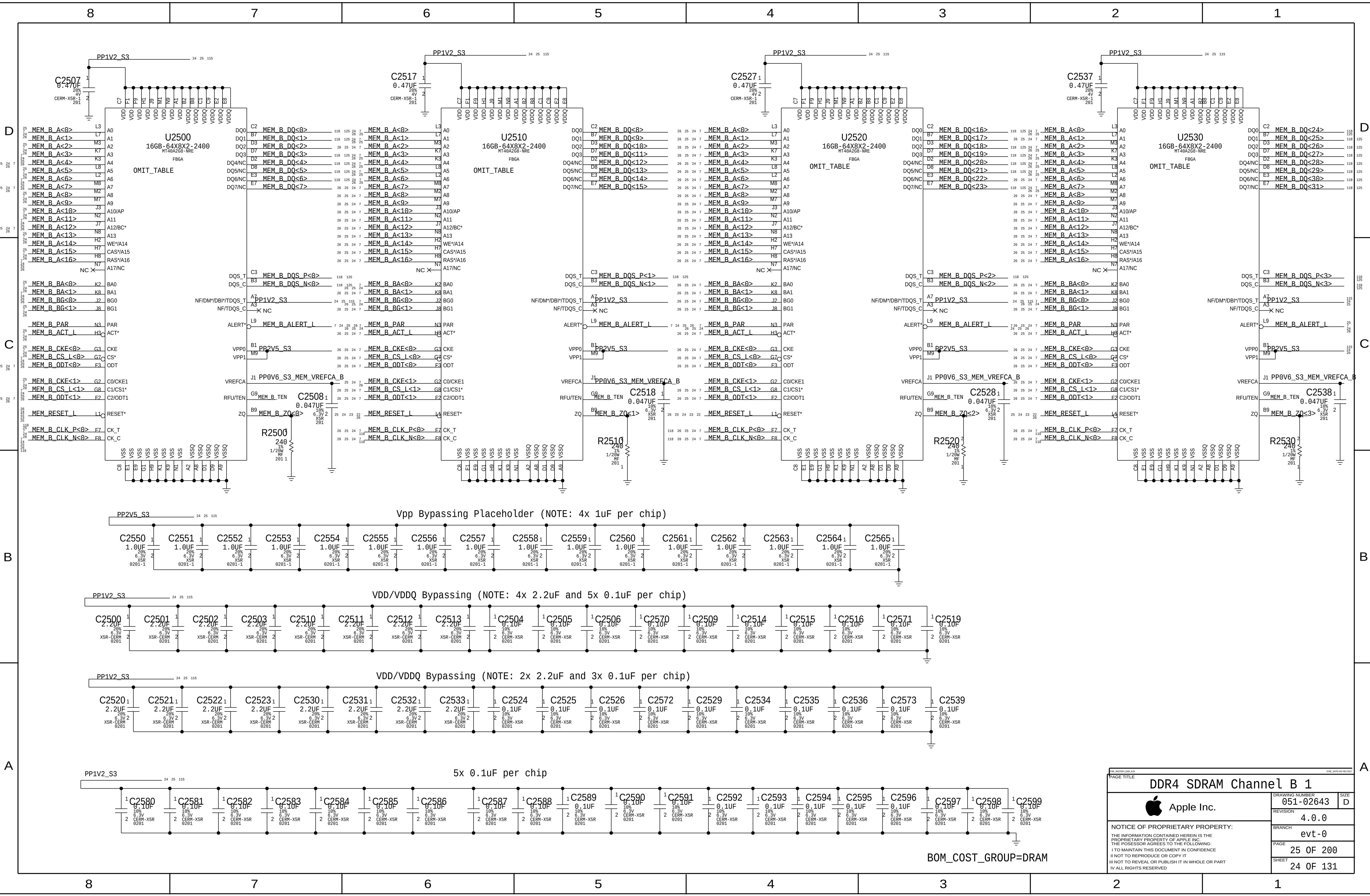


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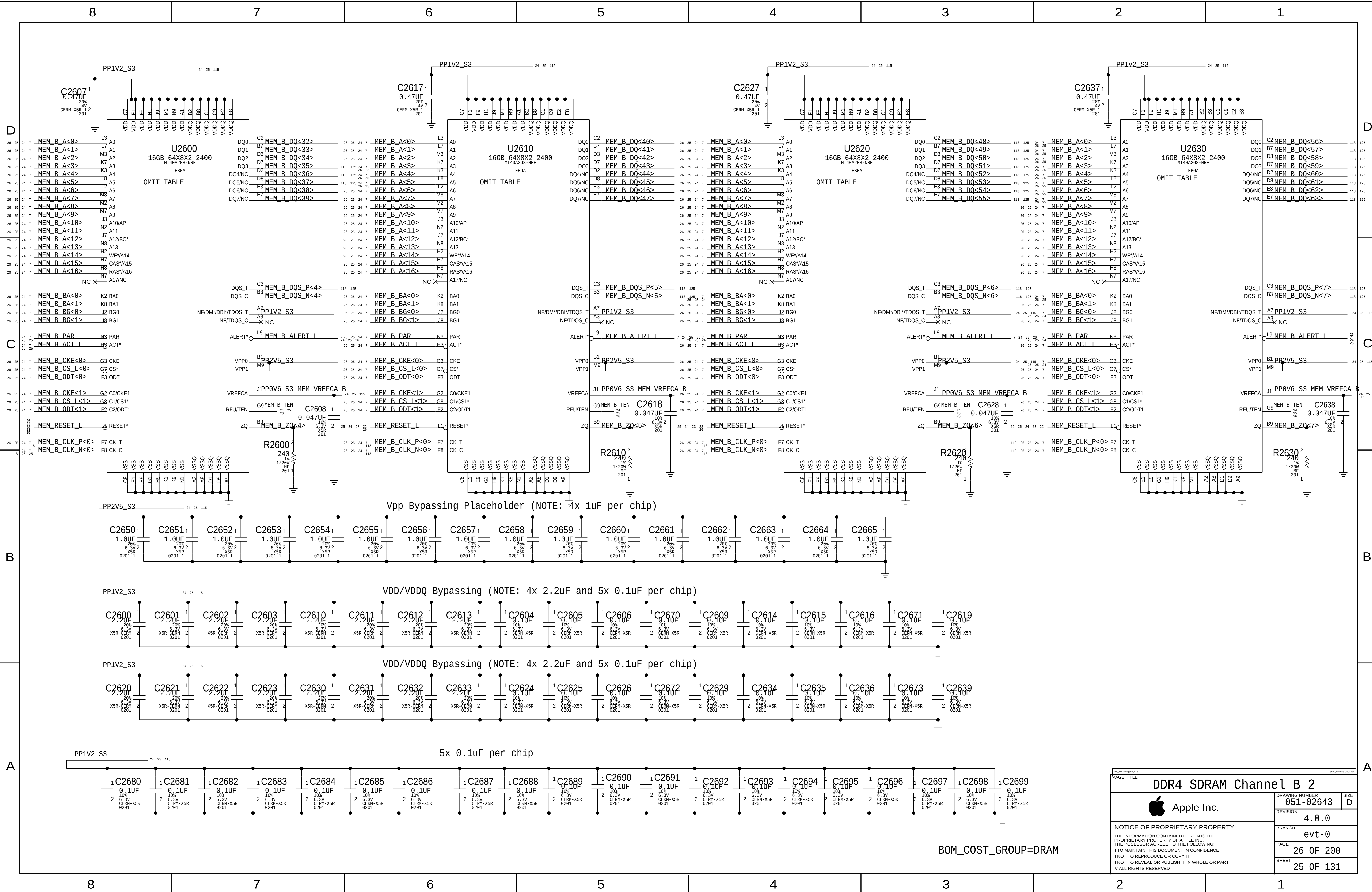
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I V ALL RIGHTS RESERVED			24 OF 200		
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			23 OF 131		

BOM_COST_GROUP=DRAM



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	PAGE	25 OF 200
	SHEET	24 OF 131

BOM_COST_GROUP=DRAM



JEDEC 4.20.18 Unbuffered SODIMM Raw Card F spec recommends 36 Ohm term to VTT for CS,CKE,ODT and 36 Ohm for BA,A,RAS,CAS,WE

D

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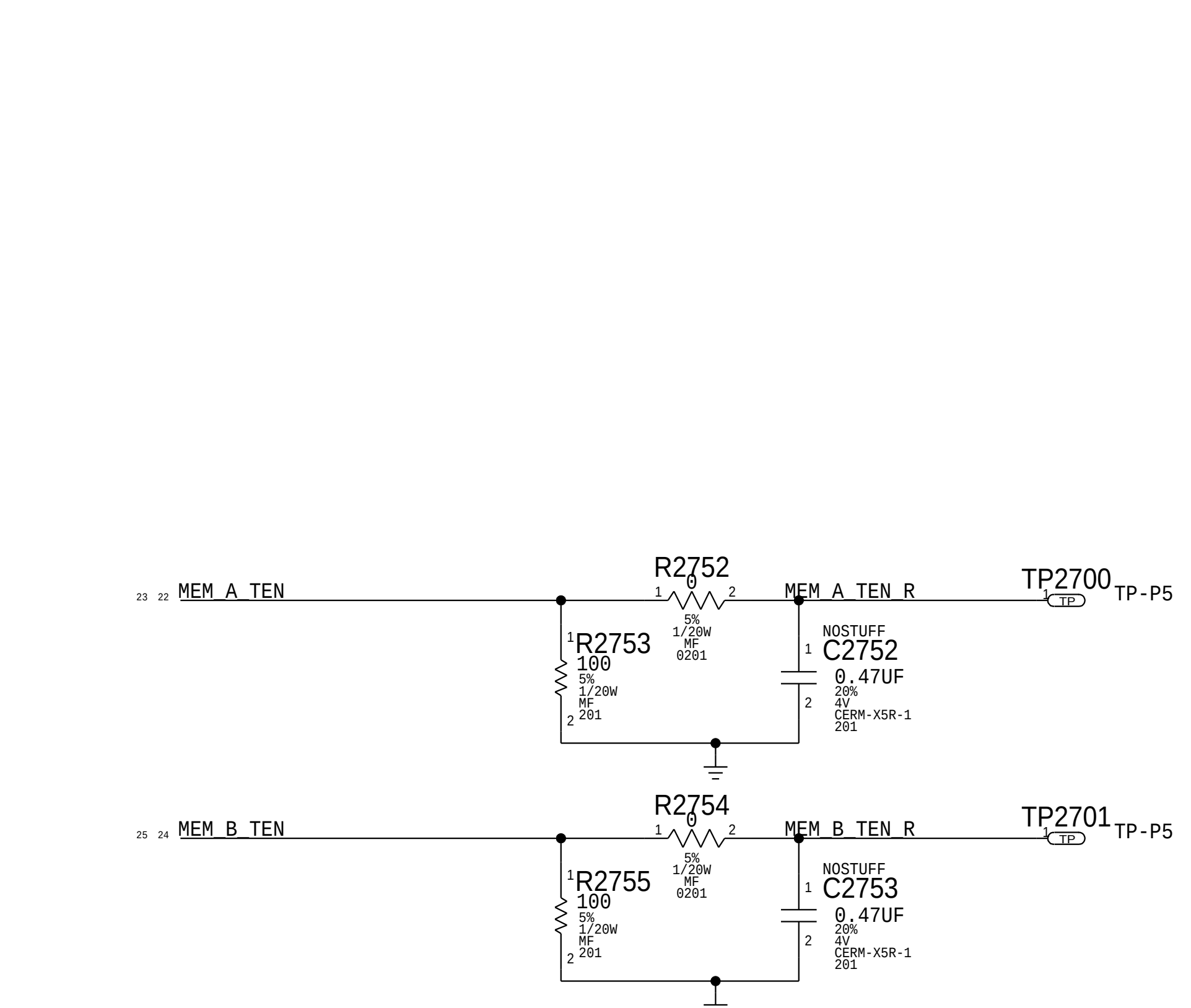
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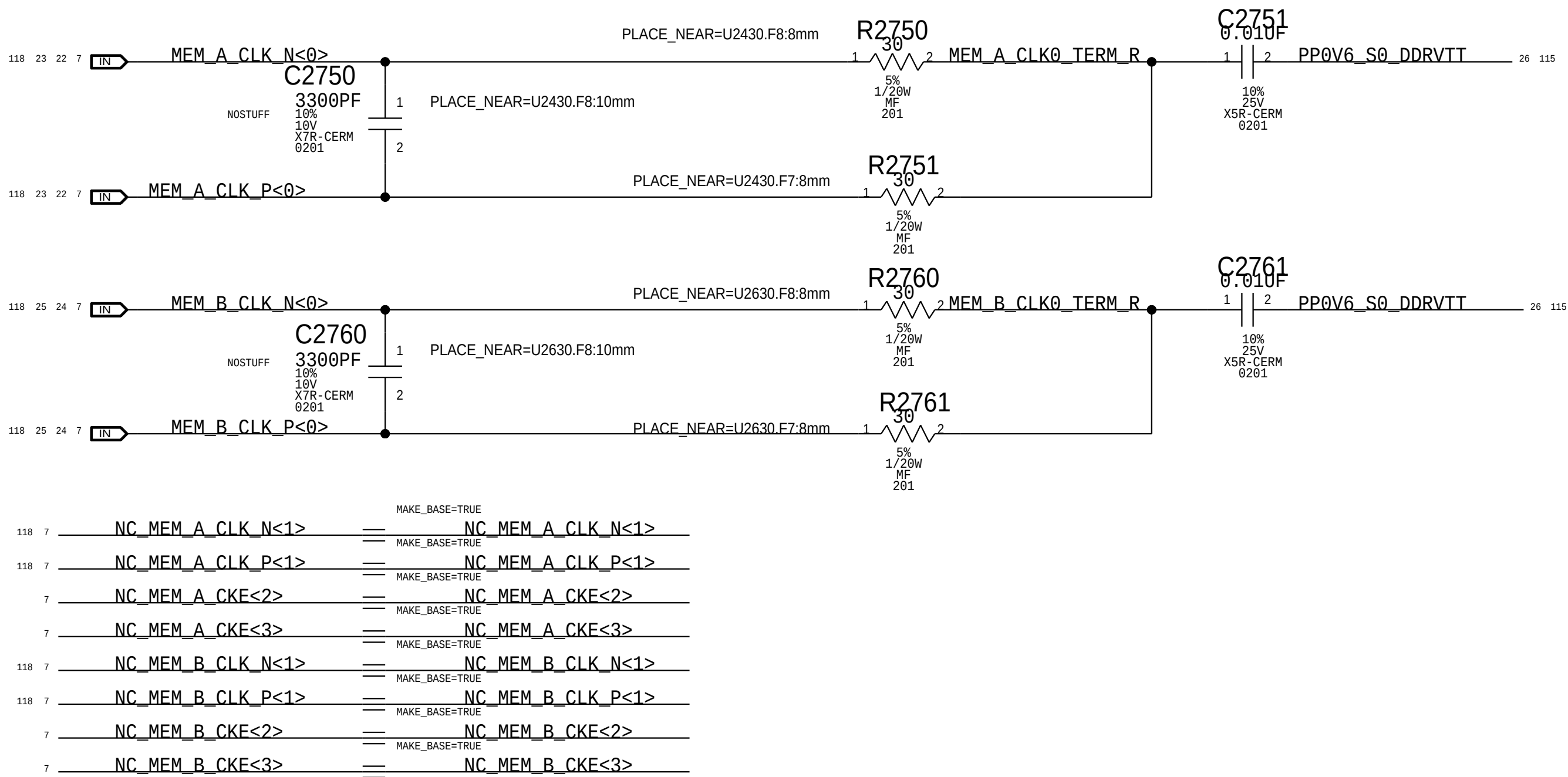
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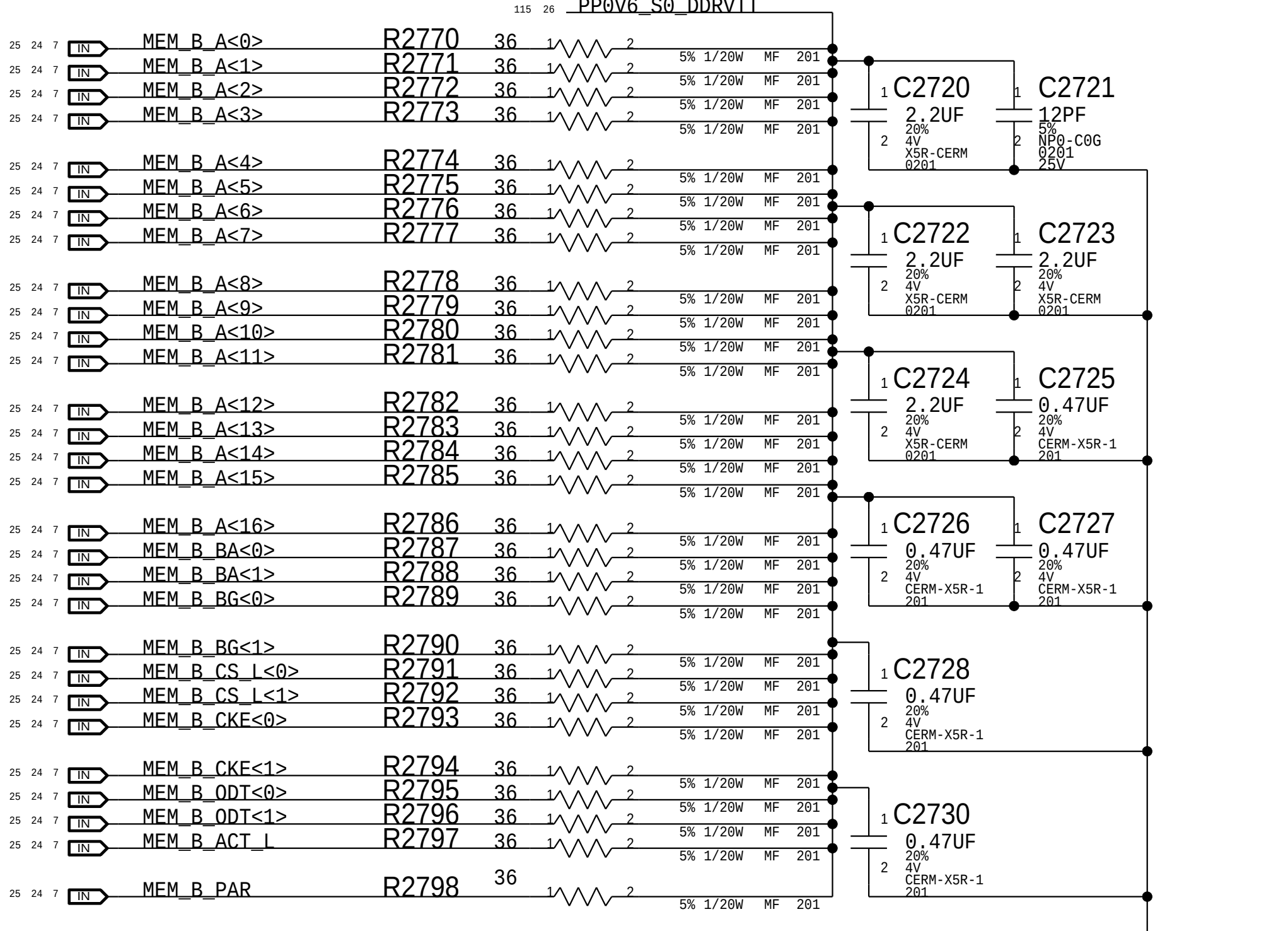
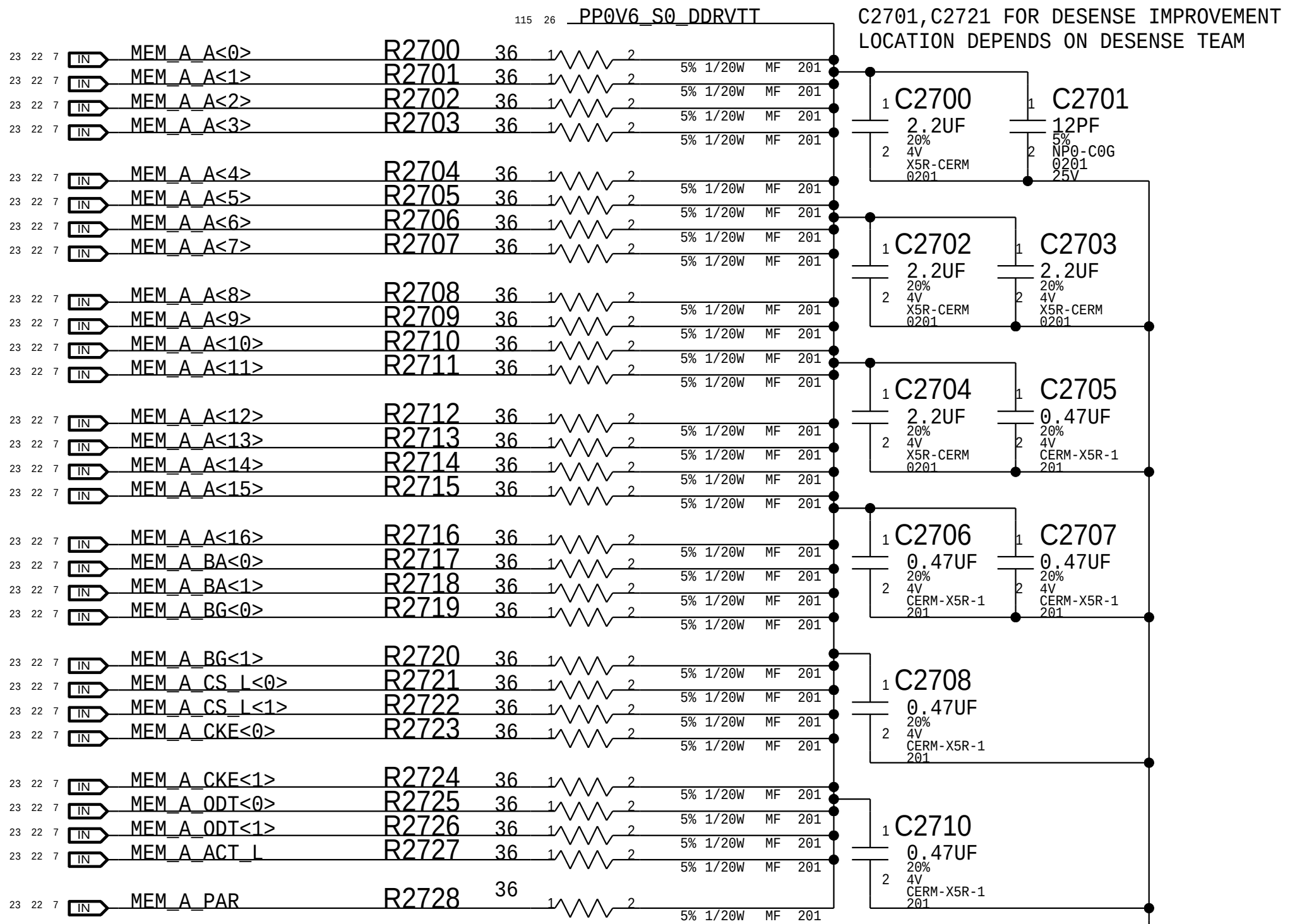
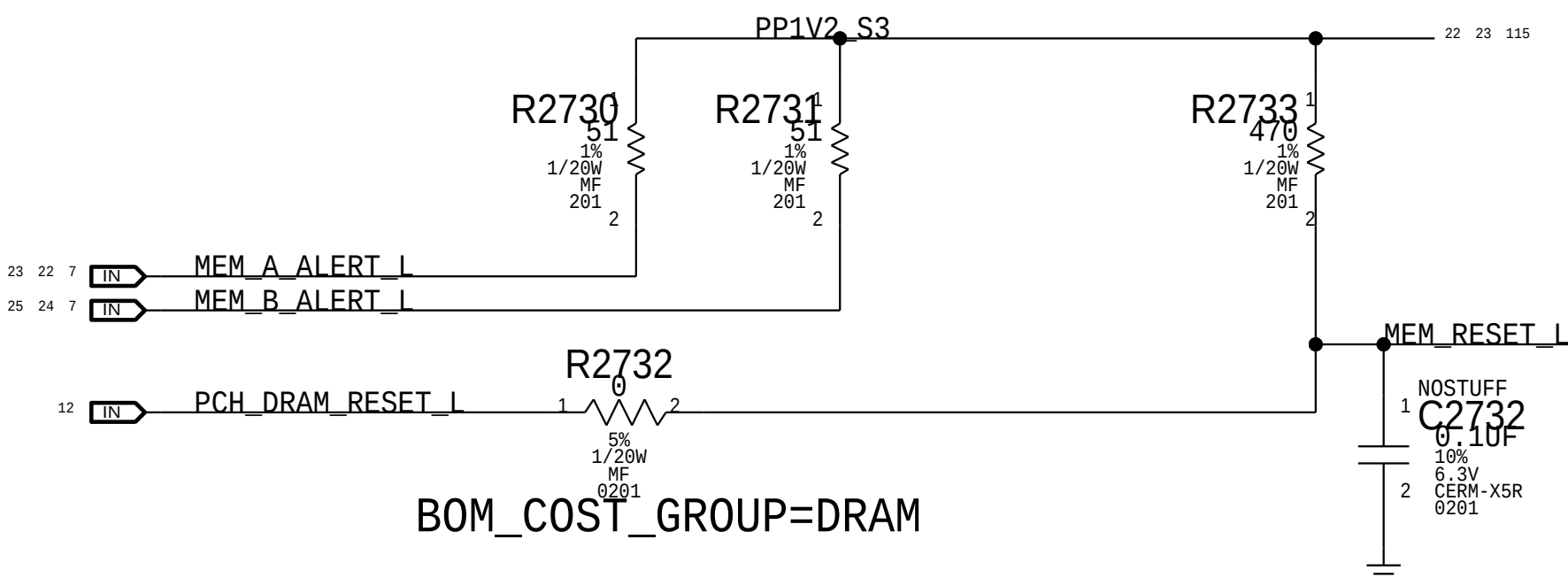


MEM Clock Termination

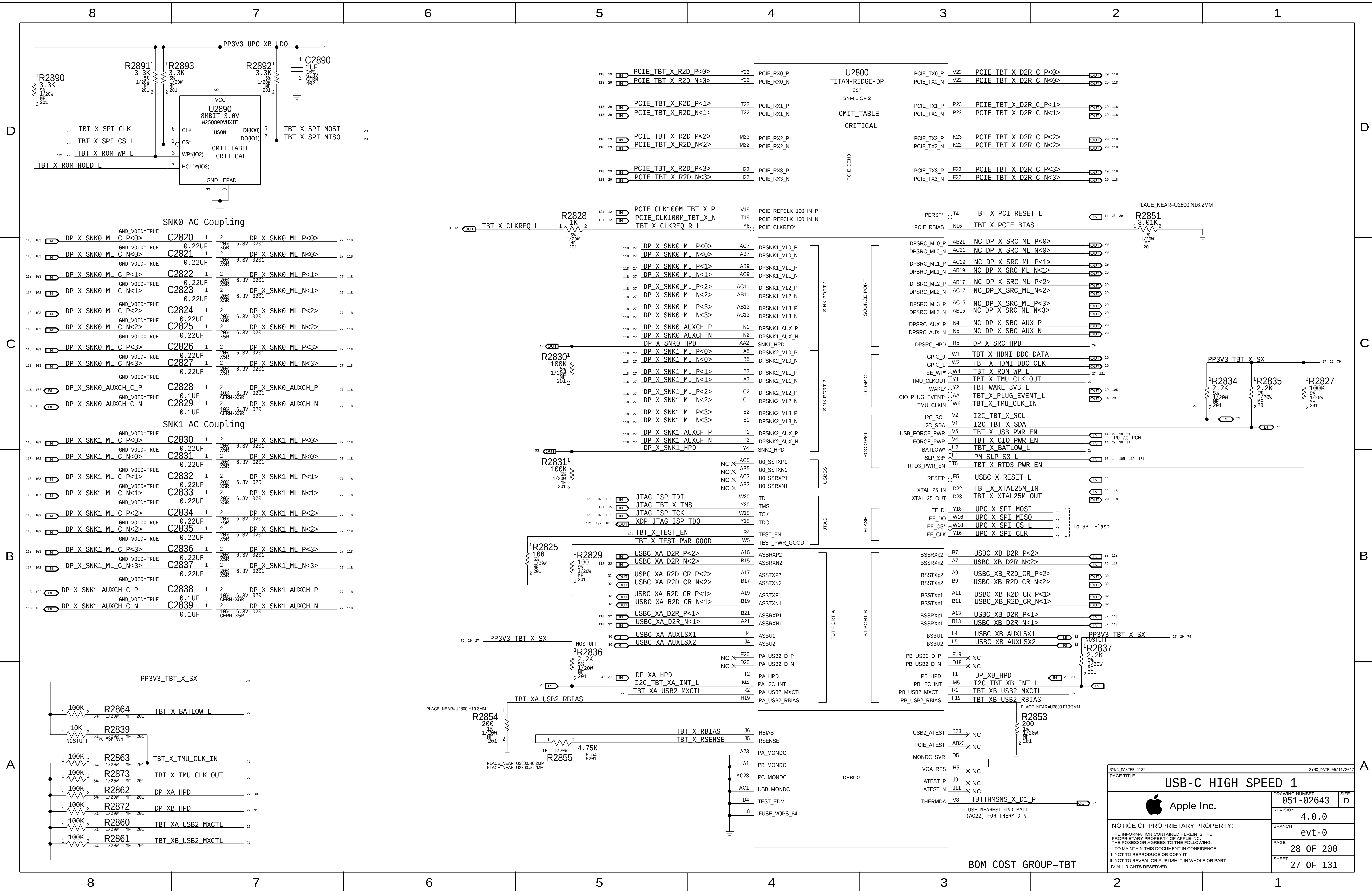
Place RC end termination after last DRAM
Place Source Cterm at neckdown at first DRAM

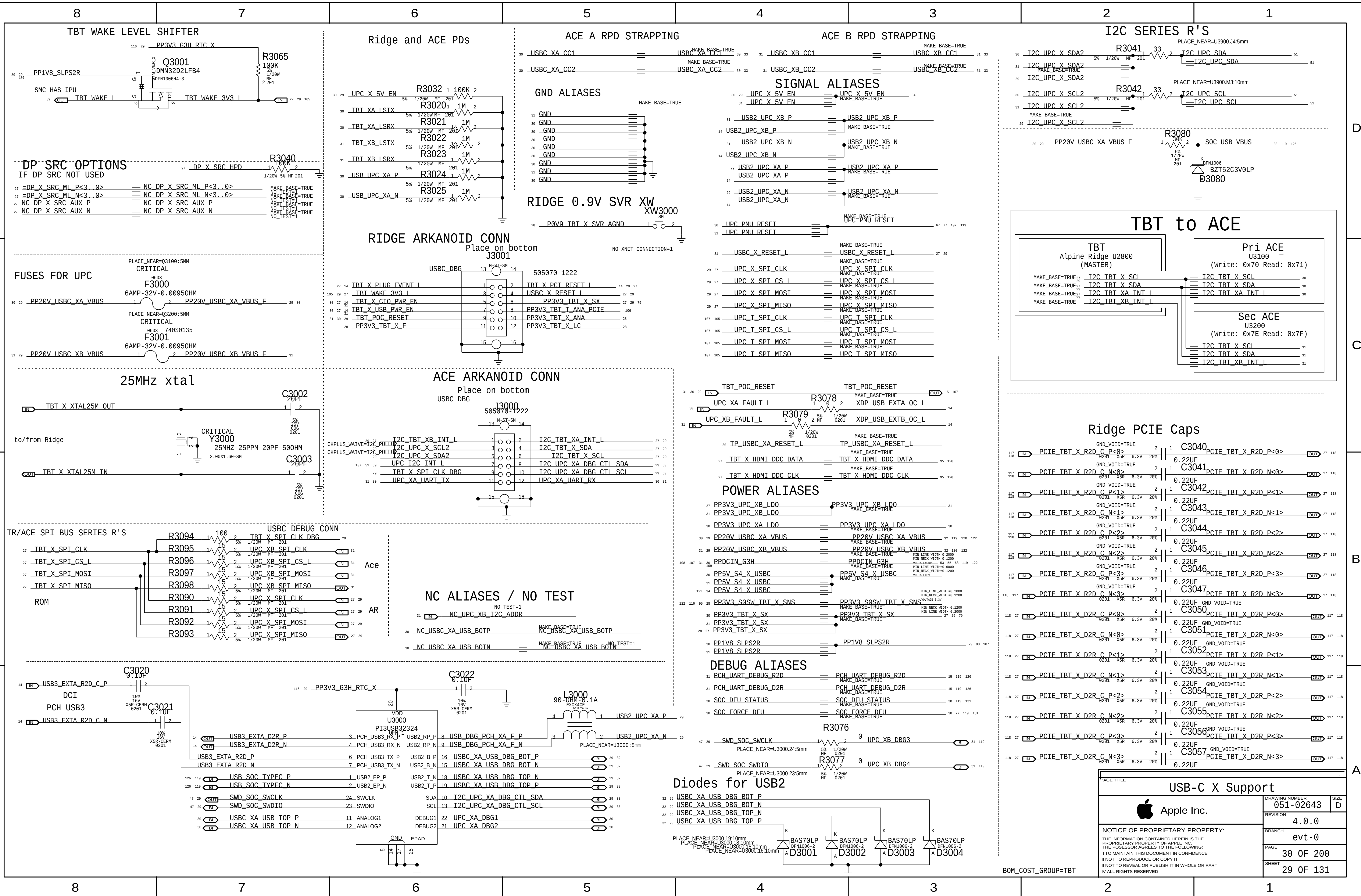


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7		NC_MEM_A_CKE<2>	==	MAKE_BASE=TRUE	NC_MEM_A_CKE<2>
7		NC_MEM_A_CKE<3>	==	MAKE_BASE=TRUE	NC_MEM_A_CKE<3>
118	7	NC_MEM_B_CLK_N<1>	==	MAKE_BASE=TRUE	NC_MEM_B_CLK_N<1>
118	7	NC_MEM_B_CLK_P<1>	==	MAKE_BASE=TRUE	NC_MEM_B_CLK_P<1>
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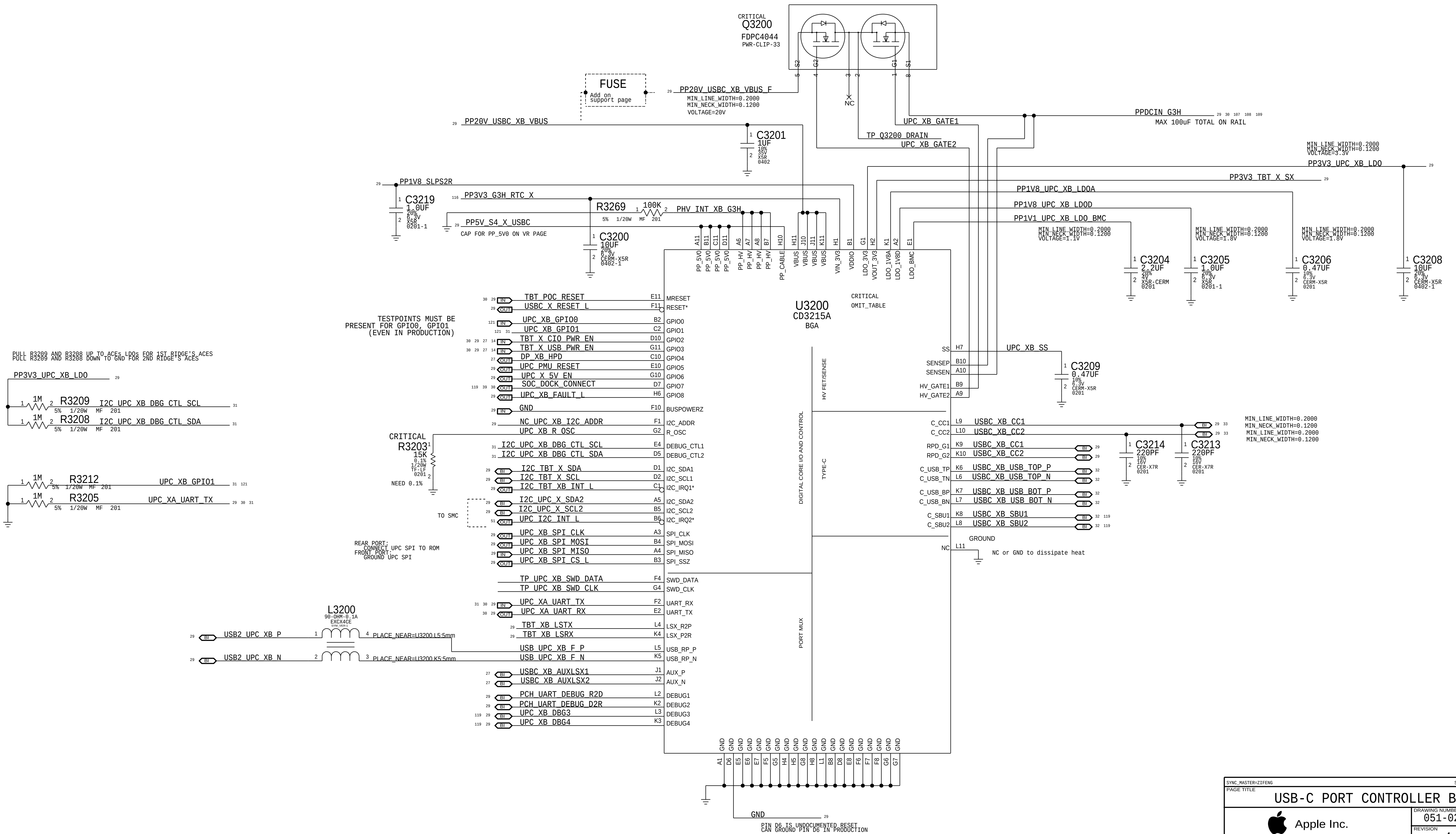
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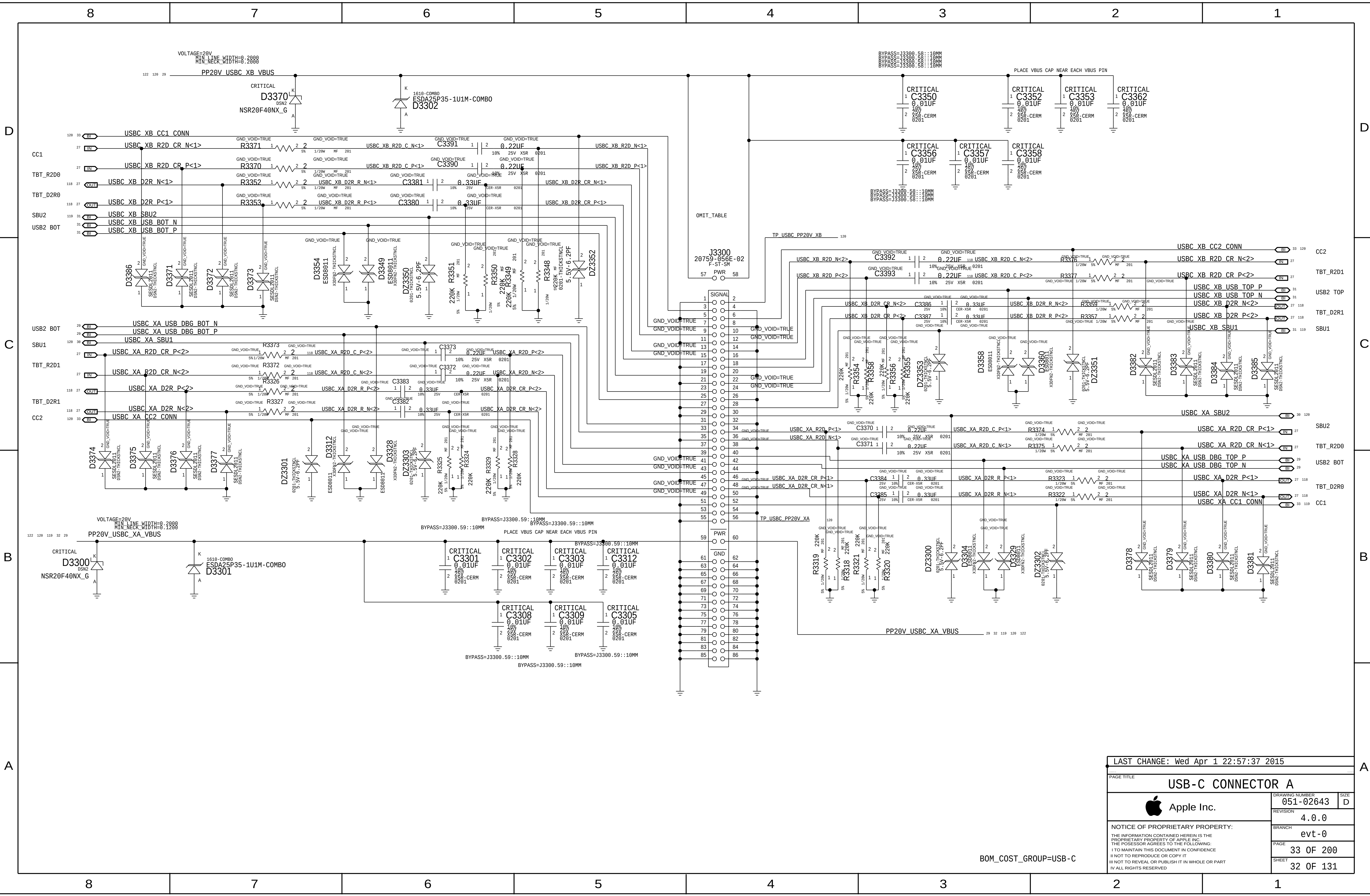
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	BRANCH	evt-0
	PAGE	31 OF 209
	SHEET	30 OF 131

SECONDARY ACE USB-C PORT CONTROLLER (UPC)



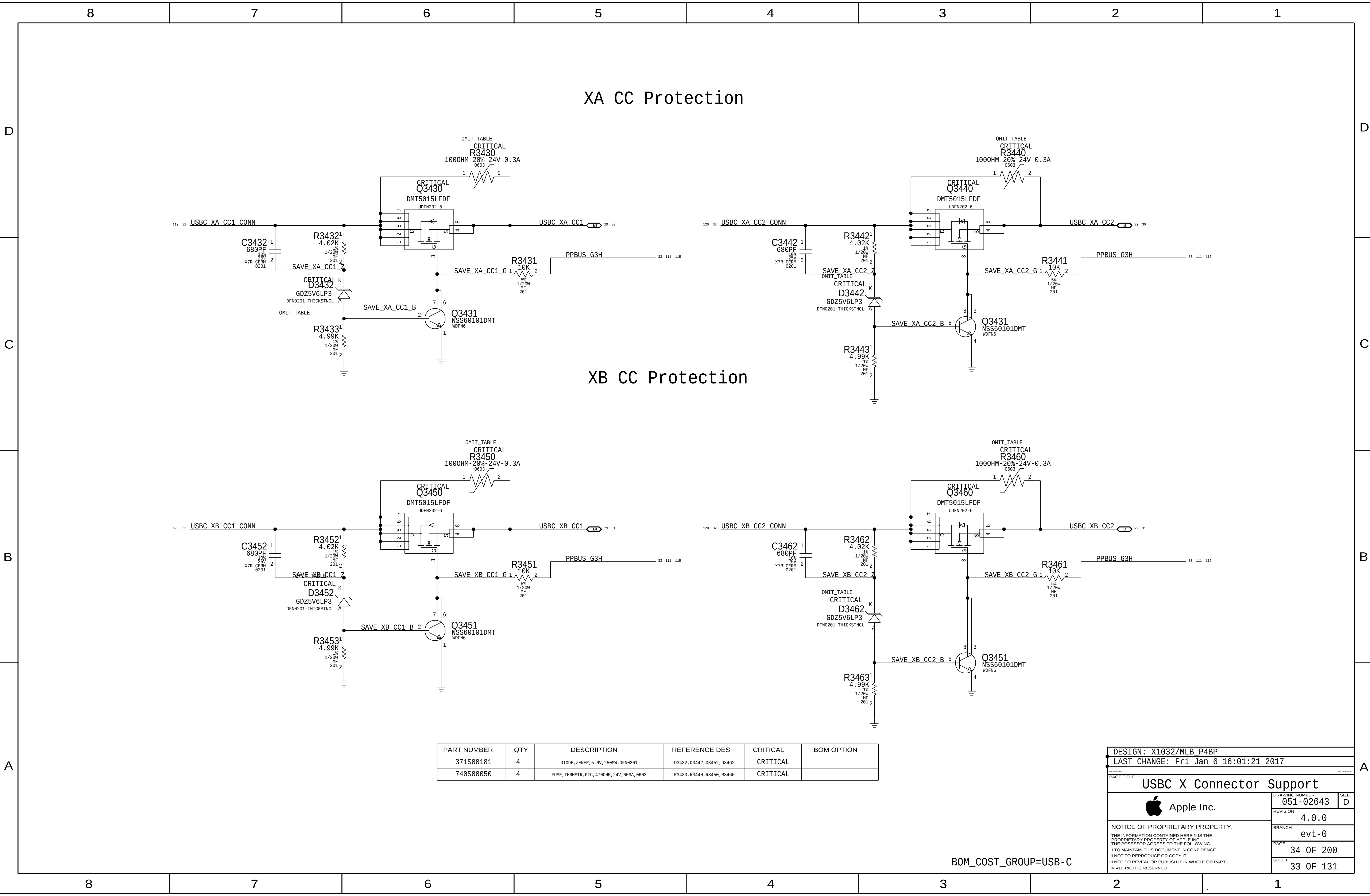
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33 OF 200		32 OF 131	

BOM_COST_GROUP=USB-C



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
371S00181	4	DIODE, ZENER, 5.6V, 250MW, DFN0201	D3432, D3442, D3452, D3462	CRITICAL	
740S00050	4	FUSE, THRMSTR, PTC, 4700HM, 24V, 60MA, 0603	R3430, R3440, R3450, R3460	CRITICAL	

DESIGN: X1032/MLB_P4BP

LAST CHANGE: Fri Jan 6 16:01:21 2017

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USBC X Connector Support

DRAWING NUMBER
051-02643

SIZE
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REVISION
4.0.0

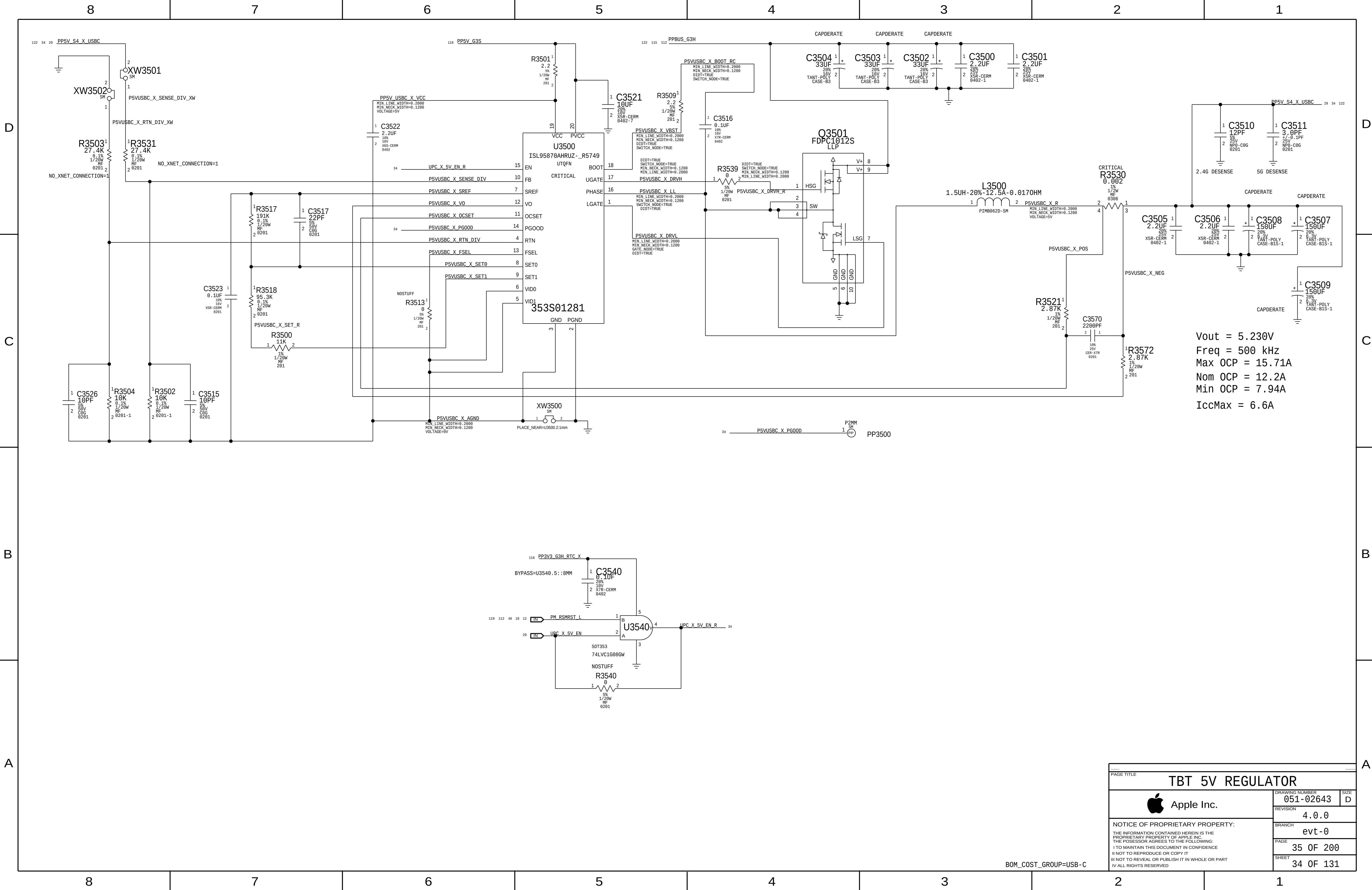
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BOM_COST_GROUP=USB-C



Vout = 5.230V
Freq = 500 kHz
Max OCP = 15.71A
Nom OCP = 12.2A
Min OCP = 7.94A
IccMax = 6.6A

PAGE TITLE		
TBT 5V REGULATOR		
 Apple Inc.	DRAWING NUMBER	051-02643
	REVISION	4.0.0
	BRANCH	evt-0
	PAGE	35 OF 200
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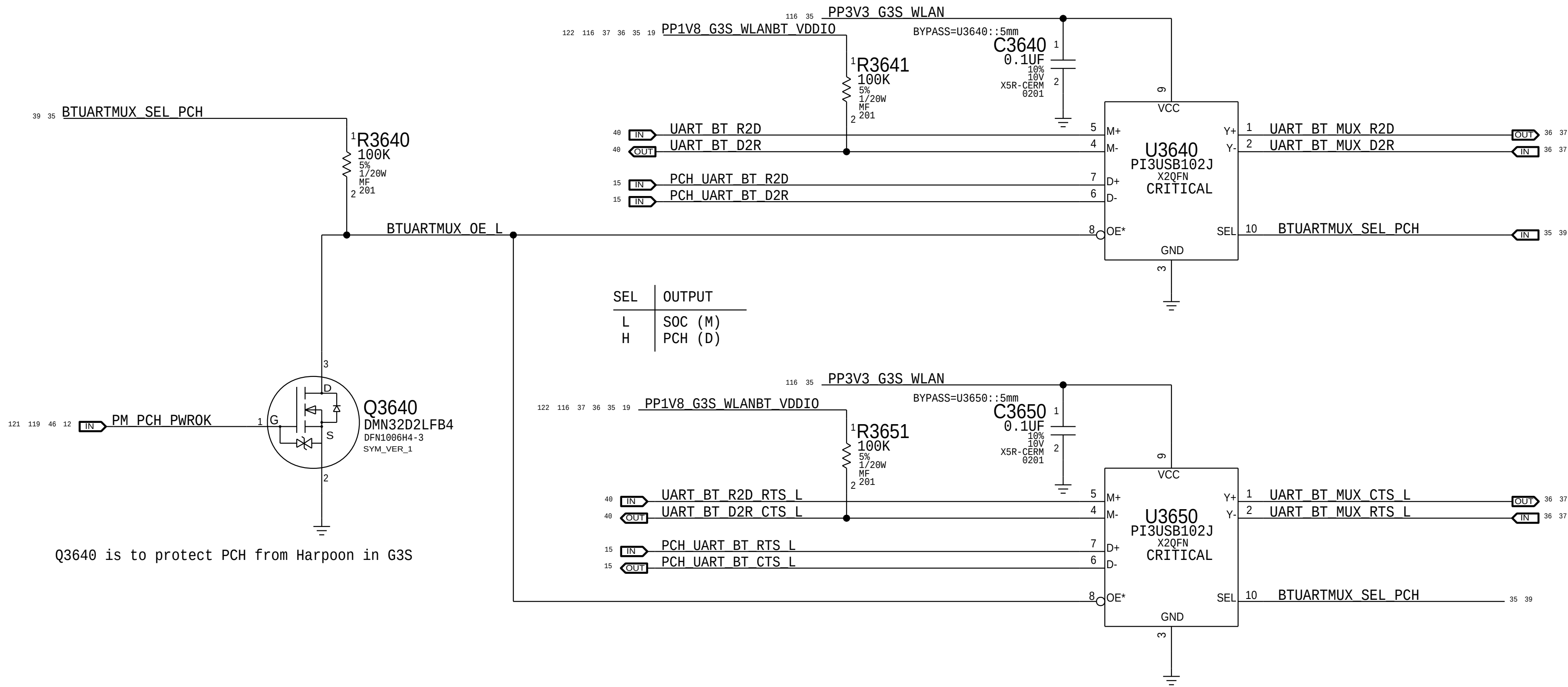
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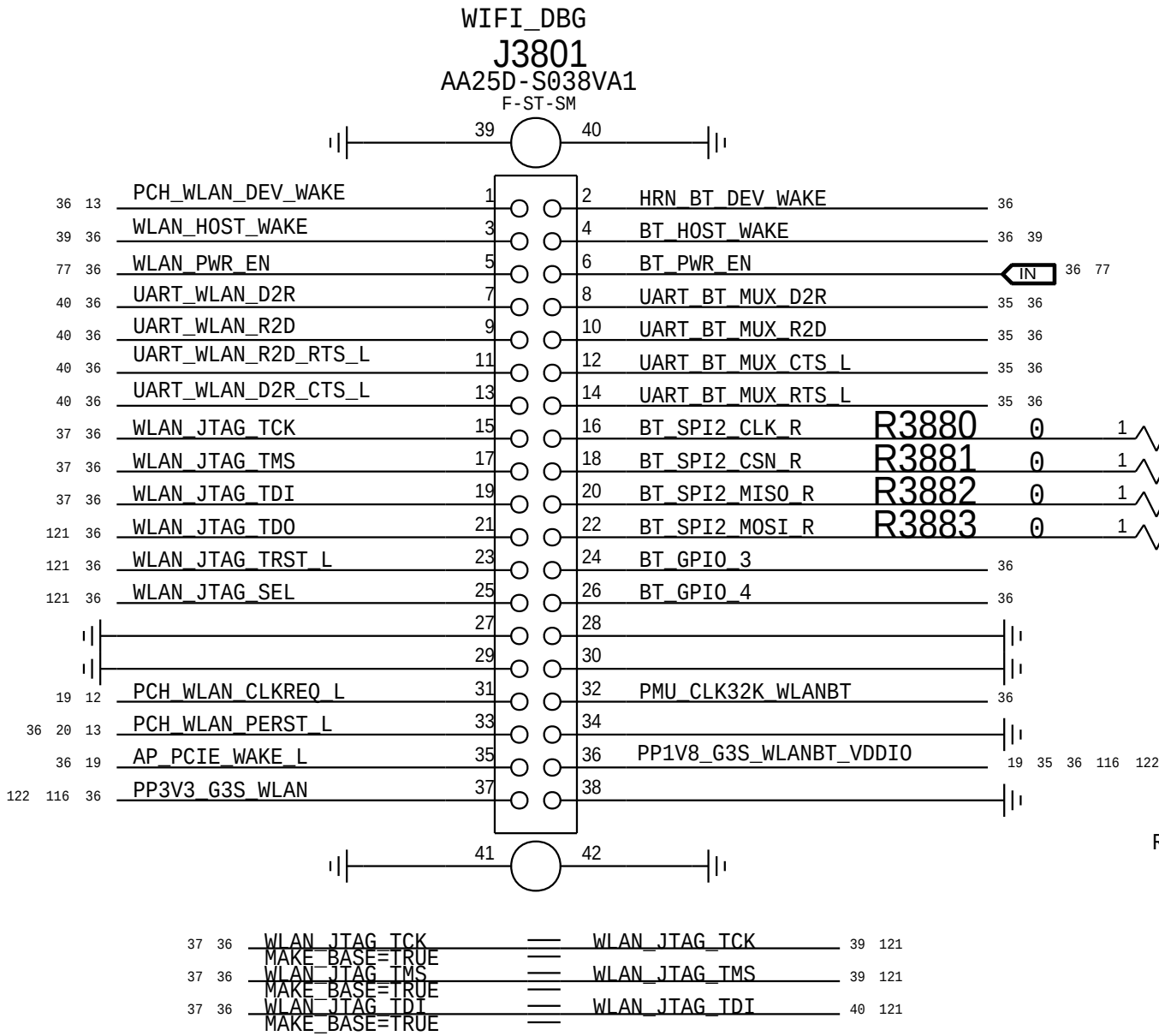
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		PAGE	36 OF 200
		SHEET	35 OF 131

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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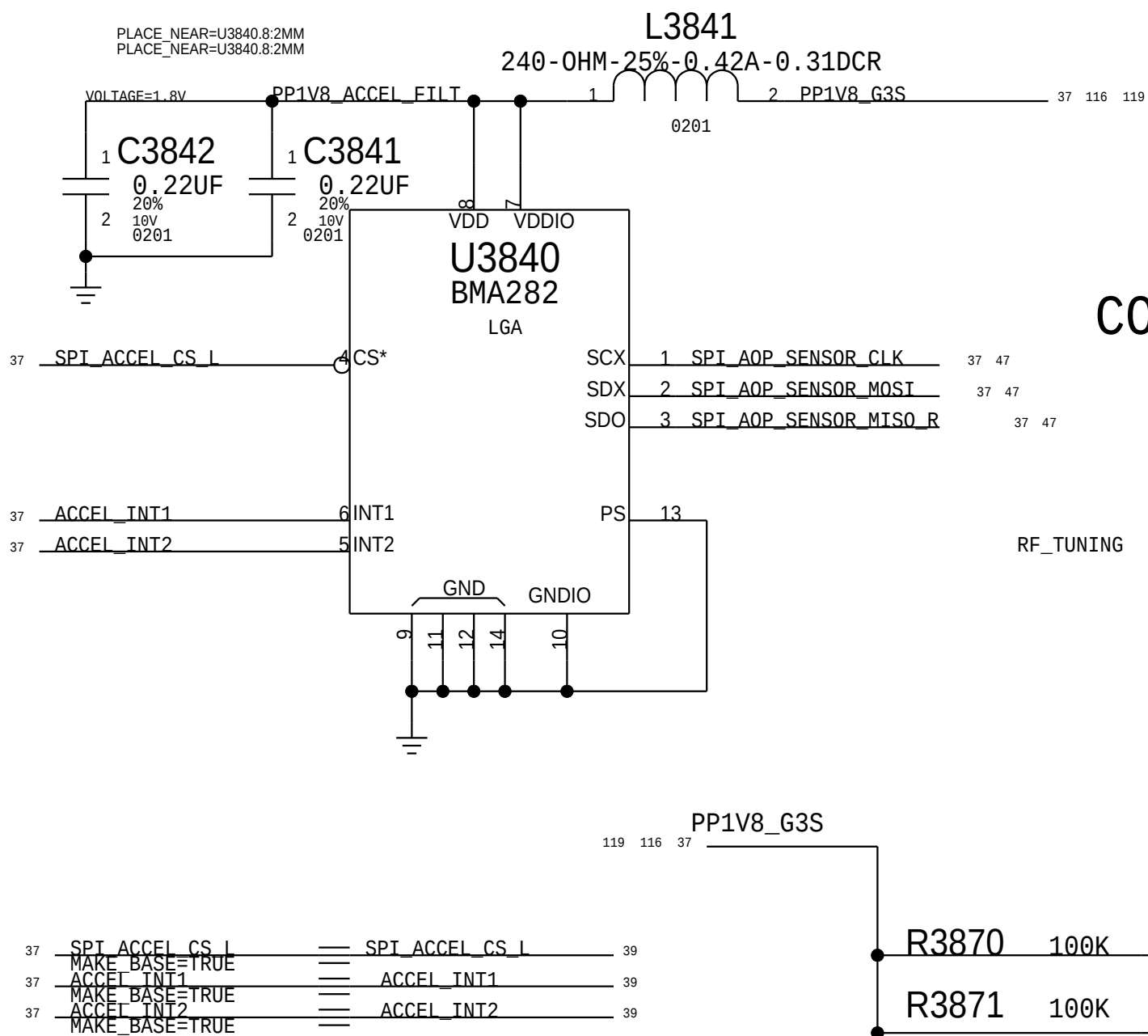
CORE0 DIPLEXER AND MATCHING NETWORK

DEBUG CONNECTOR

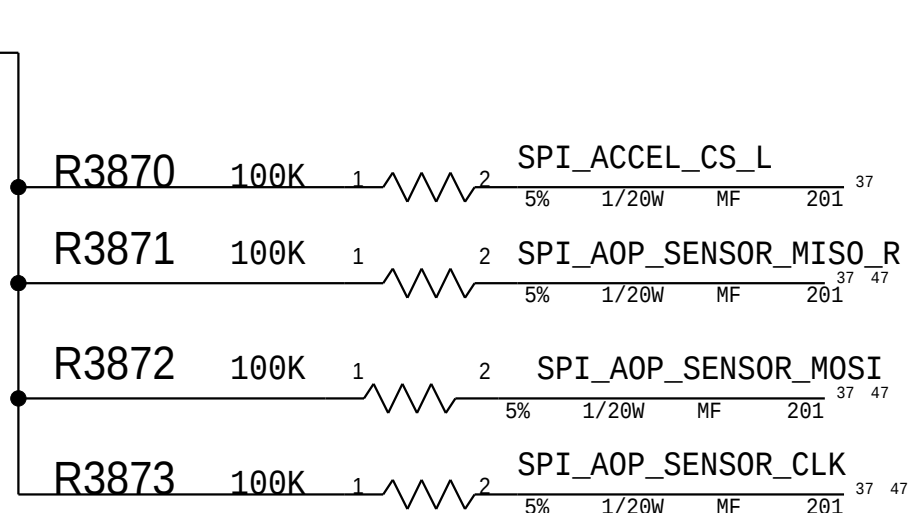


CORE1 DIPLEXER AND MATCHING NETWORK

ACCELERATION SENSOR



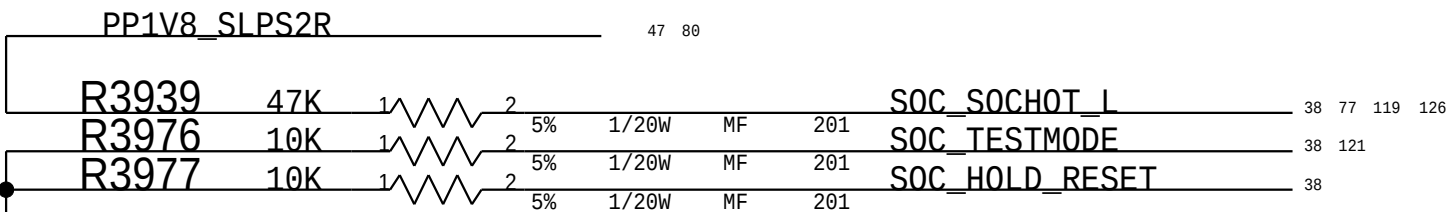
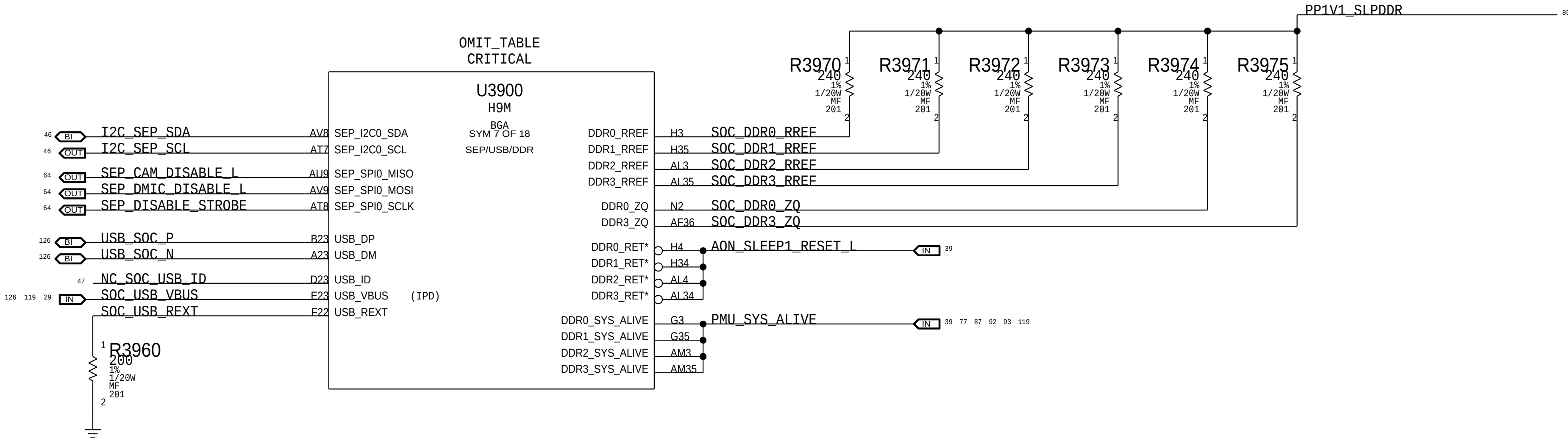
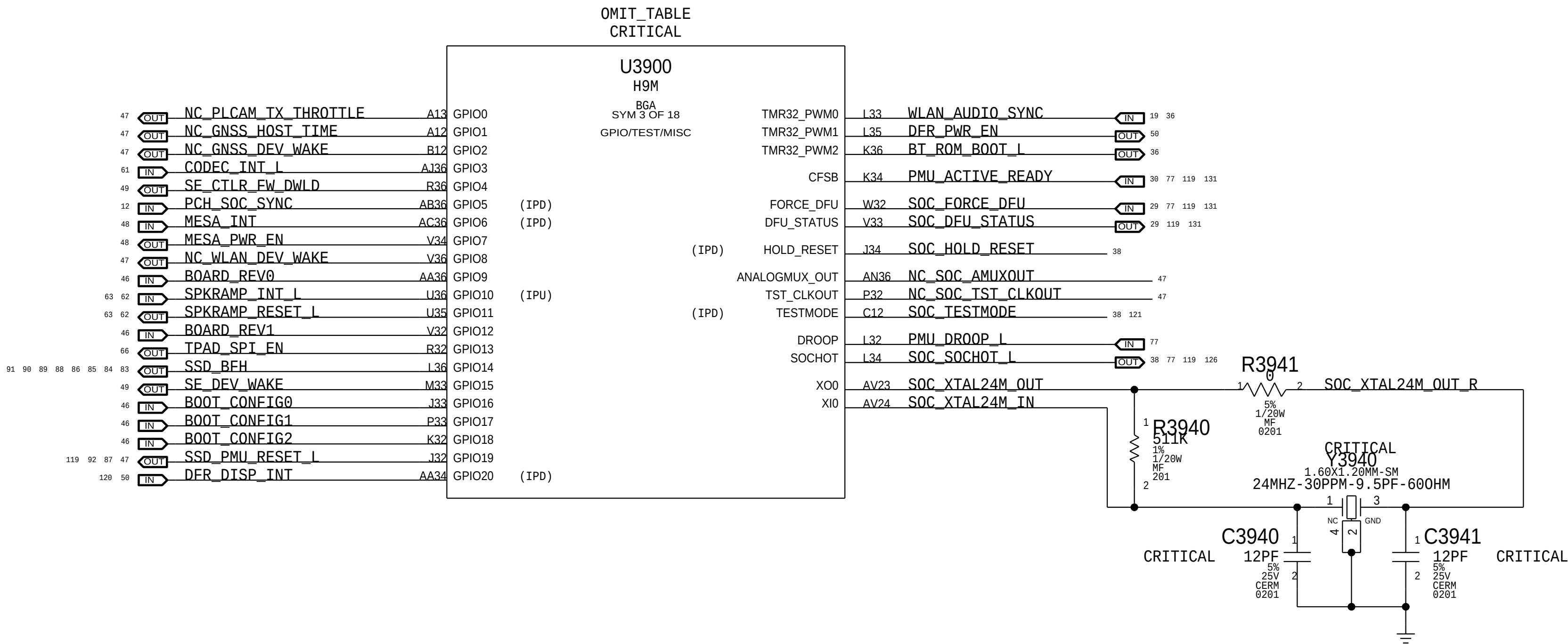
CORE2/Aux DIPLEXER AND MATCHING NETWORK



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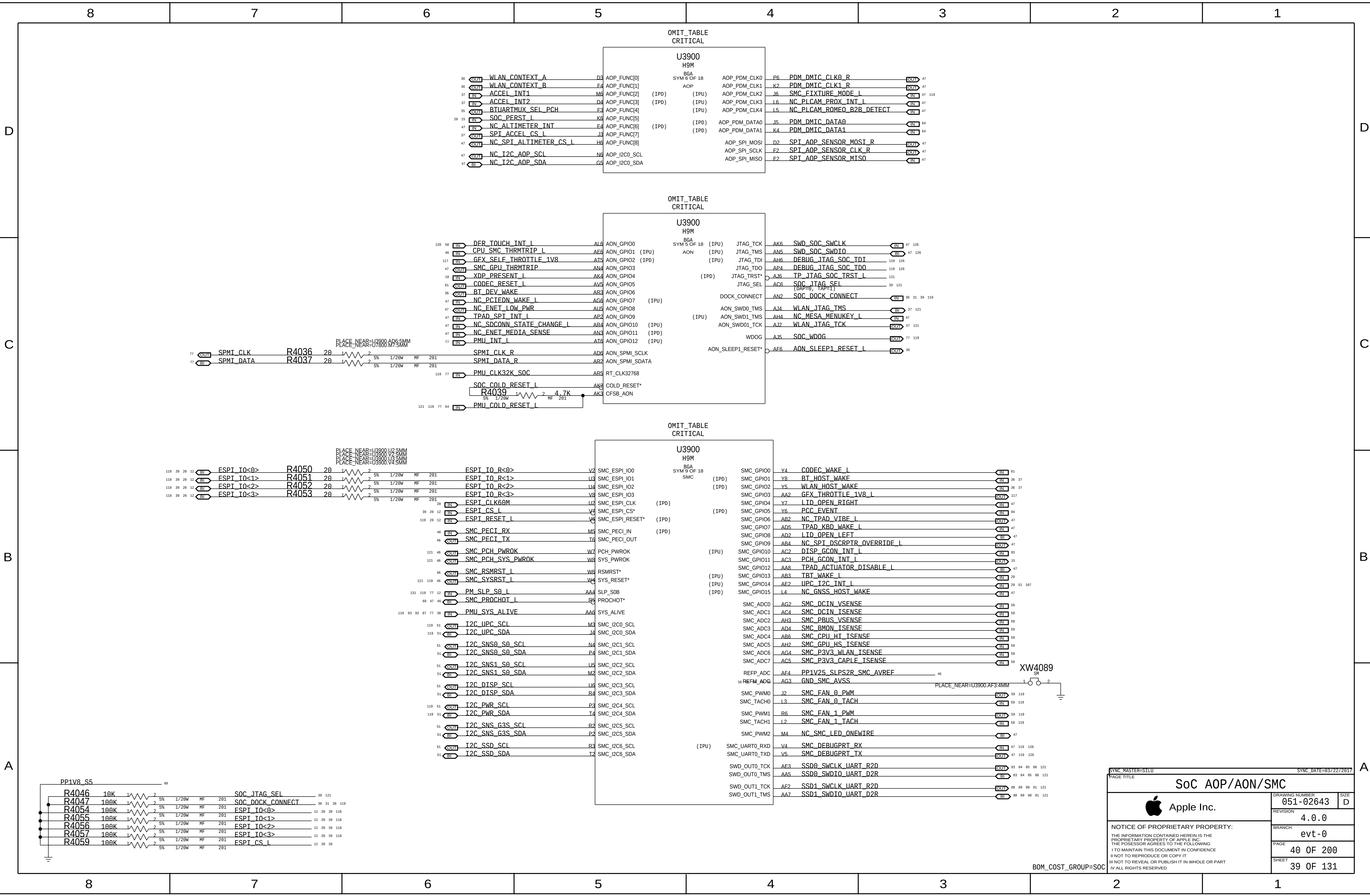
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Note 1) IPU represents SW configured state, not HW default



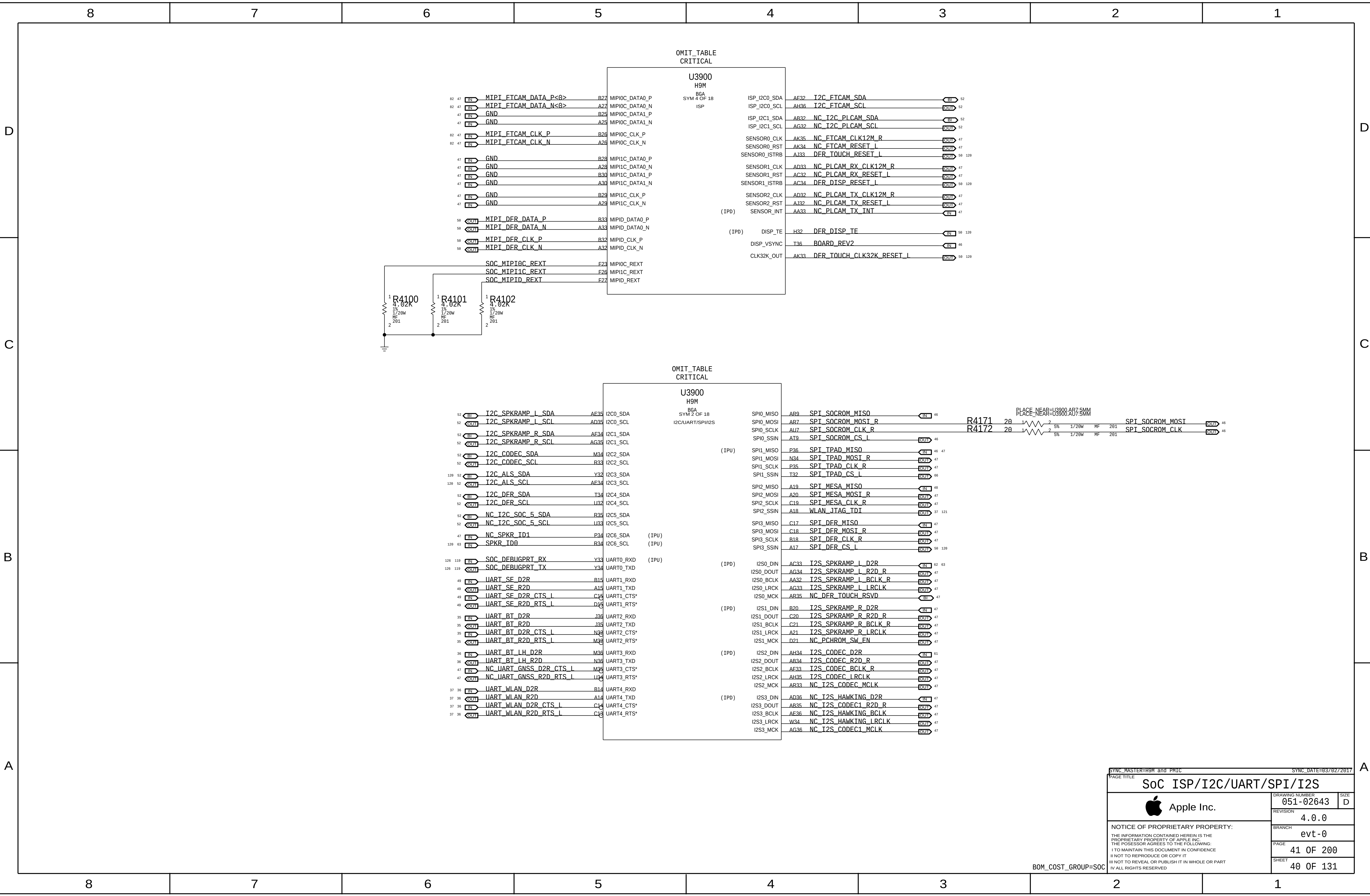
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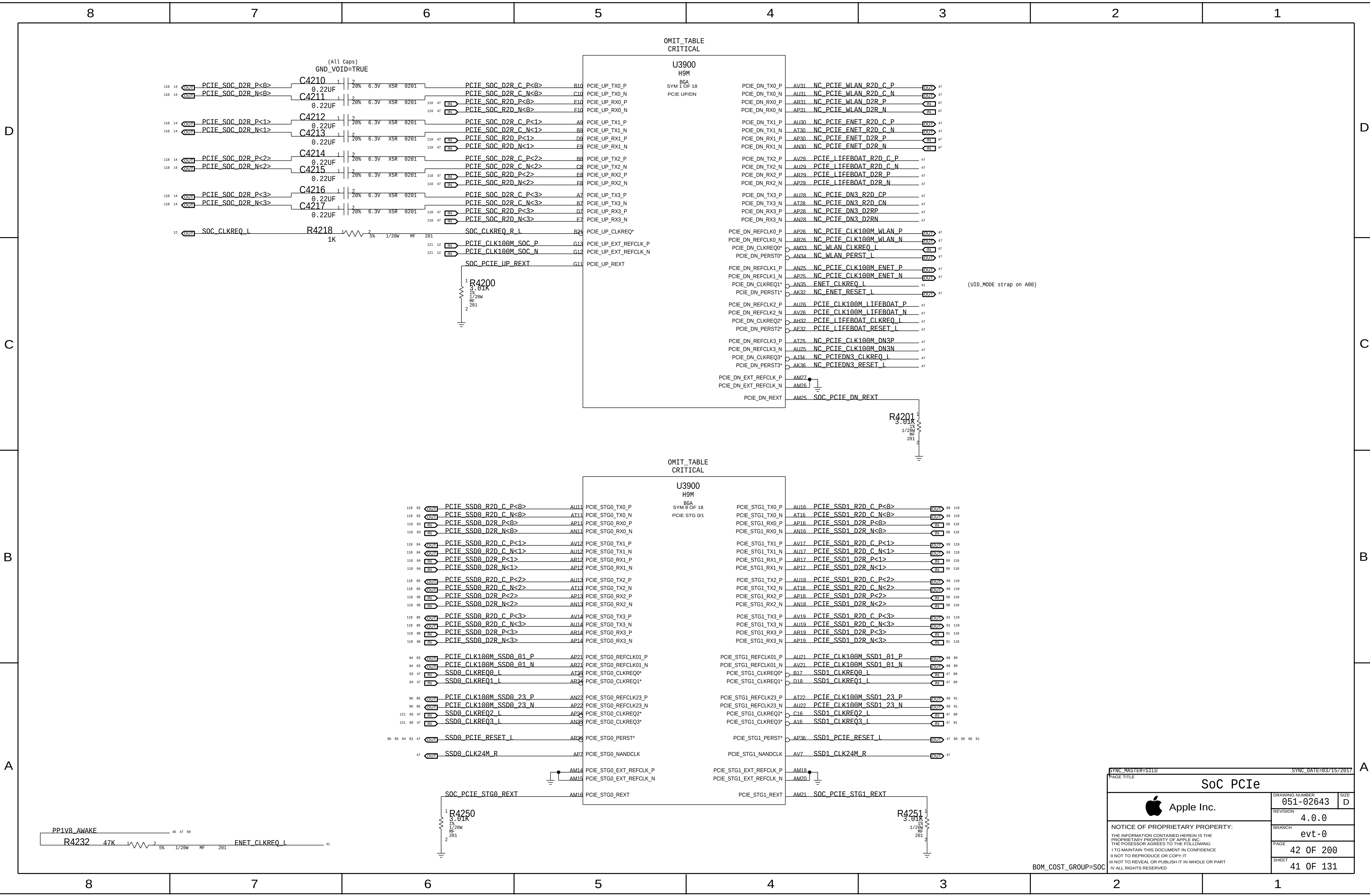
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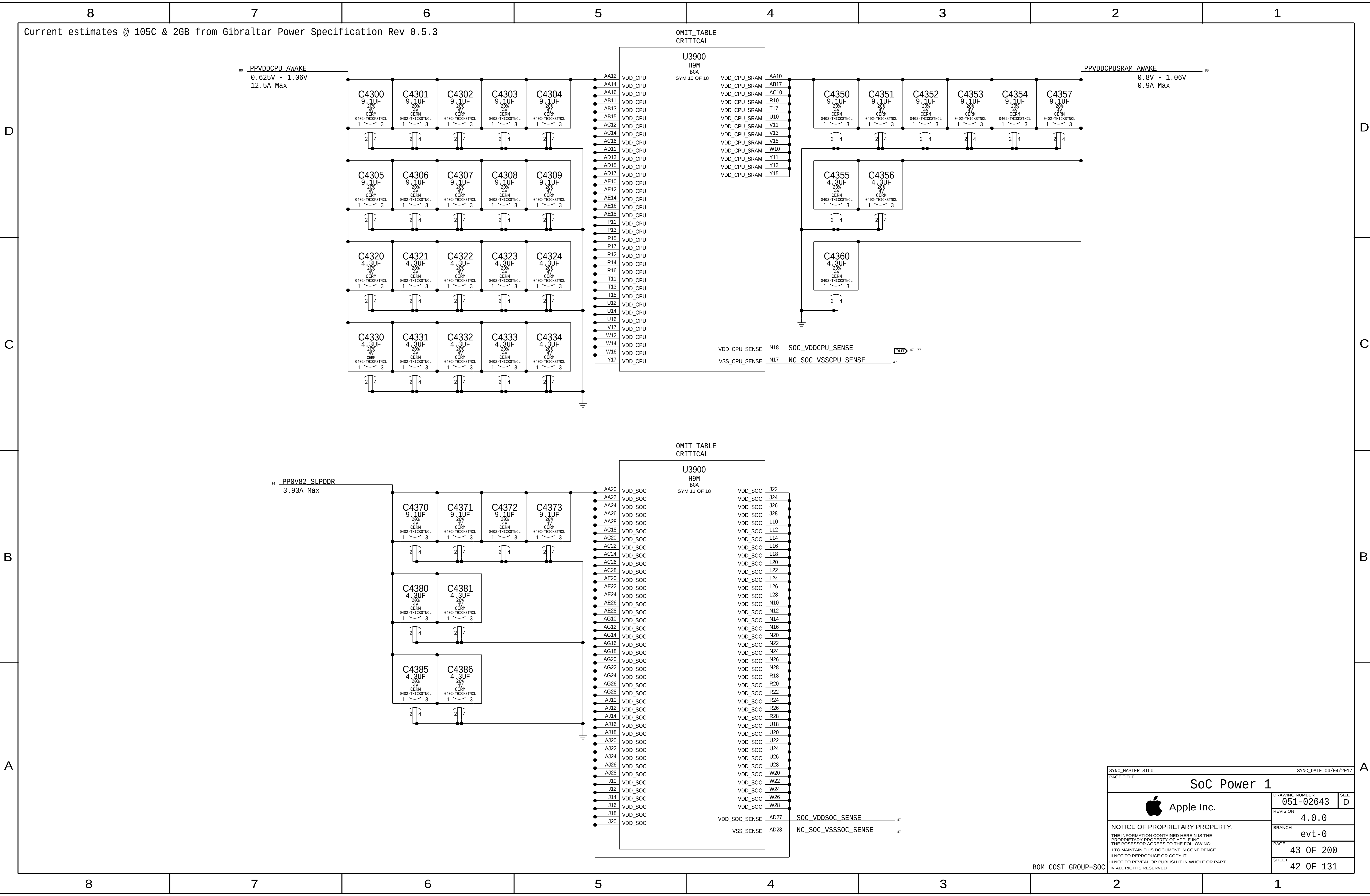
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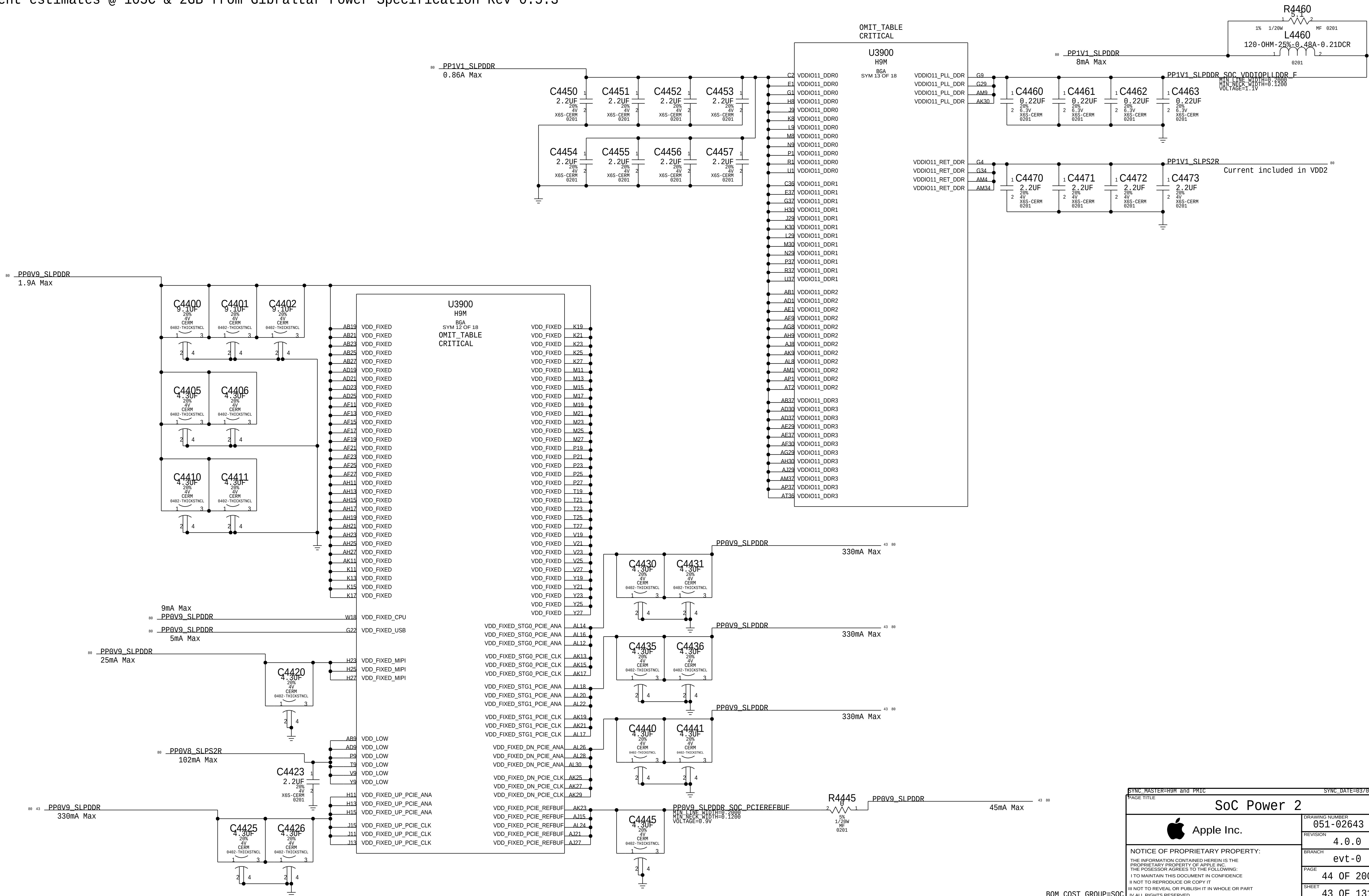
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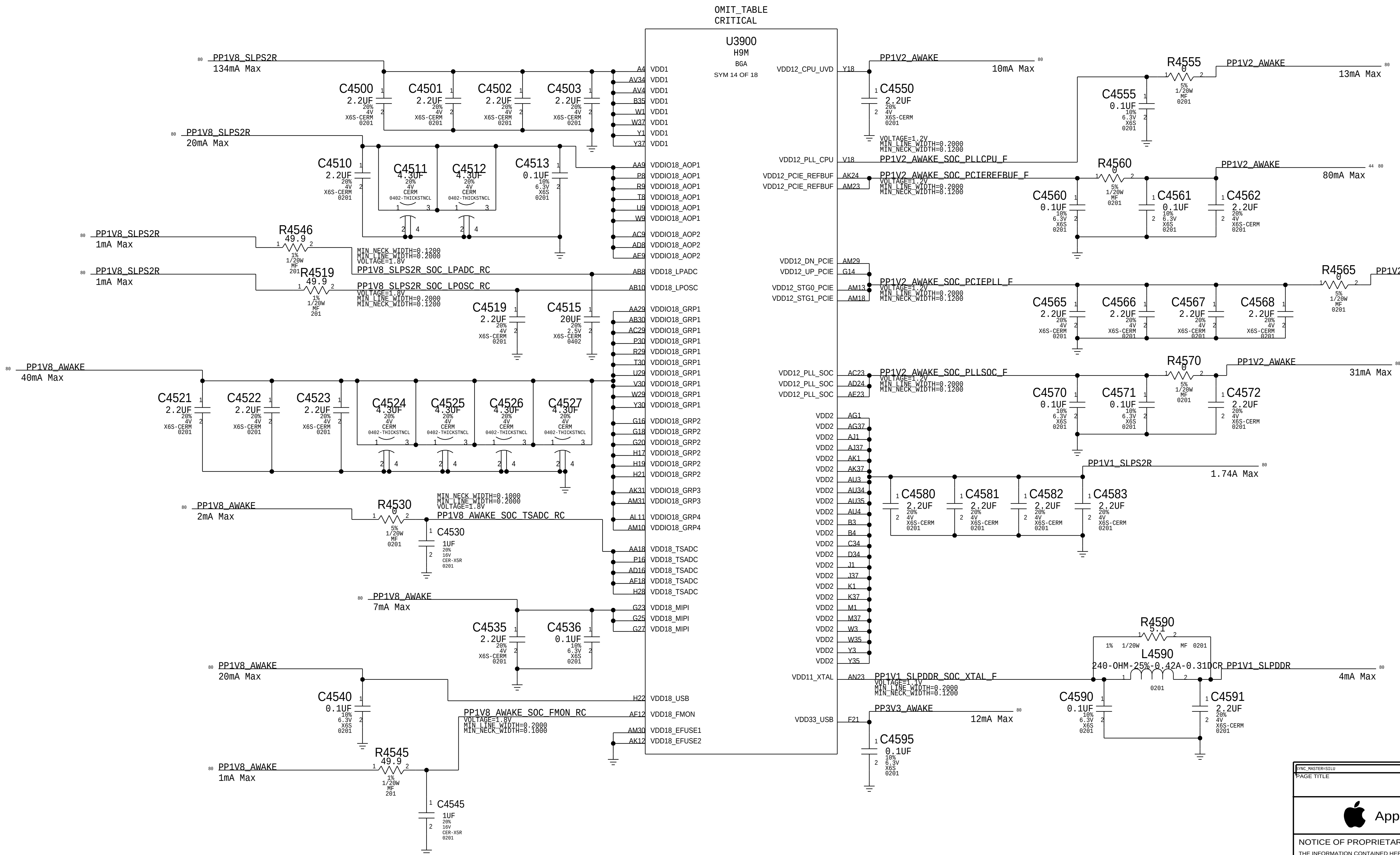


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Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3

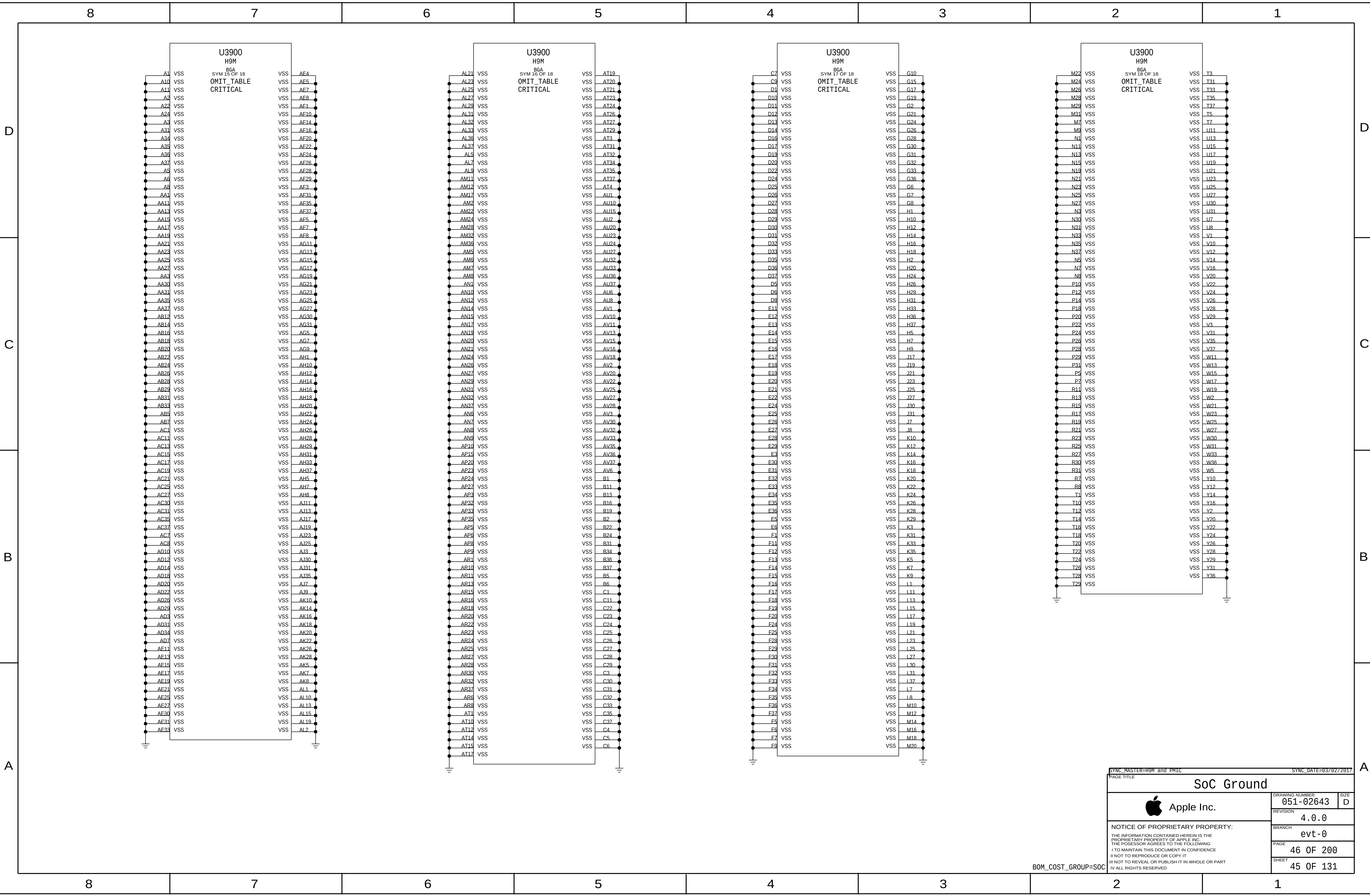


Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3



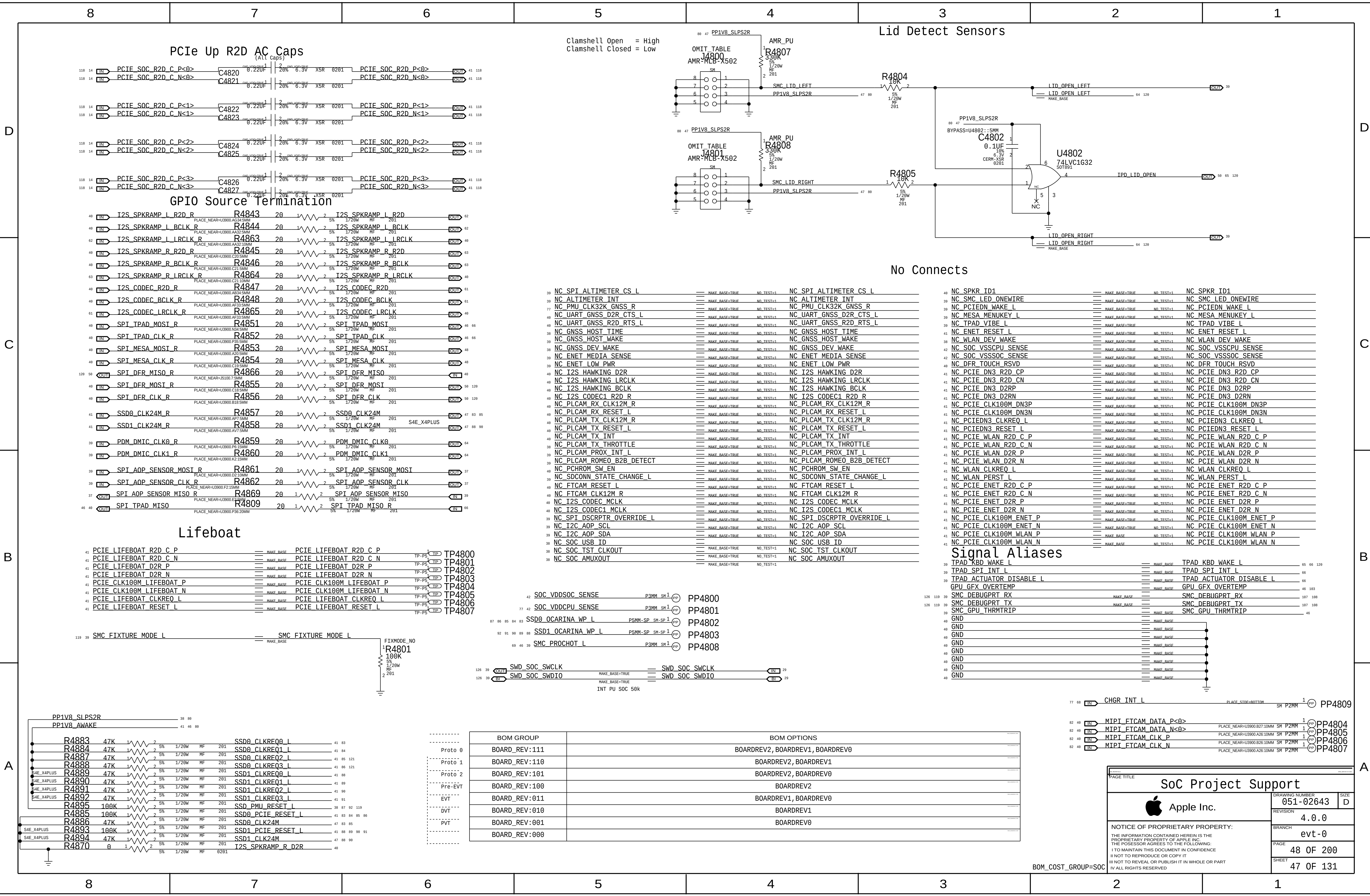
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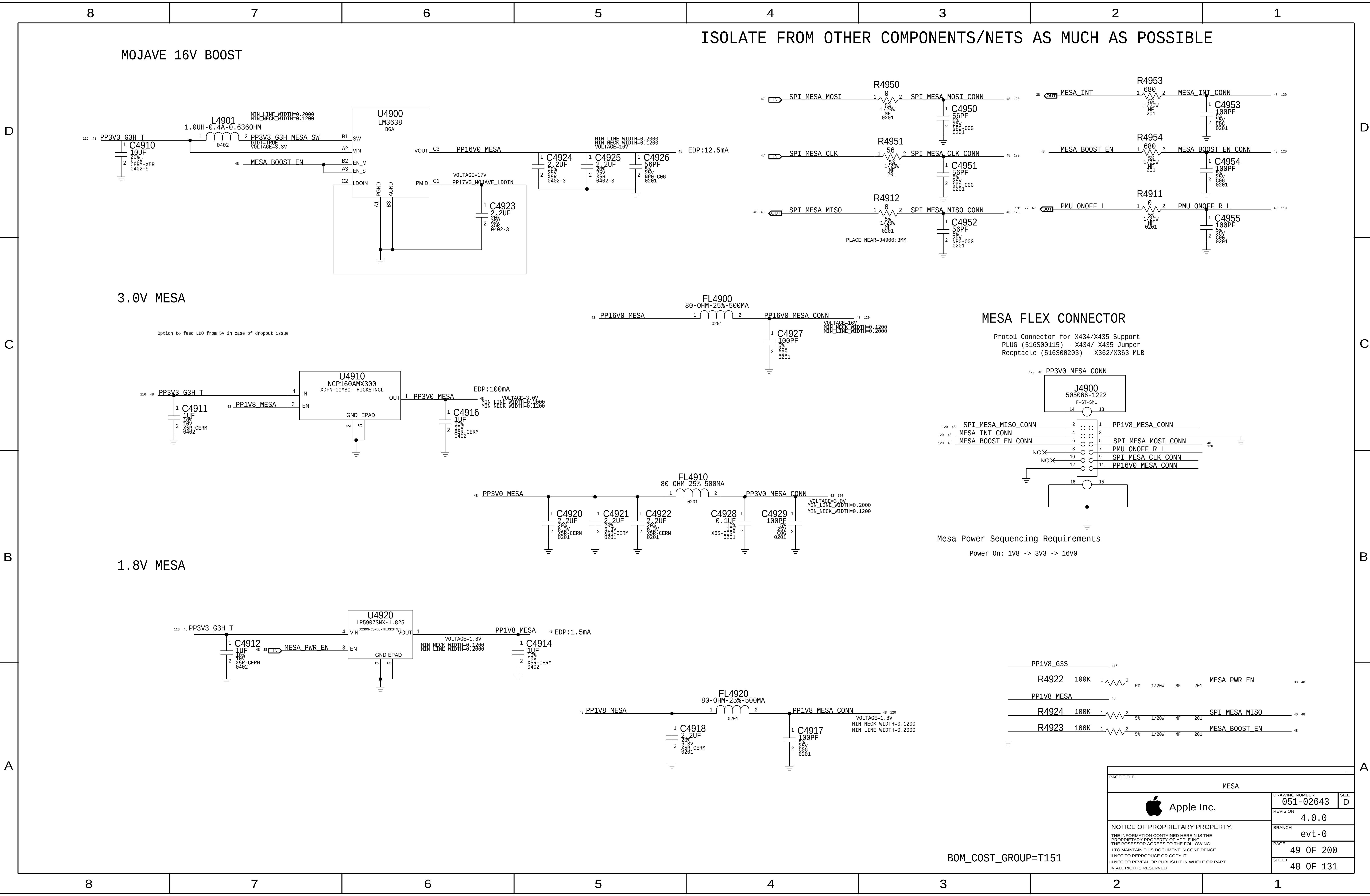
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	REVISION		4.0.0	
	BRANCH		evt-0	
	PAGE		45 OF 200	
	SHEET		44 OF 131	
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		BRANCH	evt-0
		PAGE	46 OF 200
		SHEET	45 OF 131

BOM_COST_GROUP=SOIC





MOJAVE 16V BOOST

3.0V MESA

1.8V MESA

ISOLATE FROM OTHER COMPONENTS/NETS AS MUCH AS POSSIBLE

MESA FLEX CONNECTOR

Proto1 Connector for X434/X435 Support
PLUG (516S00115) - X434/ X435 Jumper
Recptacle (516S00203) - X362/X363 MLB

Mesa Power Sequencing Requirements

Power On: 1V8 -> 3V3 -> 16V0

PAGE TITLE			
MESA			
 Apple Inc.	DRAWING NUMBER		051-02643
	REVISION		4.0.0
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		PAGE	49 OF 200
		SHEET	48 OF 131

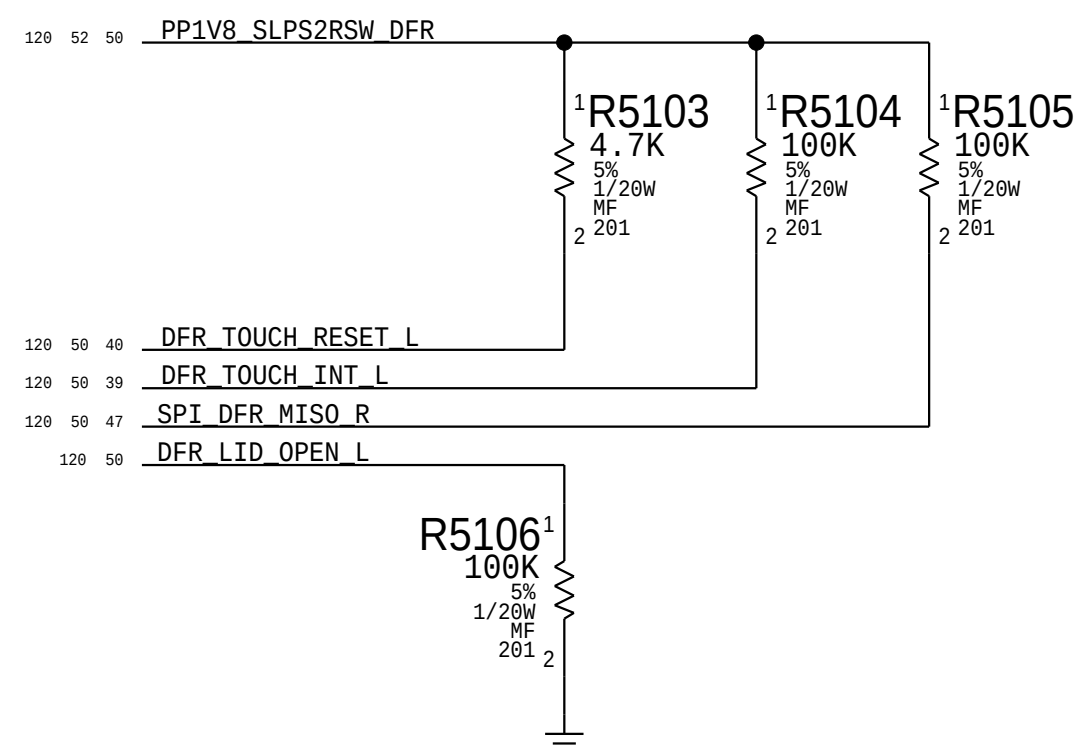
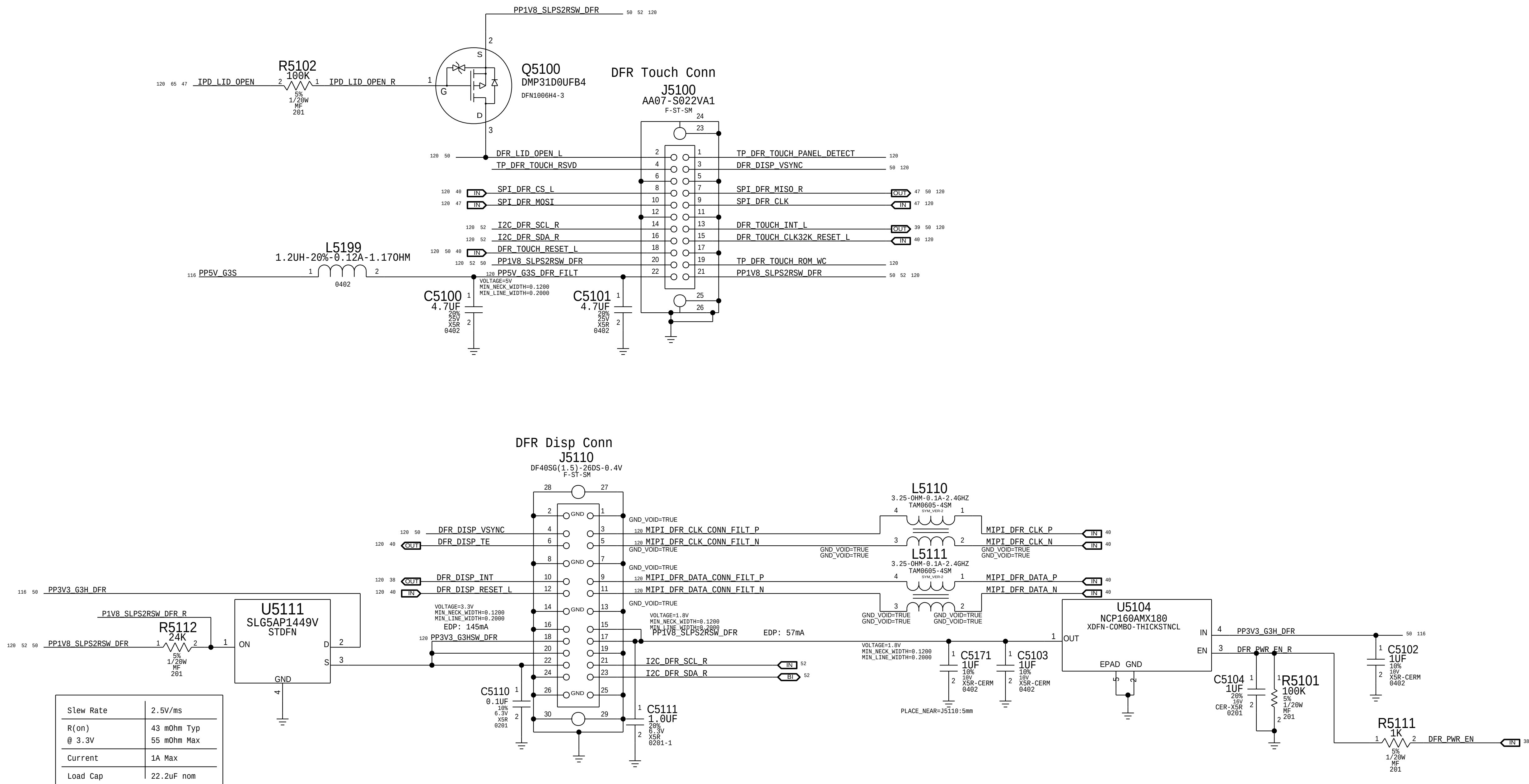
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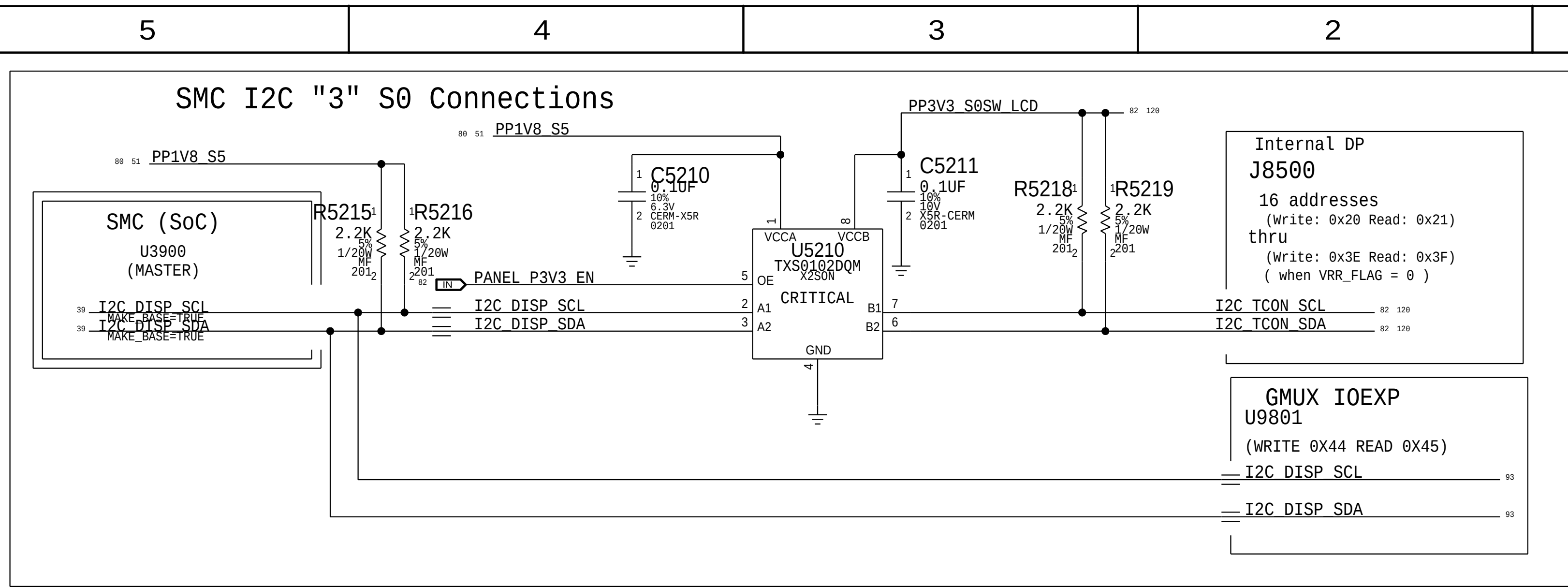
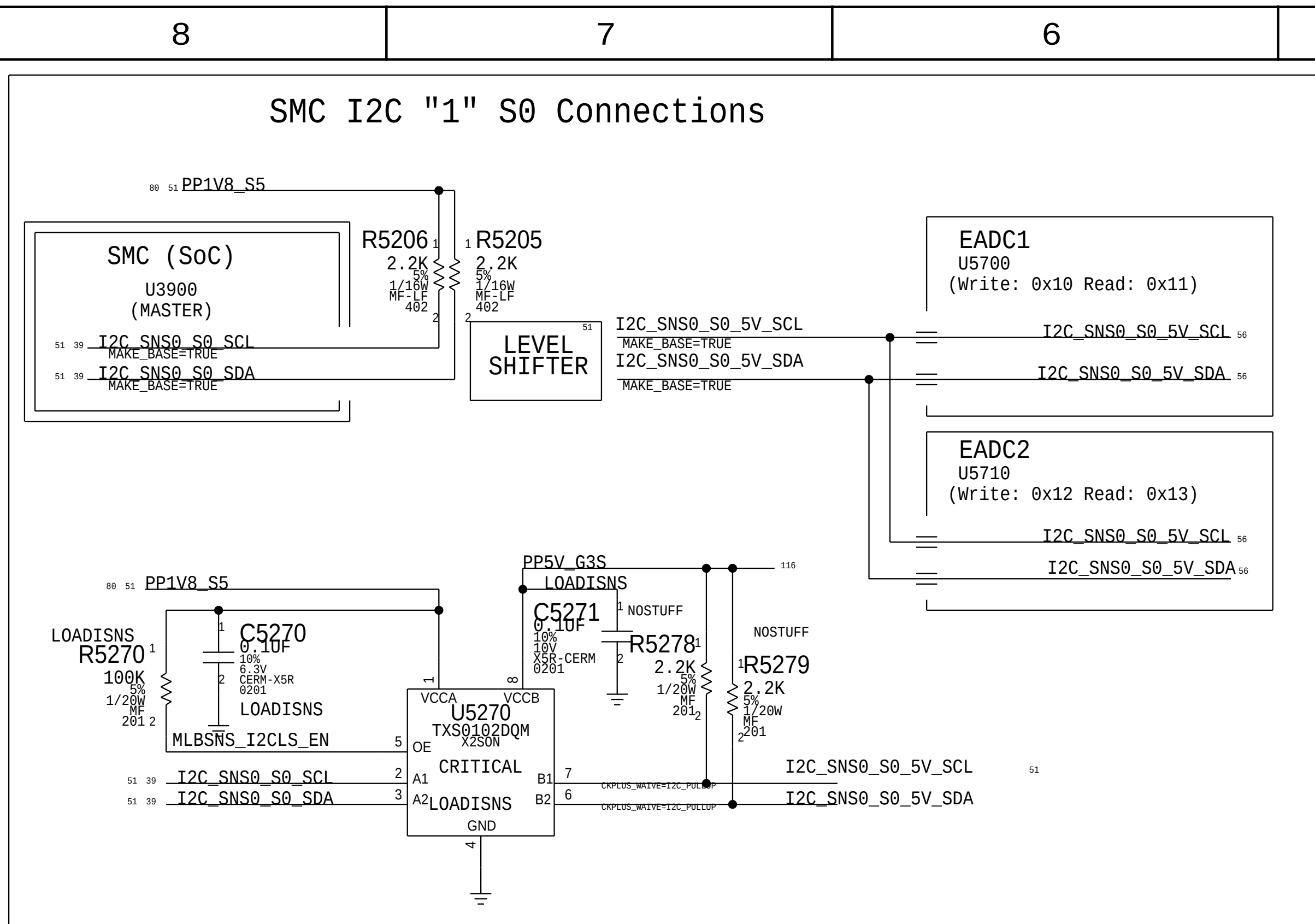
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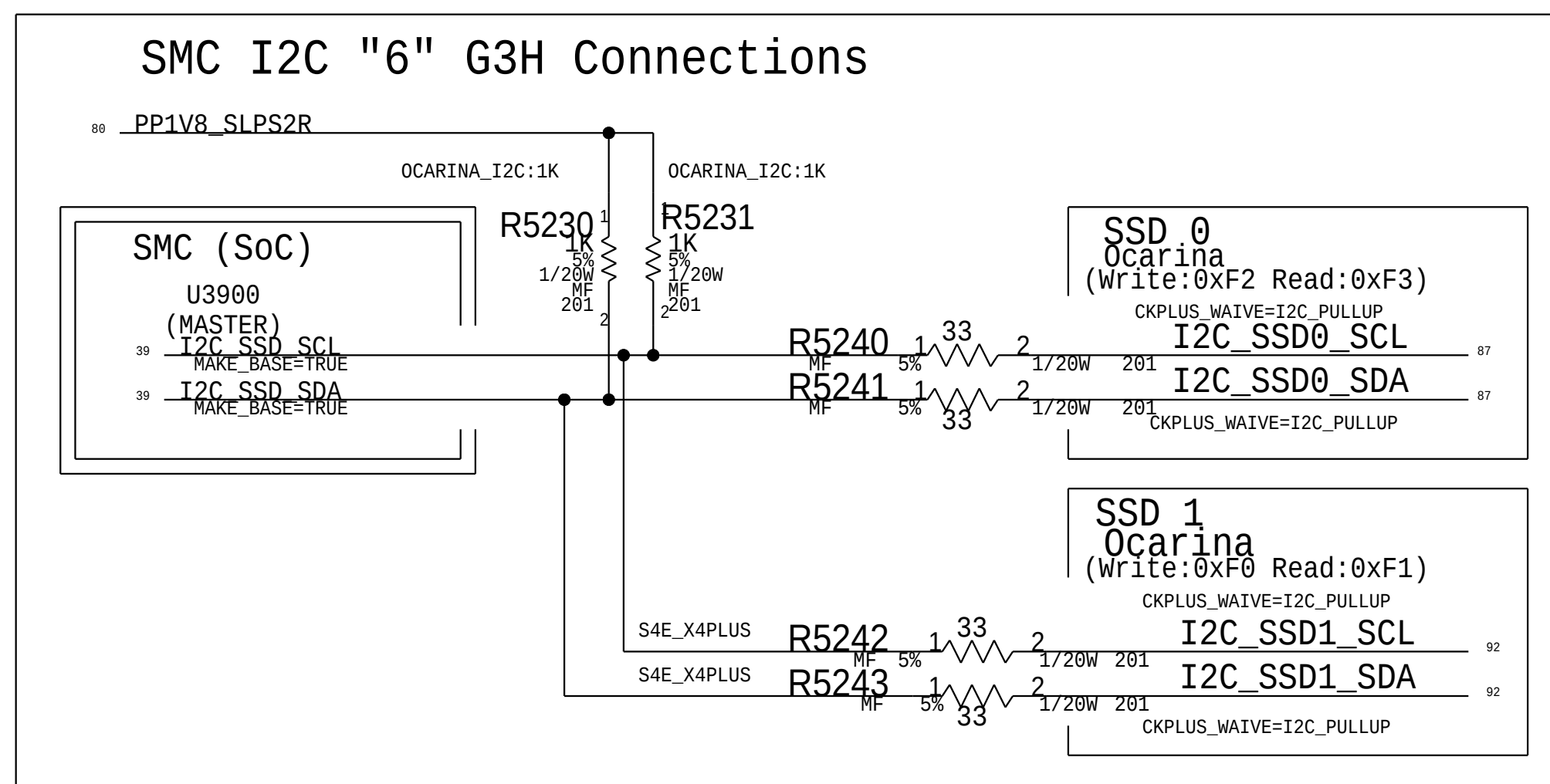
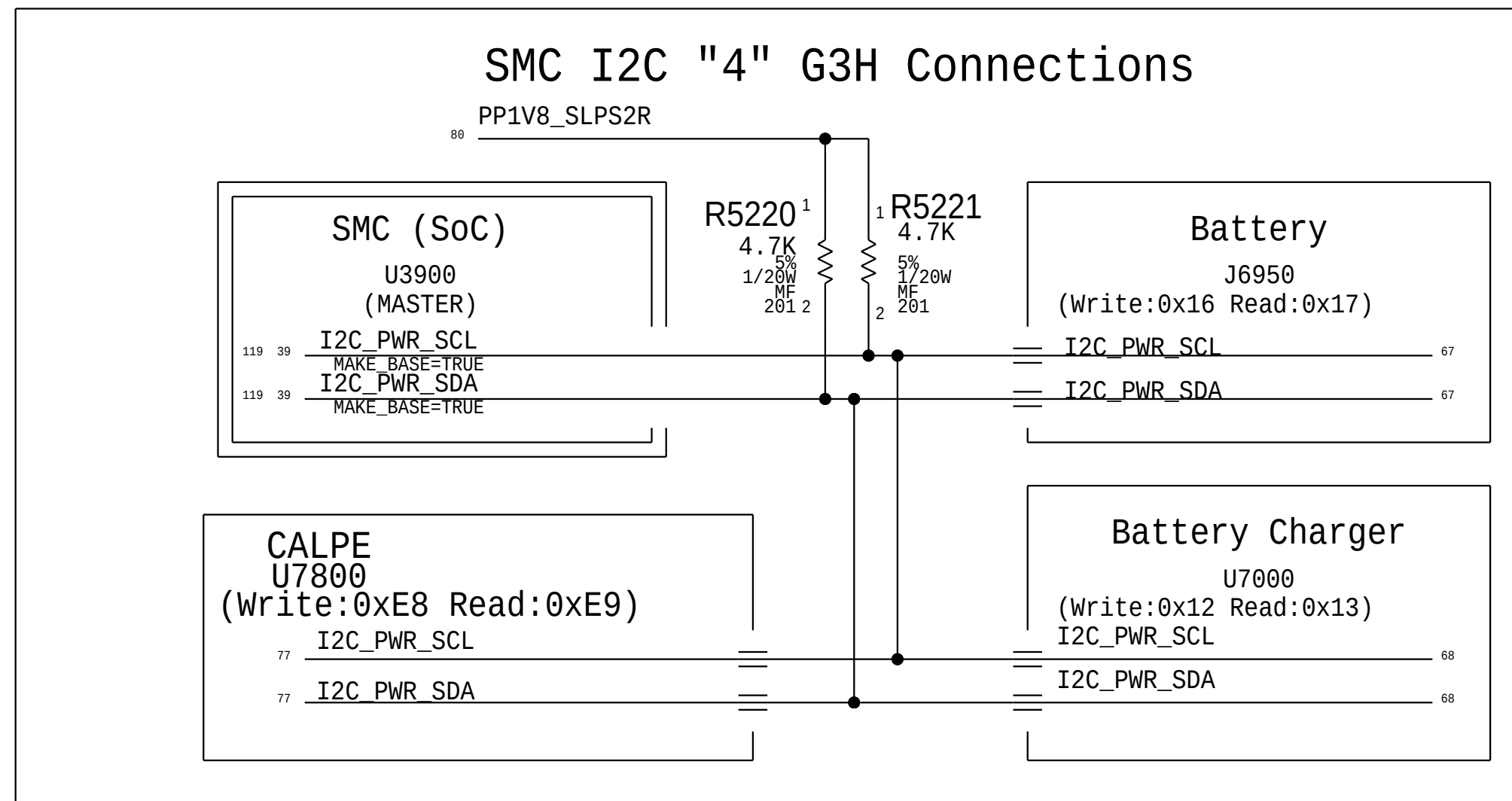
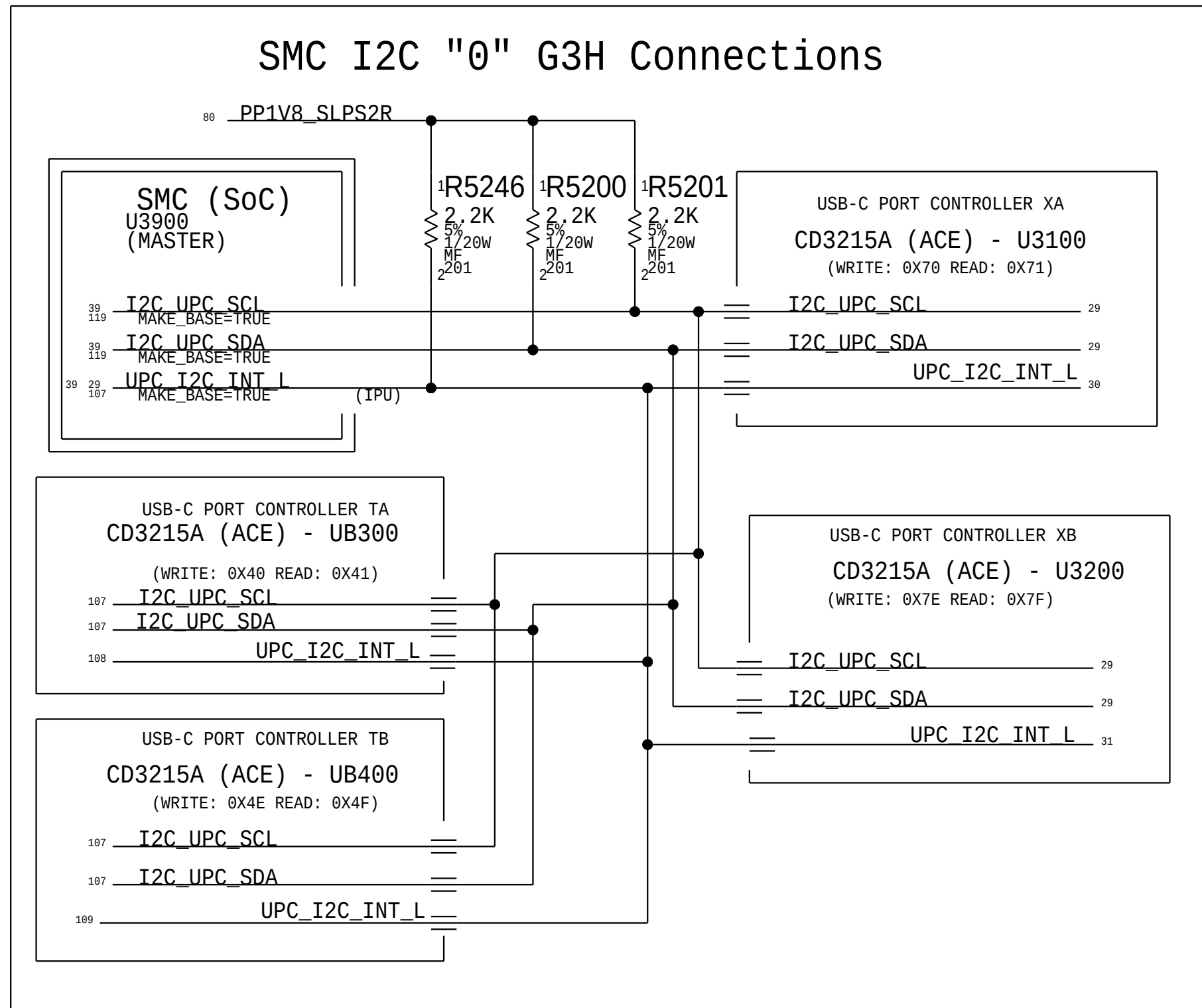
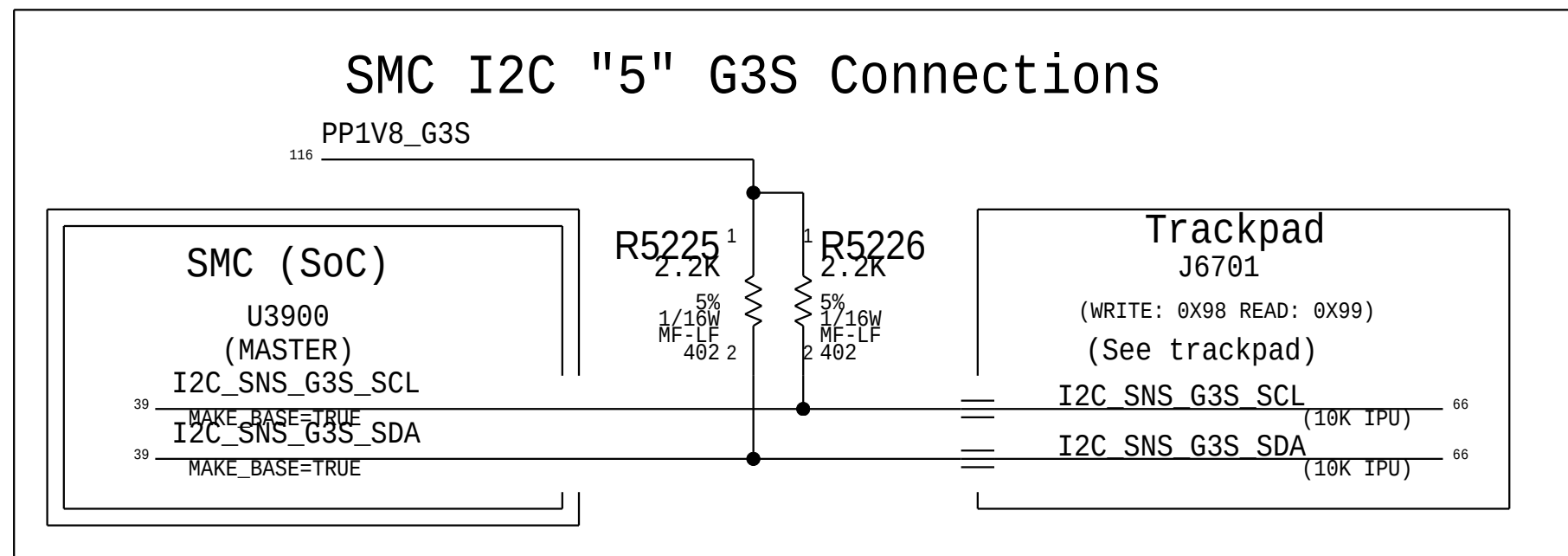
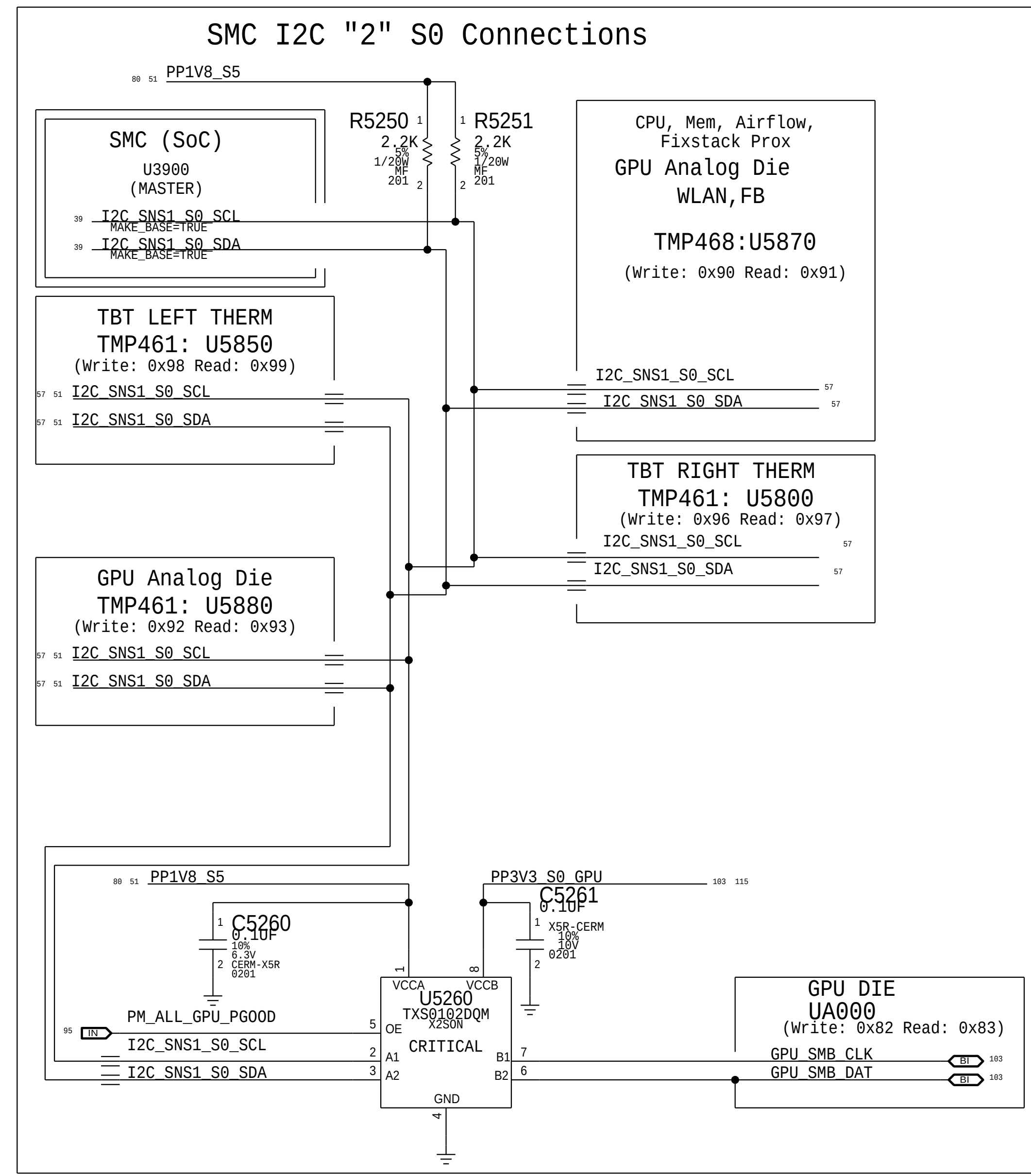
T208 Support

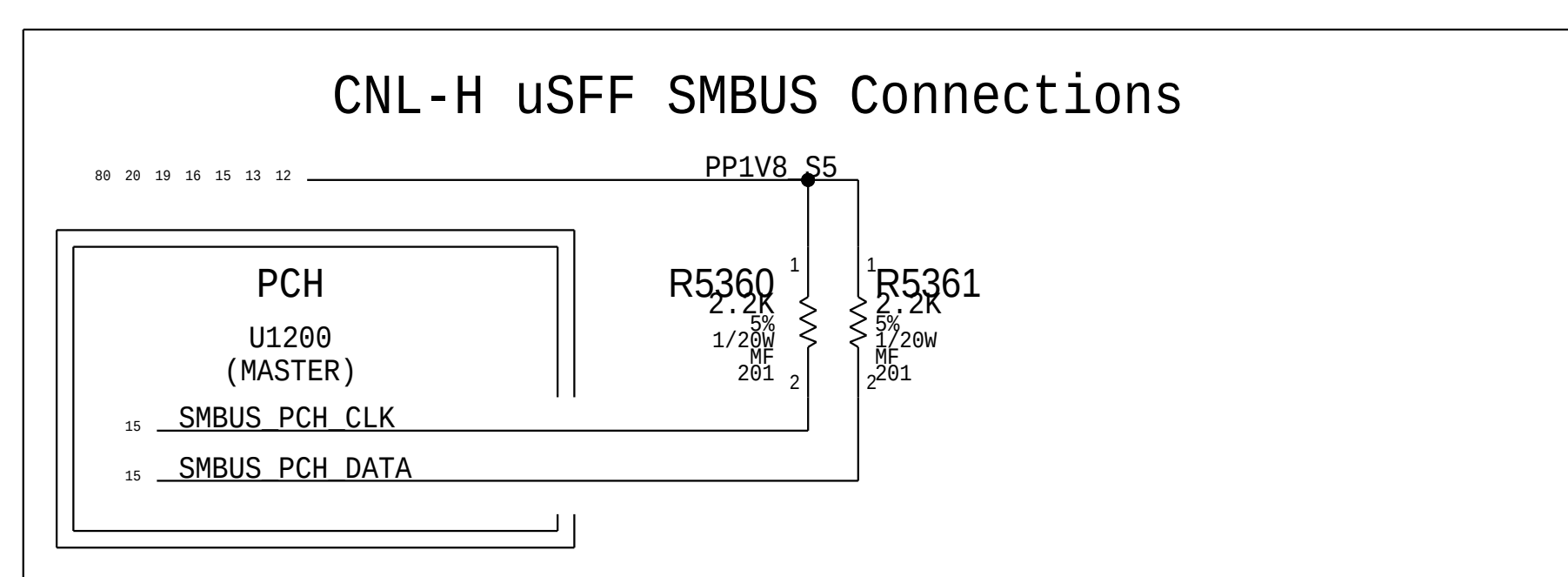
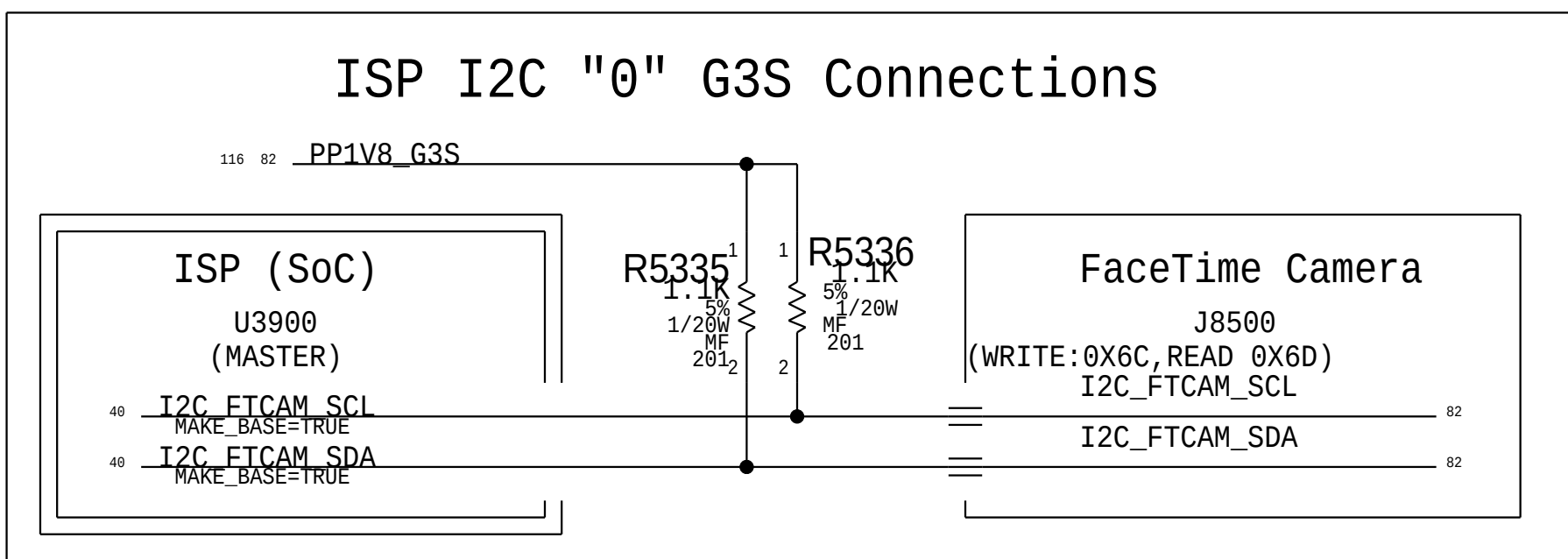
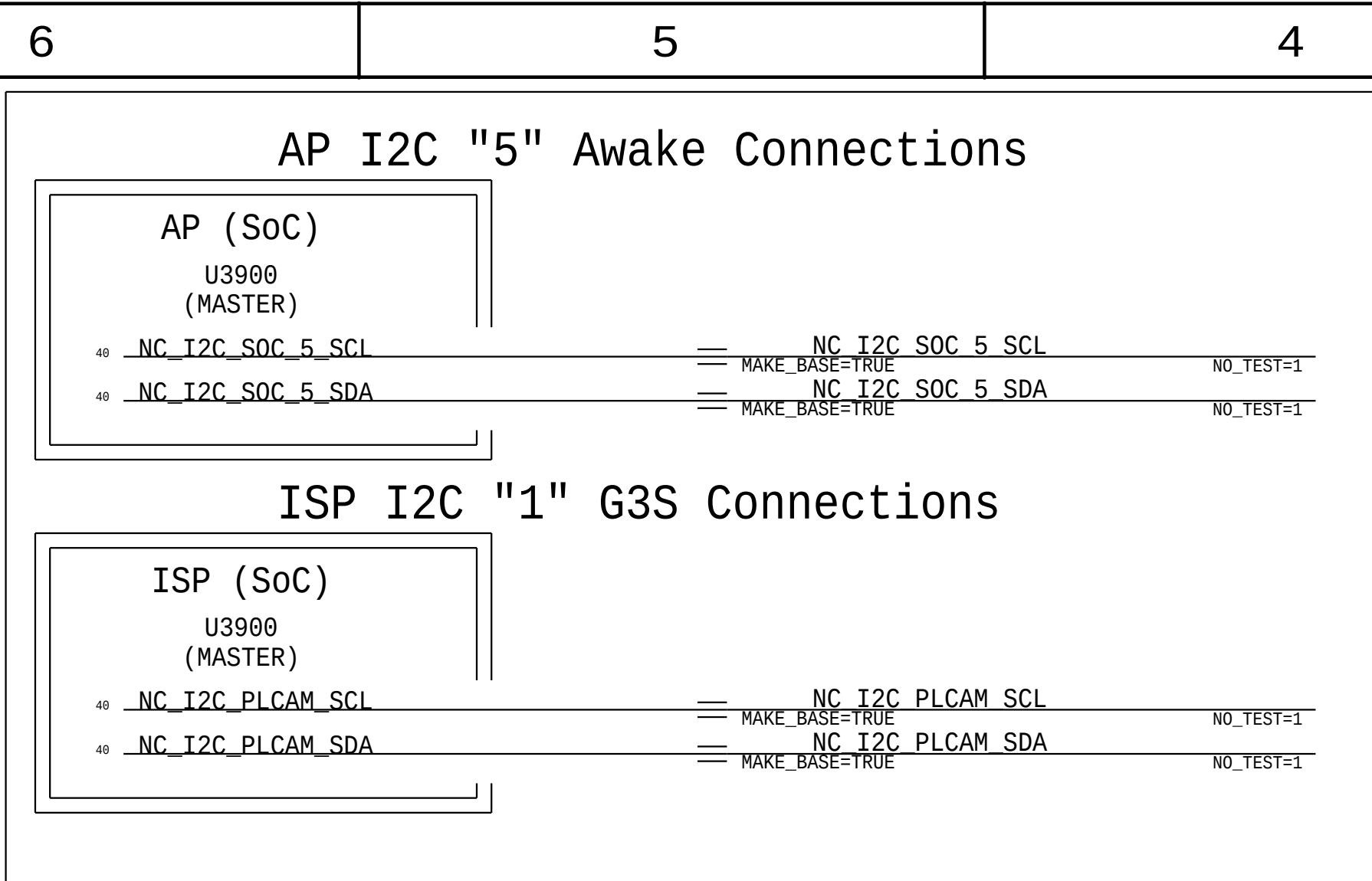
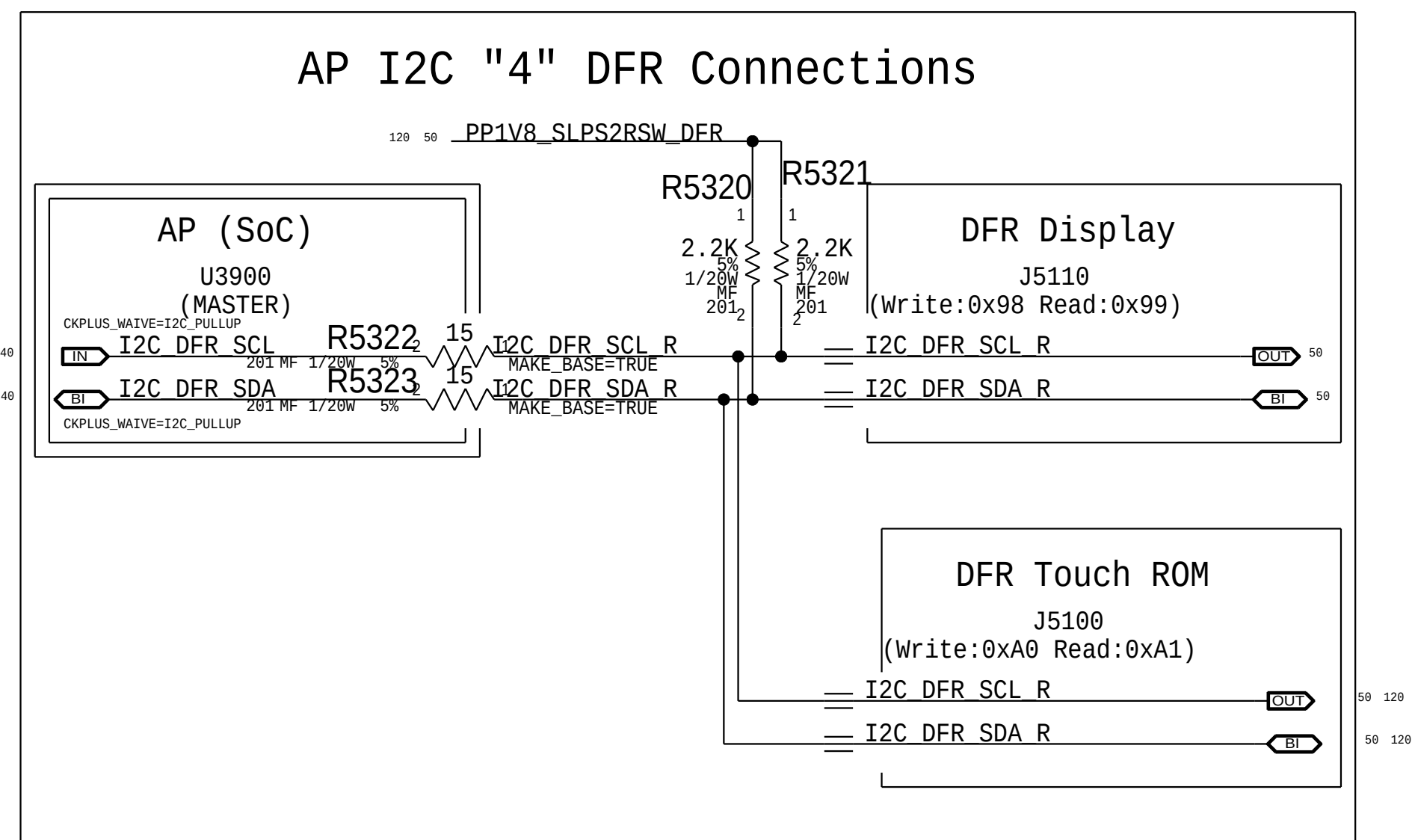
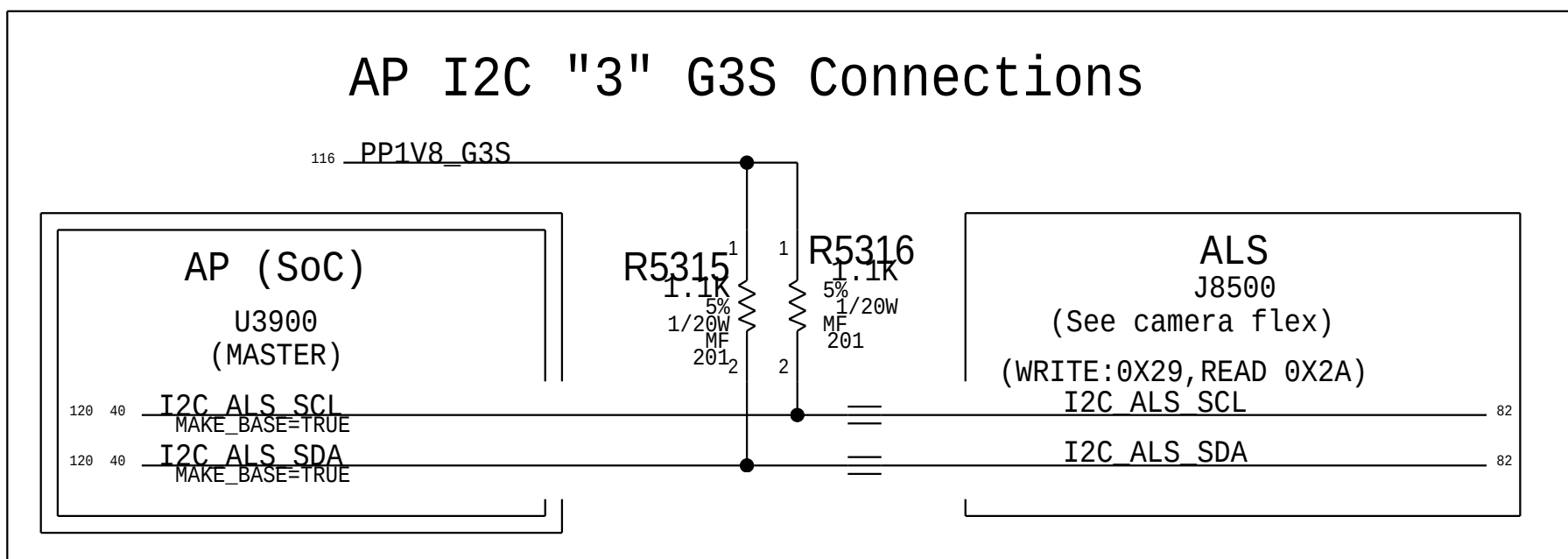
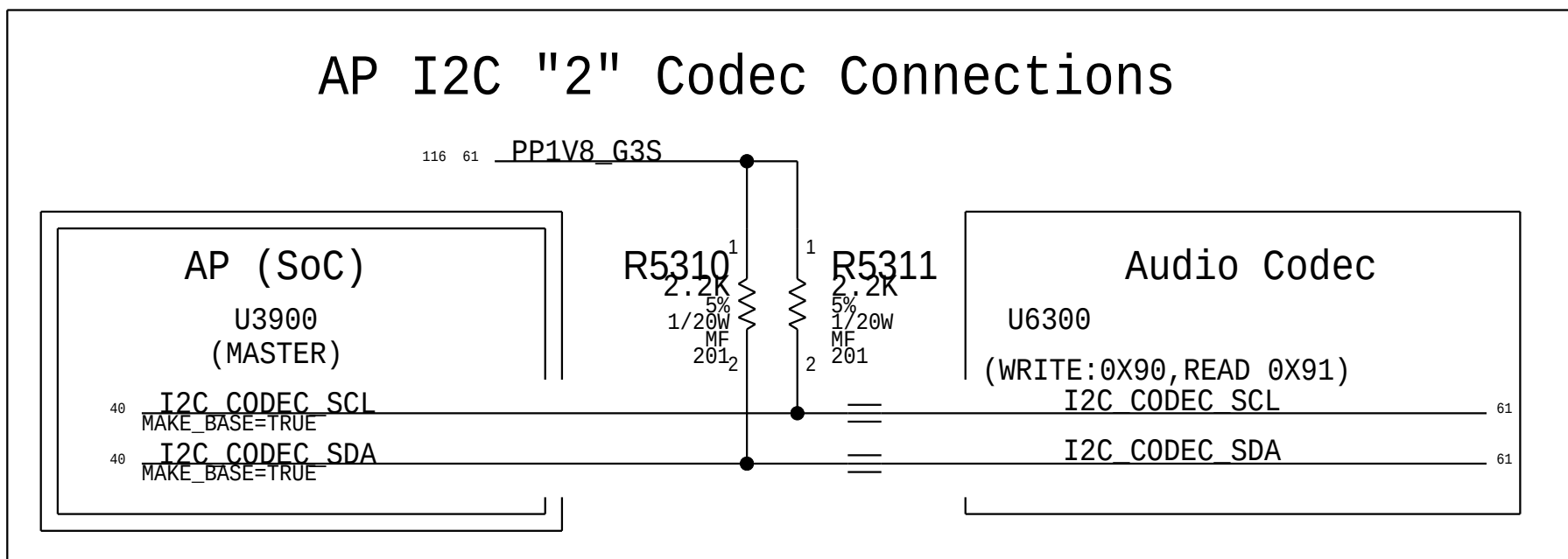
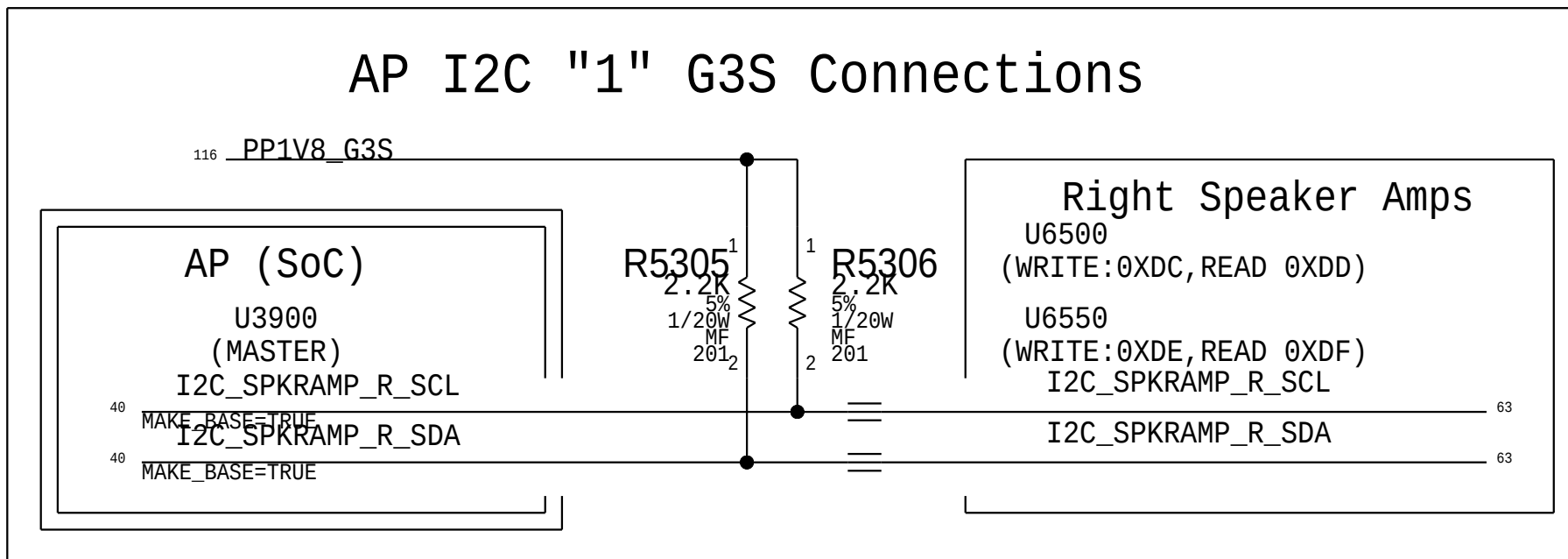
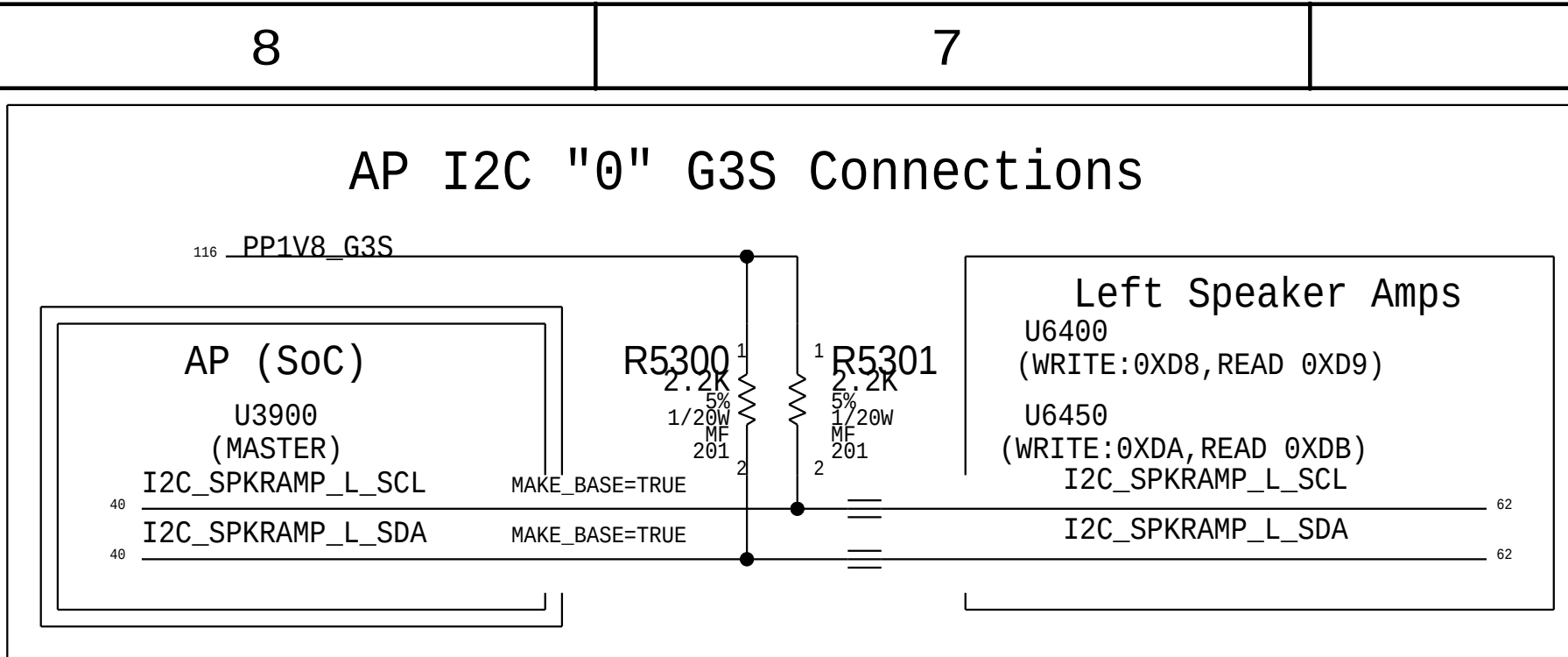


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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0059	2	RES, 2. 2K, 1/20W, 0201, SMD, MF	R5230, R5231		OCARINA_I2C:2.2K





2		1
I2C Device Address		
Device	SMC IF	ADDR. (8b)
ACE XA	I2C0	0X70/1
ACE XB	I2C0	0X7E/F
ACE TA	I2C0	0X40/1
ACE TB	I2C0	0X4E/F
EADC1	I2C1	0X10/1
EADC2	I2C1	0X12/3
Temp. Sensor Left	I2C2	0X98/9
Temp. Sensor Right	I2C2	0X96/7
CPU, MEM, WLAN Thermal	I2C2	0X90/1
GPU Analog Die Thermal	I2C2	0X92/3
GPU Digital Die Thermal	I2C2	0X82/3
TCON	I2C3	0X20-3F
GMUX IOEXP	I2C3	0X44/5
Charger	I2C4	0X12/3
Battery	I2C4	0X16/7
Calpe	I2C4	0XE8/9
Trackpad	I2C5	0X98/9
SSD0	I2C6	0XF2/3
SSD1	I2C6	0XF0/1
SoC IF		
Left Spkr Amp. (U6400)	I2C0	0XD8/9
Left Spkr Amp. (U6450)	I2C0	0XDA/B
Right Spkr Amp. (U6500)	I2C1	0XDC/D
Right Spkr Amp. (U6550)	I2C1	0XDE/F
Audio Codec	I2C2	0X90/1
ALS	I2C3	0X29/A
DFR Display	I2C4	0X98/9
DFR Touch	I2C4	0XA0/1
NC.	I2C5	
NC. Spkr ID1	I2C6_SDA	
Spkr ID0	I2C6_SCL	
ISP IF		
FT Camera	I2C0	0X6C/D
NC.	I2C1	
AOP IF		
NC.	I2C0	
PCH IF		
NC.	PULL-UP	

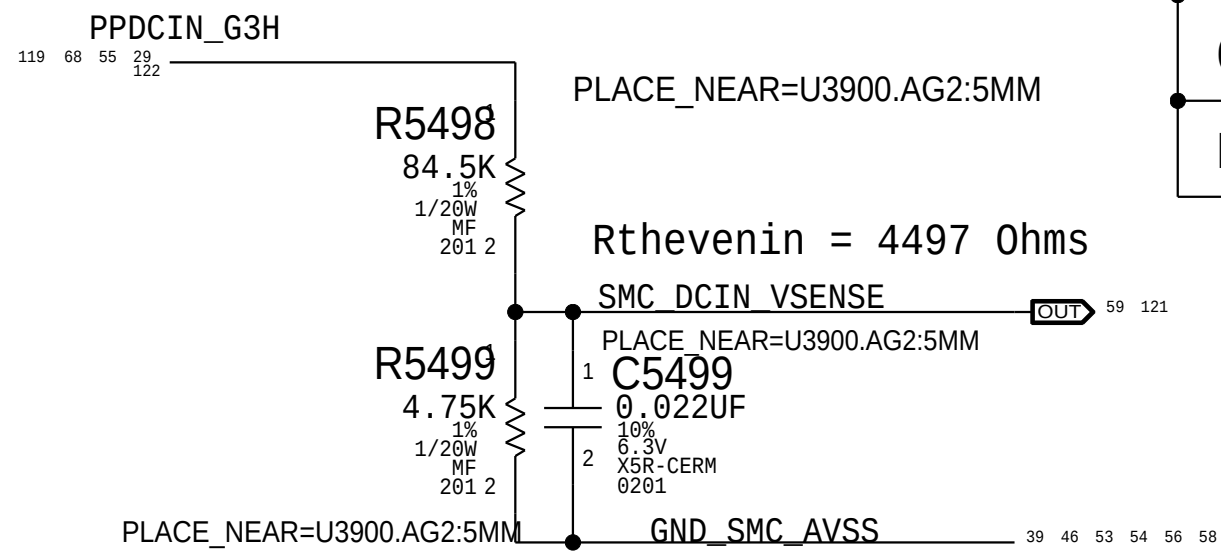
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	REVISION		
	4.0.0		
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	53	OF	200
	SHEET		
	52	OF	131

BOM_COST_GROUP=SMC

DC-IN Voltage Sense (VD0R)

Gain: 0.05322x
Vnominal: 20 V, Range: 23.49 V
SMC ADC: 00
Enables DC-In VSense divider when AC present.
2.2KHz

D

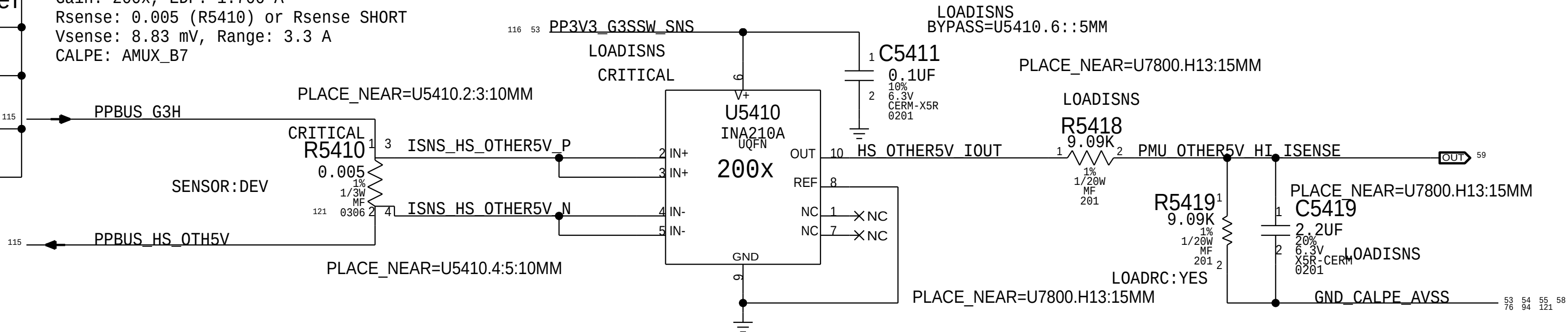


J680 SENSOR SETTINGS

CHIP	Vref(V)	Vmax	SMC sample Freq.	ADR RC filter
H9M	1.25	1.8	10khz	0.1ms
CALPE	1.5	5	100hz	10ms
EADC	2.5	5	1-2hz(10hz)	100ms

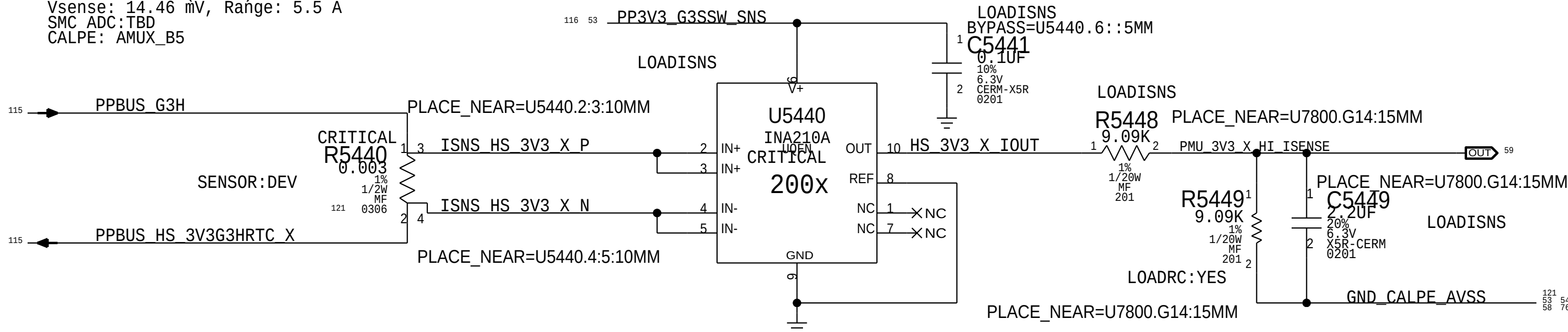
OTHER 5V High Side Current Sense (I05R)

Gain: 200x, EDP: 1.766 A
Rsense: 0.005 (R5410) or Rsense SHORT
Vsense: 8.83 mV, Range: 3.3 A
CALPE: AMUX_B7



LEFT SIDE 3.3V High Side Current Sense (I0LR)

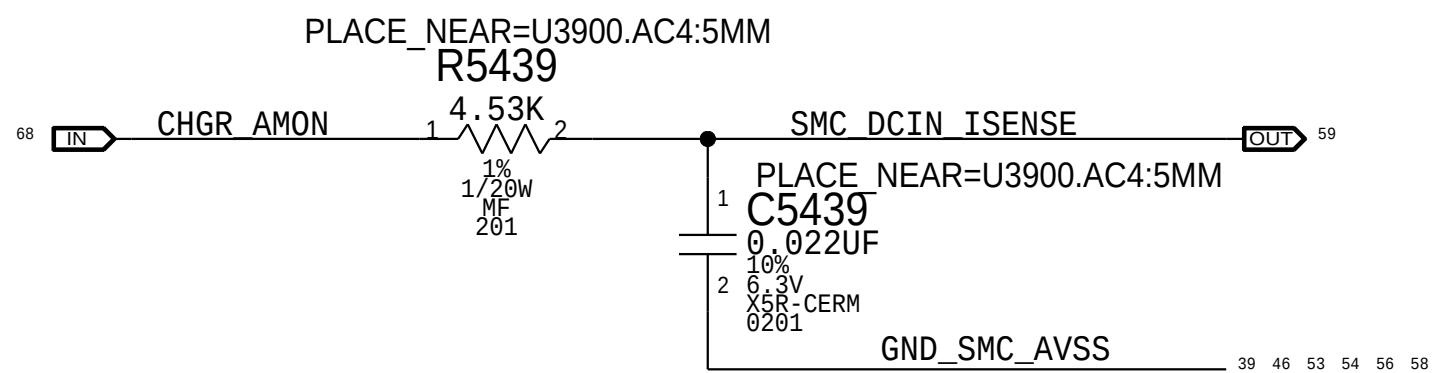
Gain: 200x, EDP: 4.82 A
Rsense: 0.003 (R5440) or Rsense SHORT
Vsense: 14.46 mV, Range: 5.5 A
SMC ADC: TBD
CALPE: AMUX_B5



DC-IN Current Sense (ID0R)

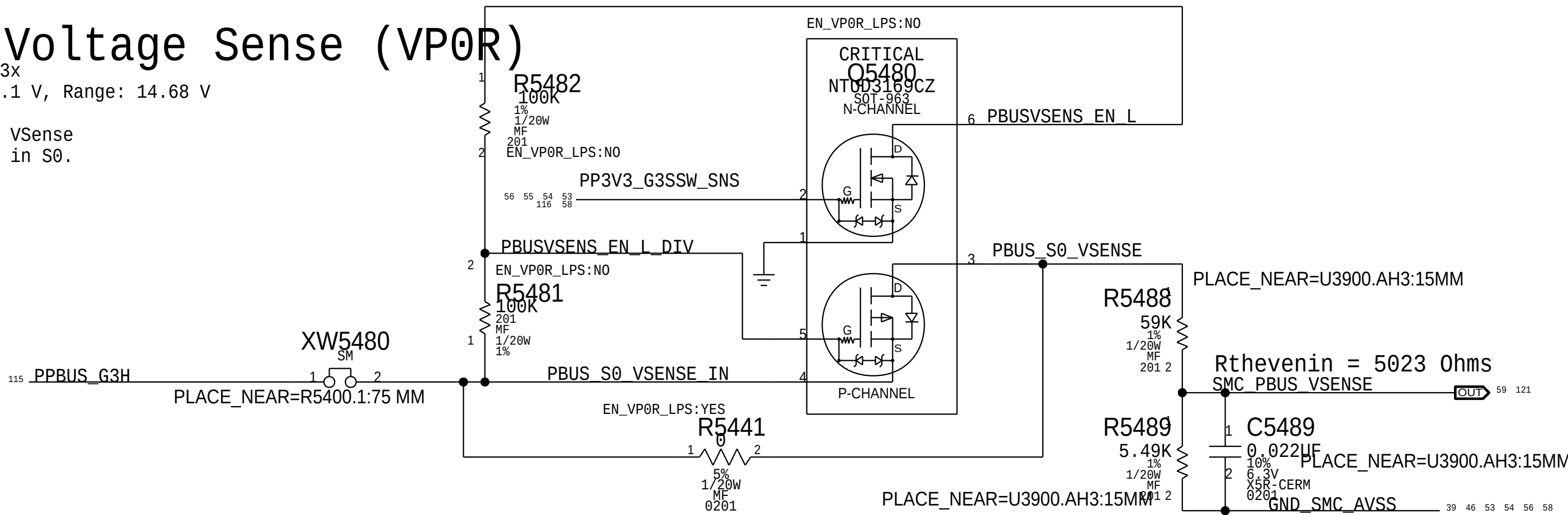
DISCharger Gain: 20x, EDP: 4.6 A
Rsense: 0.010 (R7020)
SMC ADC: 01

C



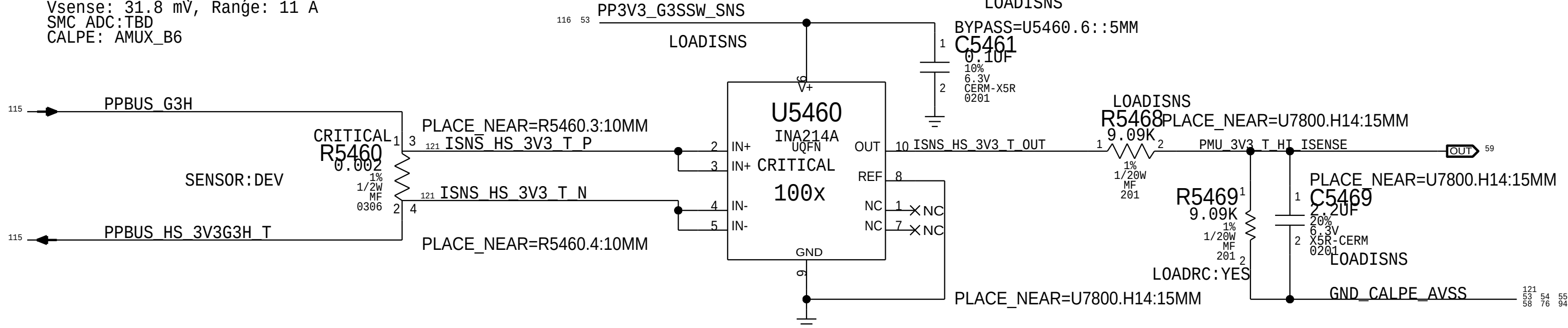
PBUS Voltage Sense (VP0R)

Gain: 0.08513x
Vnominal: 13.1 V, Range: 14.68 V
SMC ADC: 02
Enables PBUS VSense divider when in S0.



RIGHT SIDE 3.3V High Side Current Sense (I0RR)

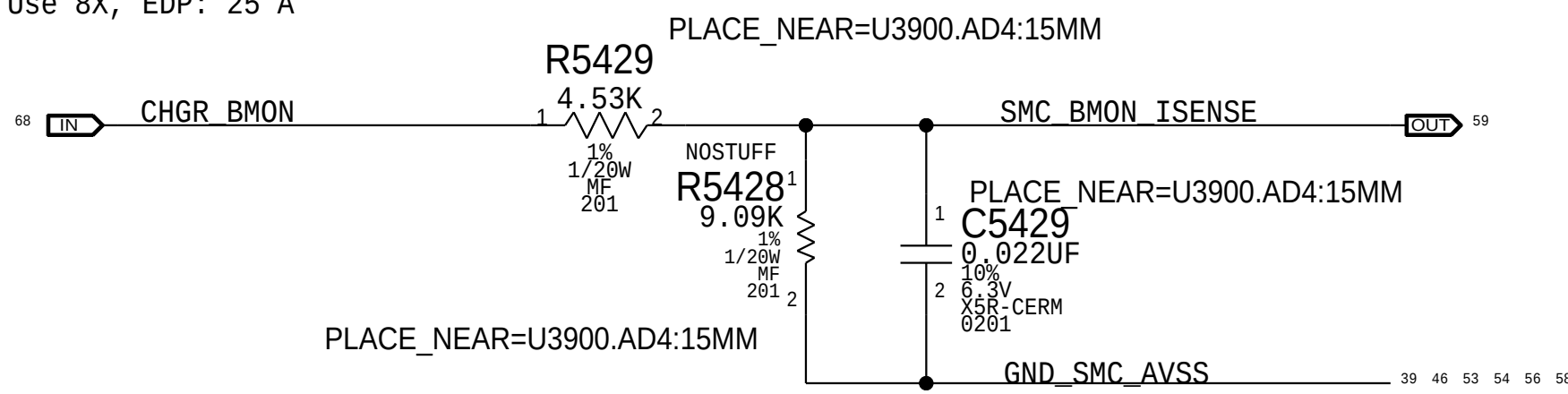
Gain: 100x, EDP: 10.6 A
Rsense: 0.003 (R5460) or Rsense SHORT
Vsense: 31.8 mV, Range: 11 A
SMC ADC: TBD
CALPE: AMUX_B6



Discharger BMON Current Sense (IPBR)

Charger Gain: 8X OR 64x, Use 8X, EDP: 25 A
Rsense: 0.005 (R7060)
SMC ADC: 03

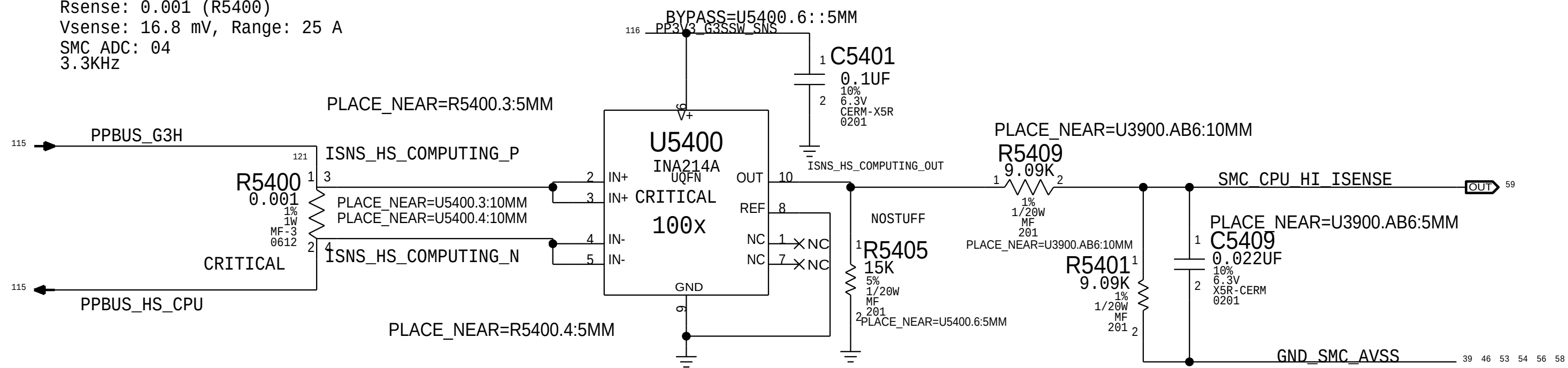
B



CPU High Side Curent Sense (IC0R)

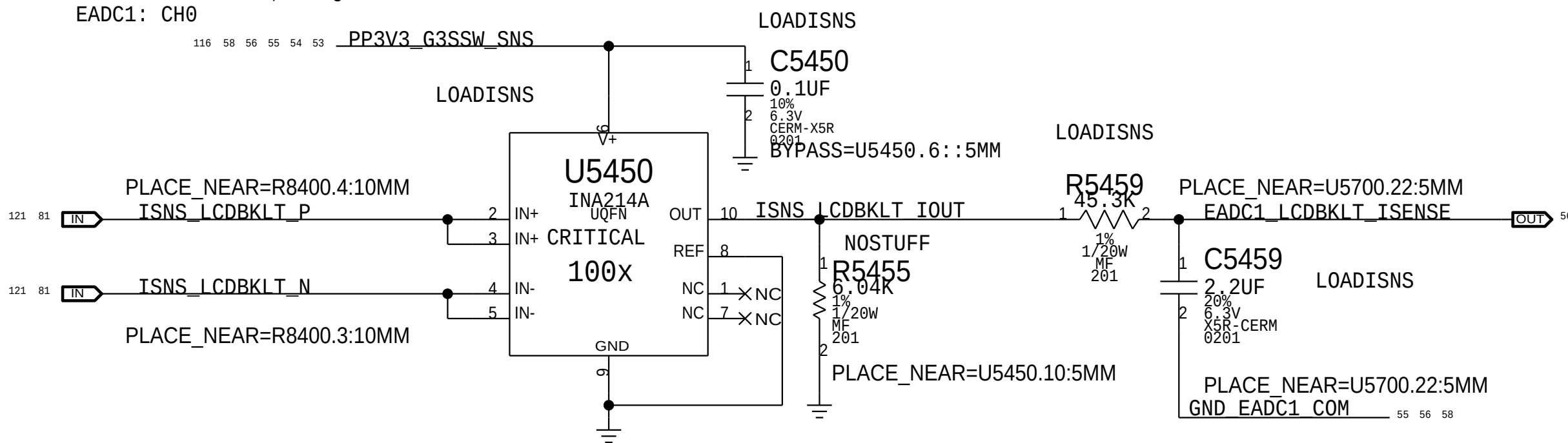
Gain: 100x, EDP: 16.8 A
Rsense: 0.001 (R5400)
Vsense: 16.8 mV, Range: 25 A
SMC ADC: 04
3.3KHz

A



LCD Backlight Current Sense (IBLR)

Gain: 100x, EDP: 0.87 A
Rsense: 0.025 (R8400)
Vsense: 21.75 mV, Range: 1.32 A
EADC1: CH0



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	3	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	R5449,R5469,R5419		LOADRC:NO

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	DRAWING NUMBER	051-02643
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		PAGE 54 OF 200
		SHEET 53 OF 131

BOM_COST_GROUP=SENSORS

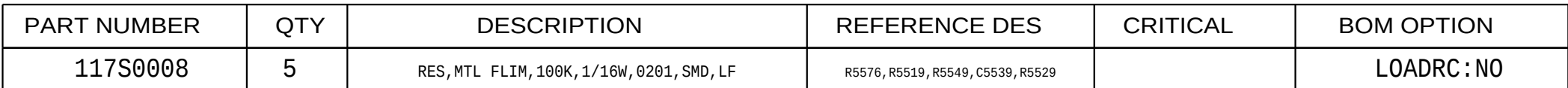
Gain: 200X, EDP: 4.7 A
Rsense: 0.001 (R5560) or XWTBD
Vsense: 14.1 mV, Range: 5.5 A
EADC2:CH1



C



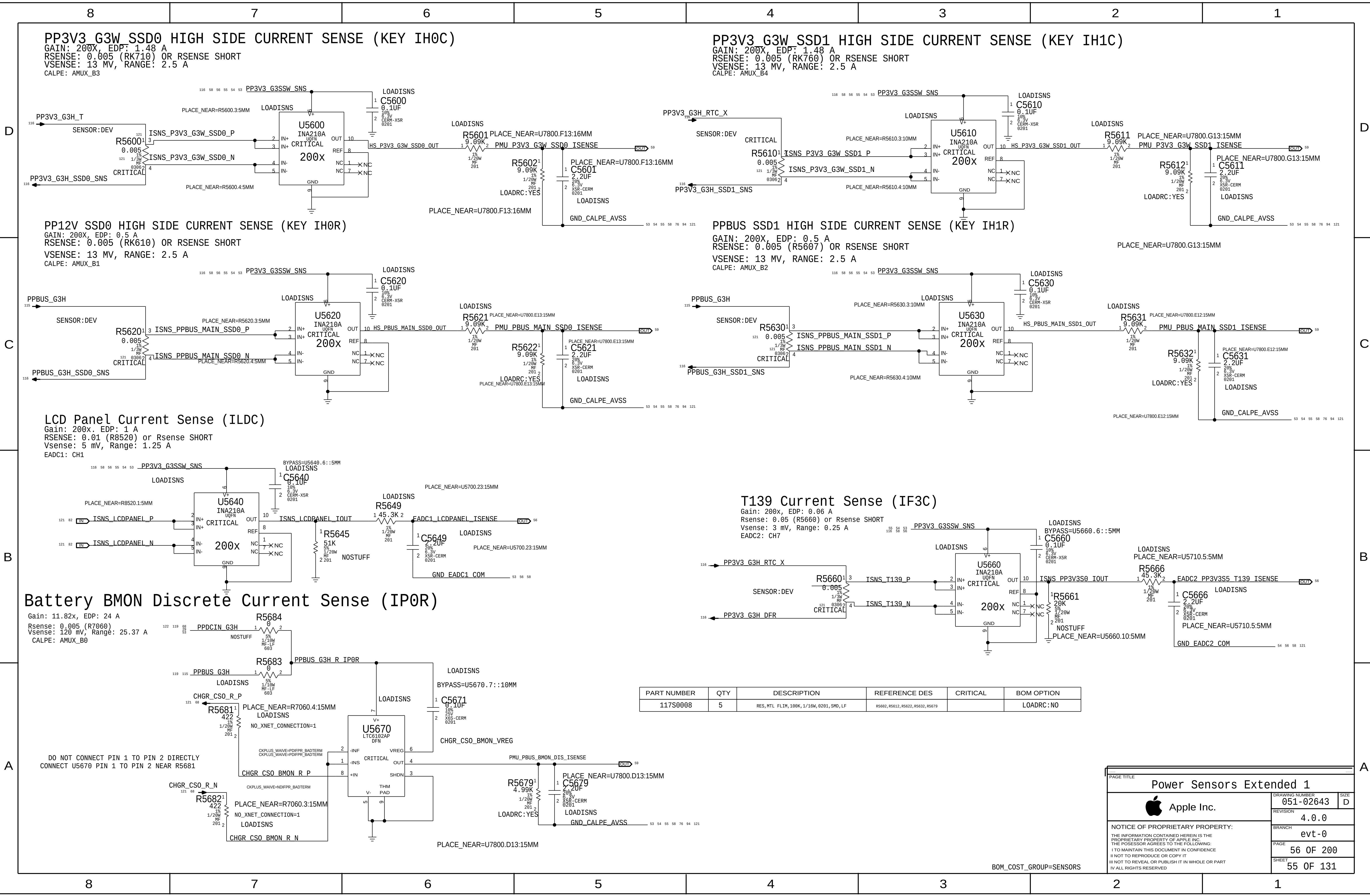
A



```
Gain: 89.48x, EDP: 128 A
Rsense: 3x of 0.00075
R7210,R7220,R7230),Rsum: 0.00025
Vsense:16.75 mV, Range: 134.11 A
CALPE AMUX: A0
```



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Power Sensors Load Side			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-02643	D	
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	PAGE	55 OF 200	
	SHEET	54 OF 131	

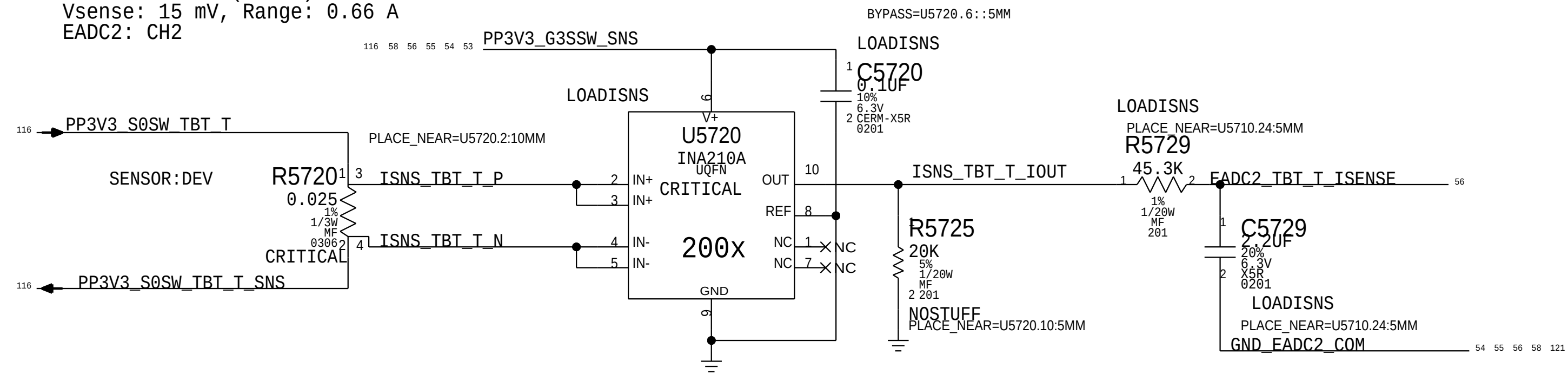


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	5	RES, MTL, FLIM, 100K, 1/16W, 0201, SMD, LF	R5682, R5612, R5622, R5632, R5679		LOADRC: NO

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Power Sensors Extended 1		
	DRAWING NUMBER	051-02643
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	PAGE	56 OF 200
	SHEET	55 OF 131

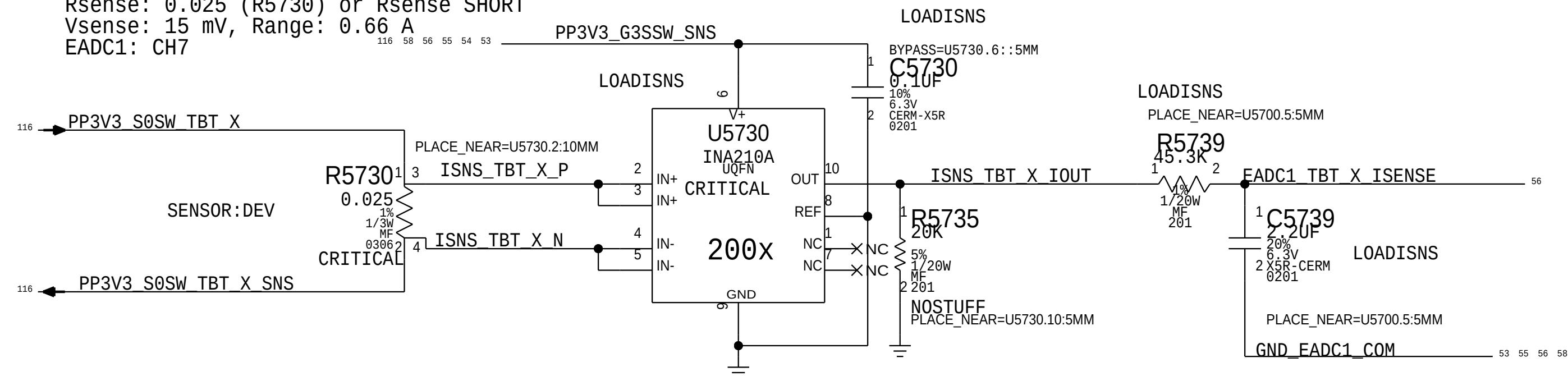
Thunderbolt TBT RIGHT Current Sense (IURC)

Gain: 200x, EDP: 0.6 A
Rsense: 0.025 (R5720) or Rsense SHORT
Vsense: 15 mV, Range: 0.66 A
EADC2: CH2



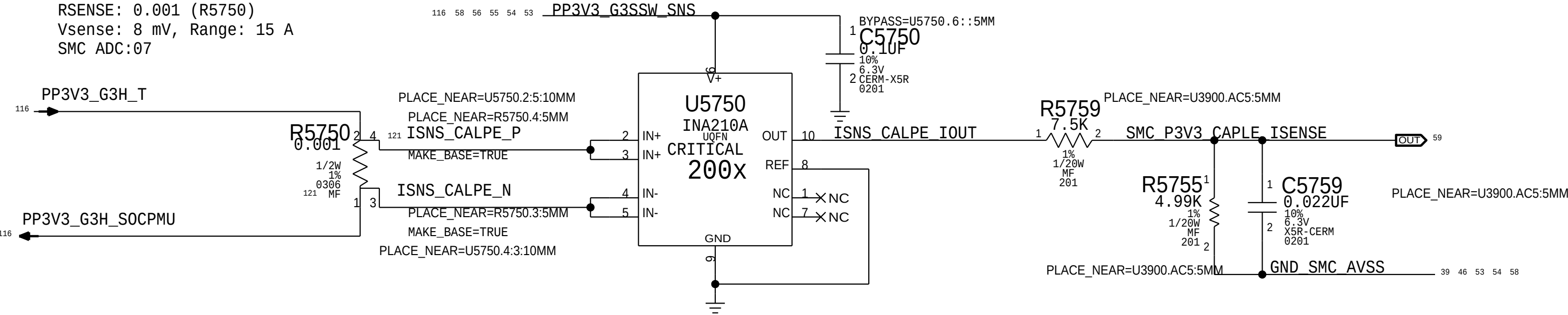
Thunderbolt TBT LEFT Current Sense (IULC)

Gain: 200x, EDP: 0.6 A
Rsense: 0.025 (R5730) or Rsense SHORT
Vsense: 15 mV, Range: 0.66 A
EADC1: CH7

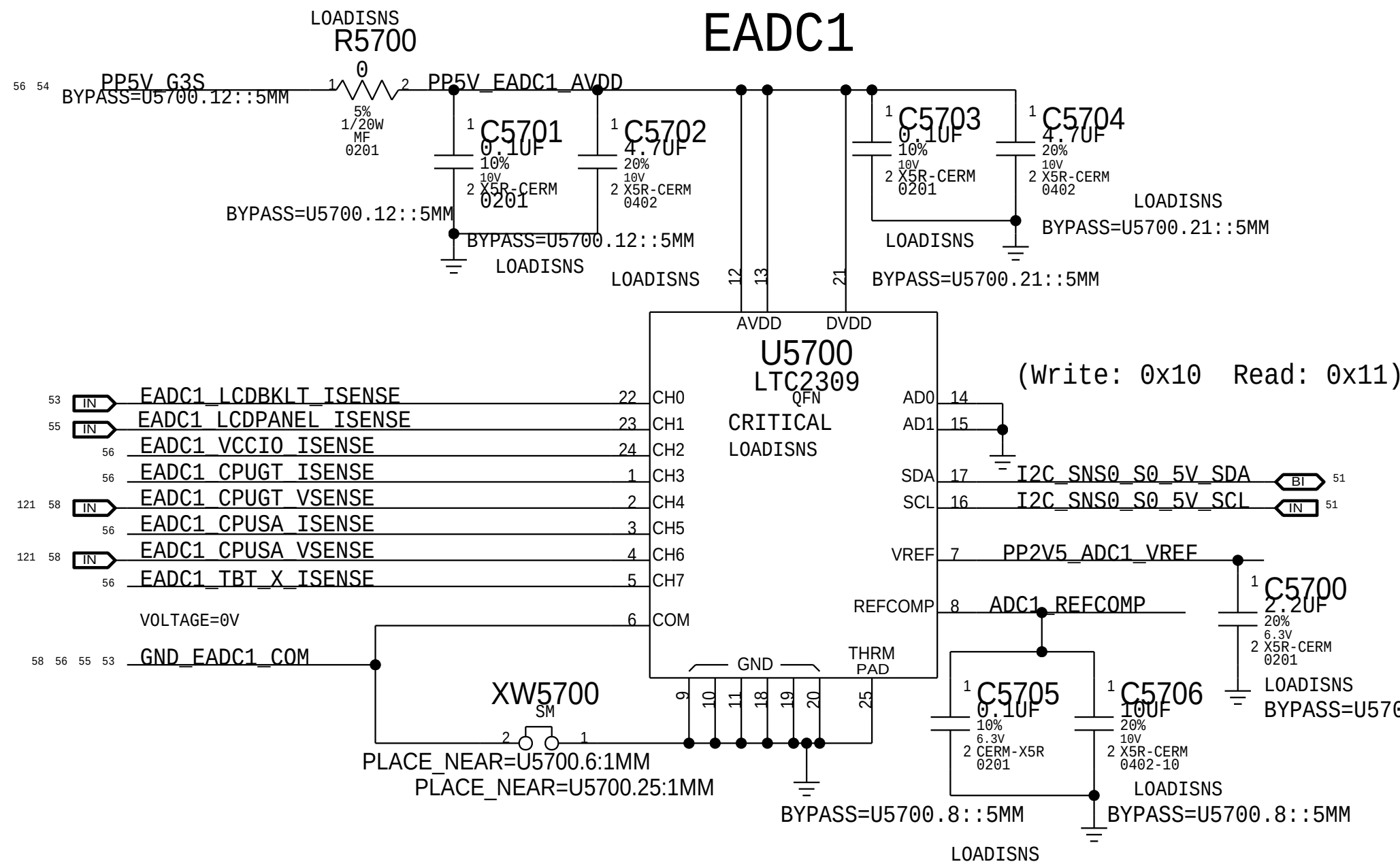


Calpe 3V3 Current Sense (ISLC)

Gain: 200x, EDP: 8 A
RSENSE: 0.001 (R5750)
Vsense: 8 mV, Range: 15 A
SMC ADC: 07

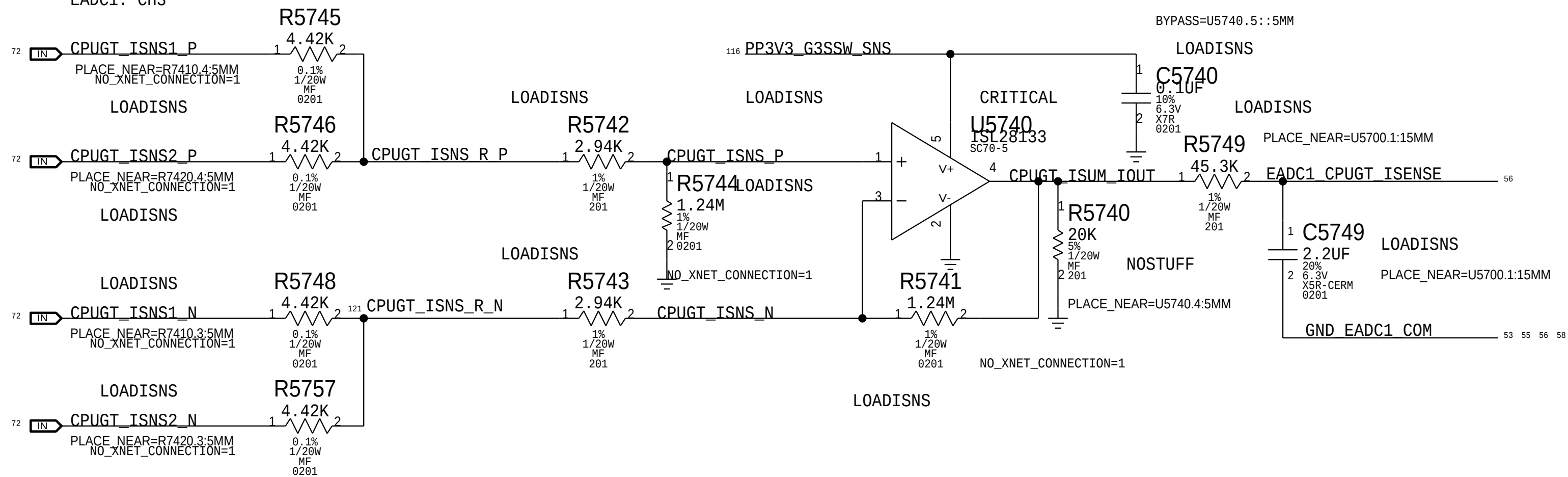


EADC1



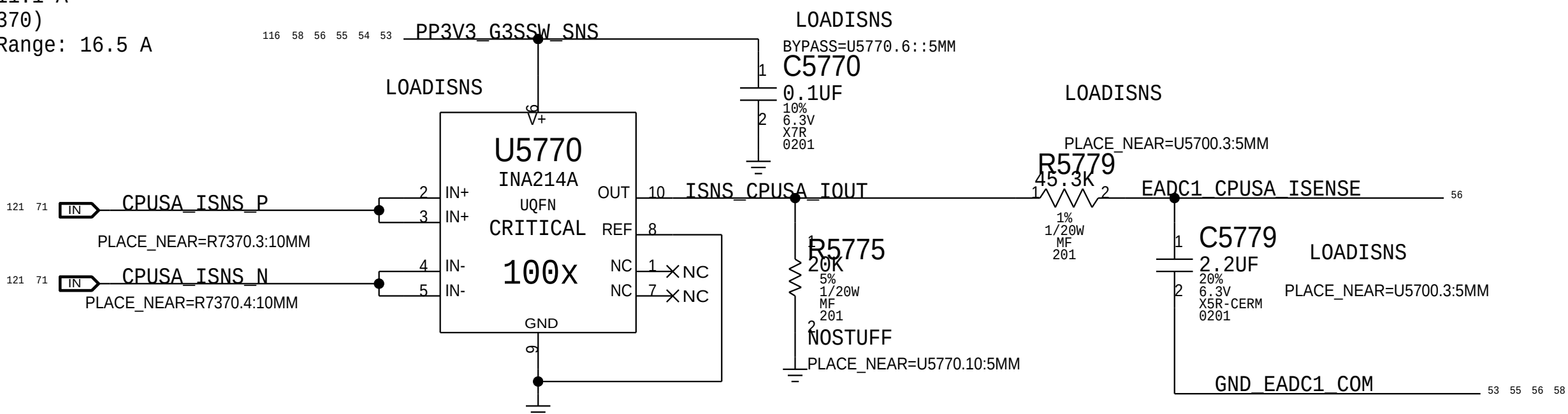
CPU GT Current Sense (ICTC)

Gain: 240.78x, EDP: 32 A
Rsense: 2x of 0.00075 (R7410, R7420), Rsum: 0.000375
Vsense: 12 mV, Range: 36.55 A
EADC1: CH3



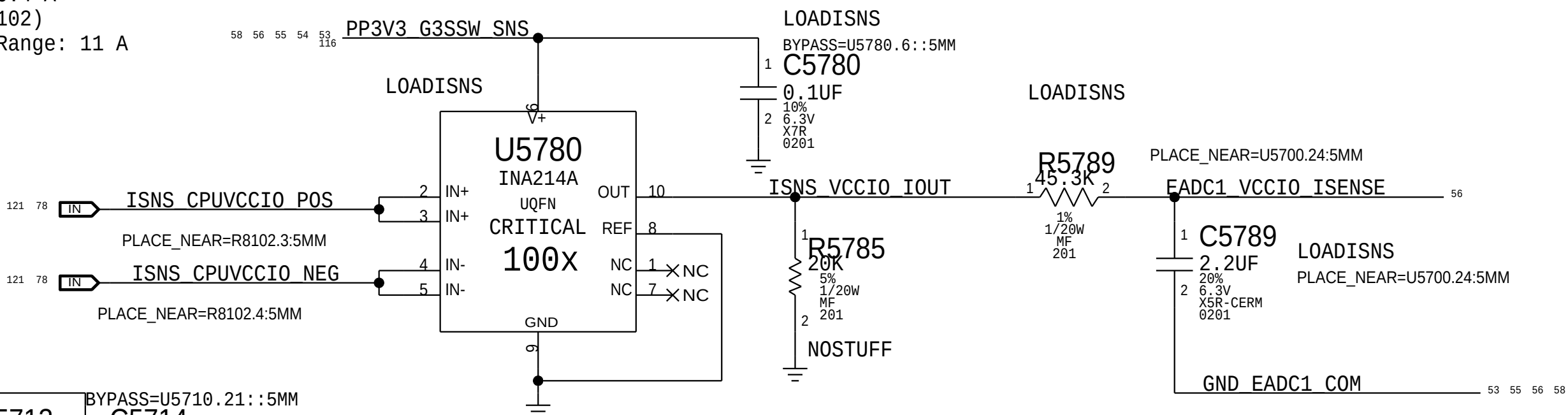
CPU SA Current Sense (ICSC)

Gain: 100x, EDP: 11.1 A
Rsense: 0.002 (R7370)
Vsense: 22.2 mV, Range: 16.5 A
EADC1: CH5

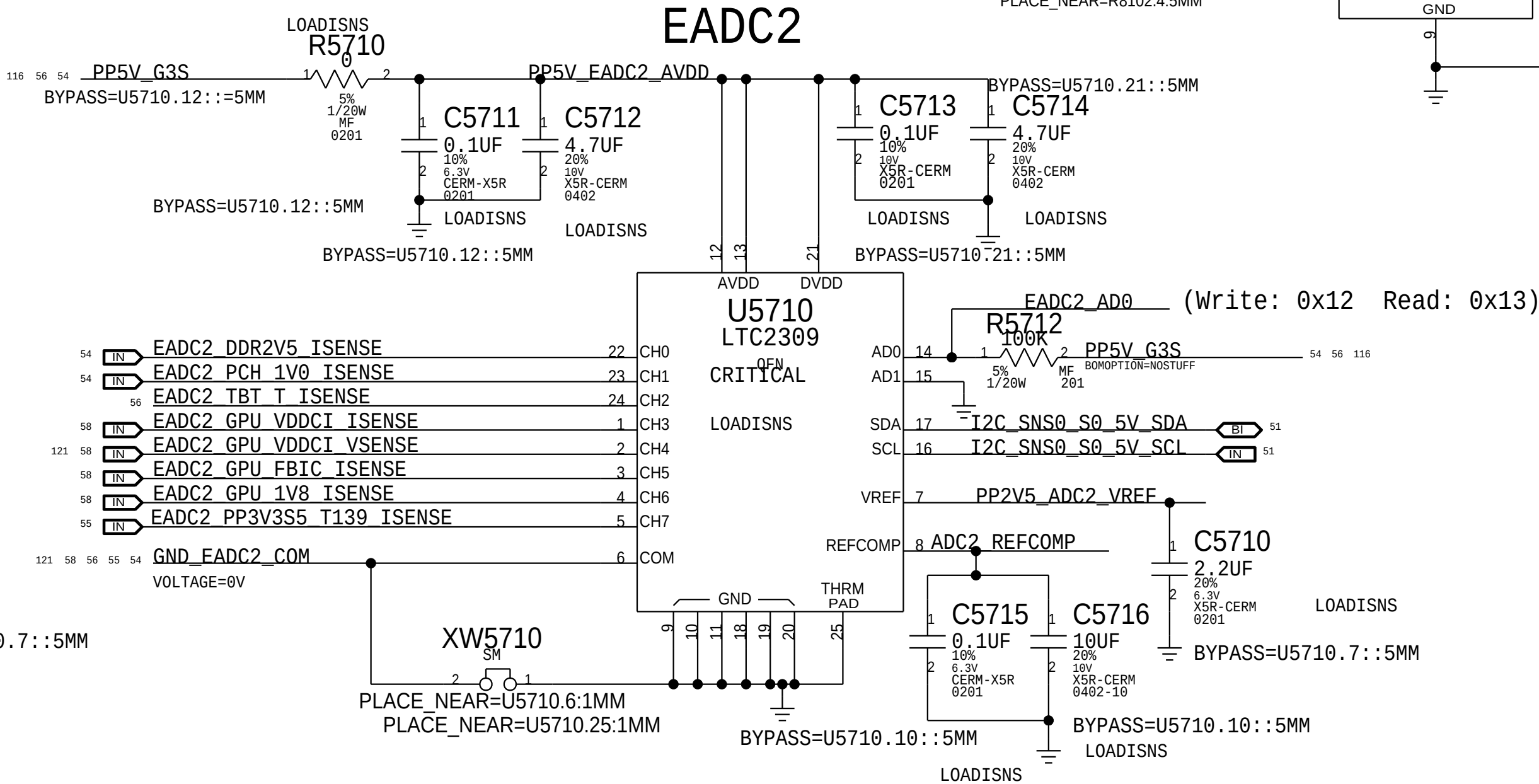


CPU VCCIO Current Sense (ICIC)

Gain: 100x, EDP: 6.4 A
Rsense: 0.003 (R8102)
Vsense: 19.2 mV, Range: 11 A
EADC1: CH2

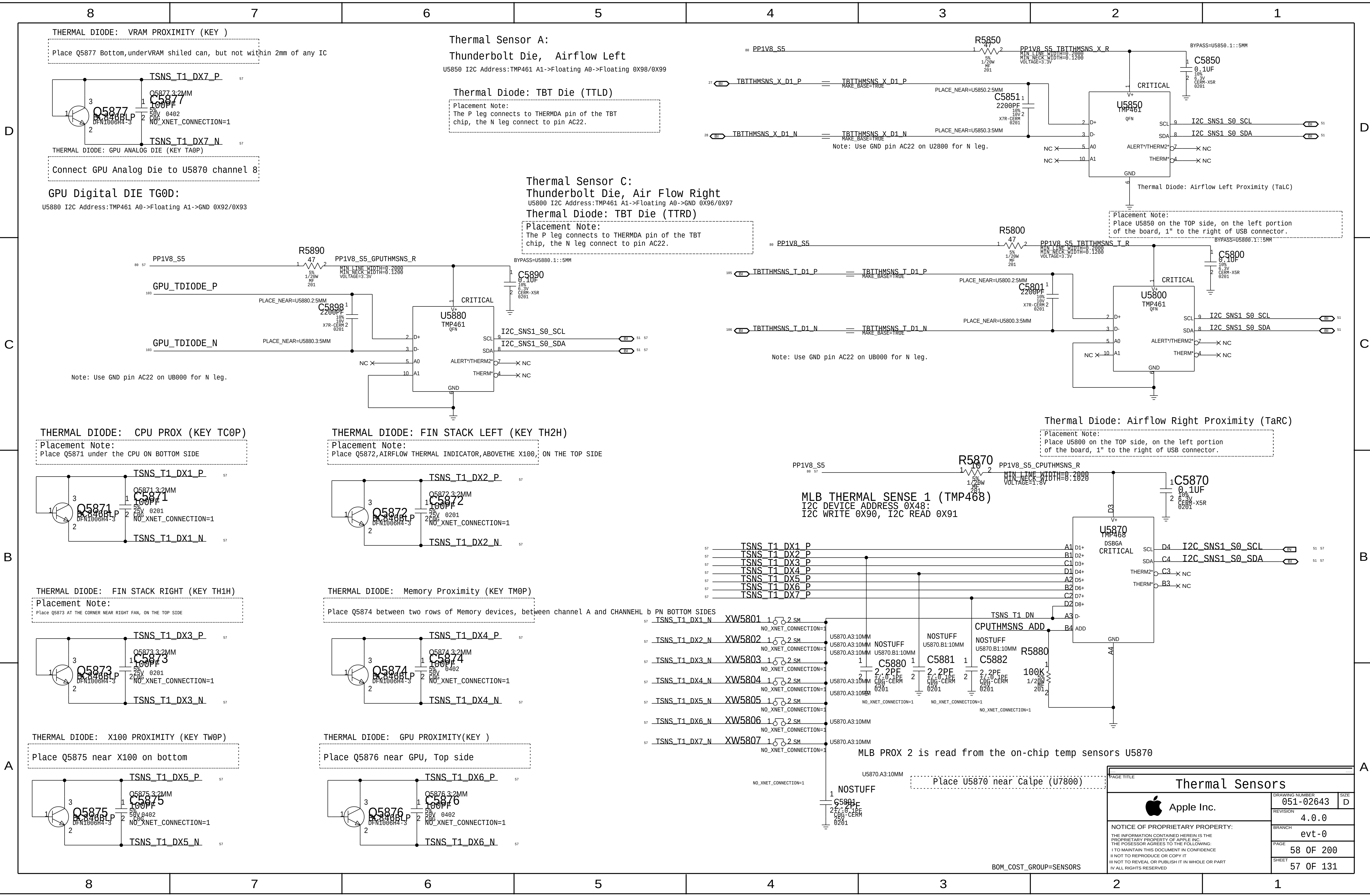


EADC2



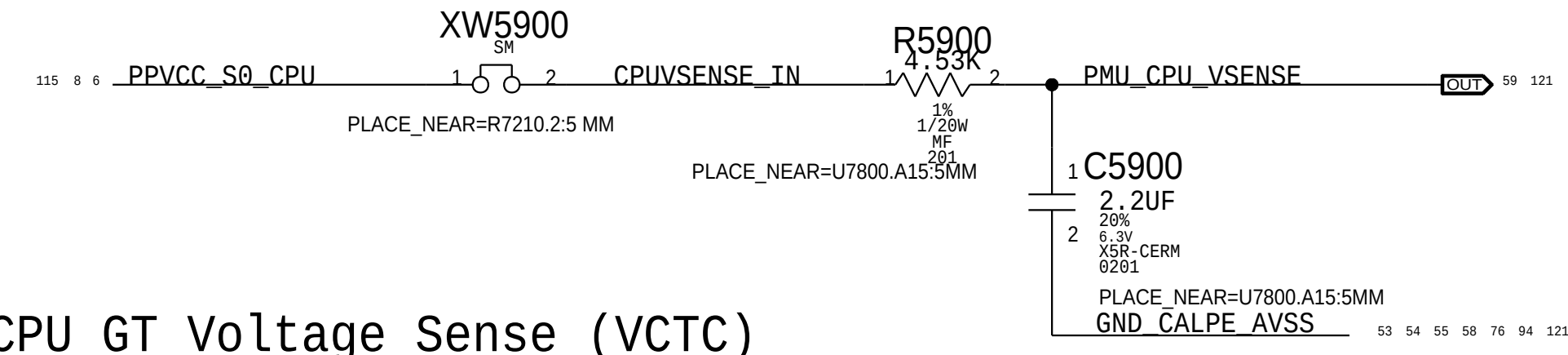
Power Sensors Extended 2			
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	BRANCH	evt-0	
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BOM_COST_GROUP=SENSORS



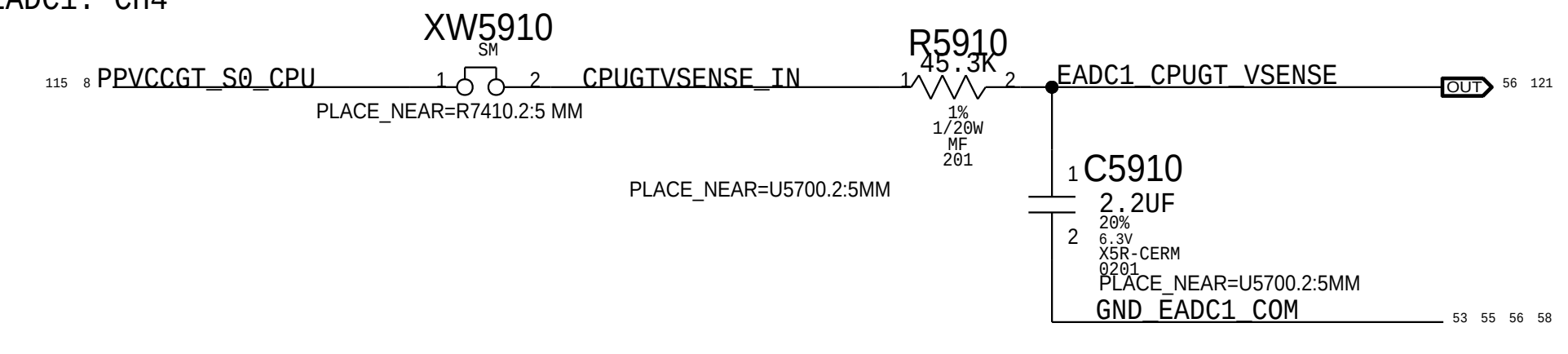
CPU Core Voltage Sense (VCAC)

CALPE : AMUX-A1



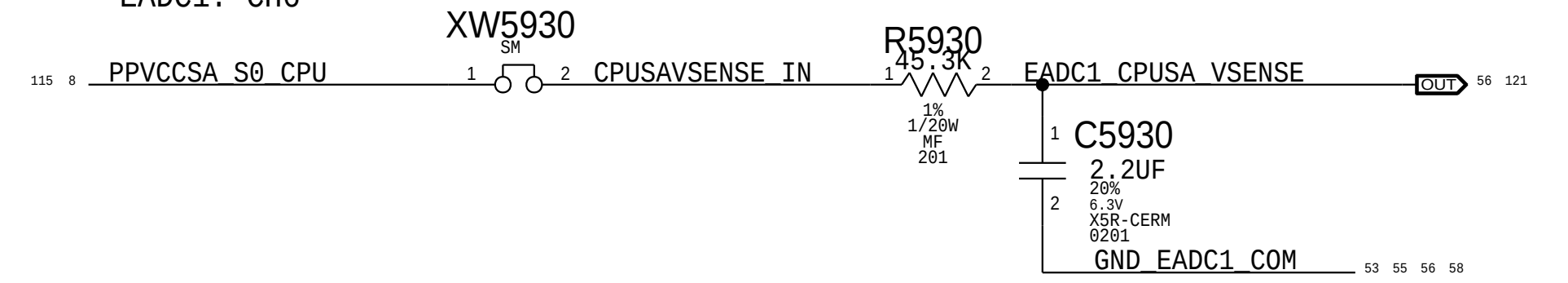
CPU GT Voltage Sense (VCTC)

EADC1: CH4



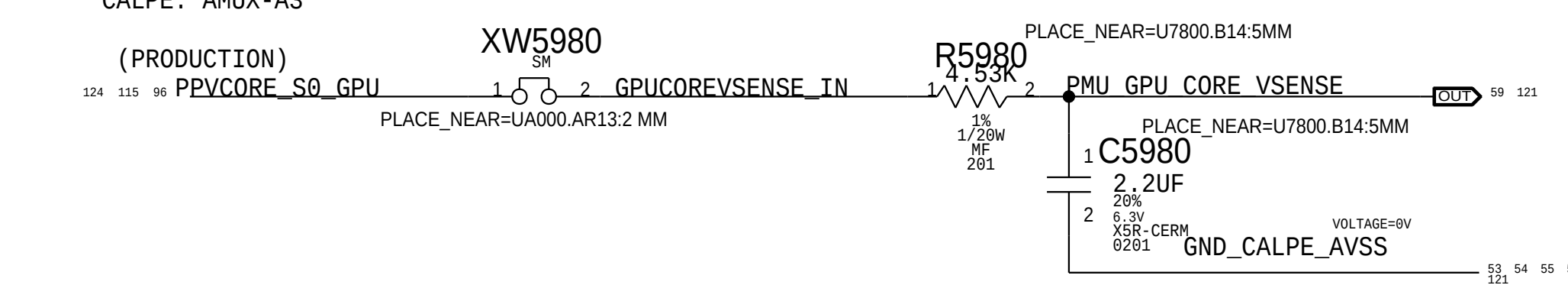
CPU SA Voltage Sense (VCSC)

EADC1: CH6



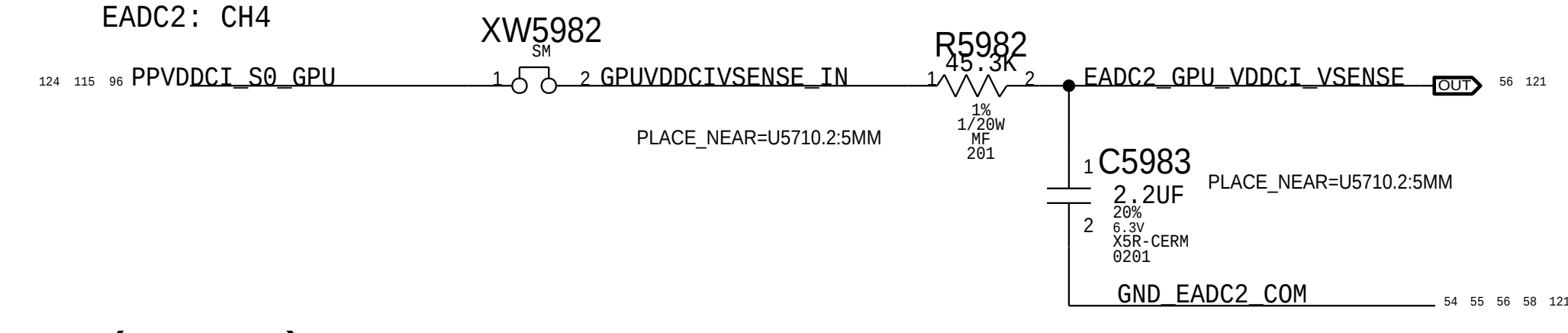
GPU CORE Voltage Sense (VG0C)

CALPE: AMUX-A3



GPU VDDCI Voltage Sense (VG2C)

EADC2: CH4



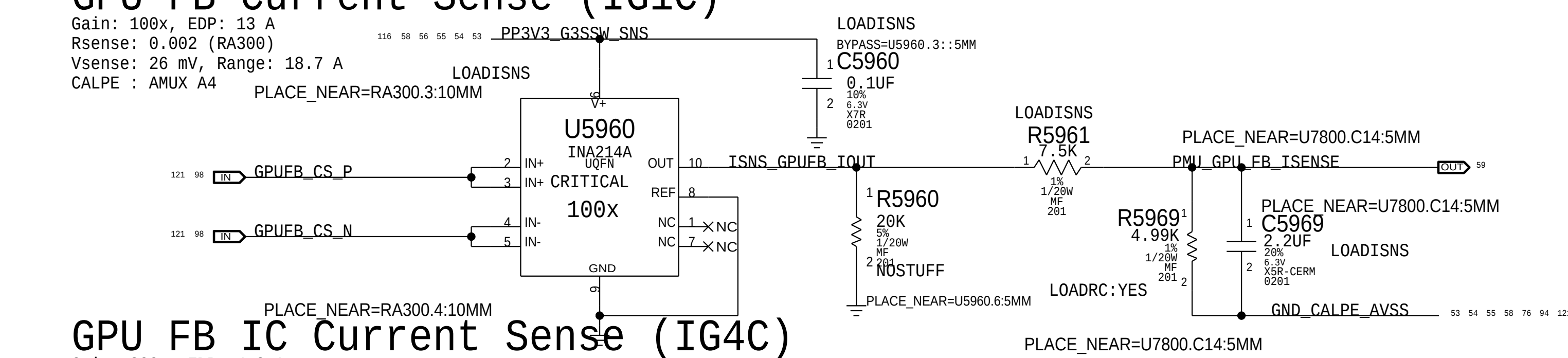
GPU CORE Current Sense (IG0C)

no longer lives on this page
it's merged with the PCC circuit
on CSA 99

Gain: 154.87x, EDP: 78 A
Rsense: 3X OF 0.00075 (RA651,RA641,RA346)
Vsense: 19.5 mV, Range: 109 A
CALPE : AMUX A2

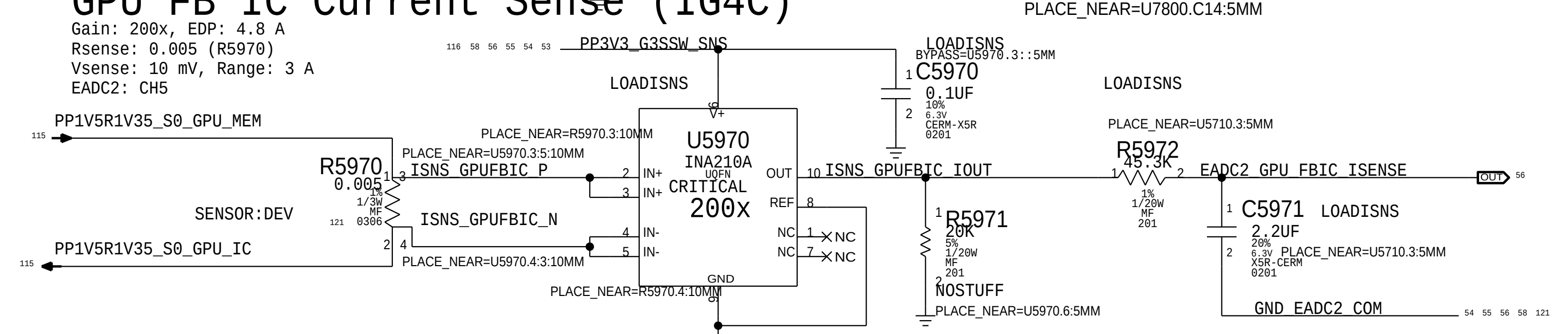
GPU FB Current Sense (IG1C)

Gain: 100x, EDP: 13 A
Rsense: 0.002 (RA300)
Vsense: 26 mV, Range: 18.7 A
CALPE : AMUX A4



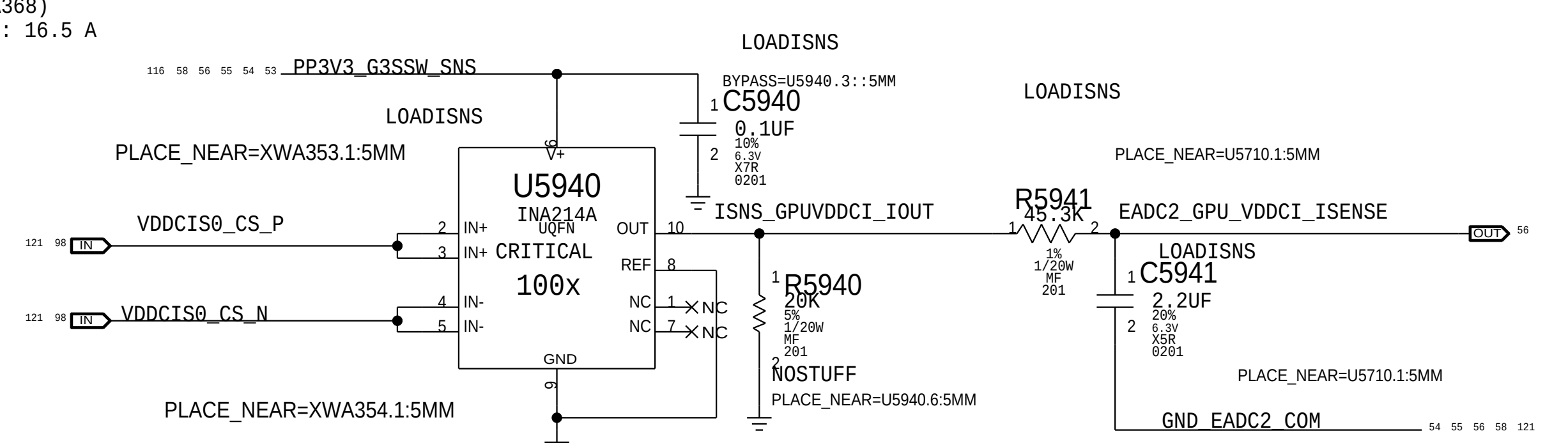
GPU FB IC Current Sense (IG4C)

Gain: 200x, EDP: 4.8 A
Rsense: 0.005 (R5970)
Vsense: 10 mV, Range: 3 A
EADC2: CH5



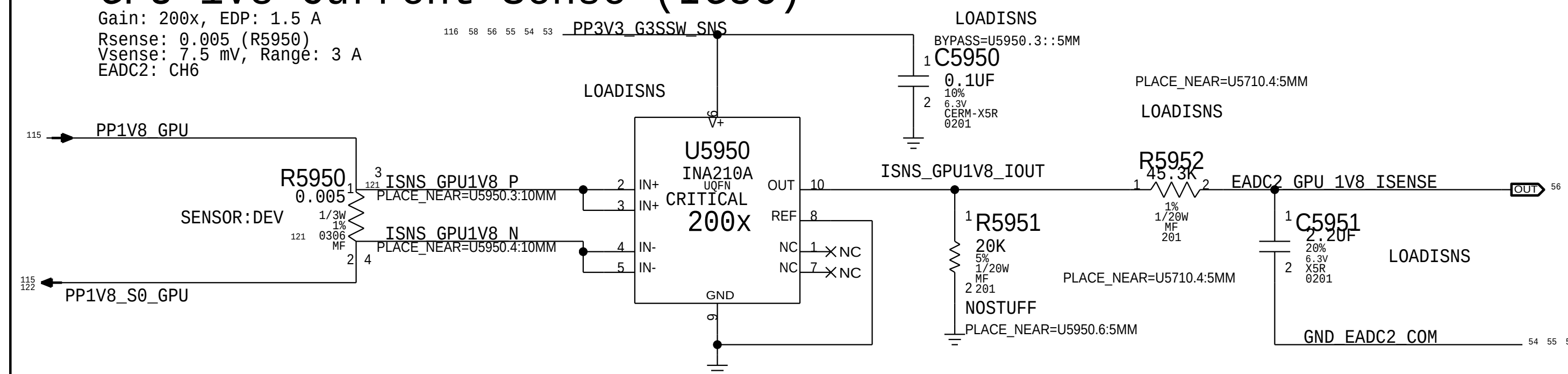
GPU VDDCI Current Sense (IG2C)

Gain: 100x, EDP: 10 A
Rsense: 0.003 (RA368)
Vsense: 20 mV, Range: 16.5 A
EADC2: CH3



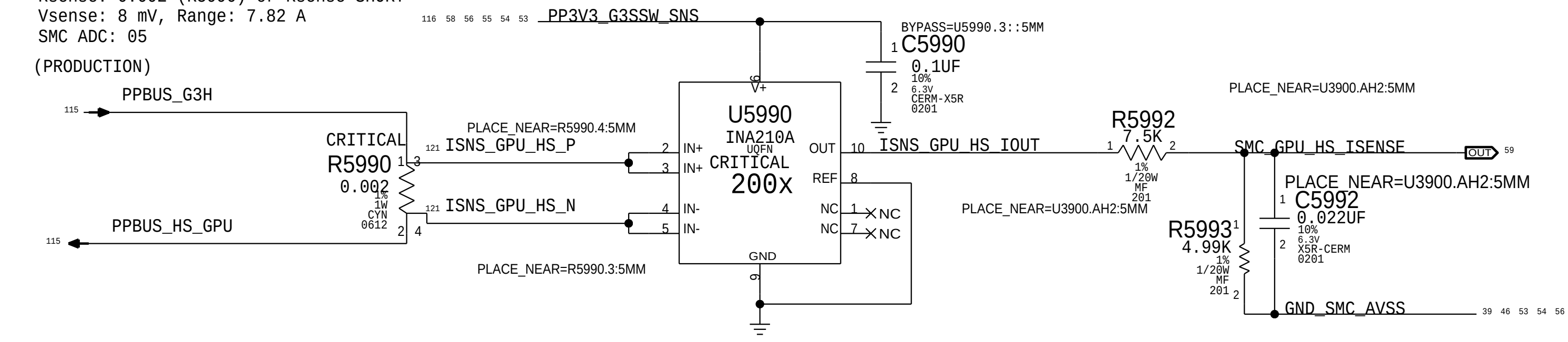
GPU 1V8 Current Sense (IG3C)

Gain: 200x, EDP: 1.5 A
Rsense: 0.005 (R5950)
Vsense: 7.5 mV, Range: 3 A
EADC2: CH6



GPU HIGH SIDE Current Sense (IG0R)

Gain: 200x, EDP: 4 A
Rsense: 0.002 (R5990) or Rsense SHORT
Vsense: 8 mV, Range: 7.82 A
SMC ADC: 05
(PRODUCTION)

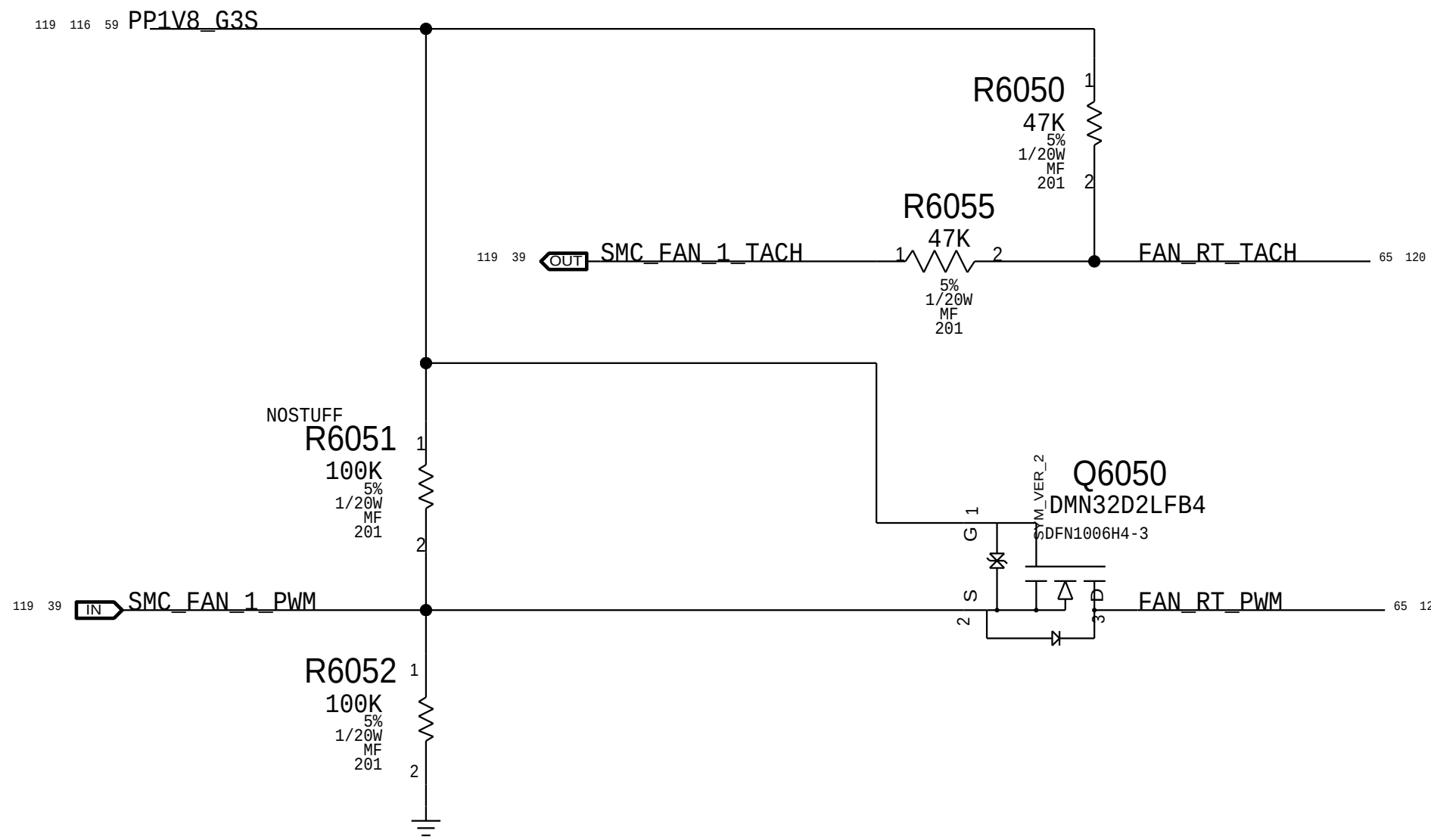
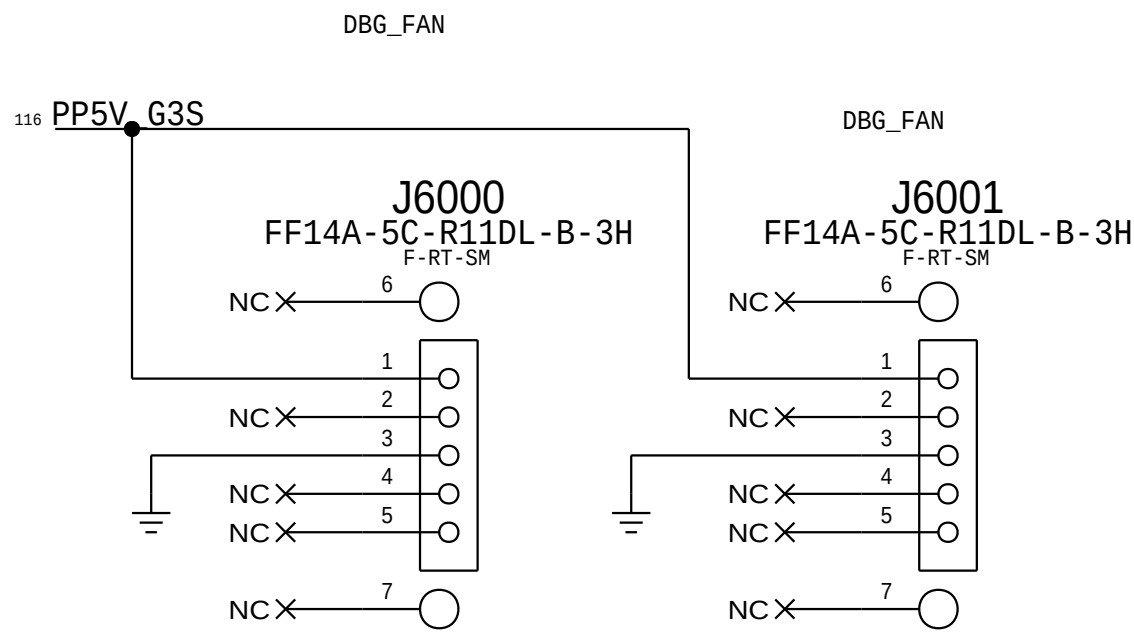
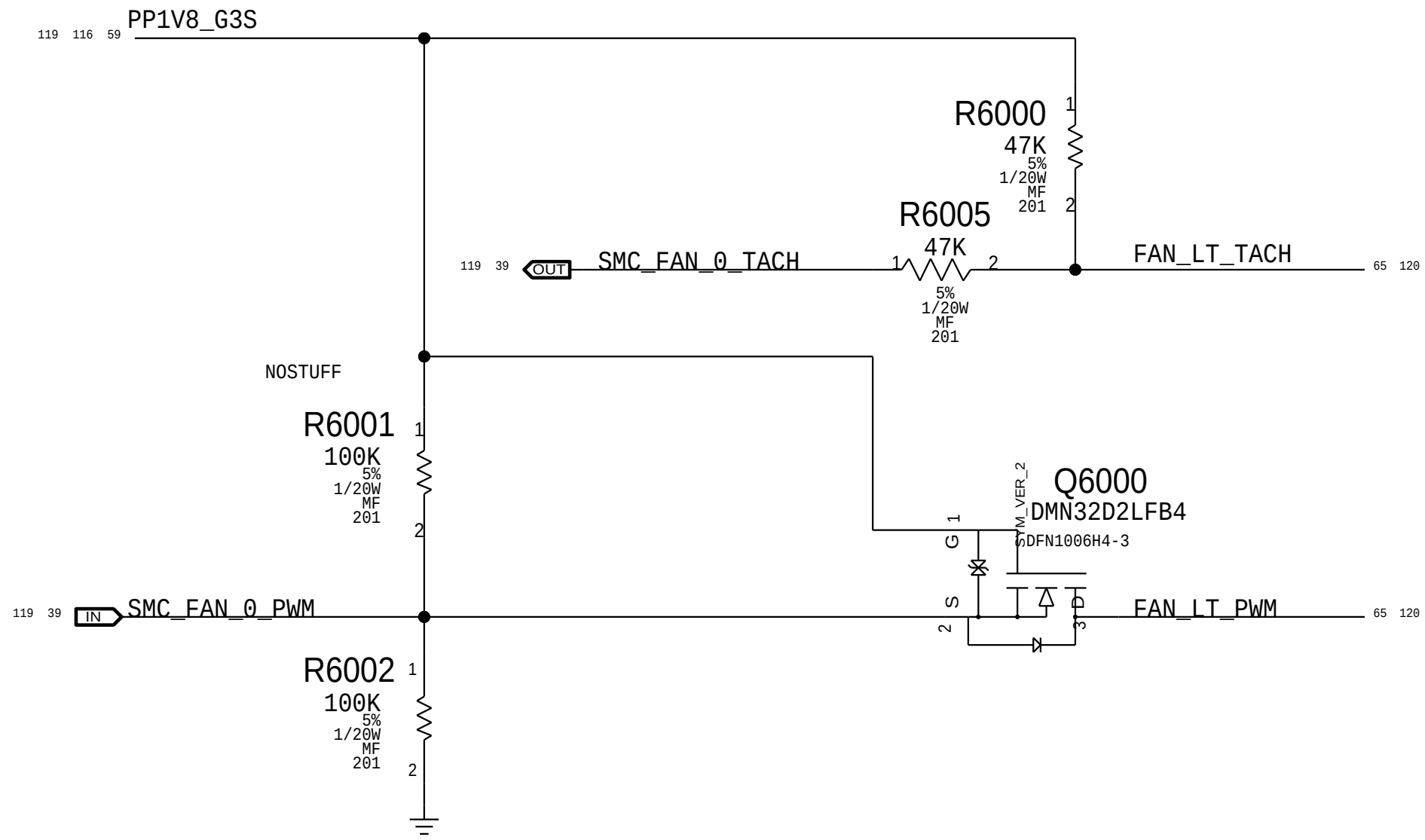


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES, MTL, FLIN, 100K, 1/16W, 0201, SMD, LF	R5969		LOADRC:NO

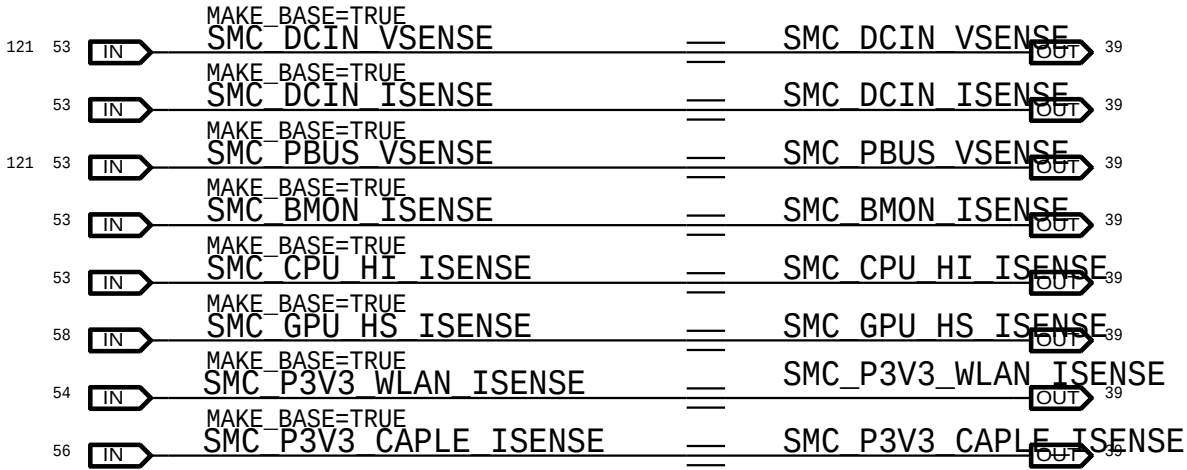
PAGE TITLE		
Power Sensor Extended 3		
	DRAWING NUMBER	051-02643
	REVISION	4.0.0
	BRANCH	evt-0
	PAGE	59 OF 200
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BOM_COST_GROUP=SENSORS

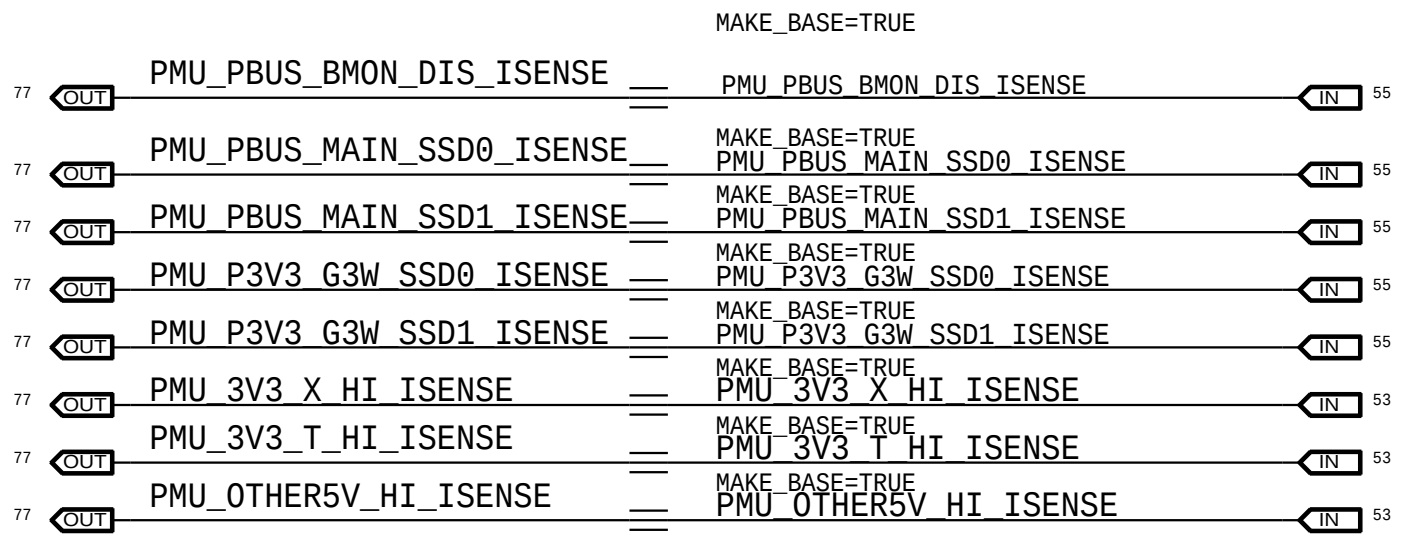
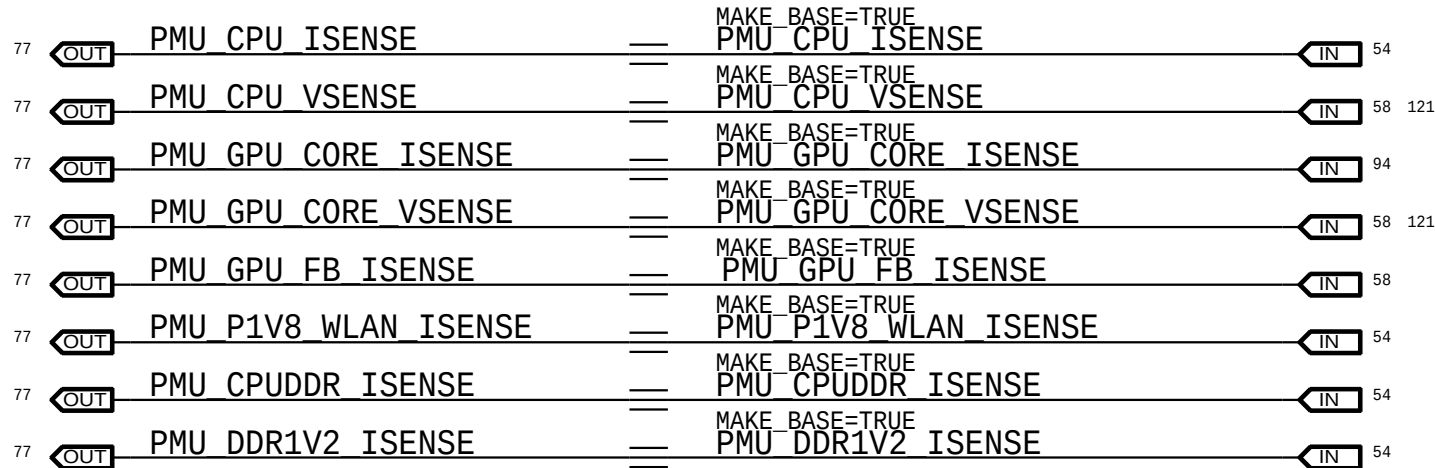
FAN CONNECTOR



H9M SMC ADC Assignments



CALPE AMUX Assignments



PAGE TITLE		
Fans/SMC/AMUX Support		
	DRAWING NUMBER	051-02643
	REVISION	4.0.0
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	PAGE	60 OF 200
	SHEET	59 OF 131

BOM_COST_GROUP=FAN

8		7		6		5		4		3		2		1			
D																D	
C																C	
B																B	
A																A	
8		7		6		5		4		3		2		1			

DESIGN: X502/DEV_MLB_U

LAST CHANGE: Wed Feb 18 17:12:24 2015

PAGE TITLE

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BRANCH

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D

BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV_MLB_U

LAST CHANGE: Wed Feb 18 17:12:24 2015

FILE: 000000001100.dwg

DATE: 2015/02/18/11:11

PAGE TITLE

Audio Placeholder

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SHEET

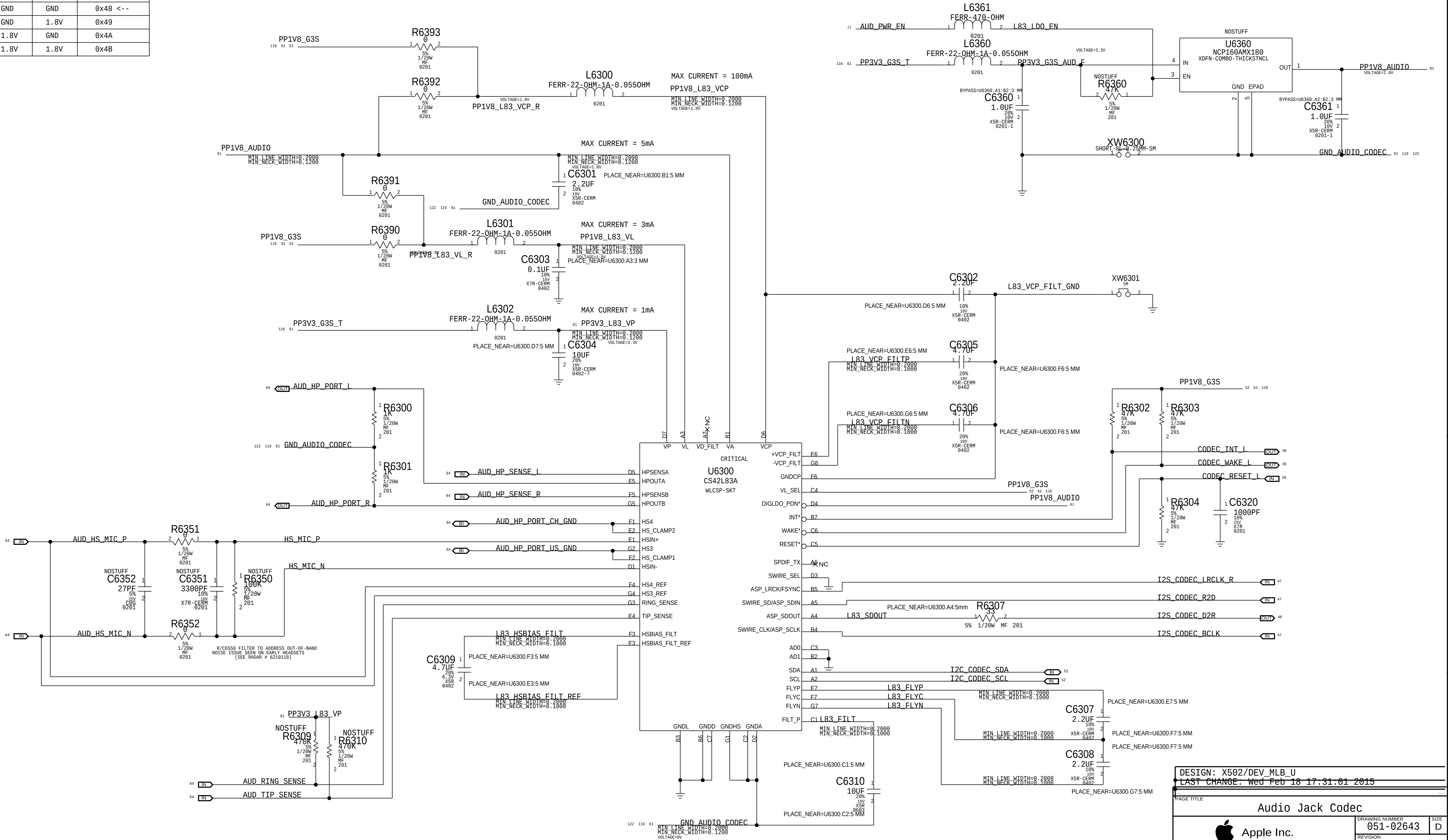
60 OF 131

SIZE

D

BOM_COST_GROUP=AUDIO

AUDIO JACK CODEC I2C ADDRESS		
AD1	AD0	ADDRESS
GND	GND	0x48 <--
1.8V	1.8V	0x49
1.8V	GND	0x4A
1.8V	1.8V	0x4B



BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV_MLB_U		
LAST CHANGE: Wed Feb 18 17:31:01 2015		
PAGE TITLE		
Audio Jack Codec		
	DRAWING NUMBER	051-02643
	REVISION	4.0.0
	BRANCH	evt-0
	PAGE	63 OF 200
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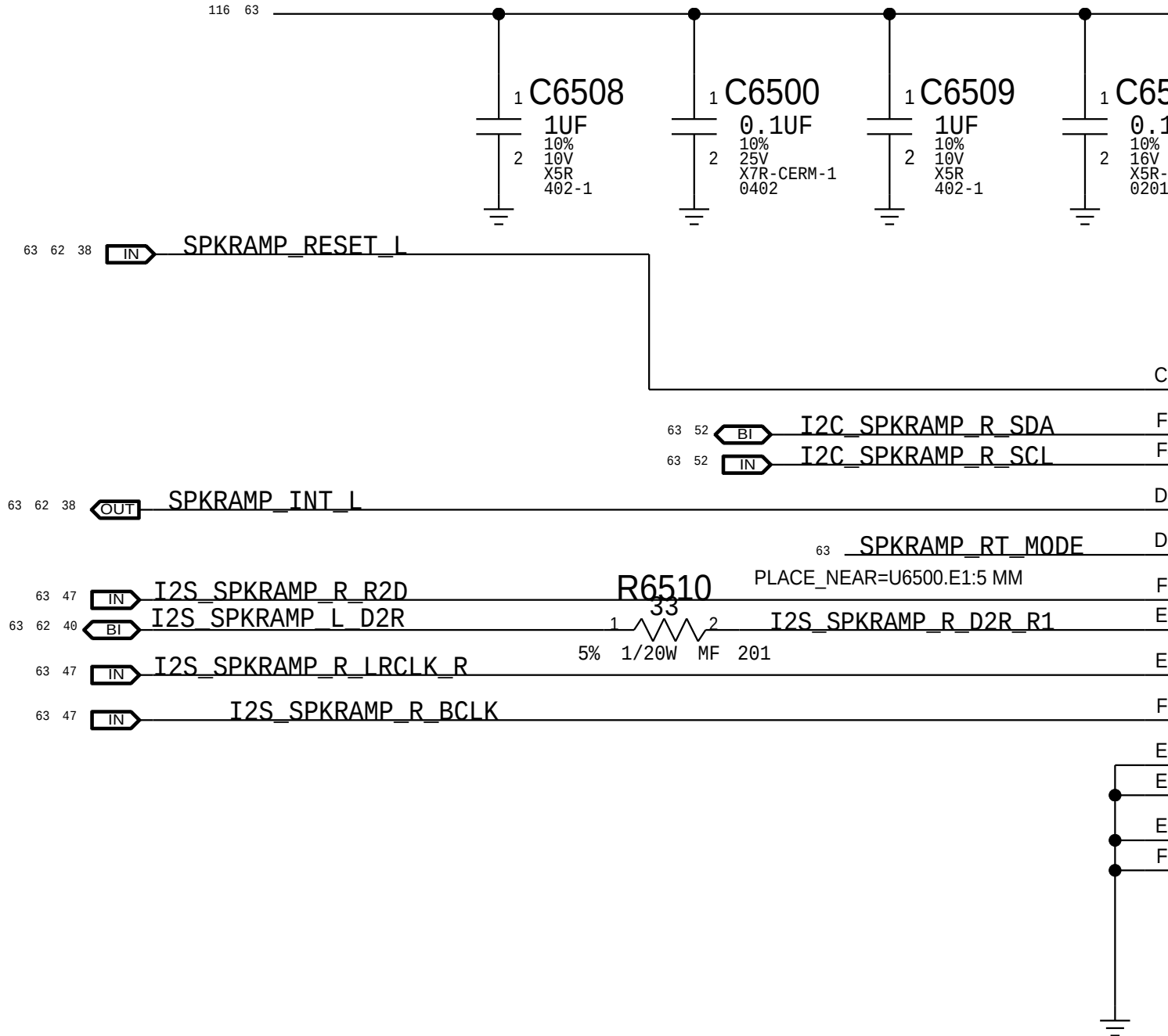
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Audio Left Amplifiers			
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	051-02643		D
	REVISION		
	4.0.0		
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		PAGE	
		64	OF 200
		SHEET	
		62	OF 131

2X MONO SPEAKER RIGHT AMPLIFIERS

APN: 353S01252
GAIN: 0DBFS = xxVRMS

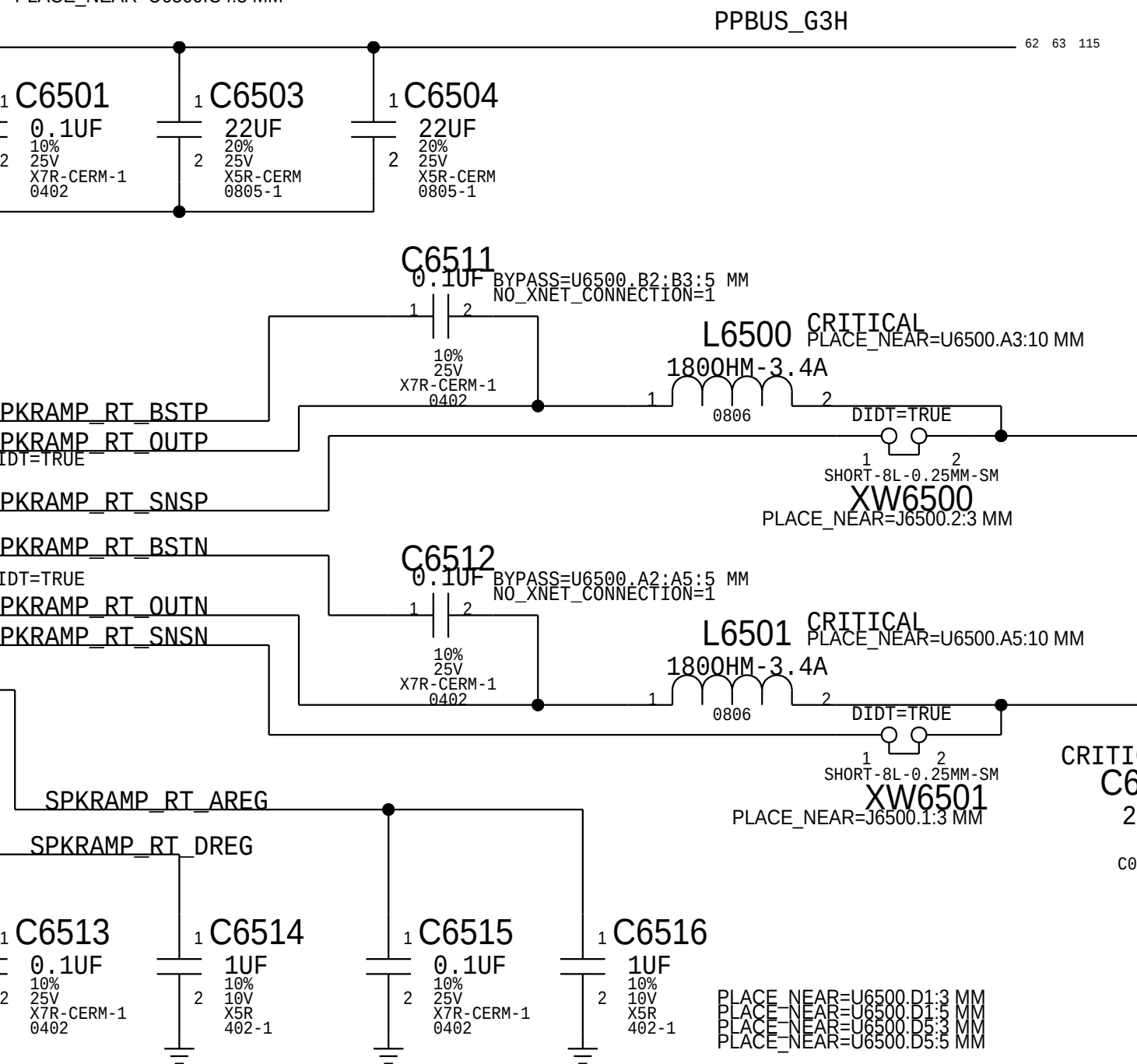
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PP1V8_G3S

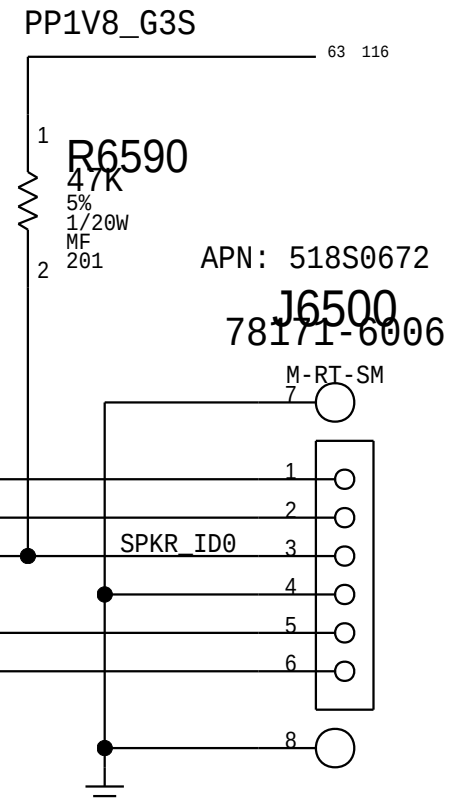


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PPBUS_G3H

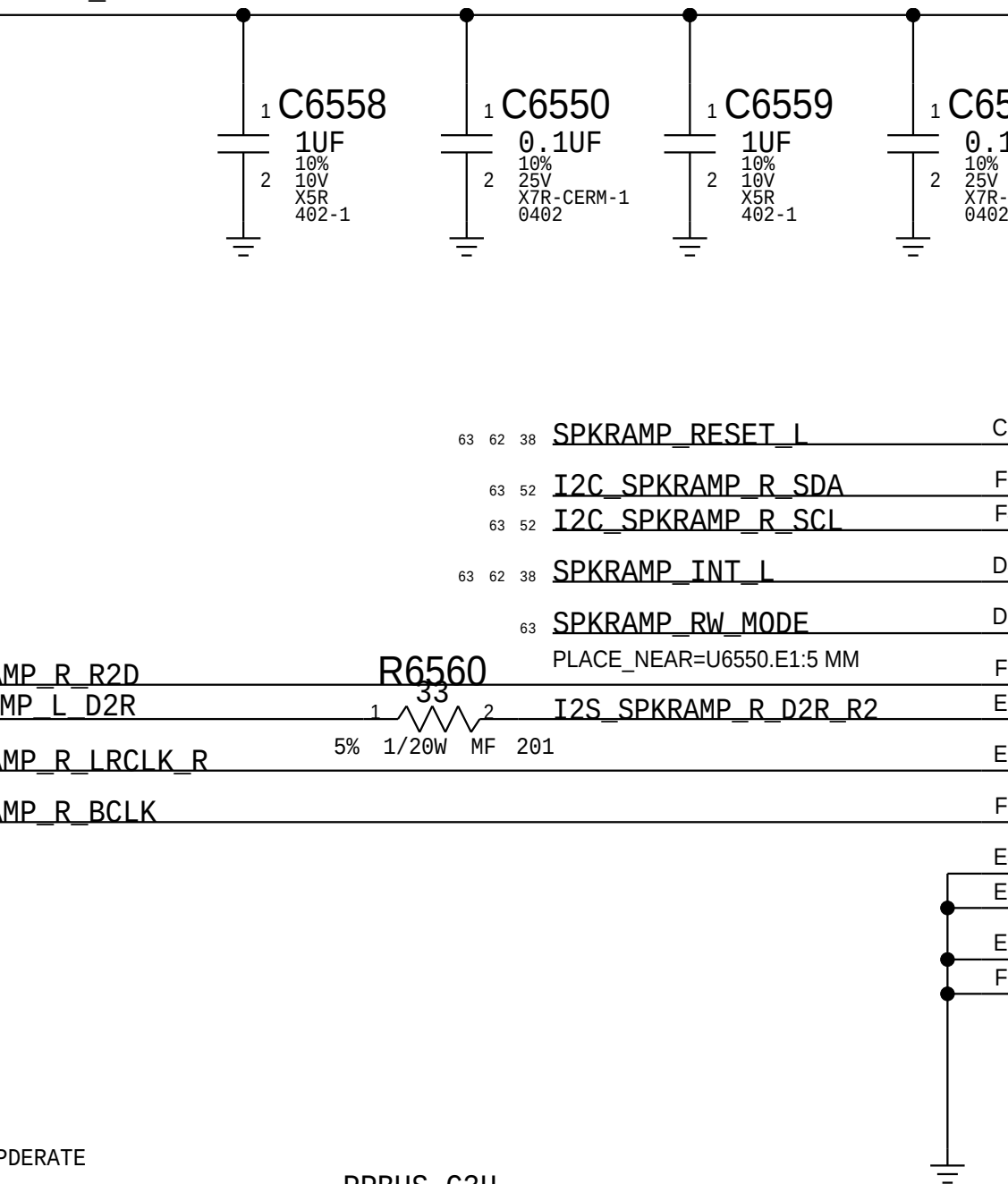


RIGHT TWEETER



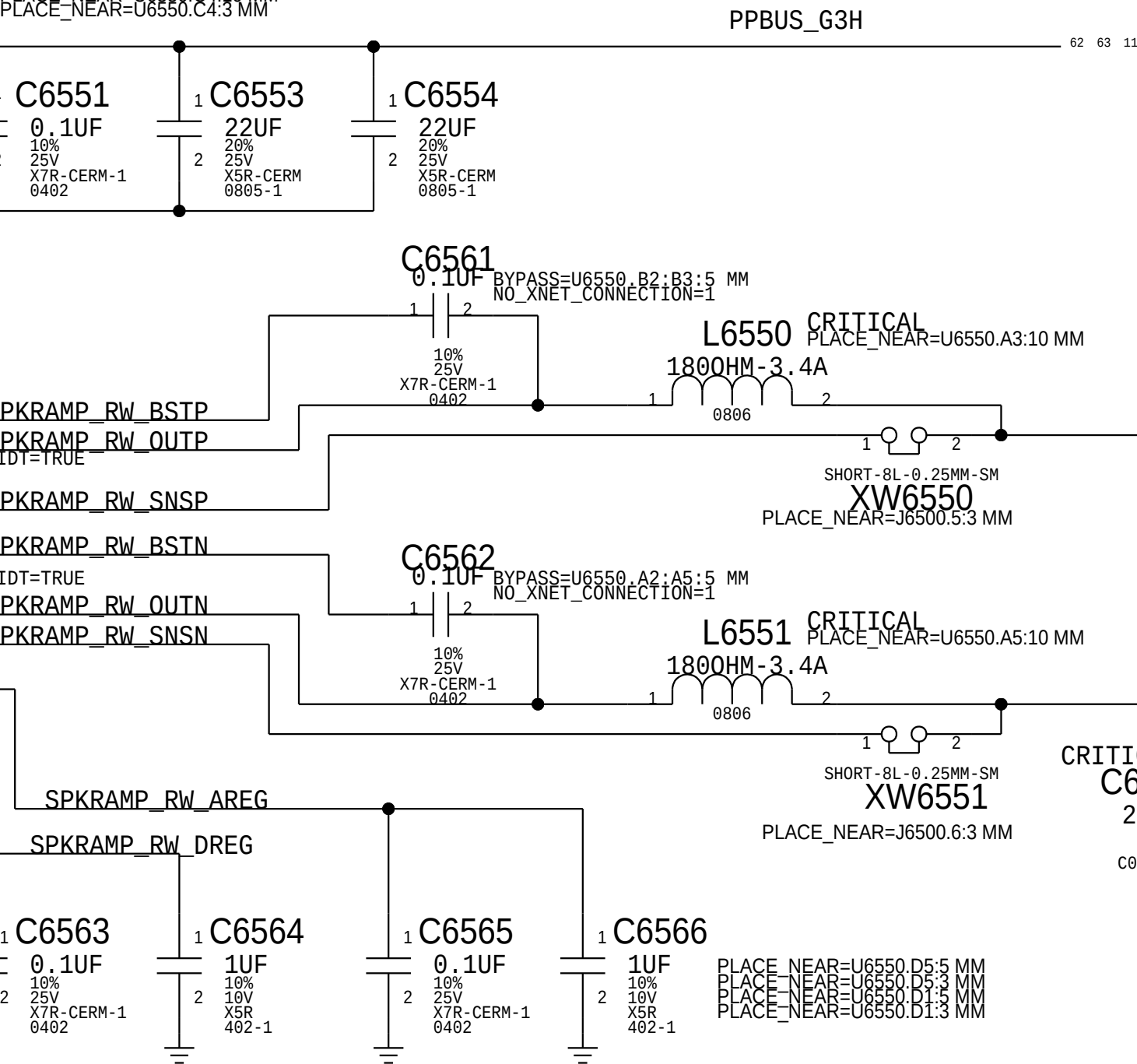
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PP1V8_G3S

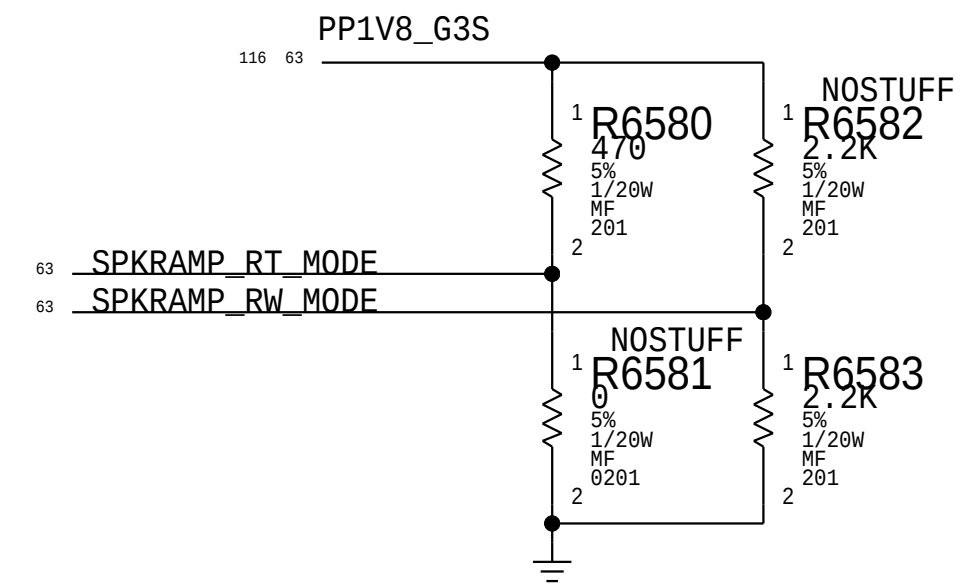
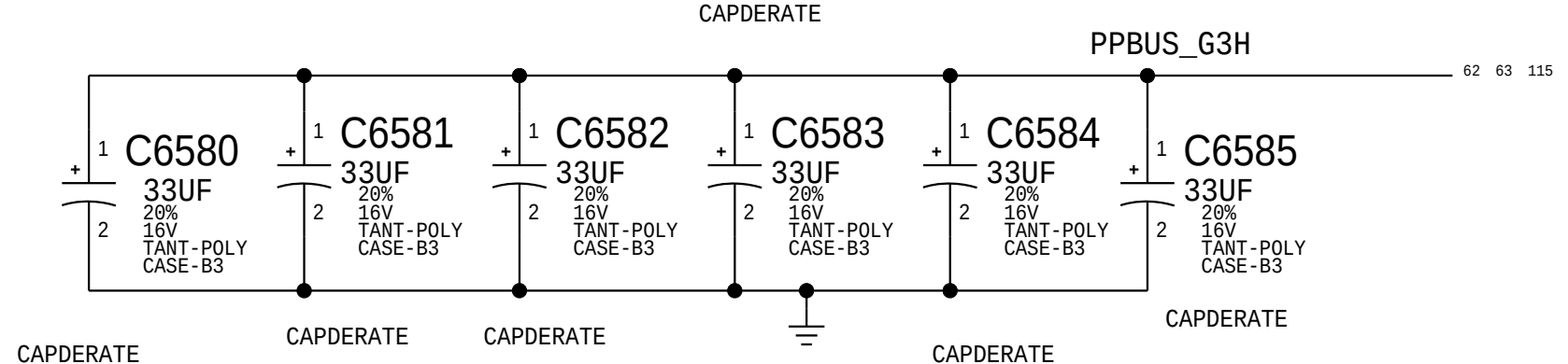


MAX CURRENT = 2A PER AMPLIFIER

PPBUS_G3H



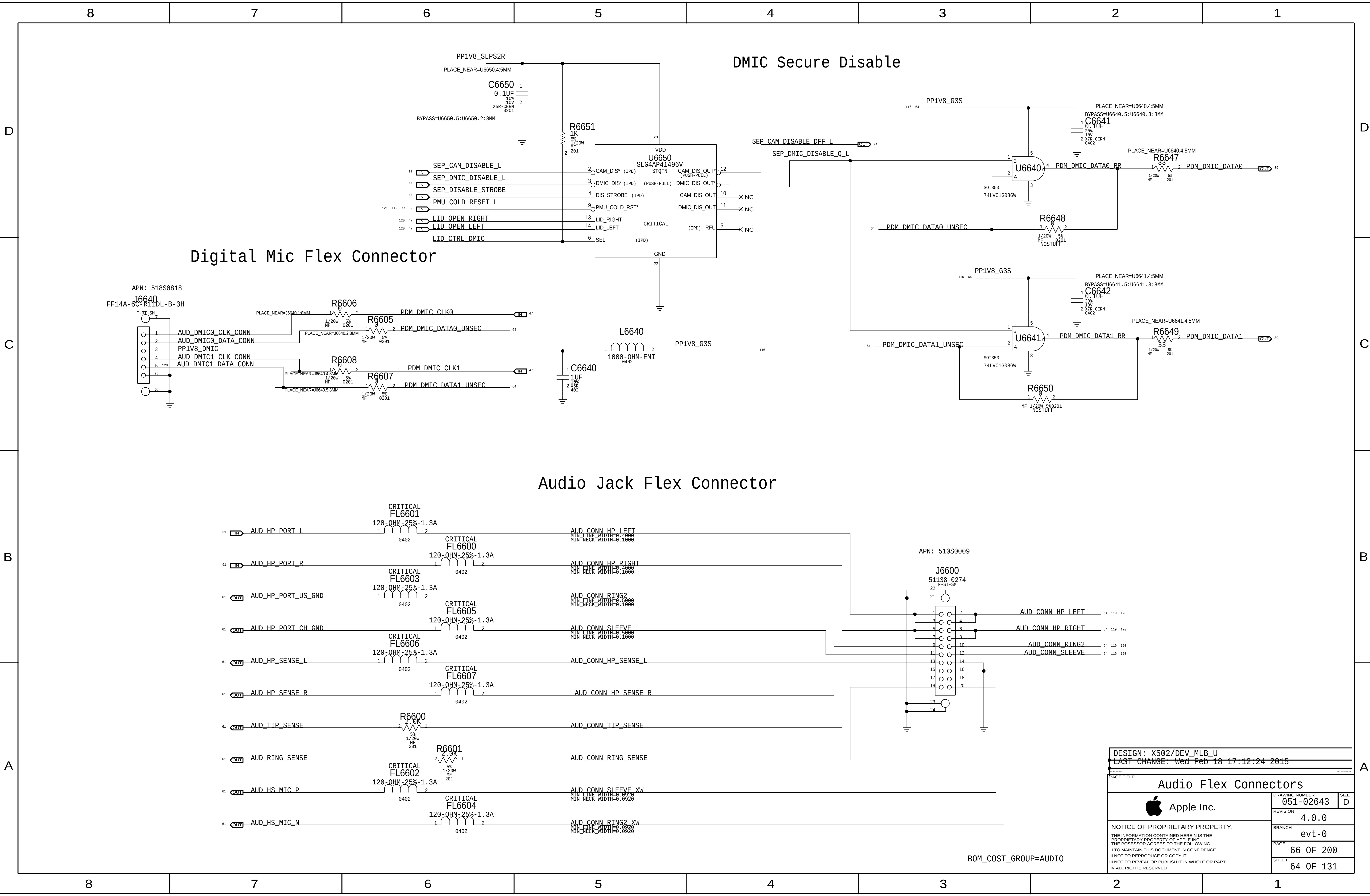
RIGHT WOOFER



I2C ADDRESS		
MODE PIN	7-BIT	CHANNEL
GND	0x31	L TW
470 to GND	0x32	L WF
470 to IOVDD	0x33	R TW
2K2 to GND	0x34	R WF
2K2 to IOVDD	0x35	
10K to GND	0x36	
10K to IOVDD	0x37	
47K to IOVDD	0x38	

BOM_COST_GROUP=AUDIO

PAGE TITLE		
Audio Right Amplifiers		
	DRAWING NUMBER	051-02643
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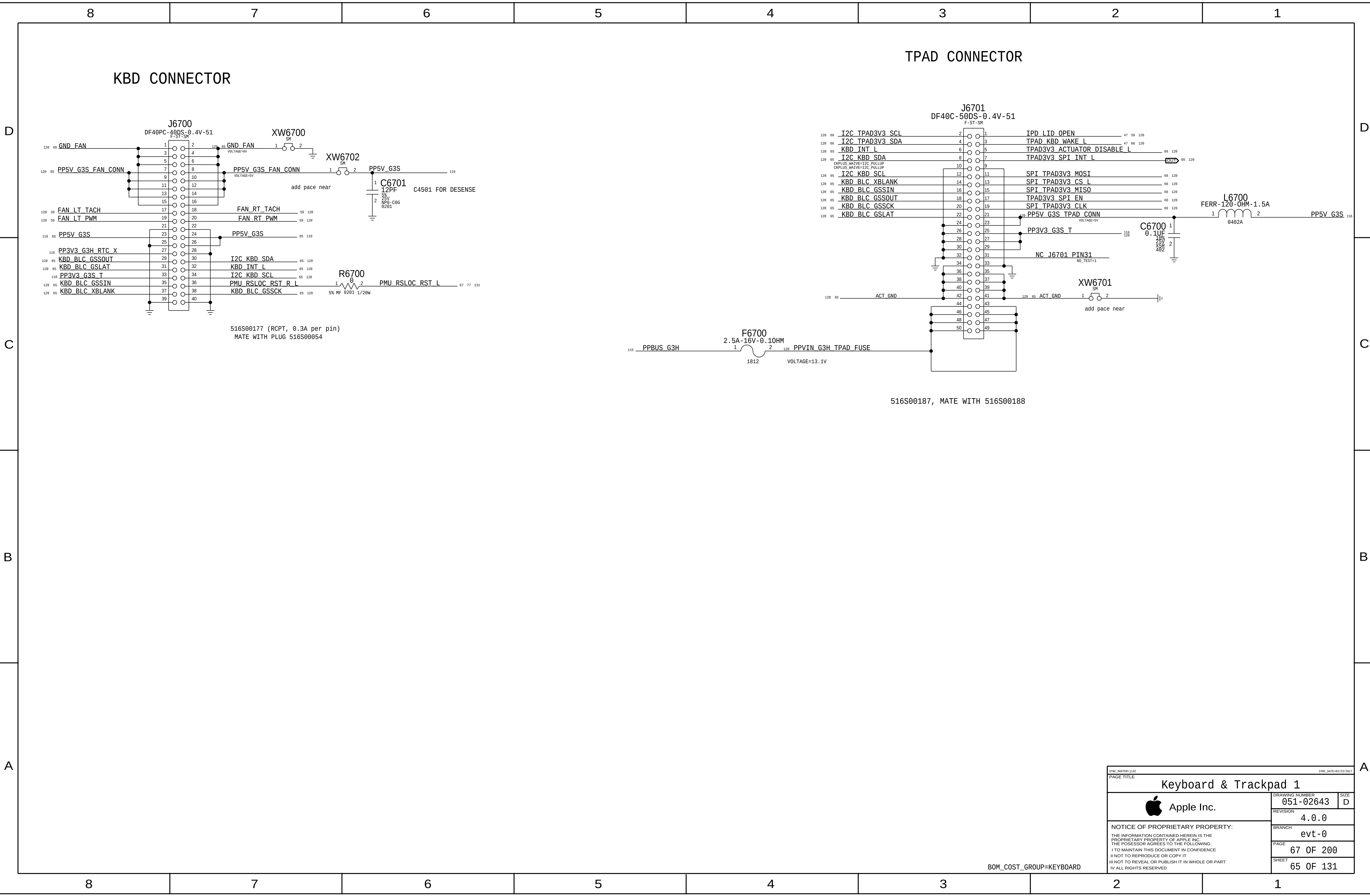


Digital Mic Flex Connector

Audio Jack Flex Connector

DESIGN: X502/DEV_MLB_U		
LAST CHANGE: wed Feb 18 17:12:24 2015		
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Audio Flex Connectors		
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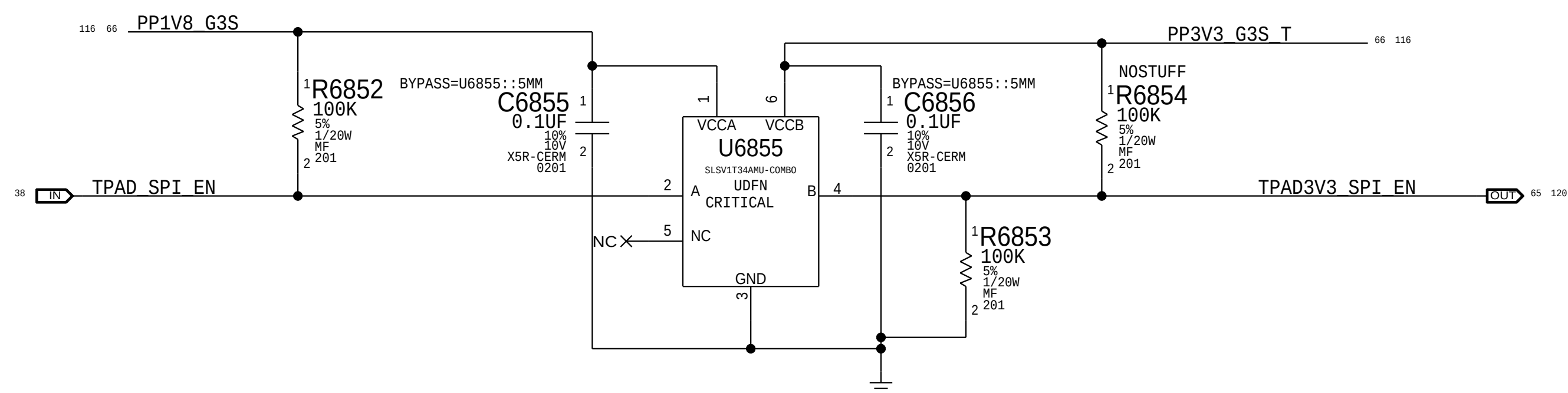
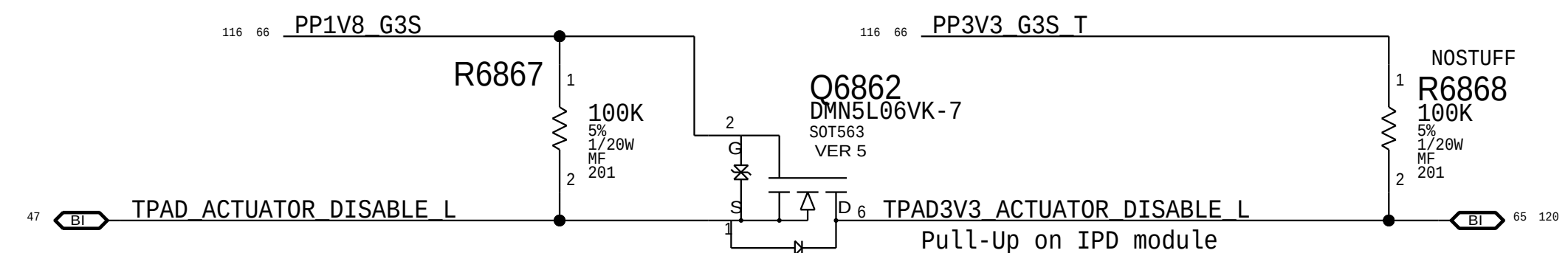
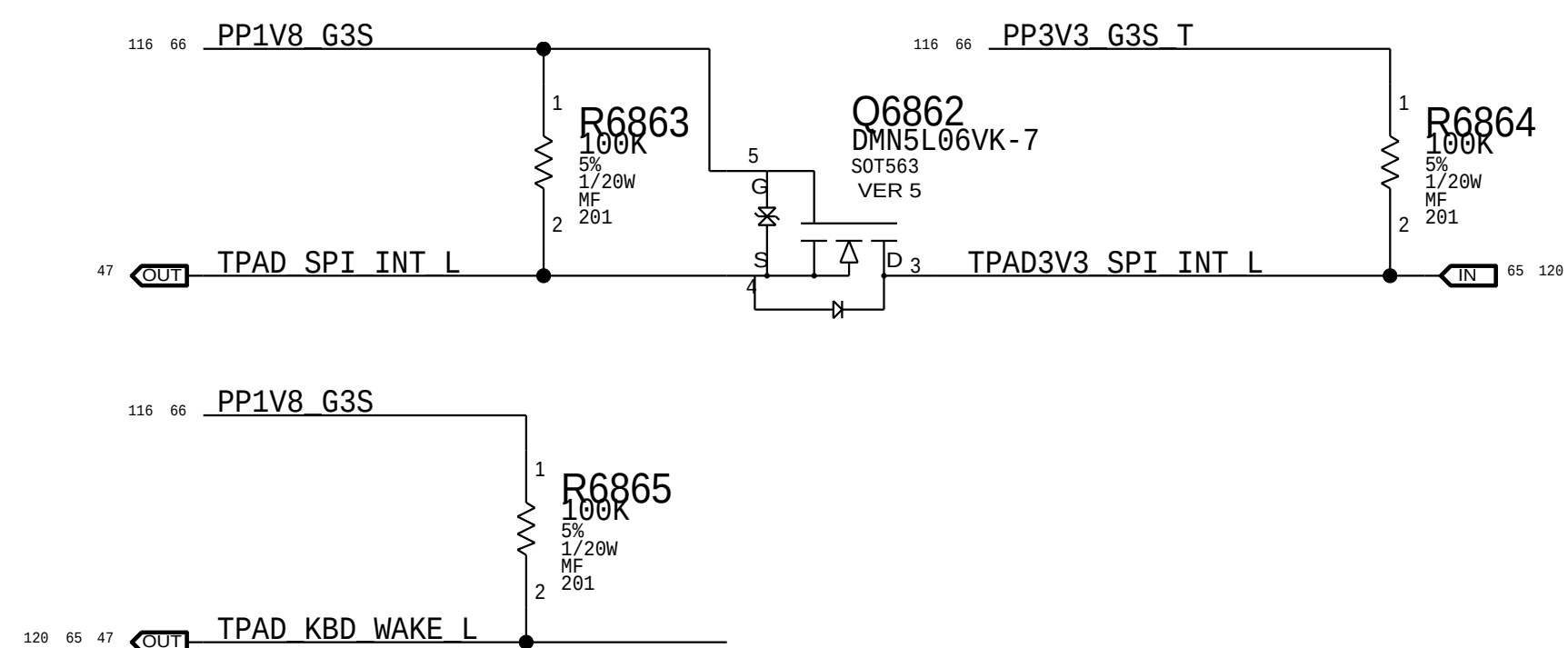
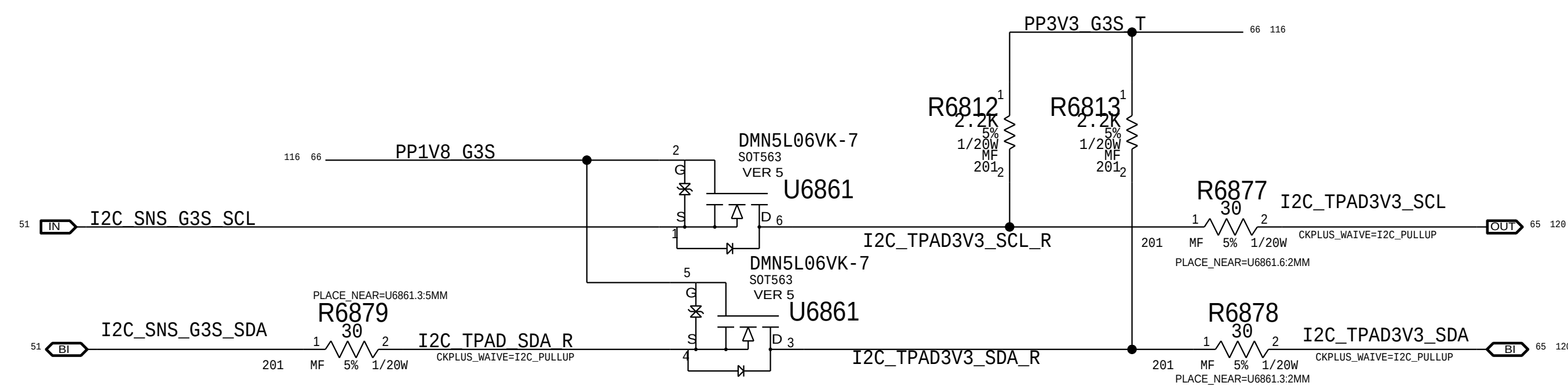
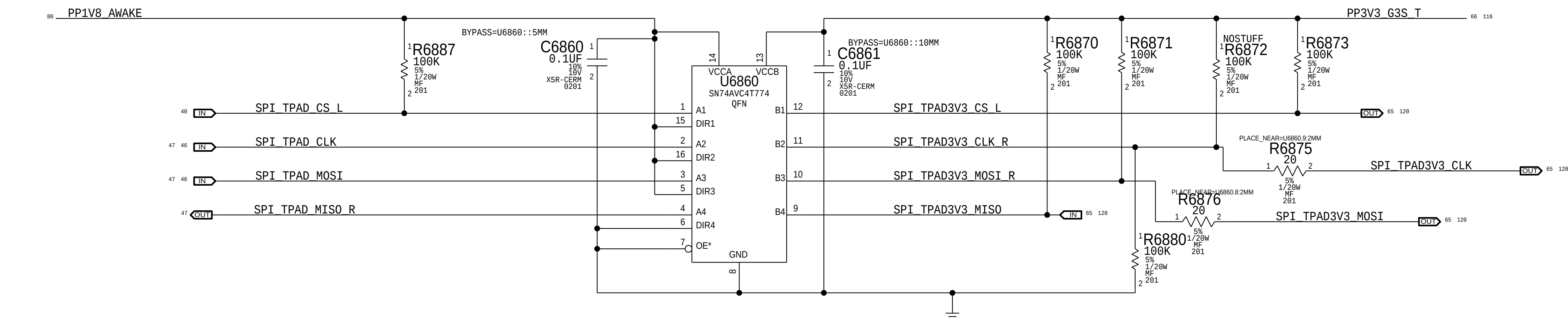
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PAGE TITLE			PAGE TITLE		
Keyboard & Trackpad 1			Keyboard & Trackpad 1		
 Apple Inc.			DRAWING NUMBER	051-02643	SIZE
			REVISION	4.0.0	D
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			SHEET	65 OF 131	

BOM_COST_GROUP=KEYBOARD

Trackpad Level Shifting



BOM_COST_GROUP=KEYBOARD

SYNC_MASTER=j132		SYNC_DATE=03/23/2017	
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		051-02643	
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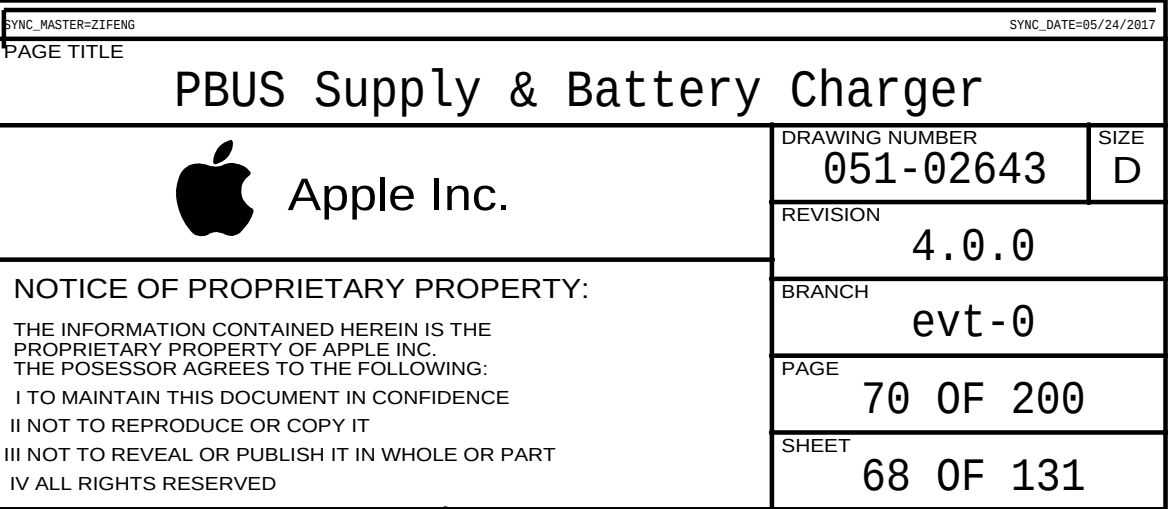
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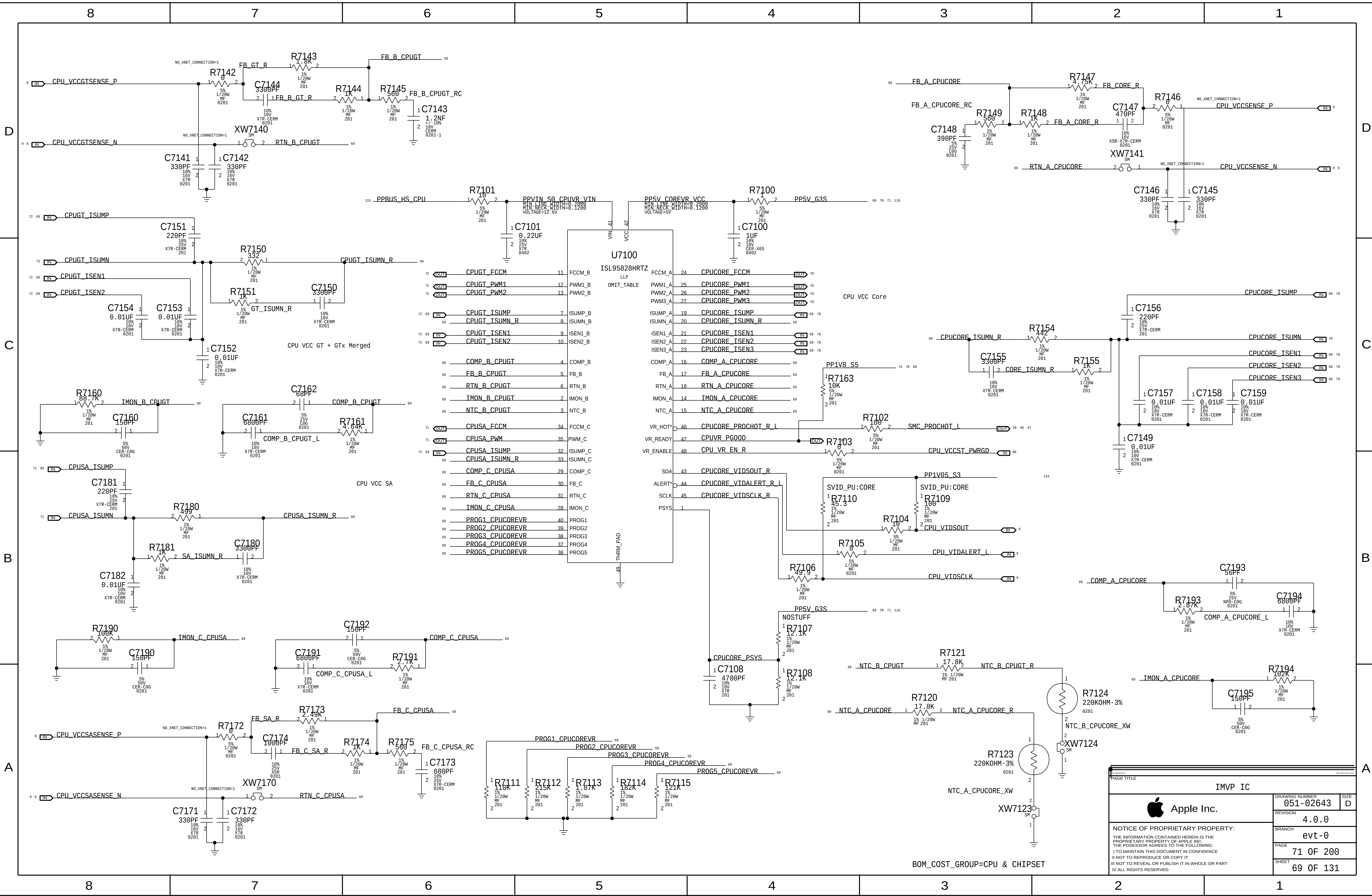


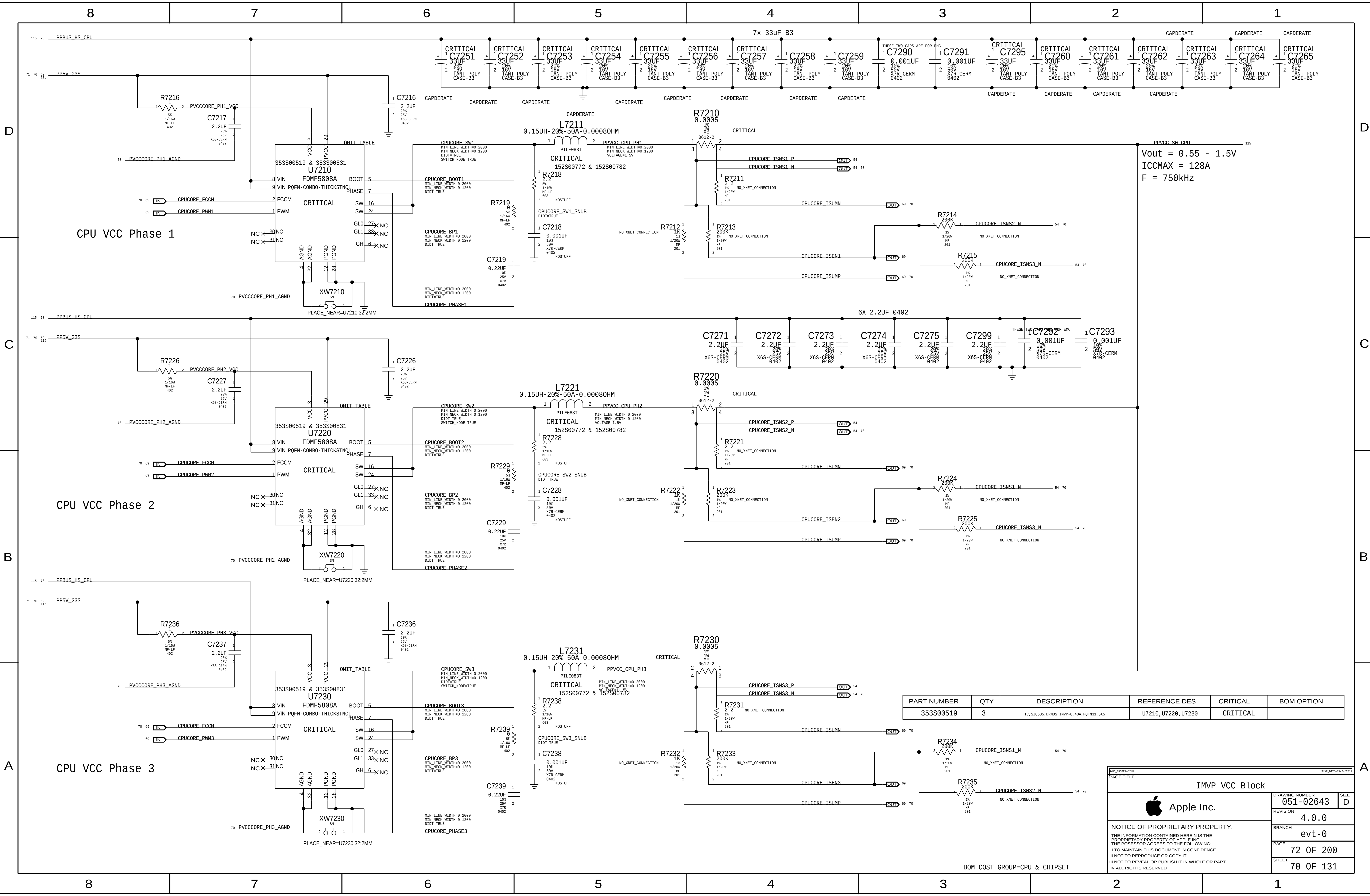
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BOM_COST_GROUP=PLATFORM POWER



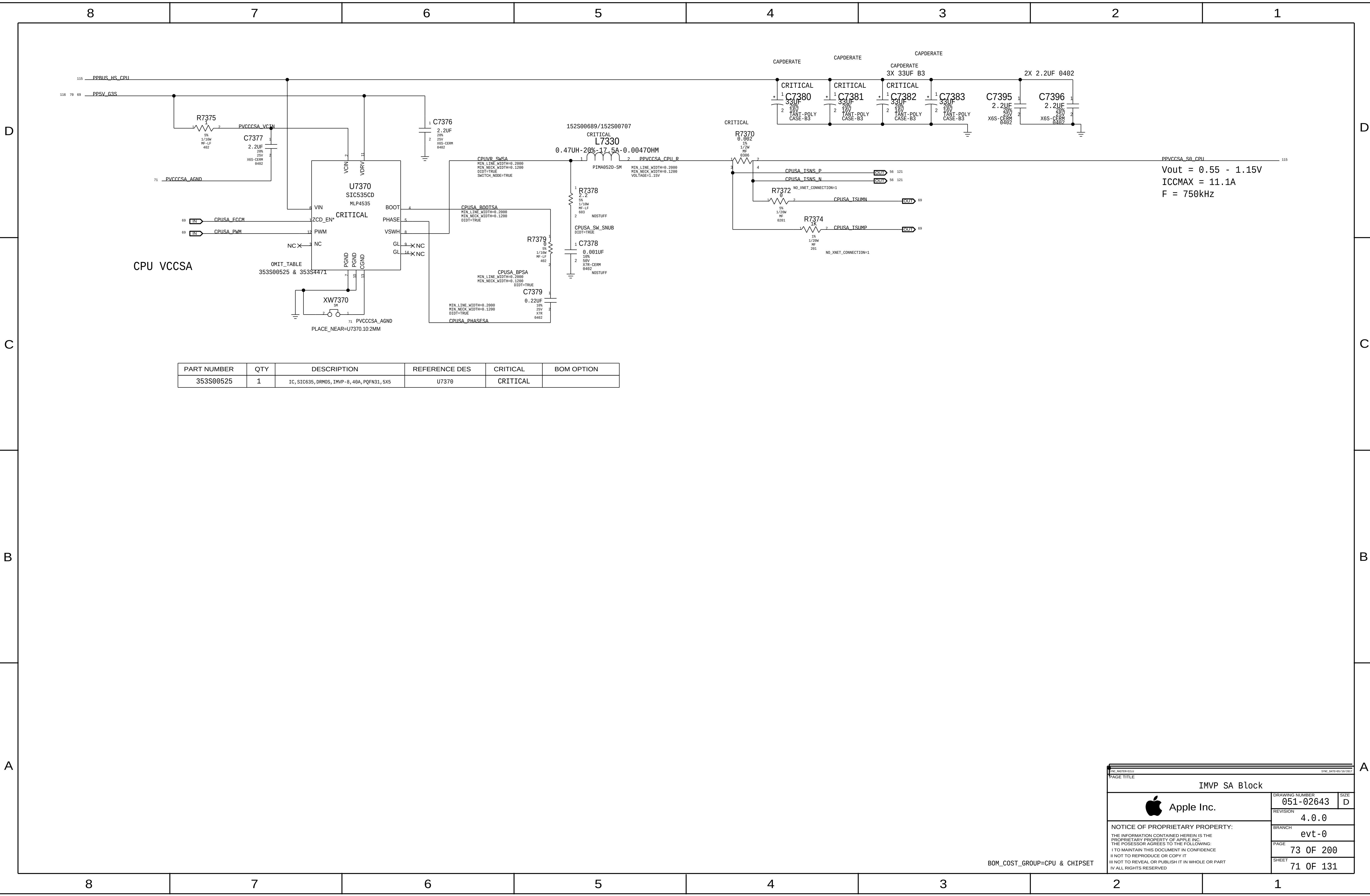




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00519	3	IC, SiC635, DRMOS, IMVP-8, 40A, PQFN31, 5X5	U7210, U7220, U7230	CRITICAL	

IMVP VCC Block			
	DRAWING NUMBER	051-02643	SIZE
	REVISION	4.0.0	D
	BRANCH	evt-0	
	PAGE	72 OF 200	
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BOM_COST_GROUP=CPU & CHIPSET



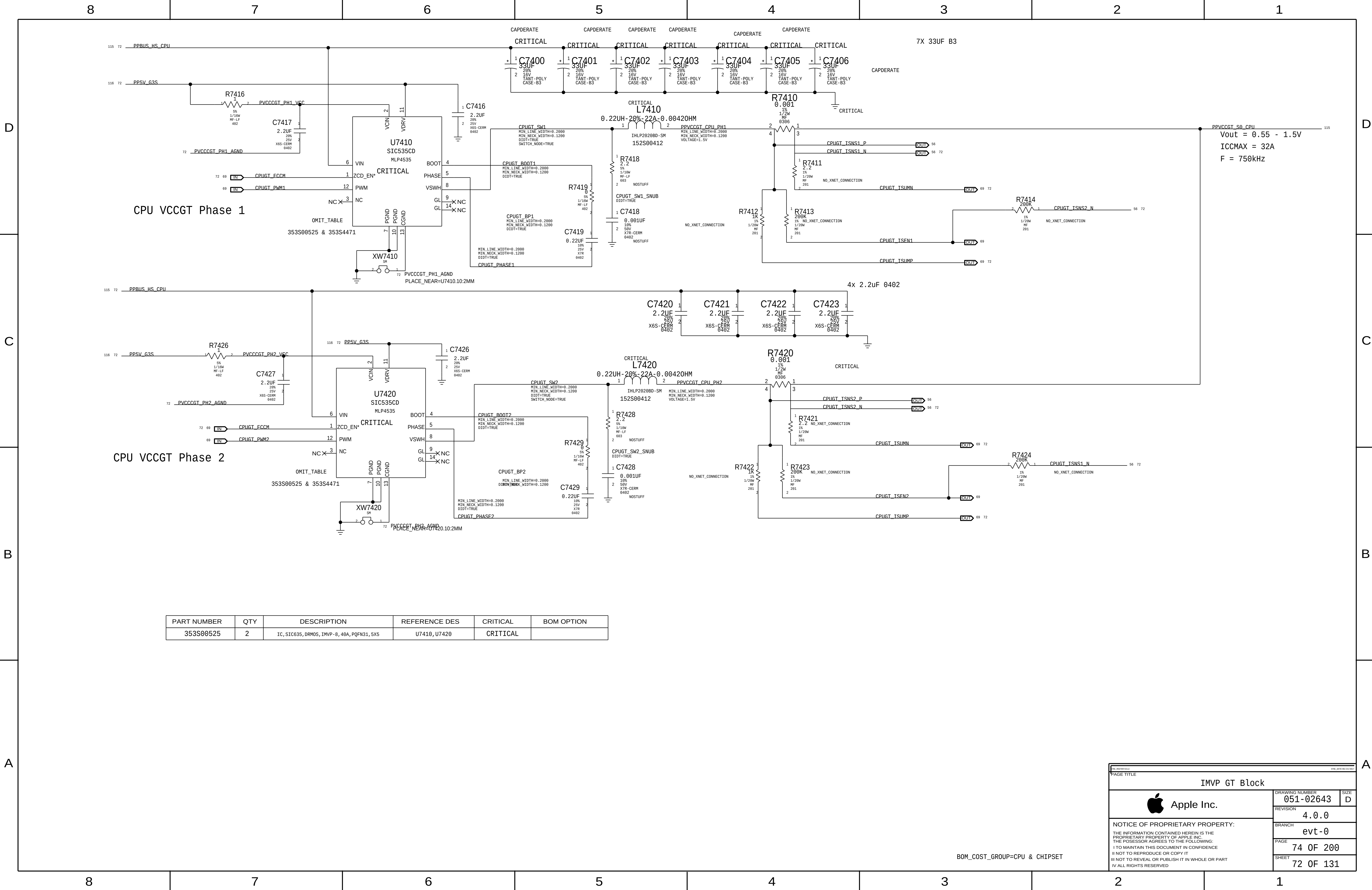
CPU VCCSA

Vout = 0.55 - 1.15V
ICCMAX = 11.1A
F = 750kHz

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00525	1	IC, SIC635, DRMOS, IMVP-8, 40A, PQFN31, 5X5	U7370	CRITICAL	

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE			PAGE TITLE		
IMVP SA Block			IMVP SA Block		
 Apple Inc.			DRAWING NUMBER	051-02643	SIZE
			REVISION	4.0.0	D
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			PAGE	73 OF 200	
			SHEET	71 OF 131	



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00525	2	IC, SIC635, DRMOS, IMVP-8, 40A, PQFN31, 5X5	U7410, U7420	CRITICAL	

PAGE TITLE		
IMVP GT Block		
	DRAWING NUMBER	051-02643
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	PAGE	74 OF 200
	SHEET	72 OF 131

BOM_COST_GROUP=CPU & CHIPSET

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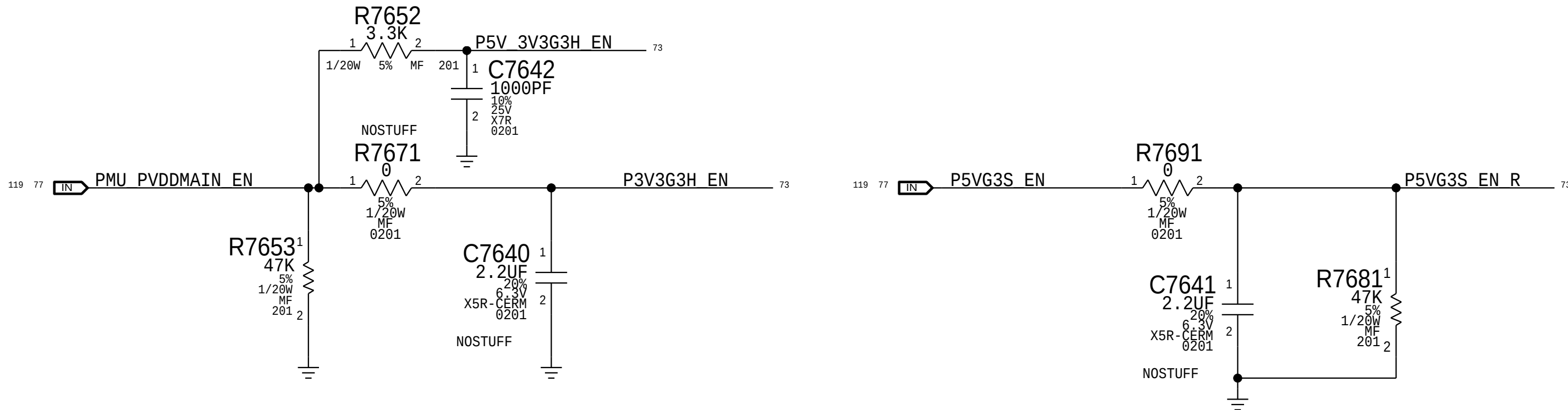
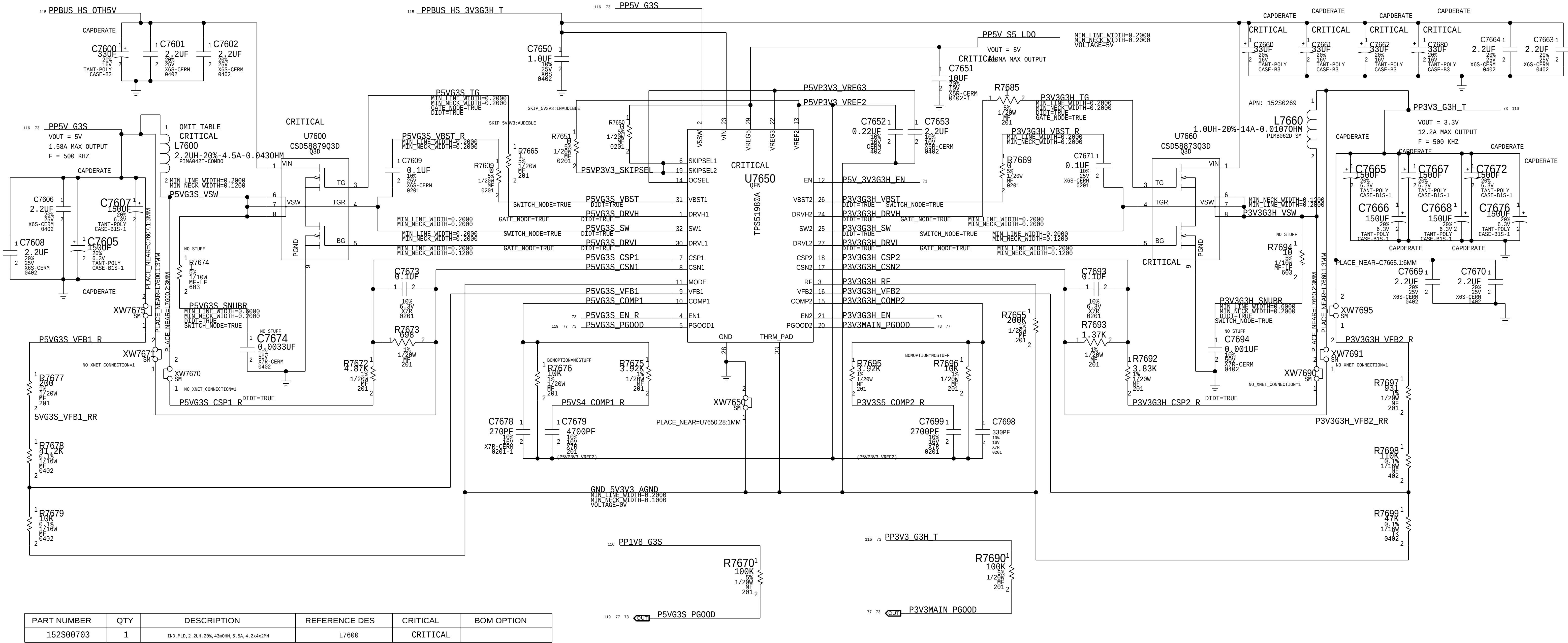
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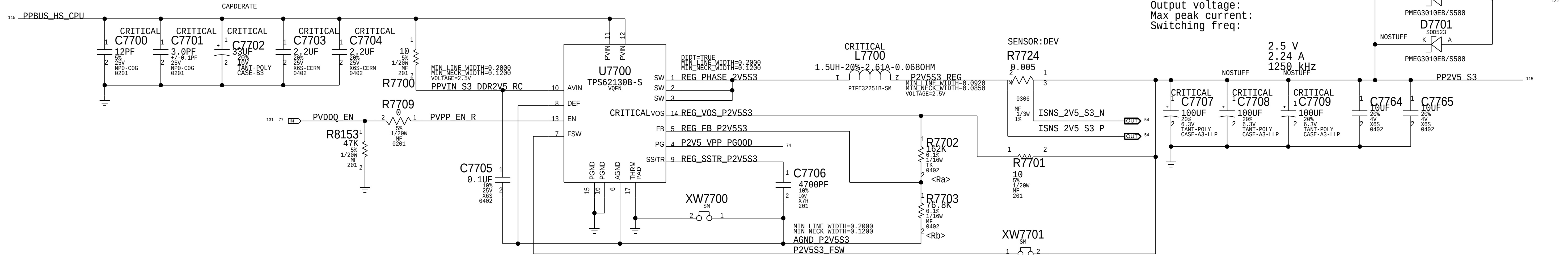
5V S0 - V5

3.3V DSW - V6

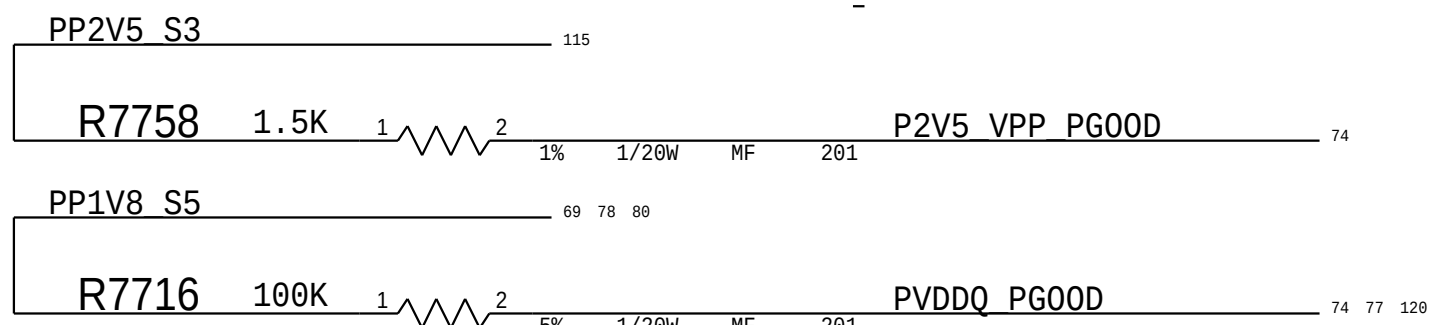
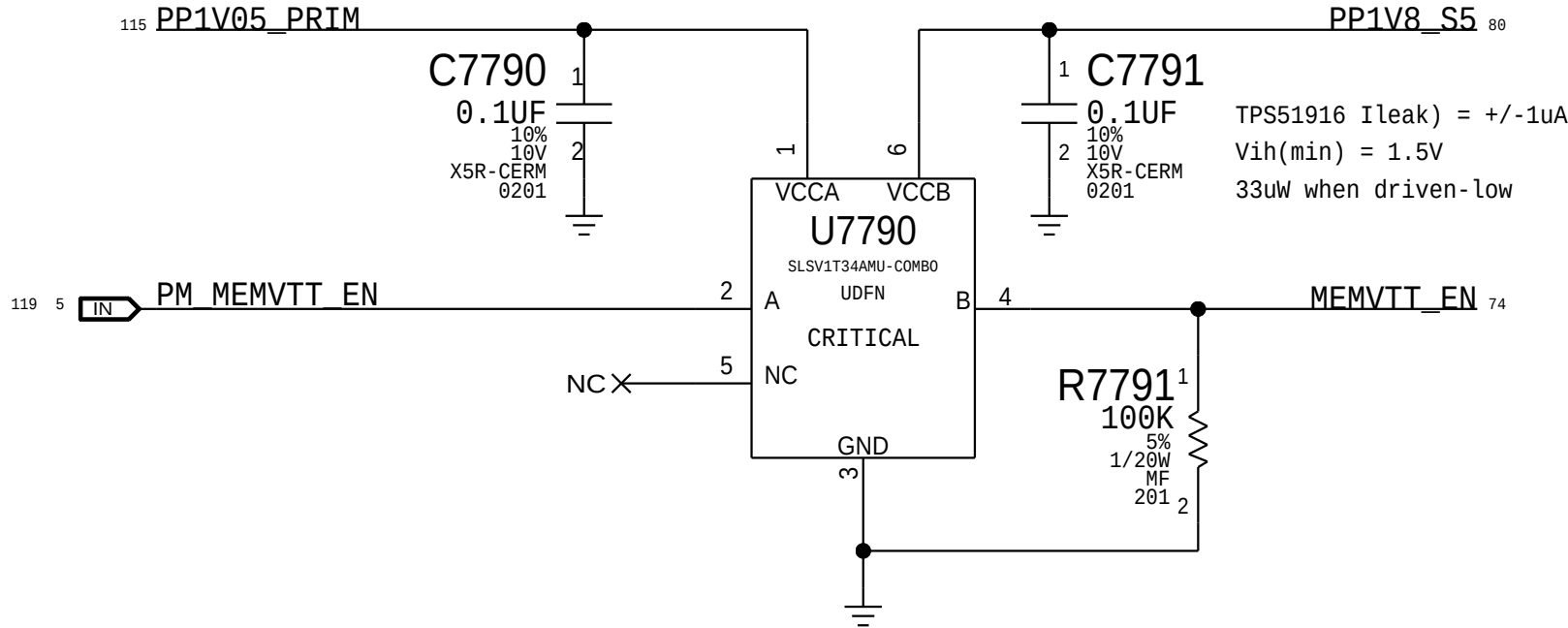
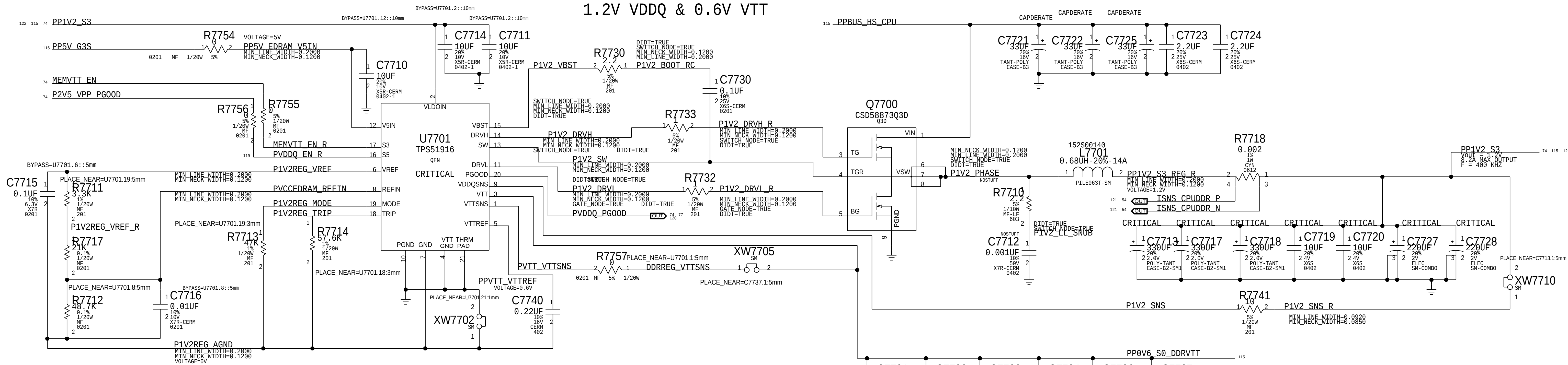


BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE		
Power - 5V 3.3V Supply		
 Apple Inc.	DRAWING NUMBER	051-02643
	REVISION	4.0.0
	BRANCH	evt-0
	PAGE	76 OF 200
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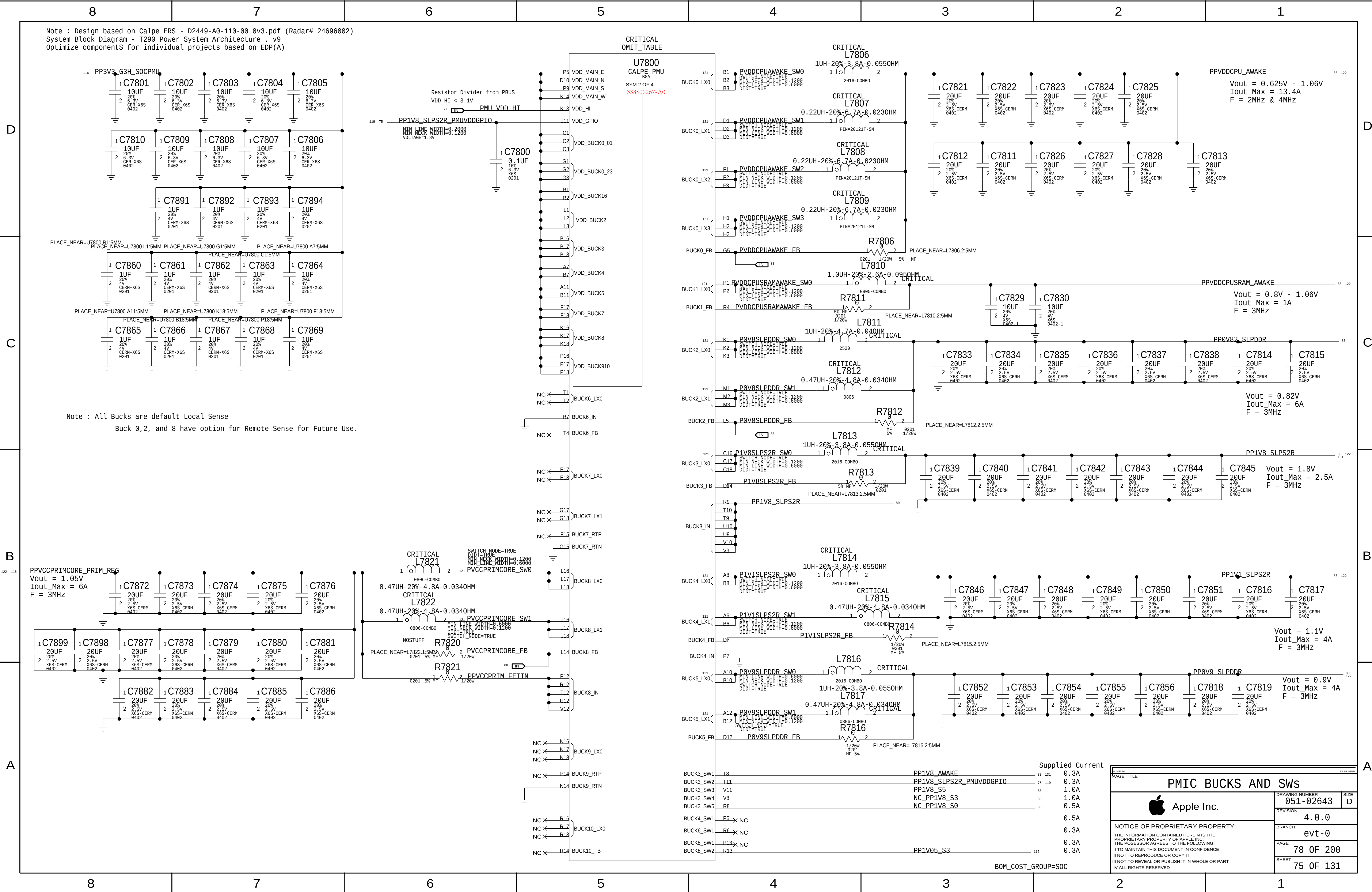


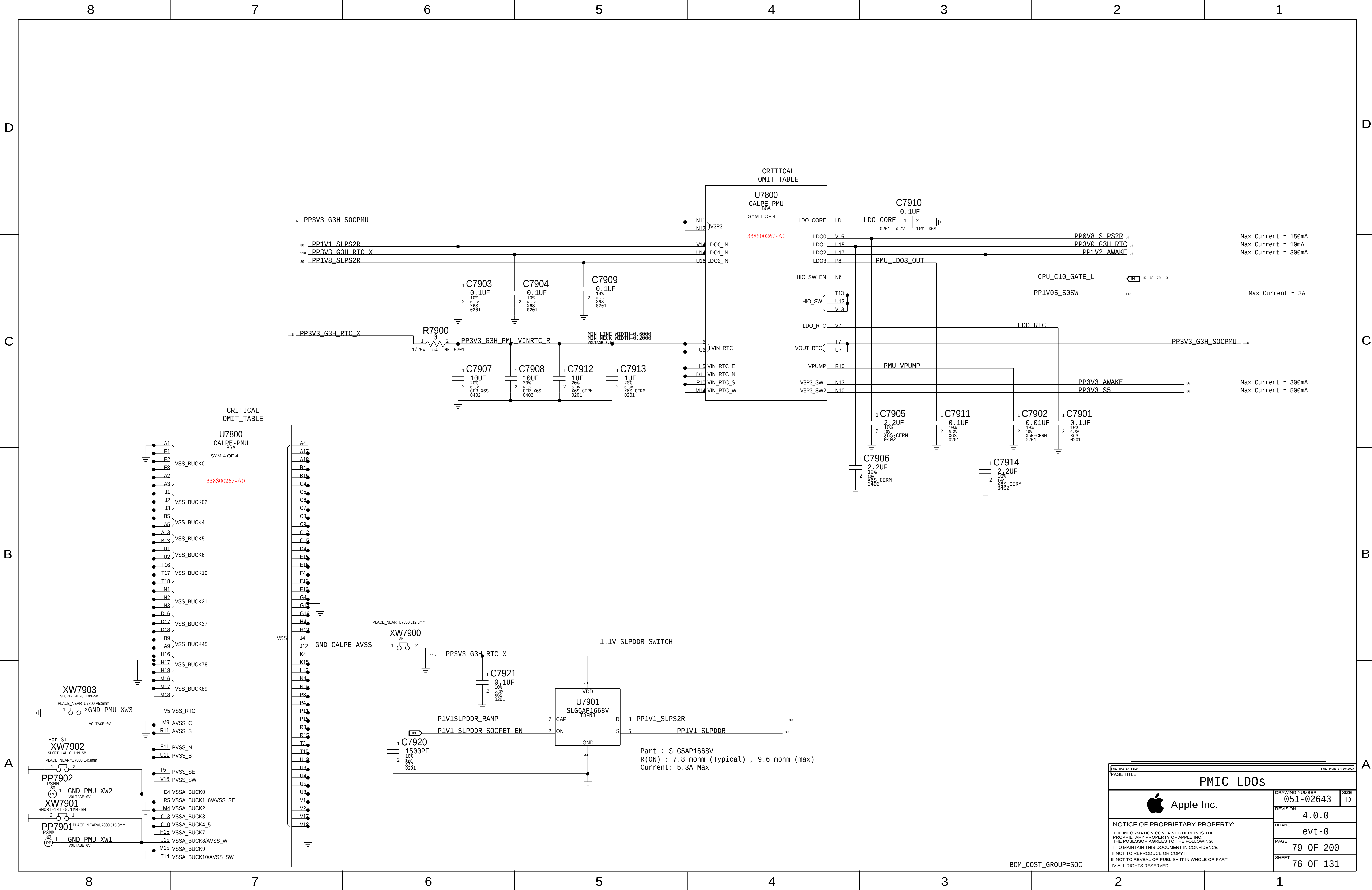
The diagram shows the output stage of the D7700 and D7701. It consists of two PMEG3010EB/S500 MOSFETs. The gates are driven by D7700 and D7701 drivers. The drains are connected to a common load (74, 115, 122) through a 3S3 diode. The sources are connected to ground through a 3S3 diode.



PAGE TITLE			
VR 2.5V & 1.2V/VTT			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-02643		D
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			4.0.0
	BRANCH		
			evt-0
	PAGE		
			77 OF 200
	SHEET		
			74 OF 131

BOM_COST_GROUP=PLATFORM POWER





PMIC LDOs		
	DRAWING NUMBER	051-02643
	REVISION	4.0.0
	BRANCH	evt-0
	PAGE	79 OF 200
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BOM_COST_GROUP=SOC

D

C

B

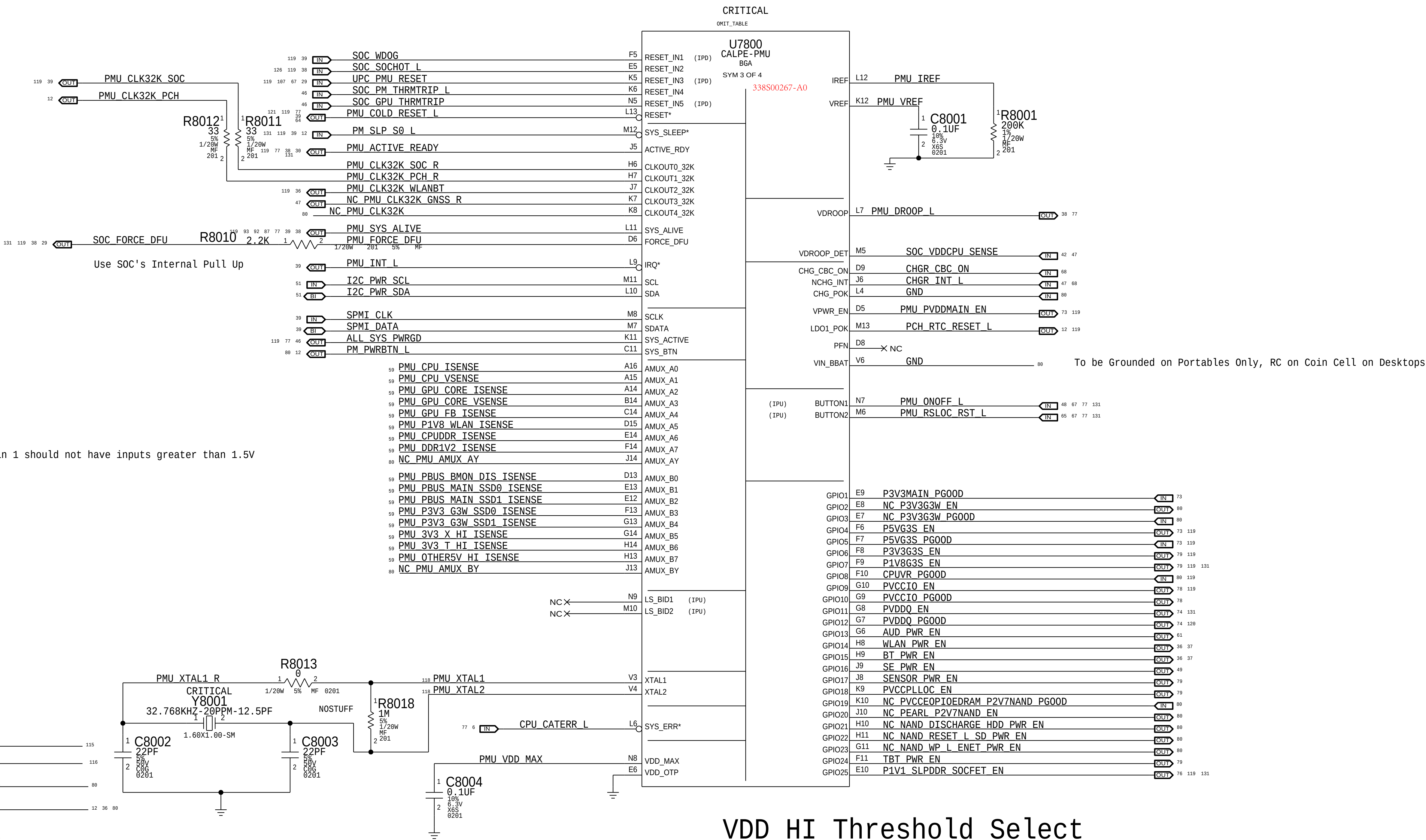
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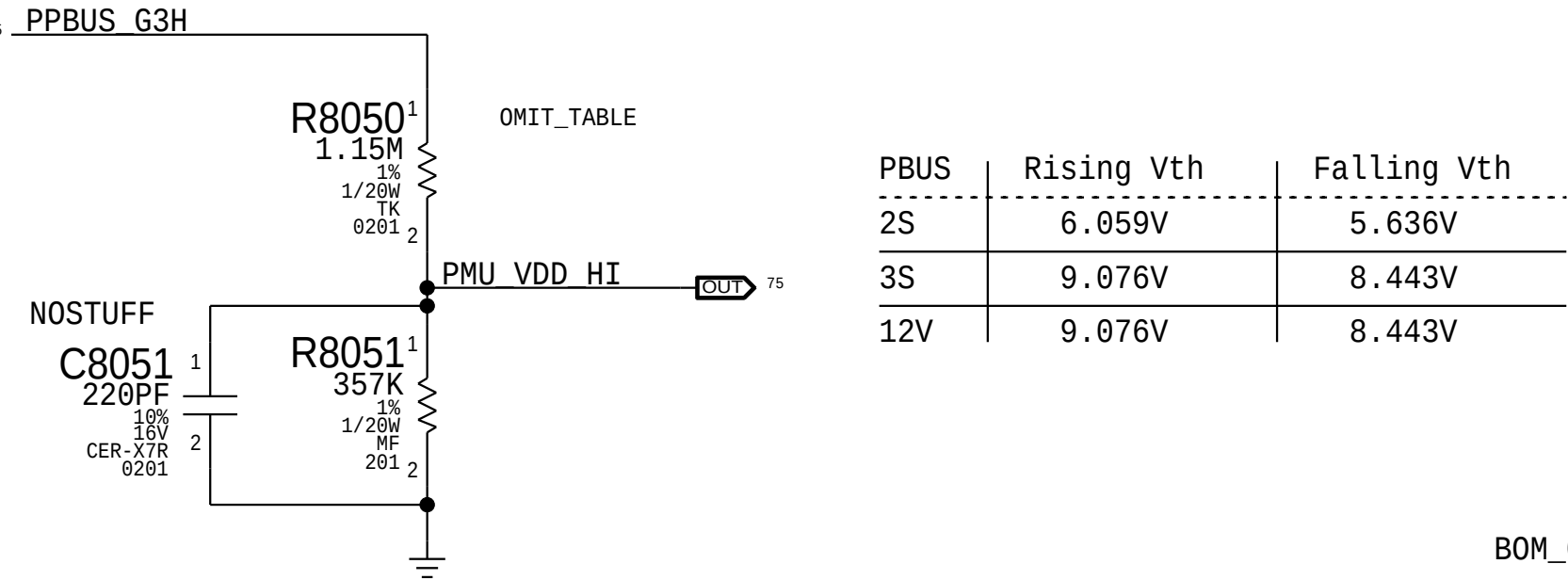
A



Caution : AMUX programmed with Gain 1 should not have inputs greater than 1.5V

VDD_HI Threshold Select

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0481	1	RES, MF, 649K, 1%, 1/20W, 0201	R8050	CRITICAL	PBUS:2S
118S00077	1	RES, MF, 1.15M, 1%, 1/16W, 0201	R8050	CRITICAL	PBUS:3S
118S00077	1	RES, MF, 1.15M, 1%, 1/16W, 0201	R8050	CRITICAL	PBUS:12V

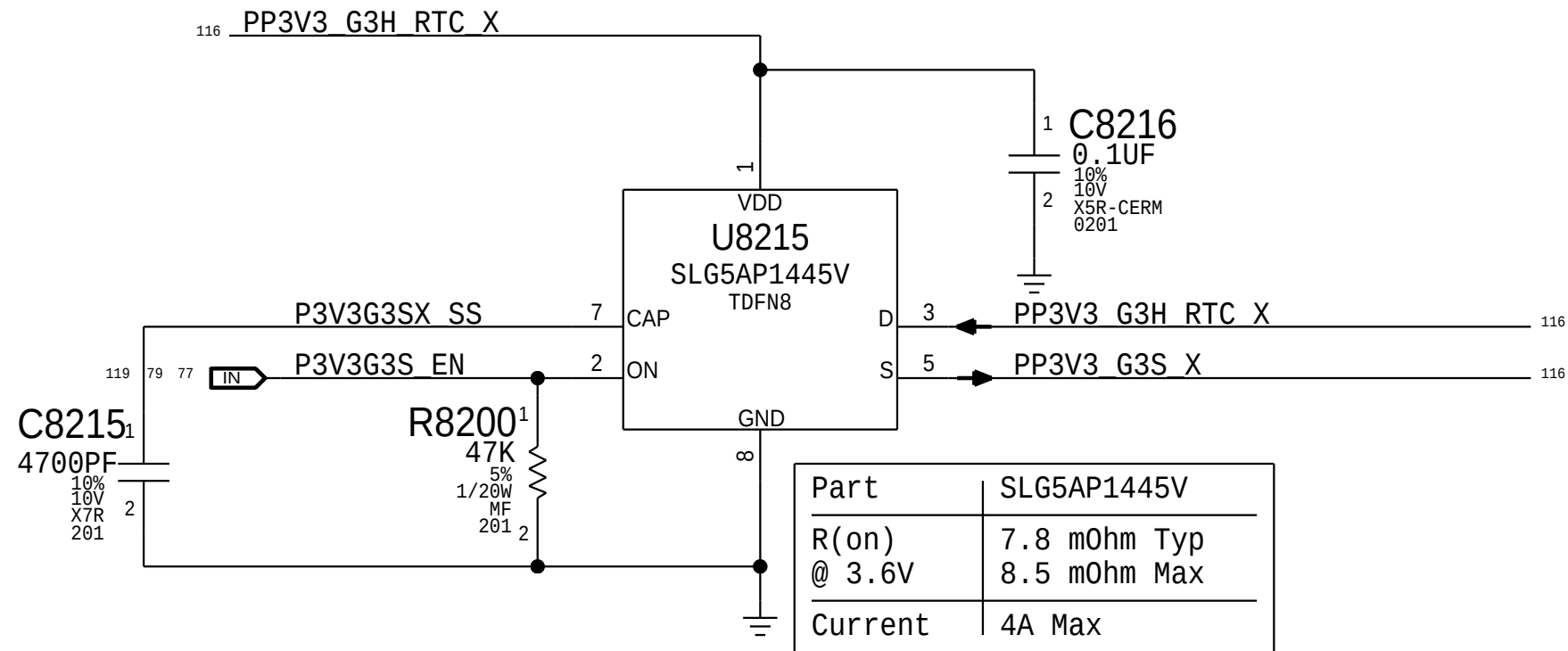


PBUS	Rising Vth	Falling Vth
2S	6.059V	5.636V
3S	9.076V	8.443V
12V	9.076V	8.443V

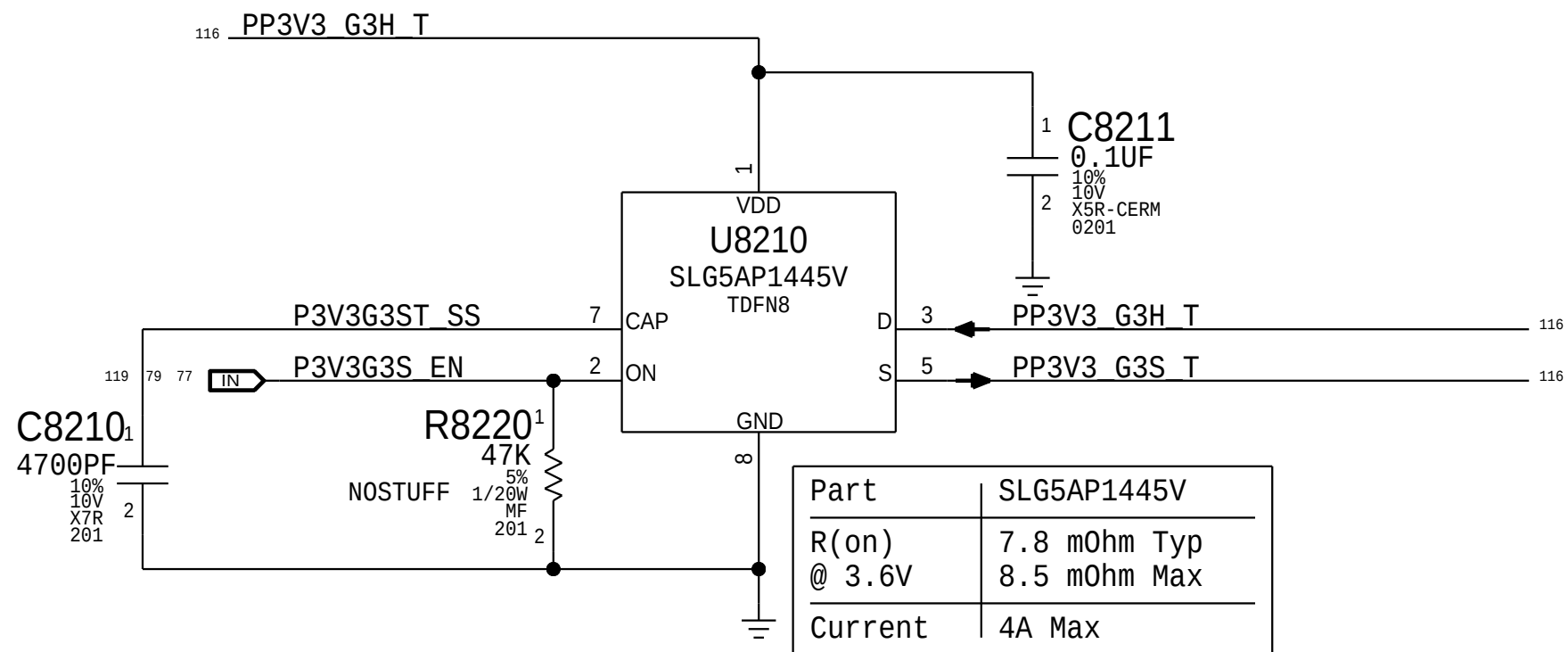
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PAGE TITLE		
PMIC GPIOs & Control		
	DRAWING NUMBER	051-02643
	REVISION	4.0.0
	BRANCH	evt-0
	PAGE	80 OF 200
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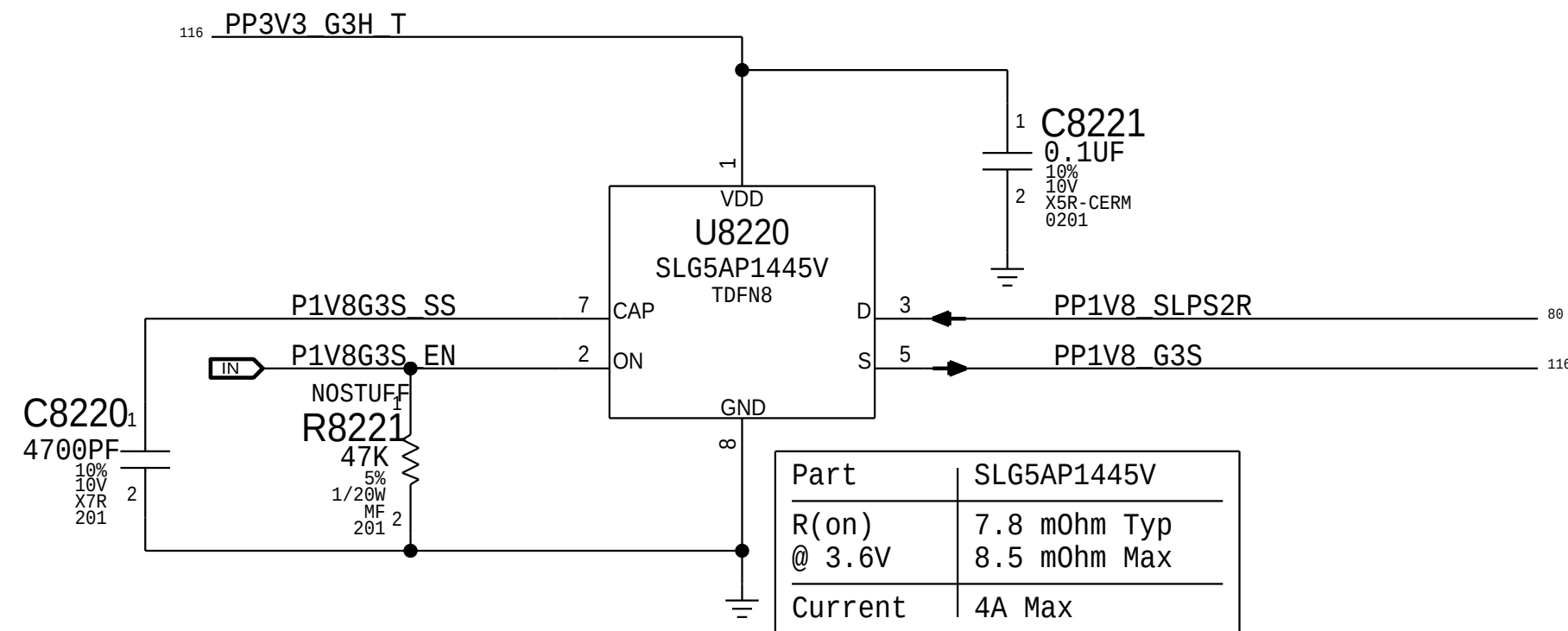
3.3V G3 Standby Switch X



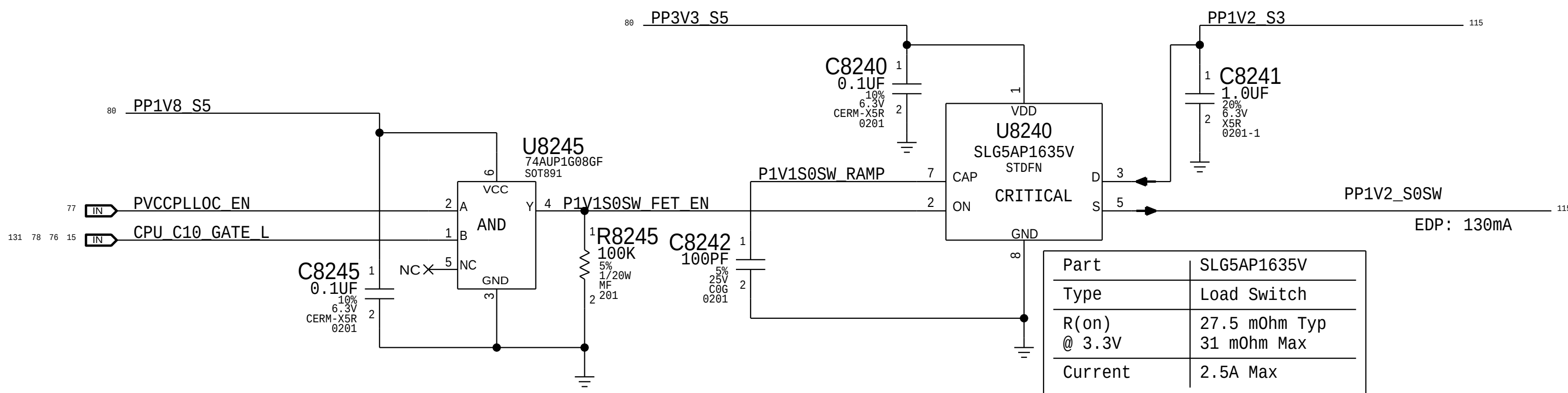
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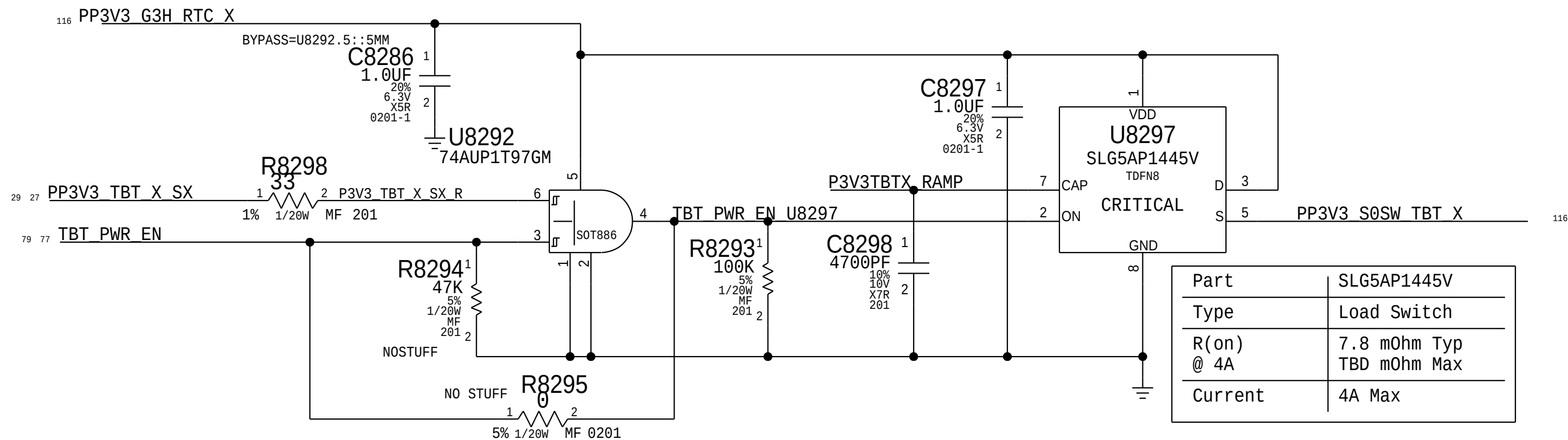
1.8V G3 Standby Switch



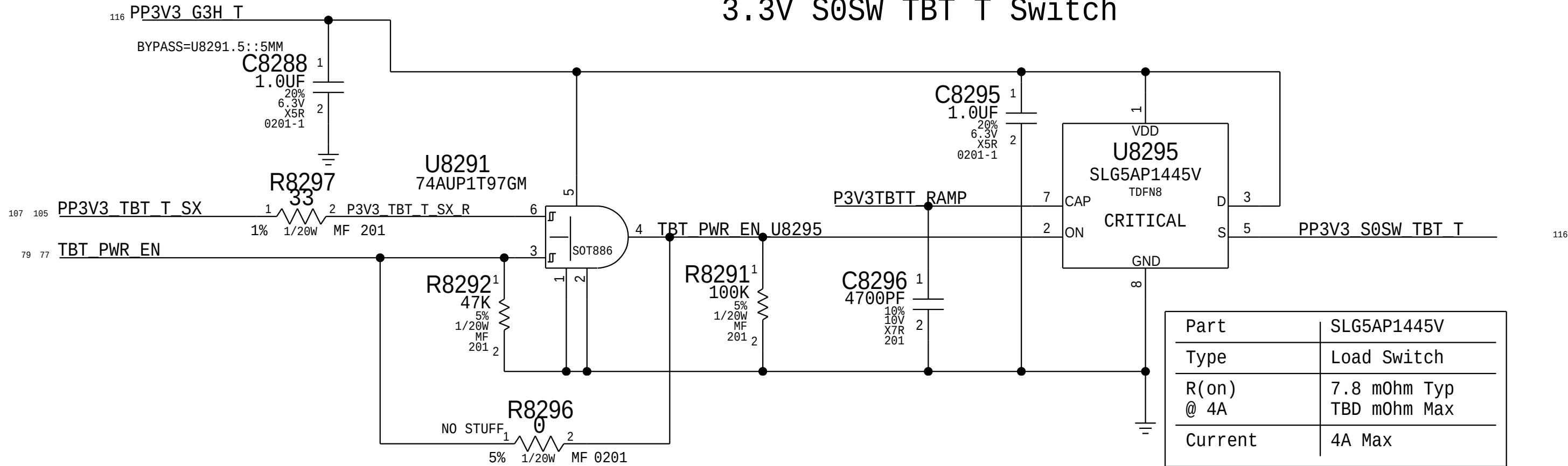
1.1V S0SW VCCPLL_OC Switch



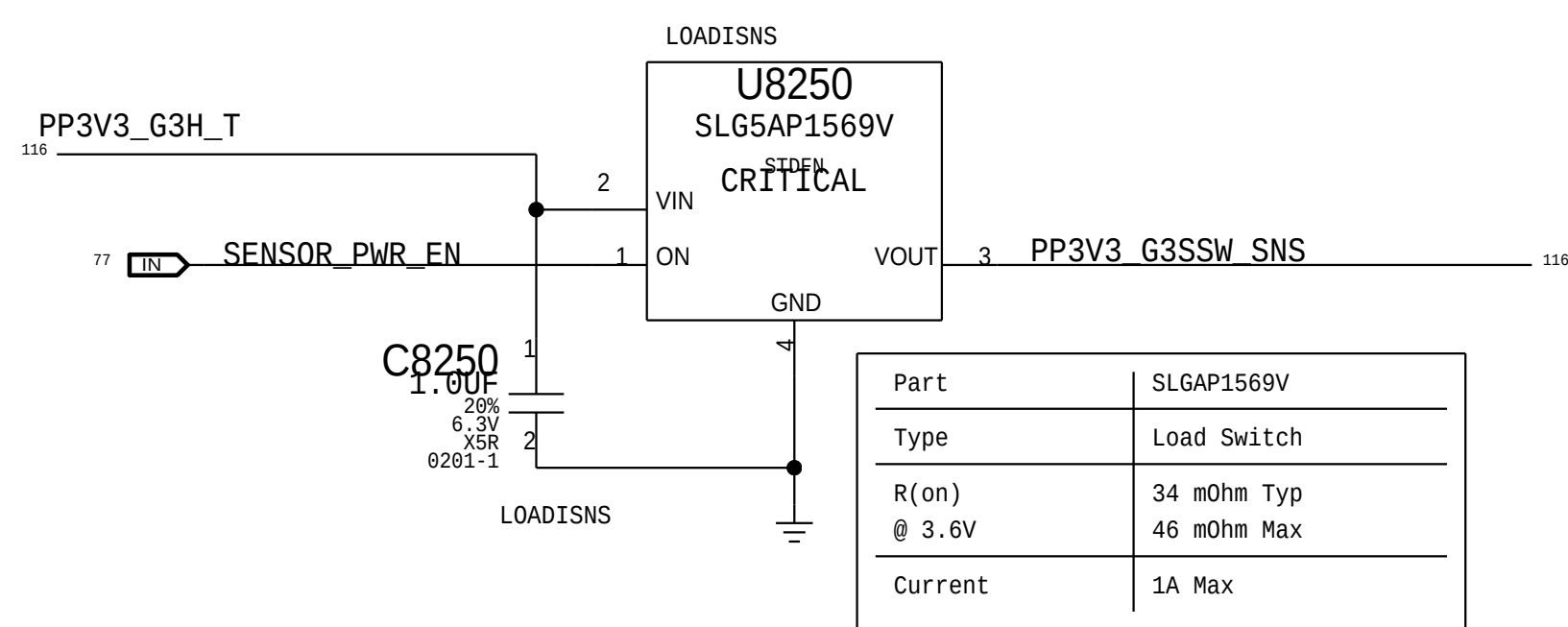
3.3V S0SW TBT X Switch



3.3V S0SW TBT T Switch



3.3V Sensor Switch



Power FETs



Apple Inc.

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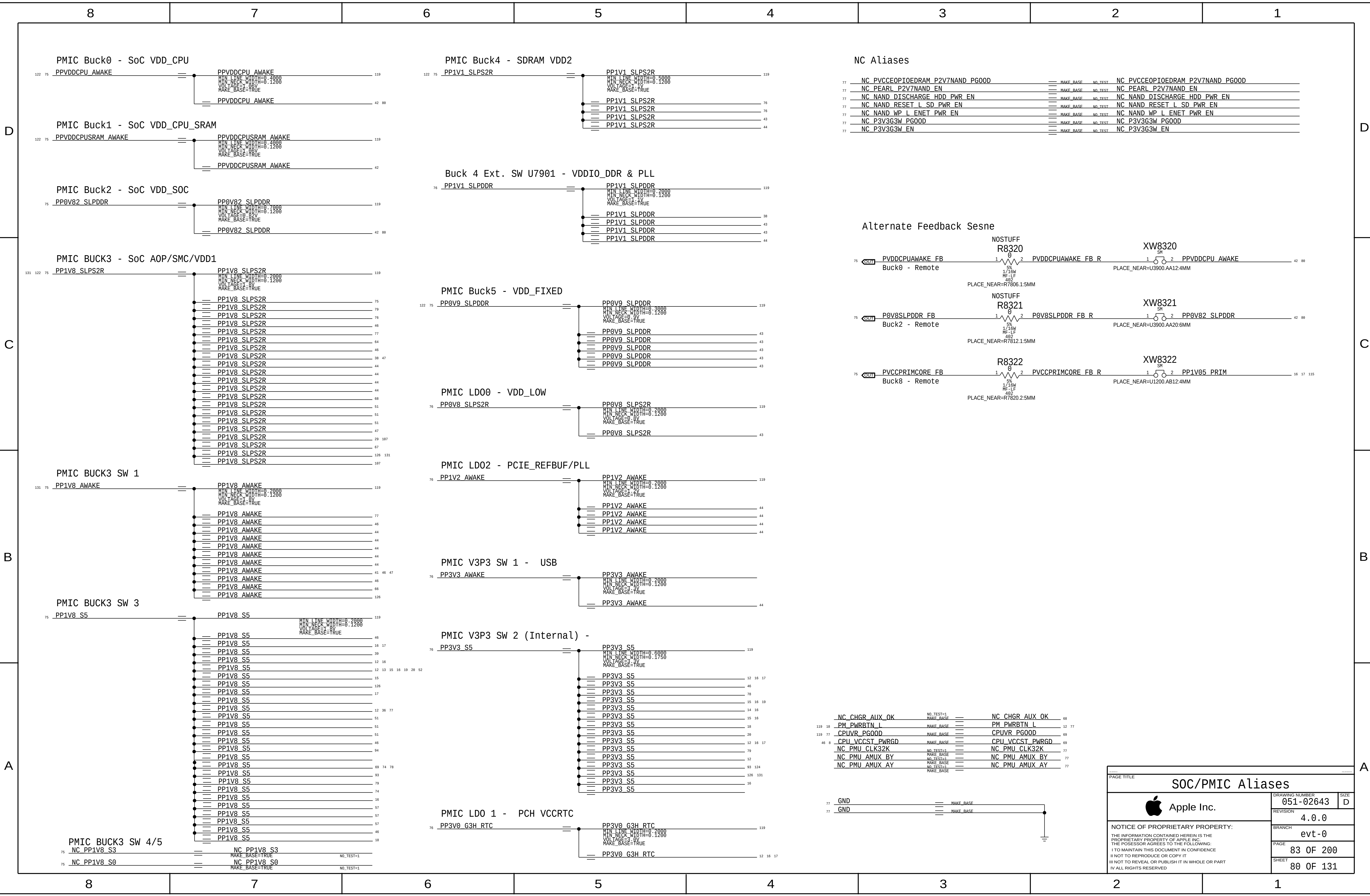
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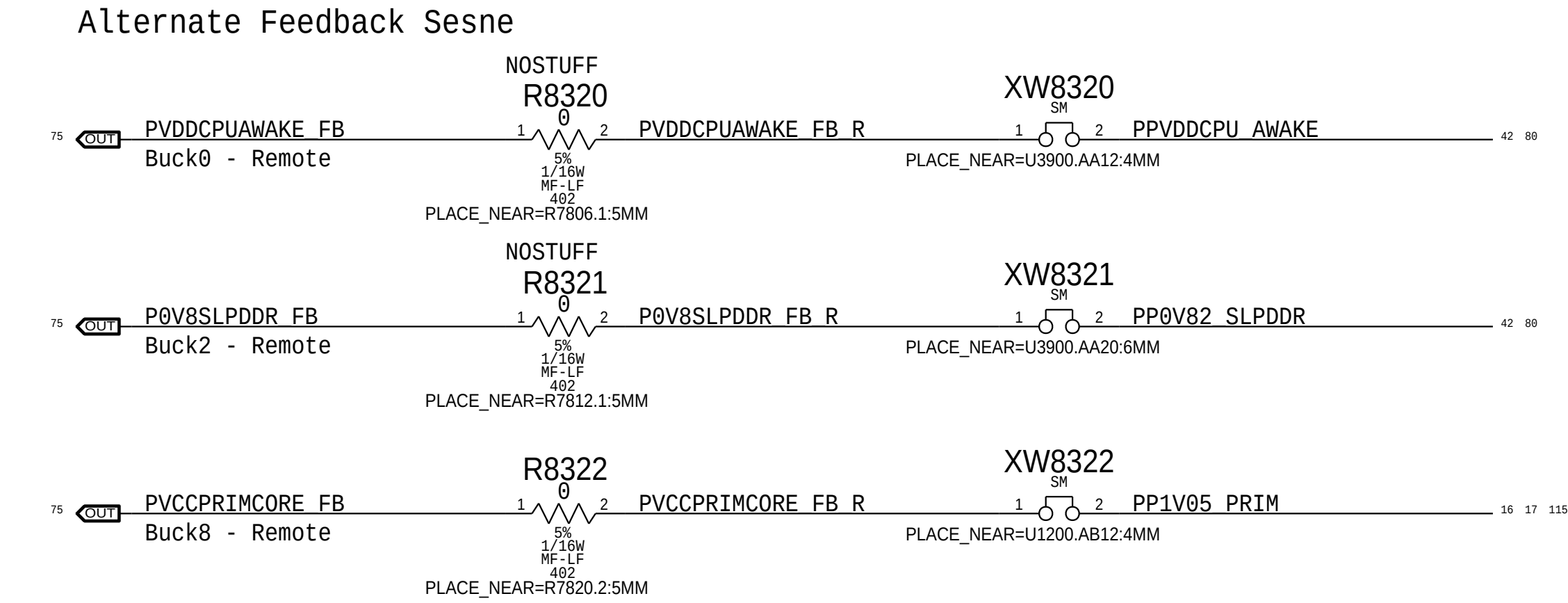
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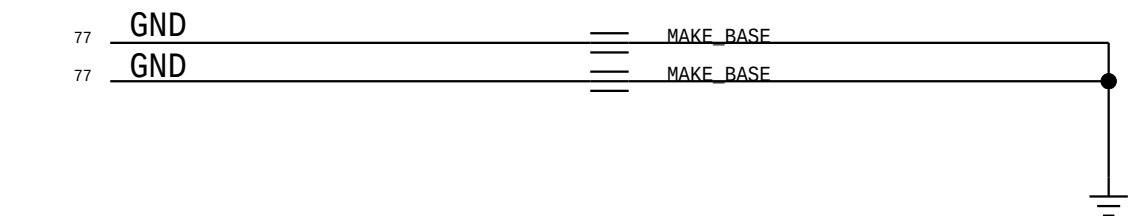
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NC Aliases		
77 NC_PVCCEOPIOEDRAM_P2V7NAND_PG00D	MAKE_BASE NO_TEST	NC_PVCCEOPIOEDRAM_P2V7NAND_PG00D
77 NC_PEARL_P2V7NAND_EN	MAKE_BASE NO_TEST	NC_PEARL_P2V7NAND_EN
77 NC_NAND_DISCHARGE_HDD_PWR_EN	MAKE_BASE NO_TEST	NC_NAND_DISCHARGE_HDD_PWR_EN
77 NC_NAND_RESET_L_SD_PWR_EN	MAKE_BASE NO_TEST	NC_NAND_RESET_L_SD_PWR_EN
77 NC_NAND_WP_L_ENET_PWR_EN	MAKE_BASE NO_TEST	NC_NAND_WP_L_ENET_PWR_EN
77 NC_P3V3G3W_PG00D	MAKE_BASE NO_TEST	NC_P3V3G3W_PG00D
77 NC_P3V3G3W_EN	MAKE_BASE NO_TEST	NC_P3V3G3W_EN



77 NC_CHGR_AUX_OK	NO_TEST=1 MAKE_BASE	NC_CHGR_AUX_OK	68
119 18 PM_PWRBTN_L	MAKE_BASE	PM_PWRBTN_L	12 77
119 77 CPUVR_PG00D	MAKE_BASE	CPUVR_PG00D	68
48 8 CPU_VCCST_PWRGD	MAKE_BASE	CPU_VCCST_PWRGD	68
NC_PMU_CLK32K	NO_TEST=1 MAKE_BASE	NC_PMU_CLK32K	77
NC_PMU_AMUX_BY	MAKE_BASE	NC_PMU_AMUX_BY	77
NC_PMU_AMUX_AY	MAKE_BASE	NC_PMU_AMUX_AY	77



SOC/PMIC Aliases

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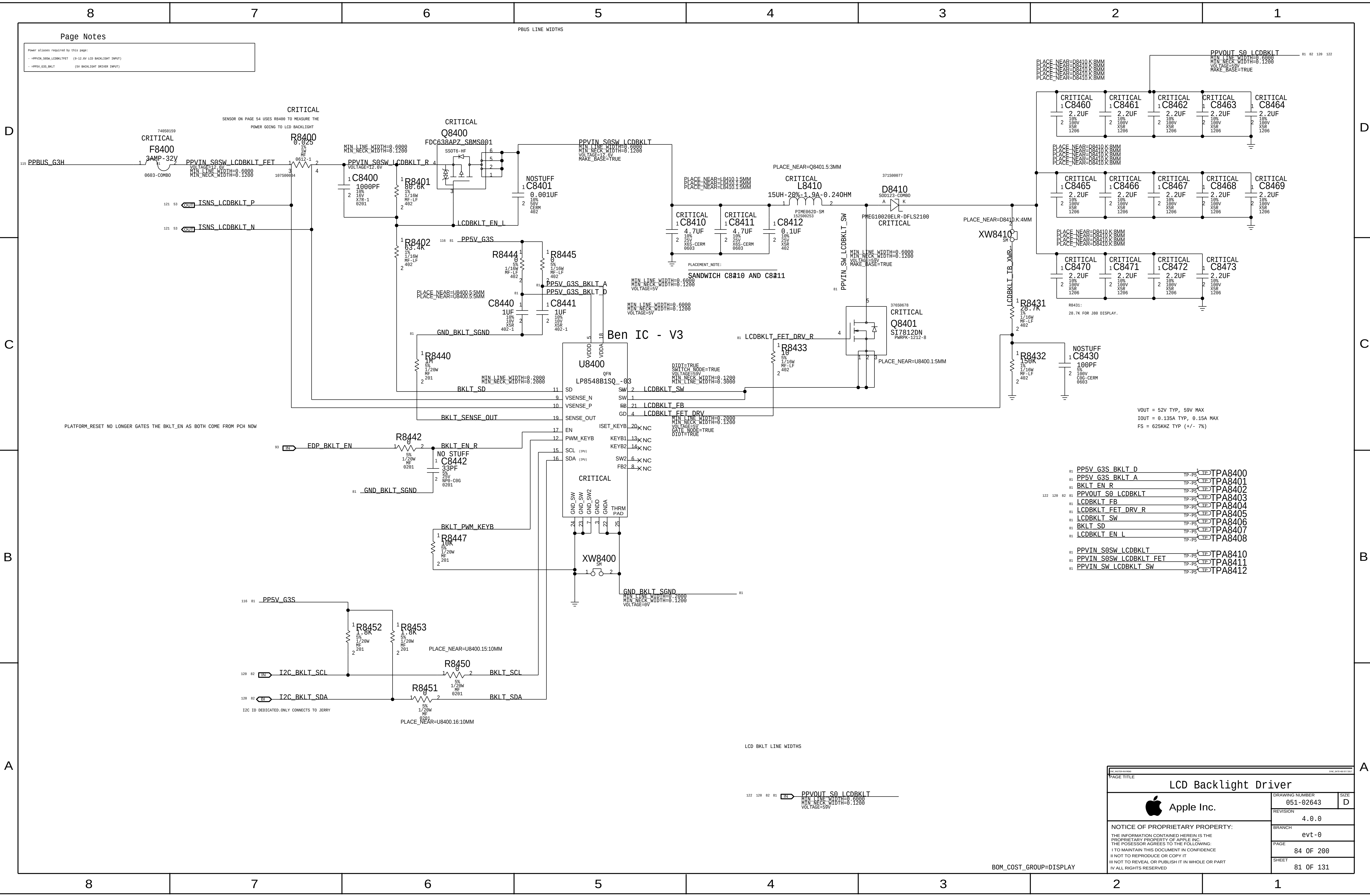
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Page Notes

Power aliases required by this page:
- PP5V_G3S_BKLT (5V BACKLIGHT DRIVER INPUT)
- PP5V_G3S_BKLT (5V BACKLIGHT DRIVER INPUT)

PLATFORM_RESET NO LONGER GATES THE BKLT_EN AS BOTH COME FROM PCH NOW

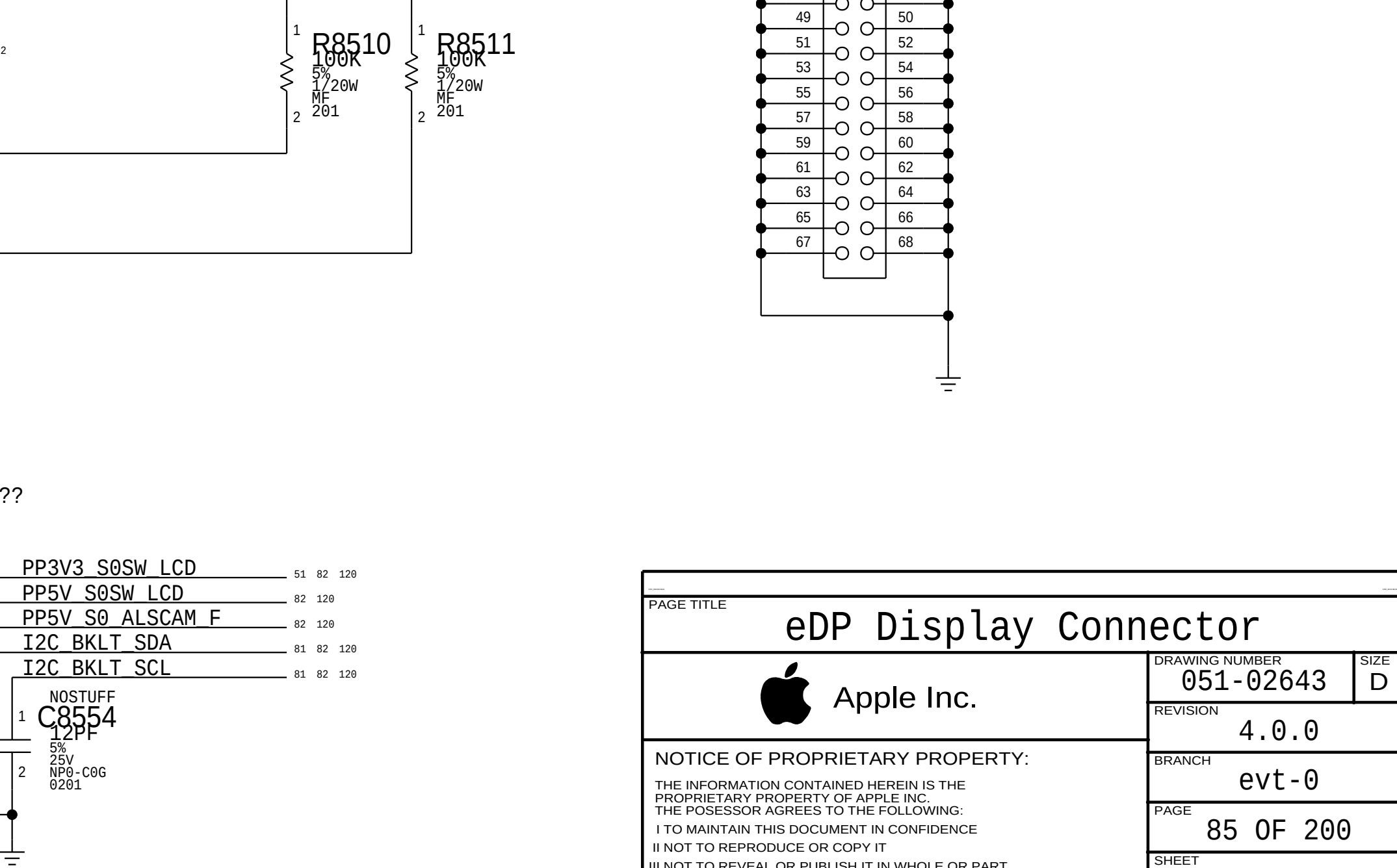
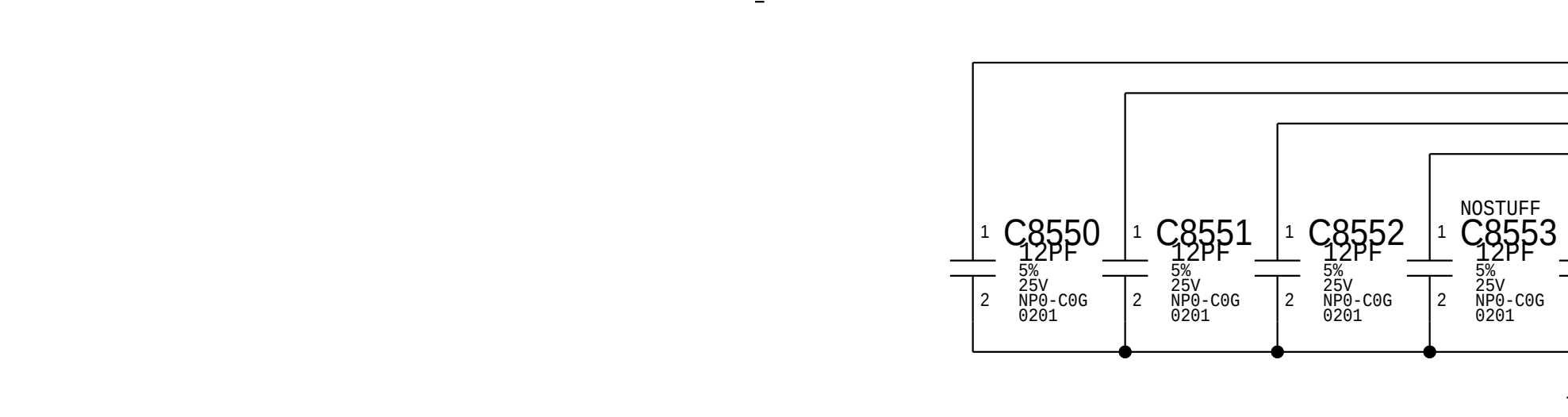
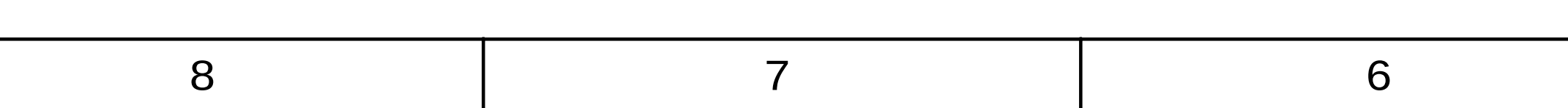
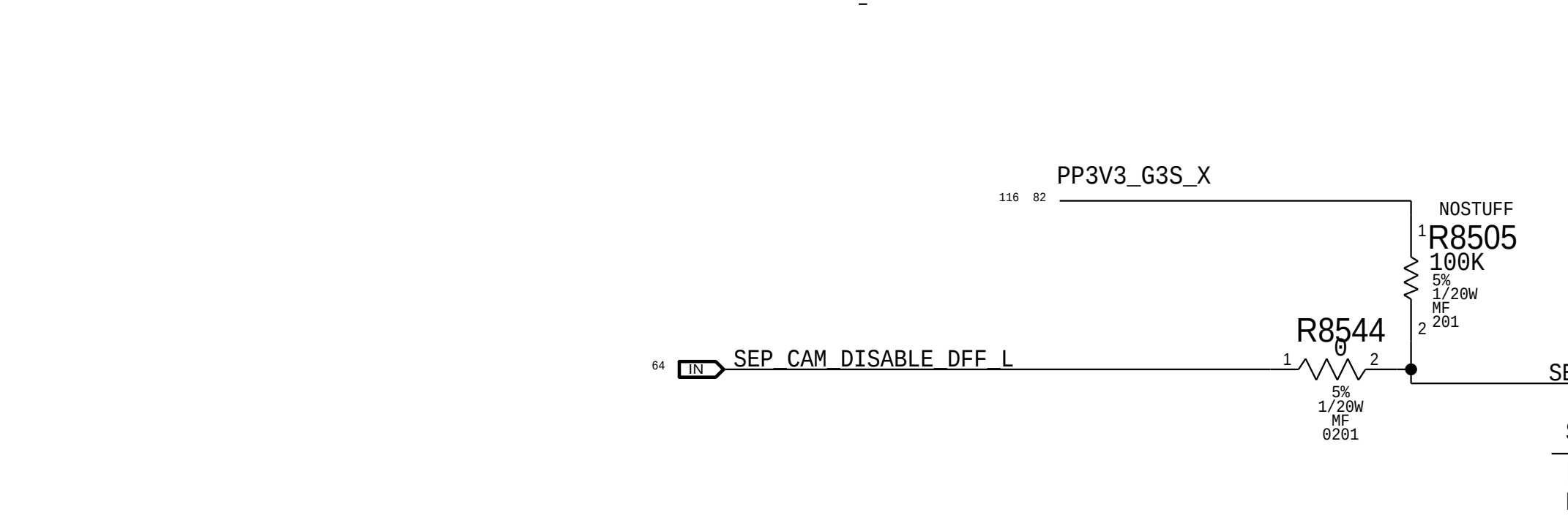
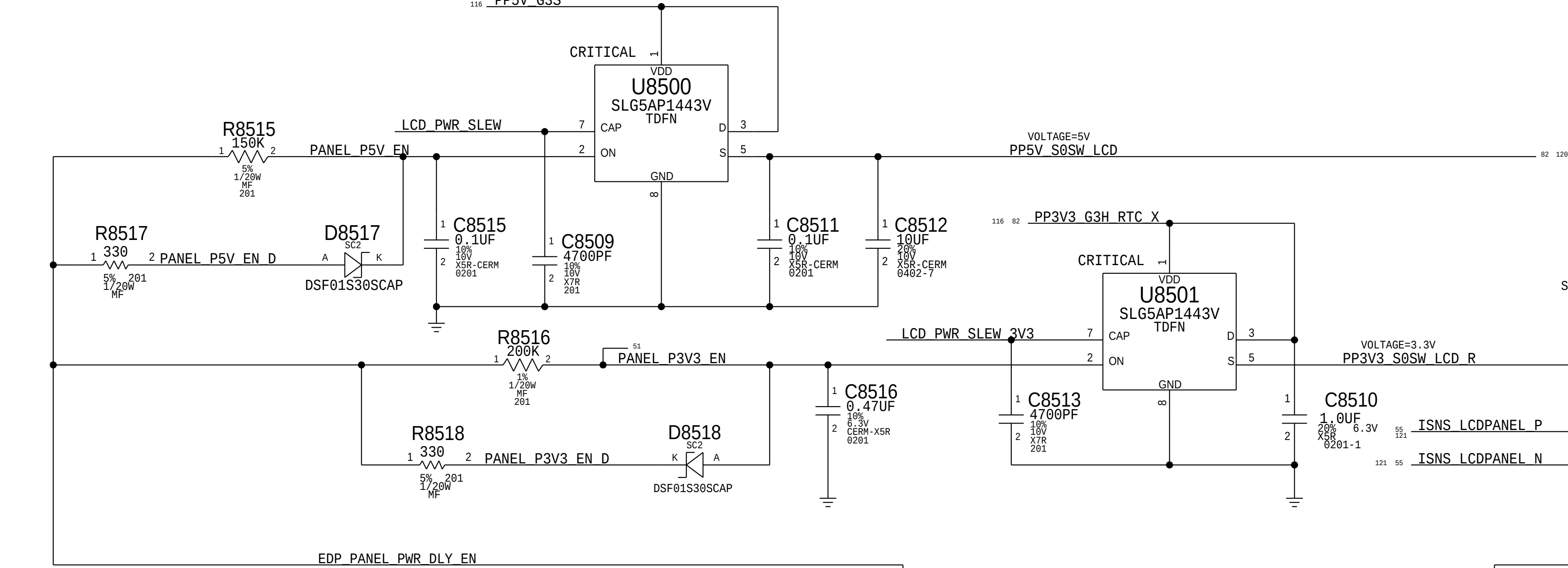
VOUT = 52V TYP, 59V MAX
IOUT = 0.135A TYP, 0.15A MAX
FS = 625KHZ TYP (+/- 7%)

PP5V_G3S_BKLT_D	TP-P5	TPA8400
PP5V_G3S_BKLT_A	TP-P5	TPA8401
BKLT_EN_R	TP-P5	TPA8402
PPVOUT_S0_LCDBKLT	TP-P5	TPA8403
LCDBKLT_FB	TP-P5	TPA8404
LCDBKLT_FET_DRV_R	TP-P5	TPA8405
LCDBKLT_SW	TP-P5	TPA8406
BKLT_SD	TP-P5	TPA8407
LCDBKLT_EN_L	TP-P5	TPA8408

PPVIN_S0SW_LCDBKLT	TP-P5	TPA8410
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PPVIN_SW_LCDBKLT_SW	TP-P5	TPA8412

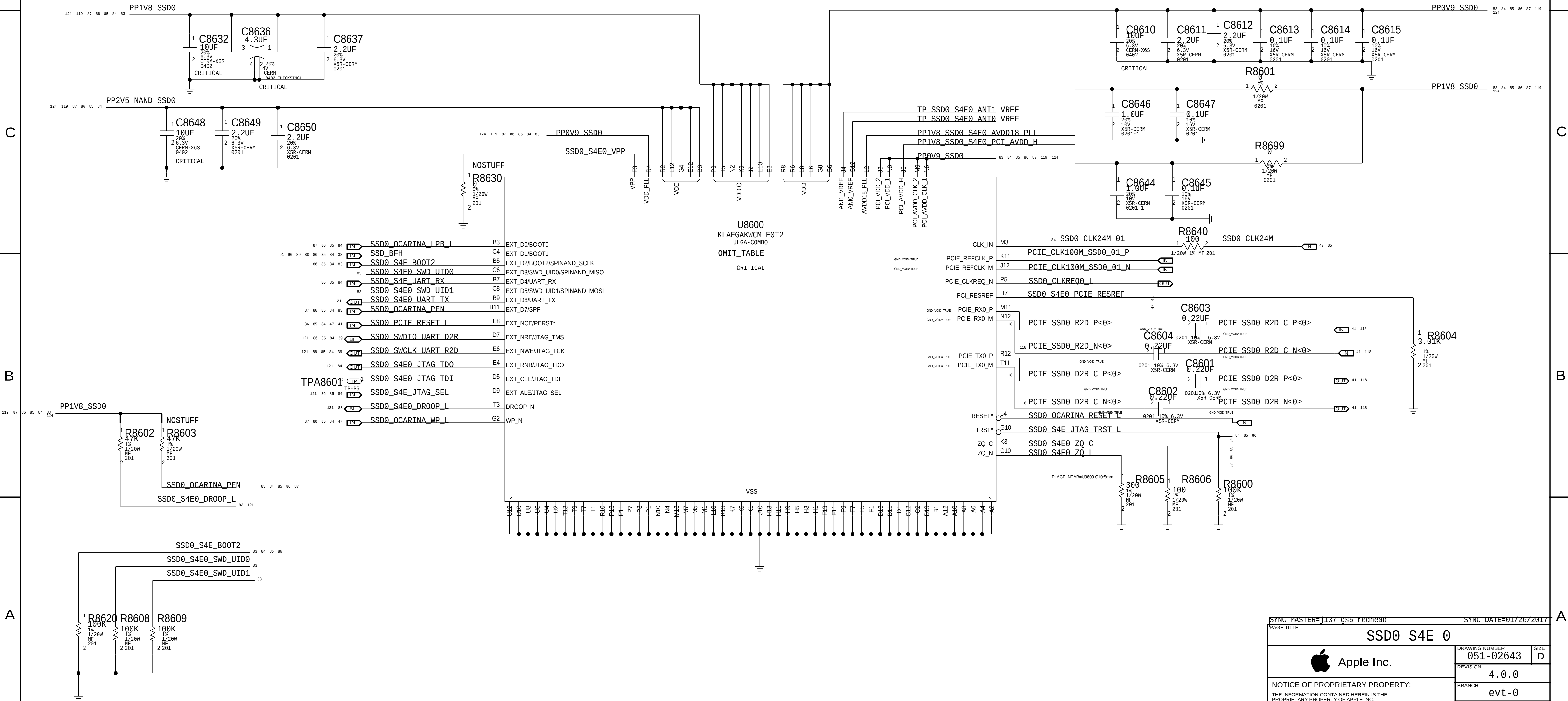
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LCD Backlight Driver		
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BOM_COST_GROUP=DISPLAY



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eDP Display Connector		
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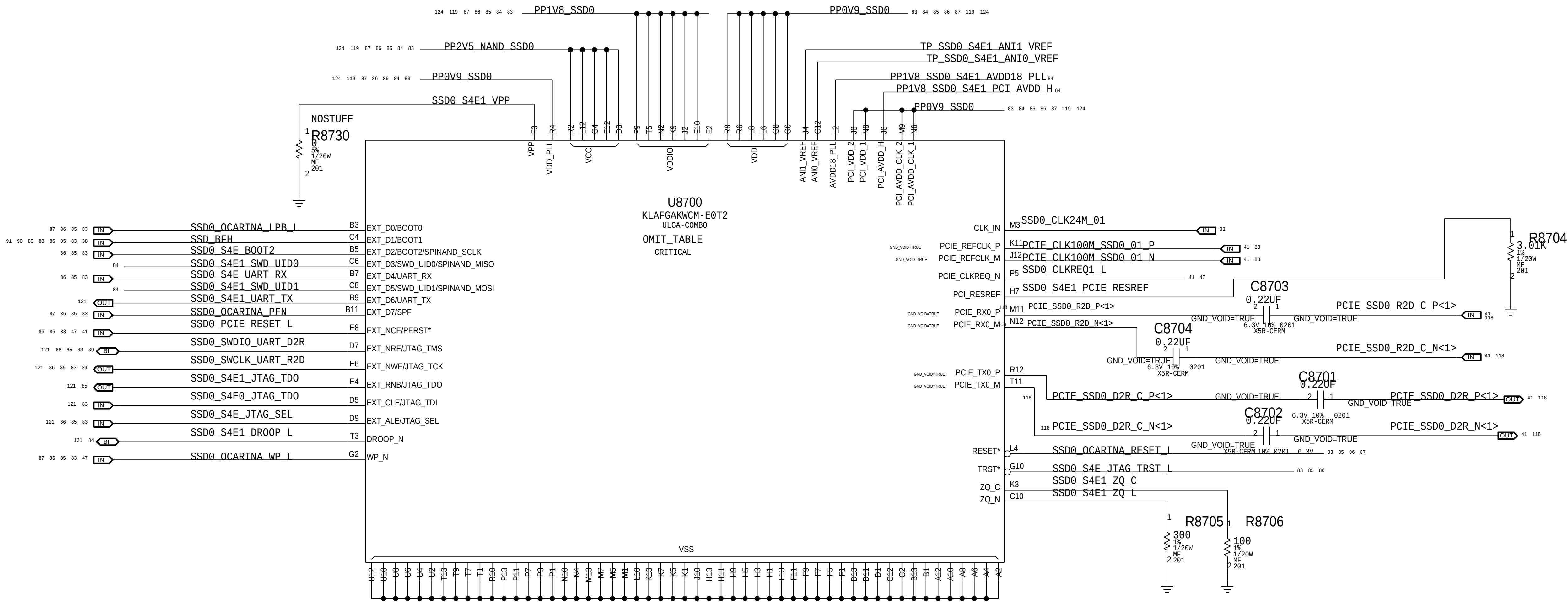
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S4E1



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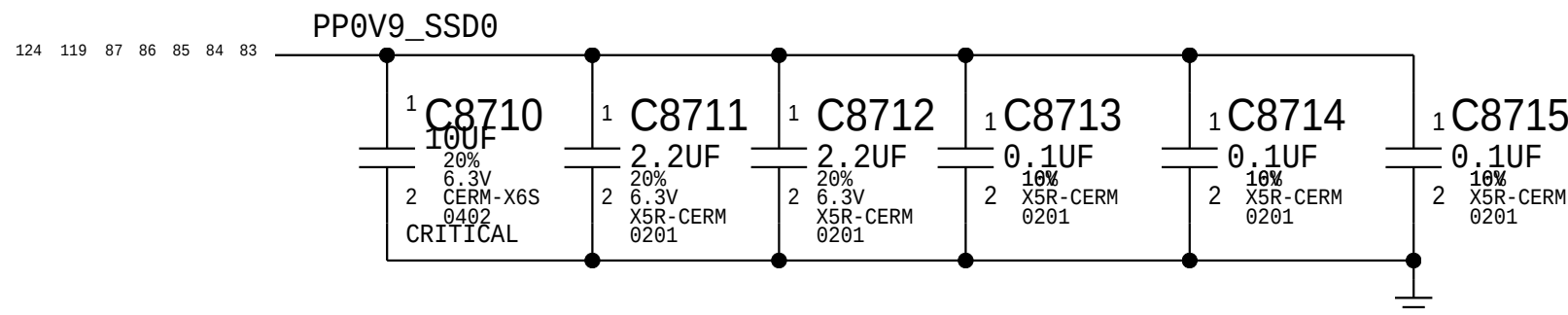
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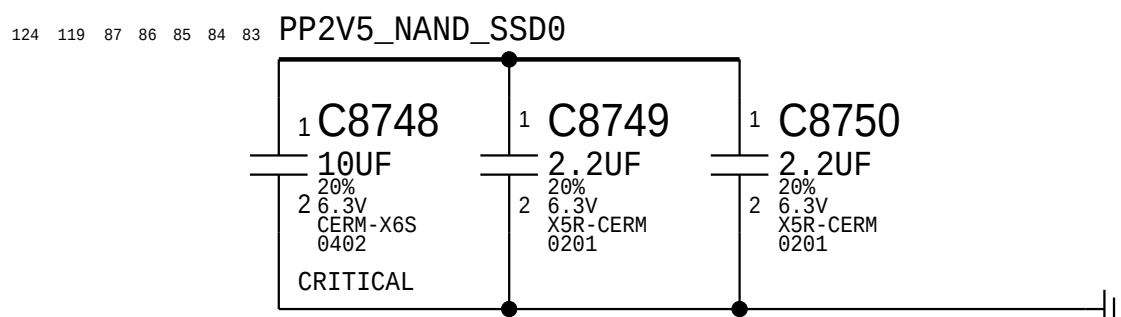
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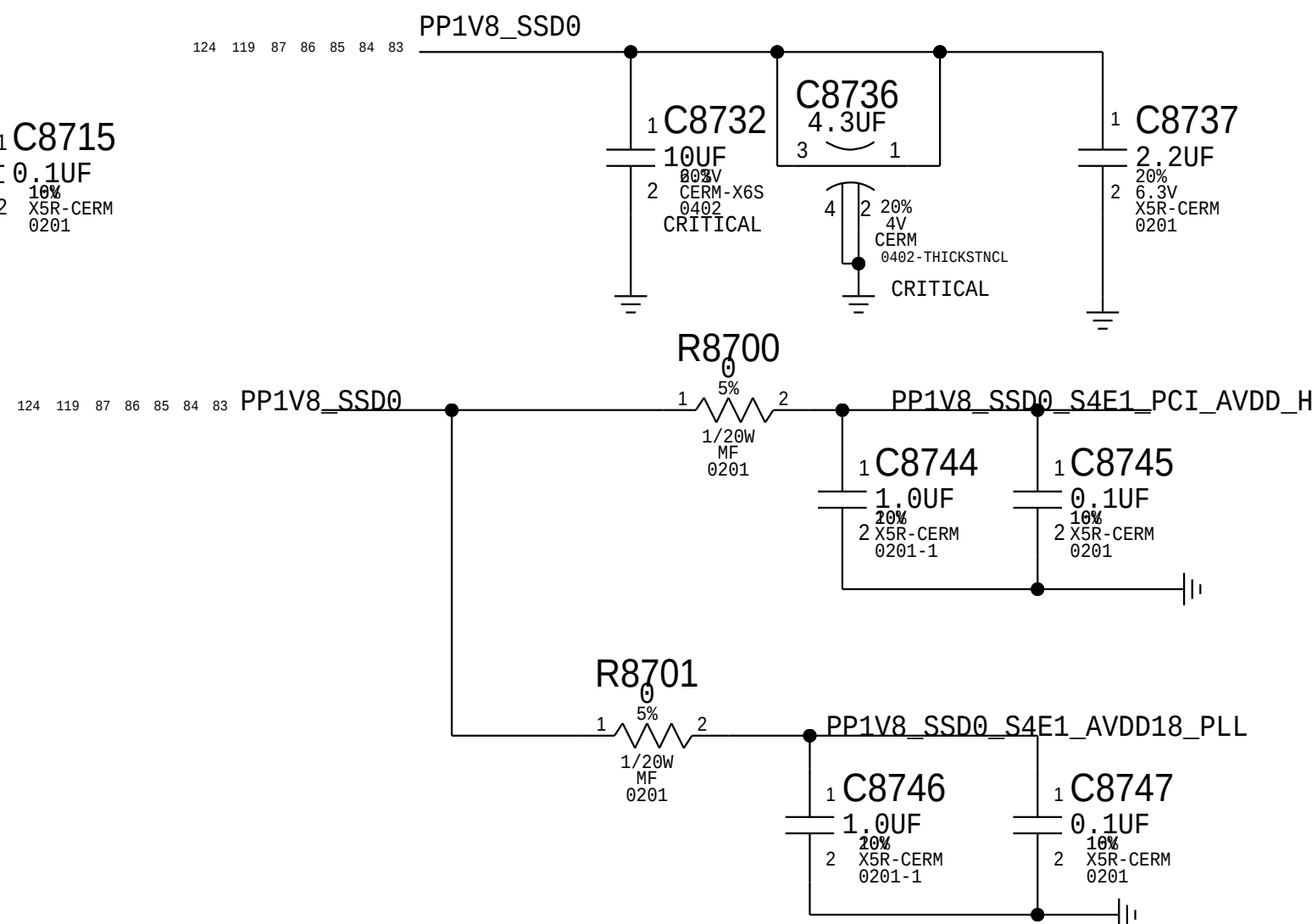
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VCC CAP



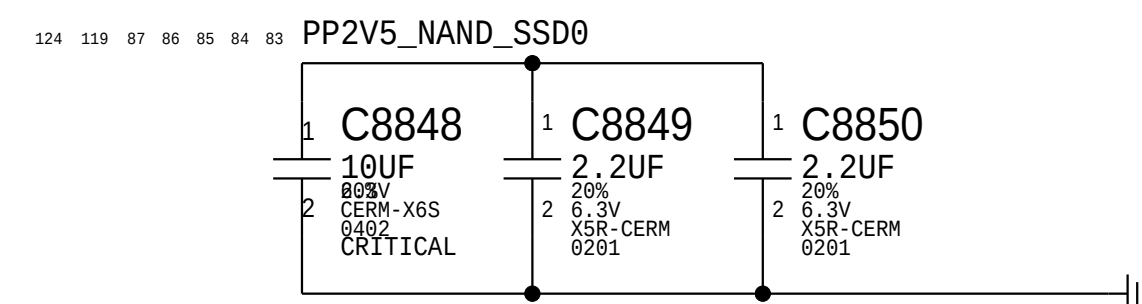
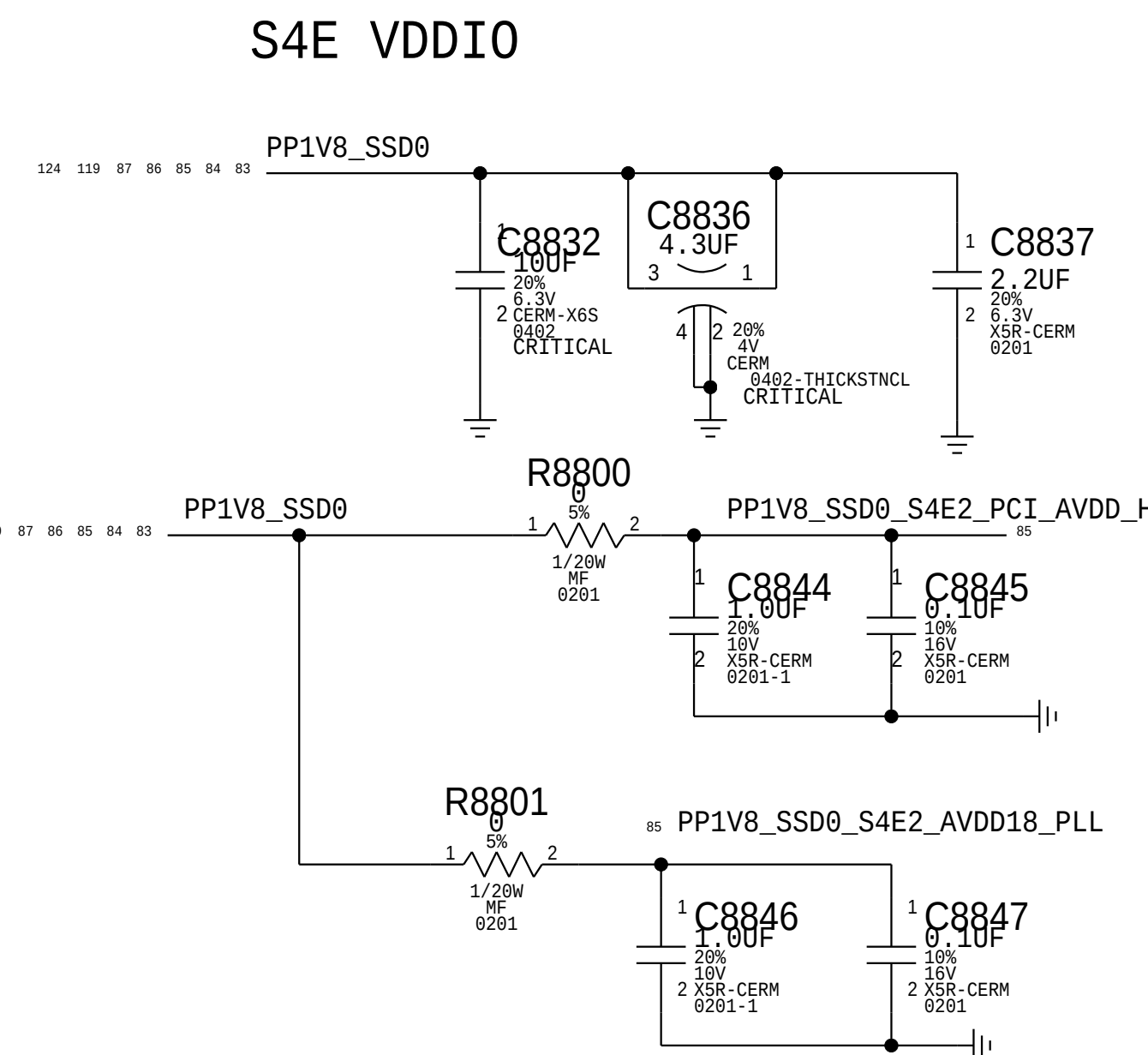
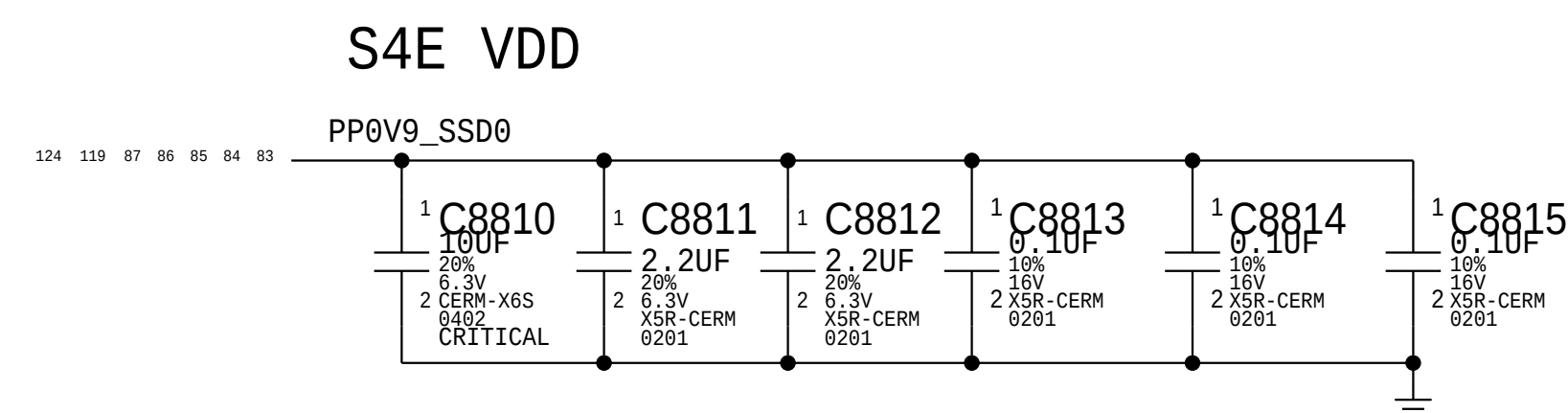
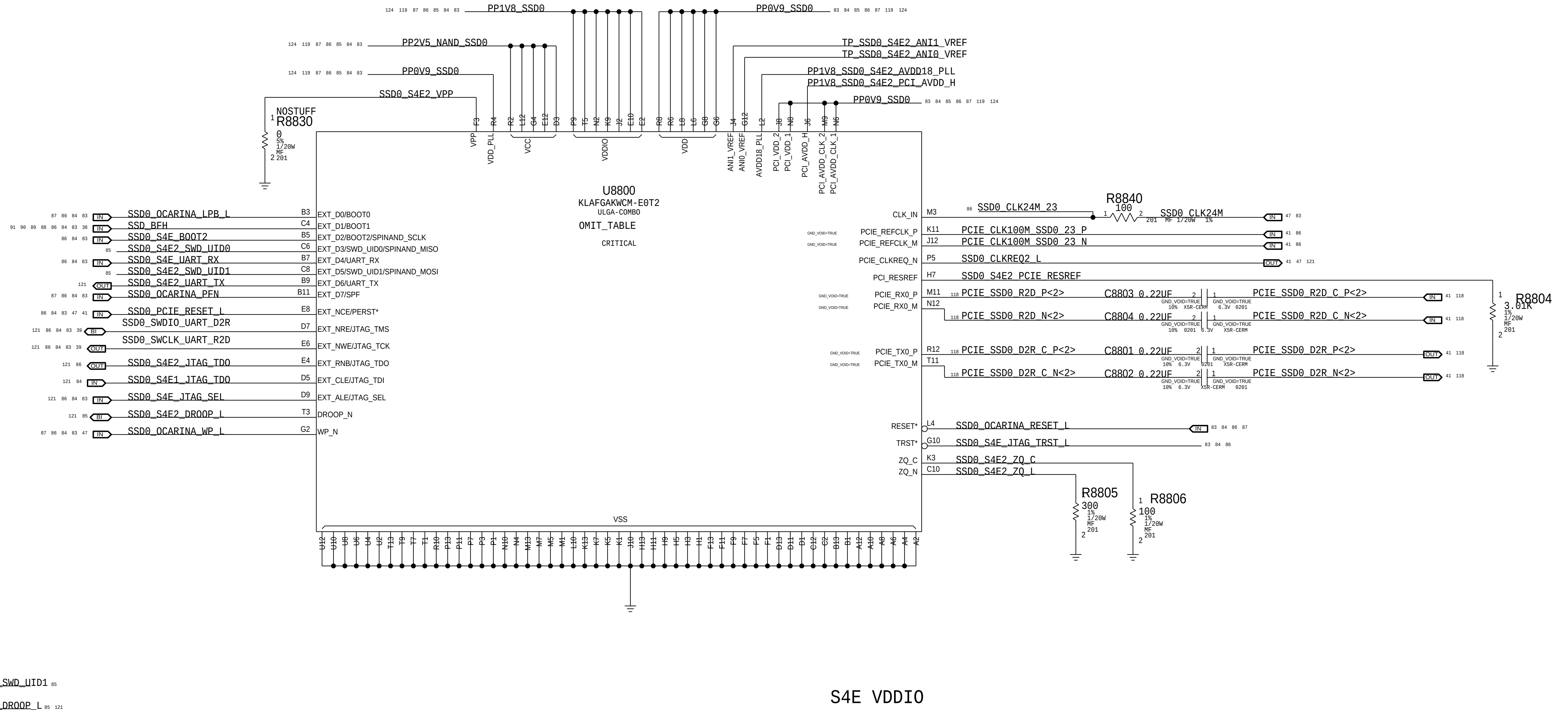
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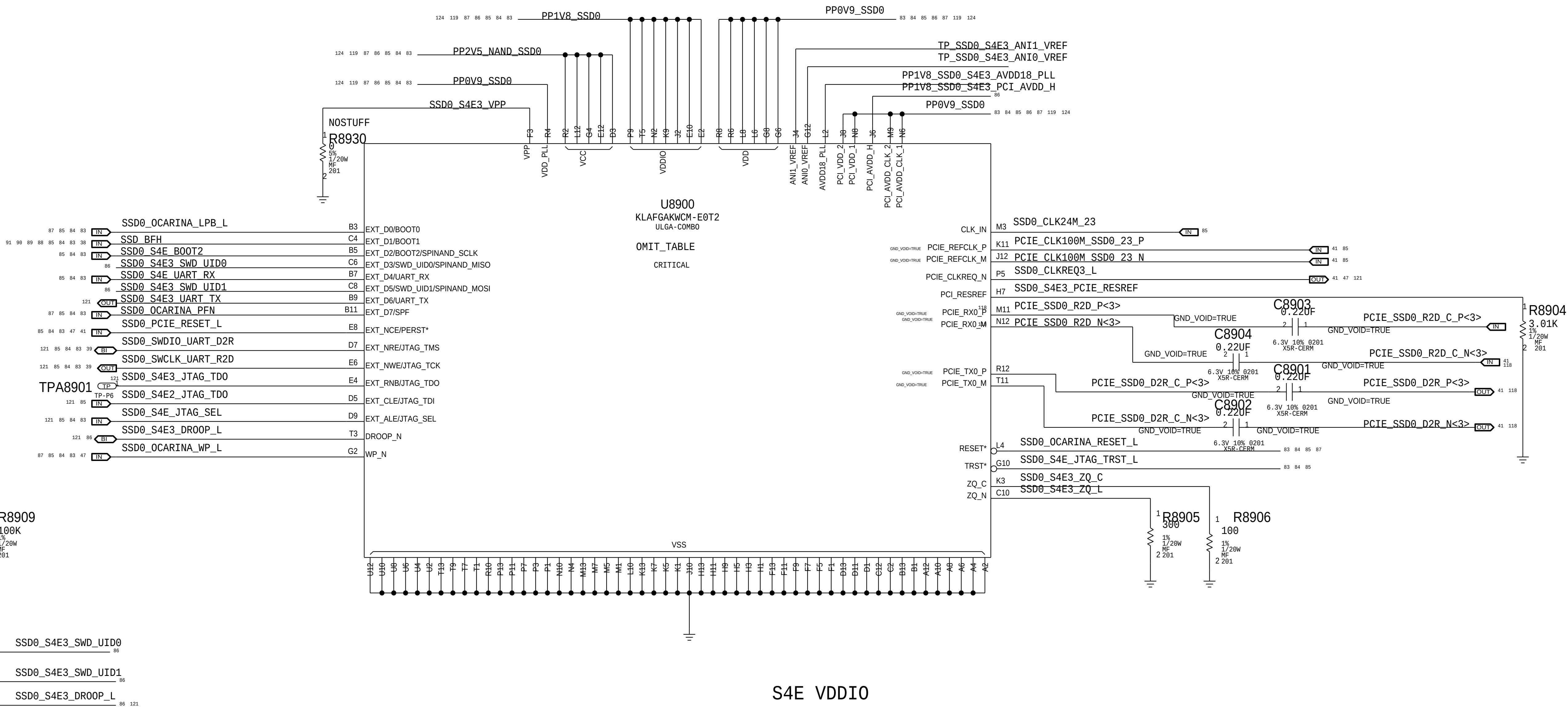
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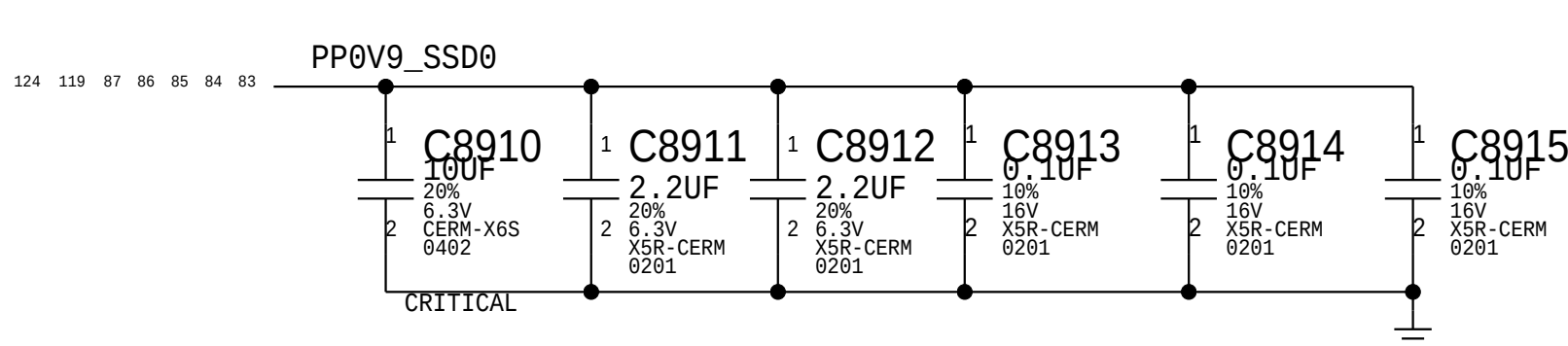


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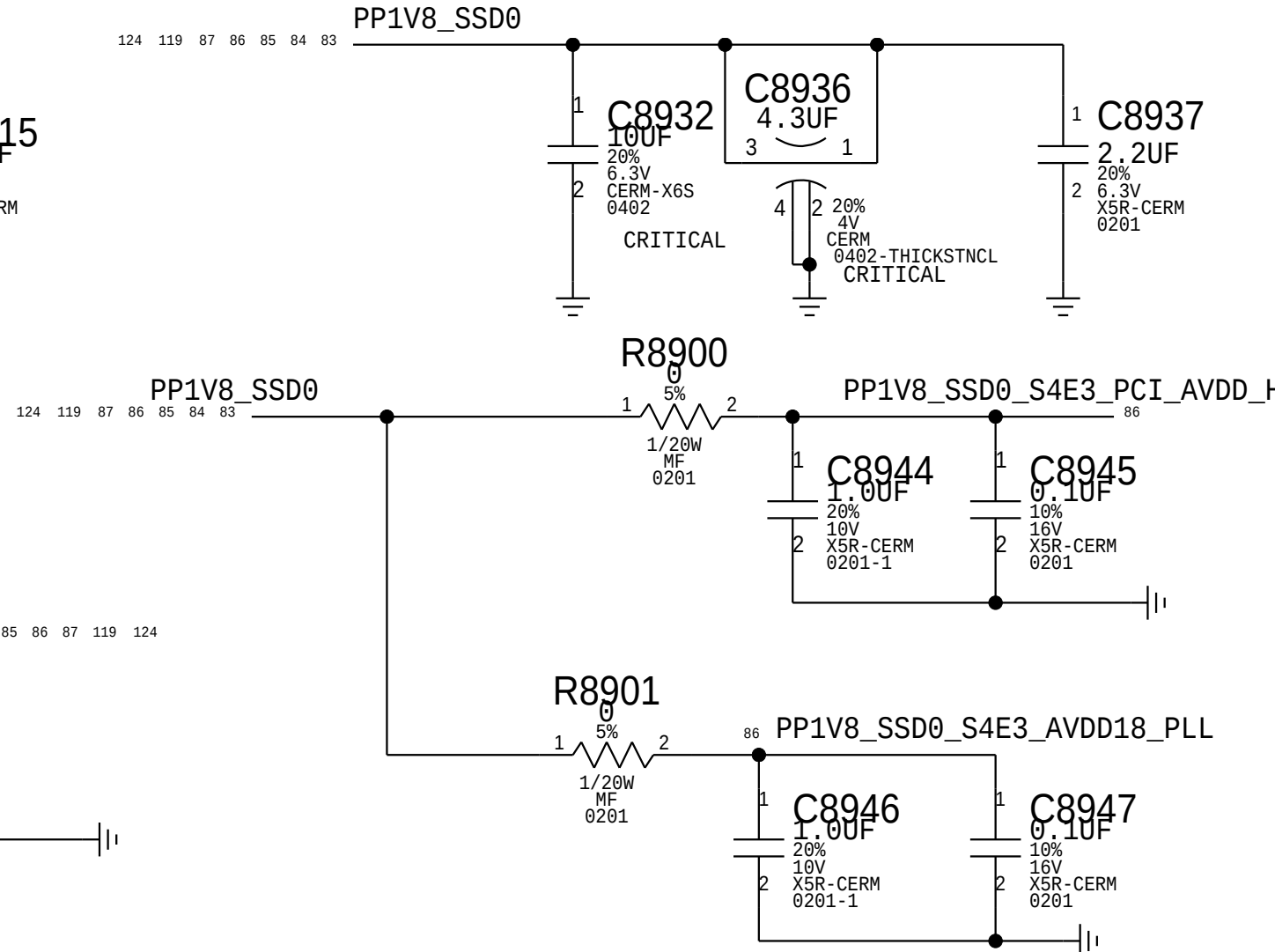
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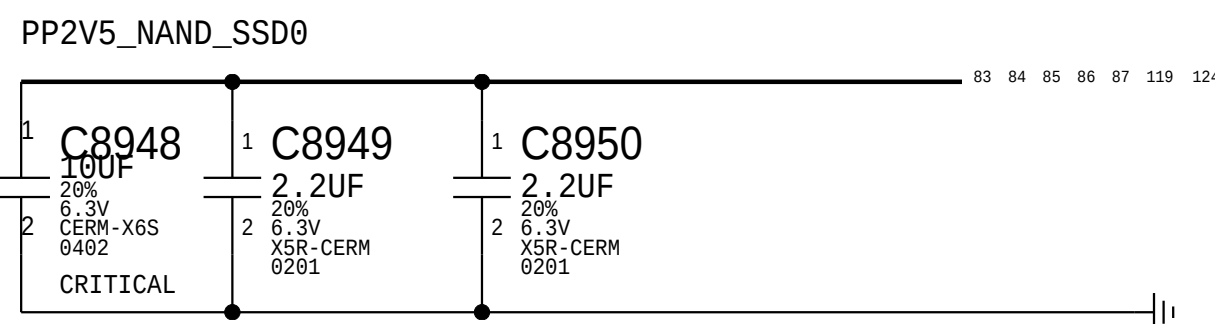
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S4E VDDIO

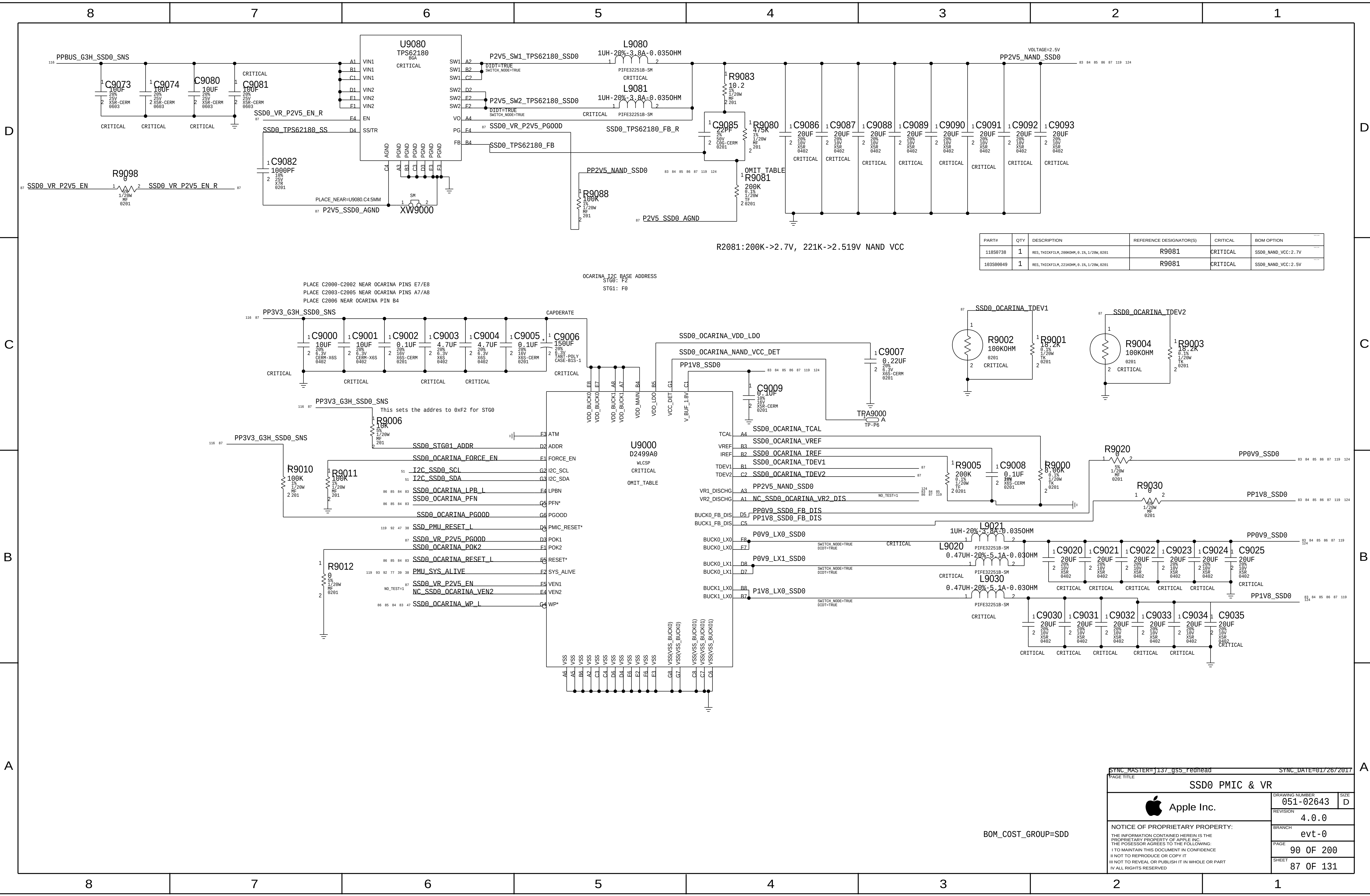


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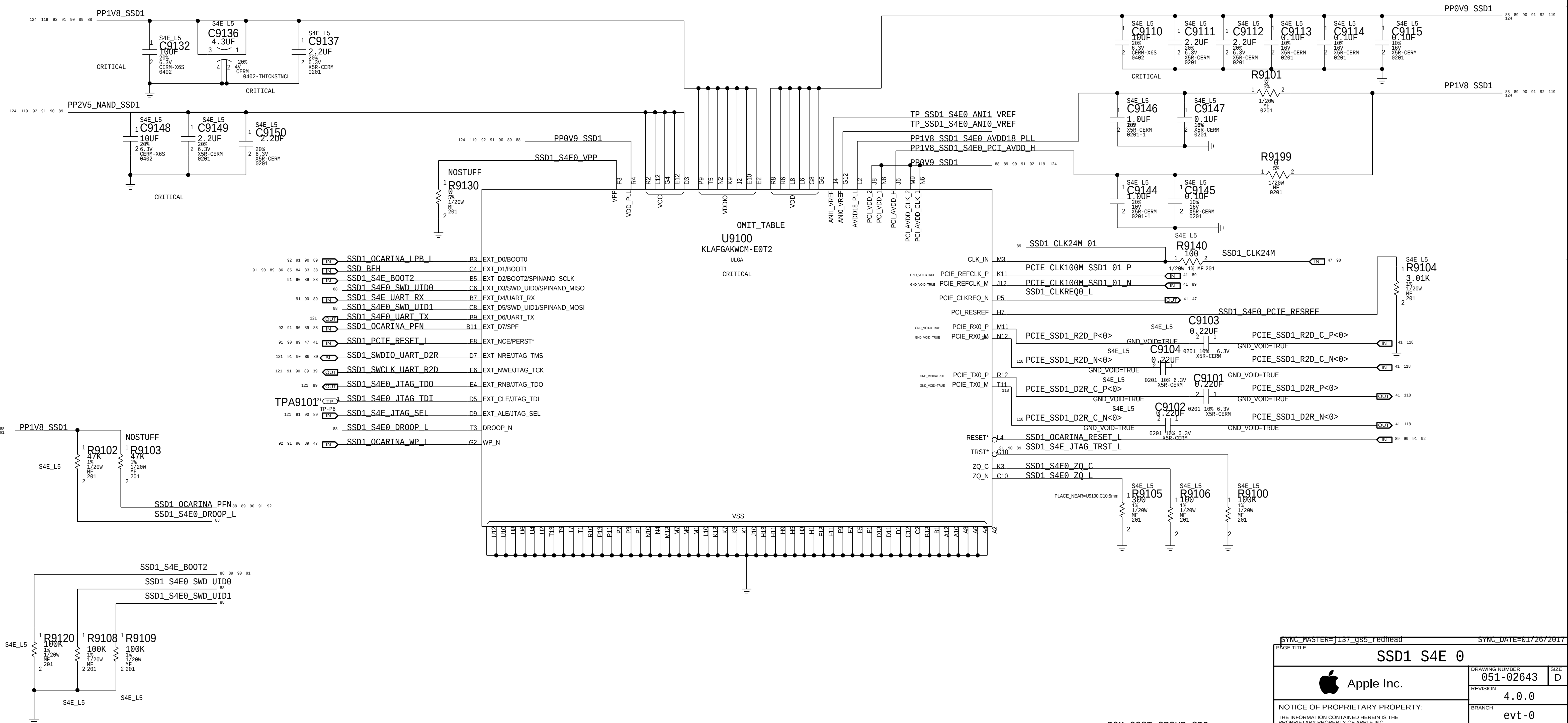
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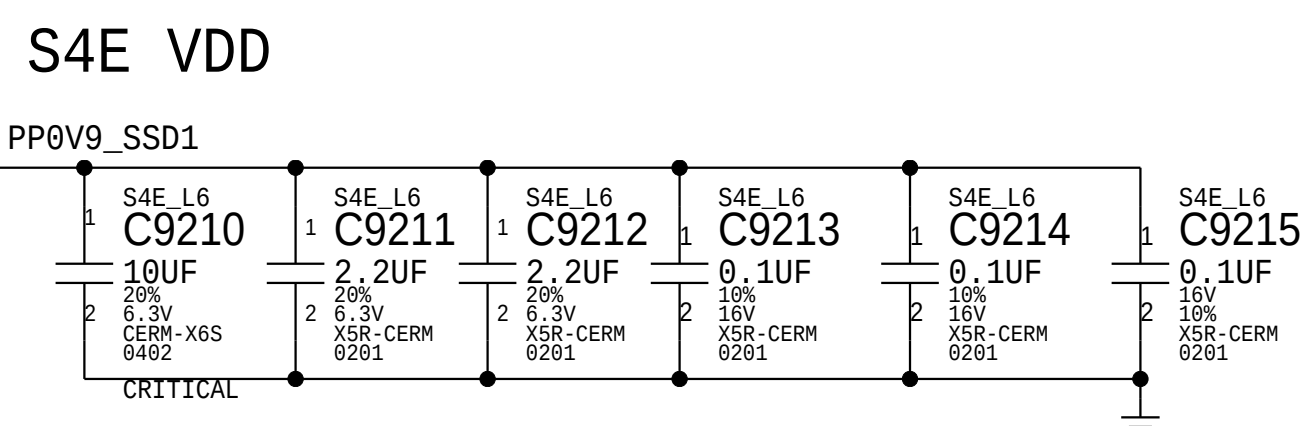
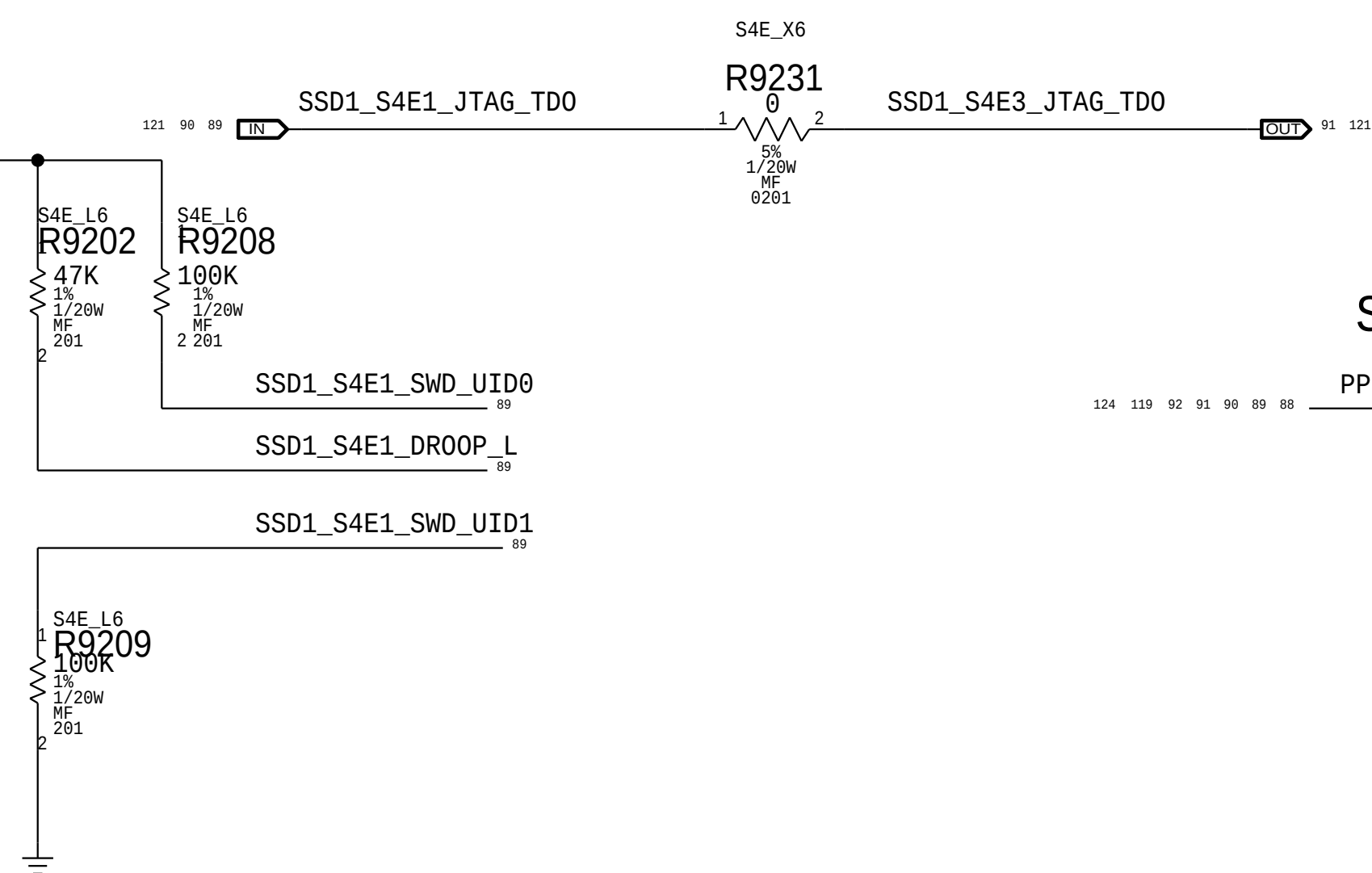
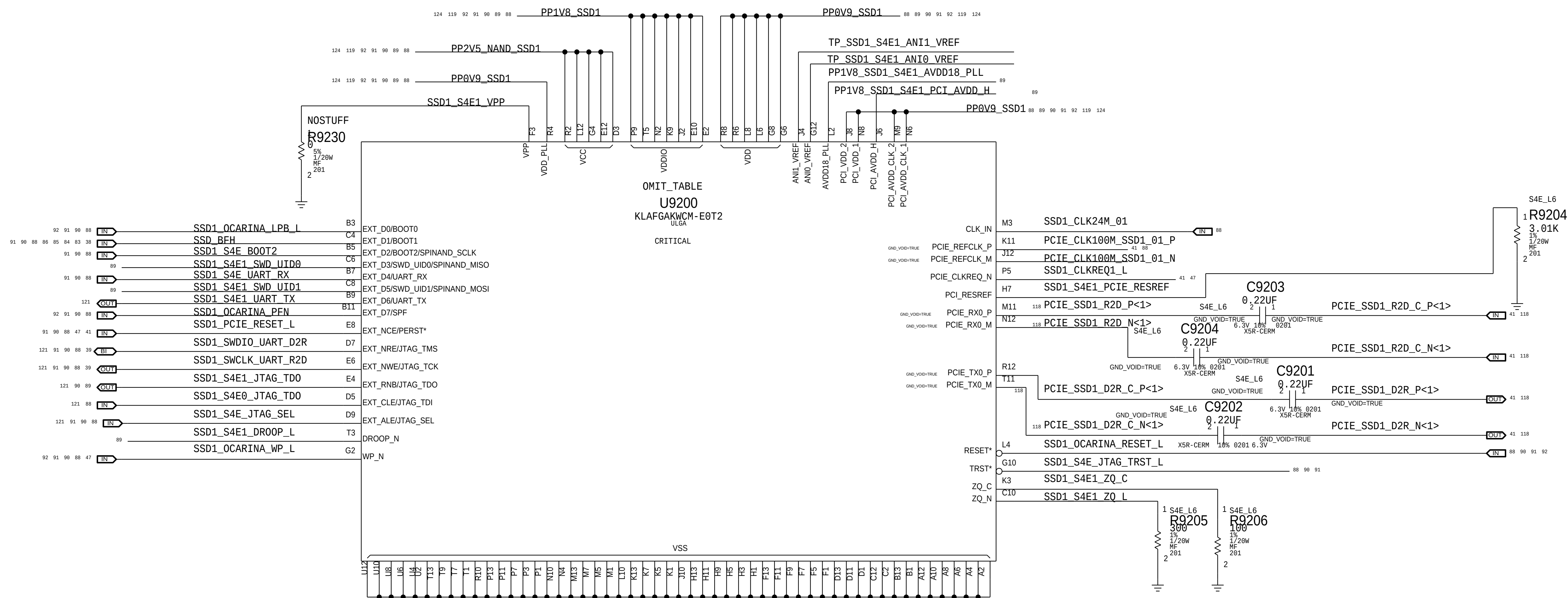
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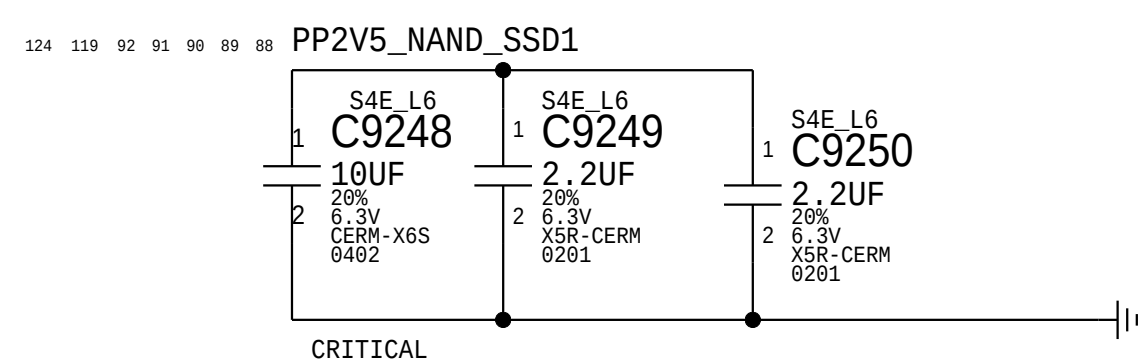
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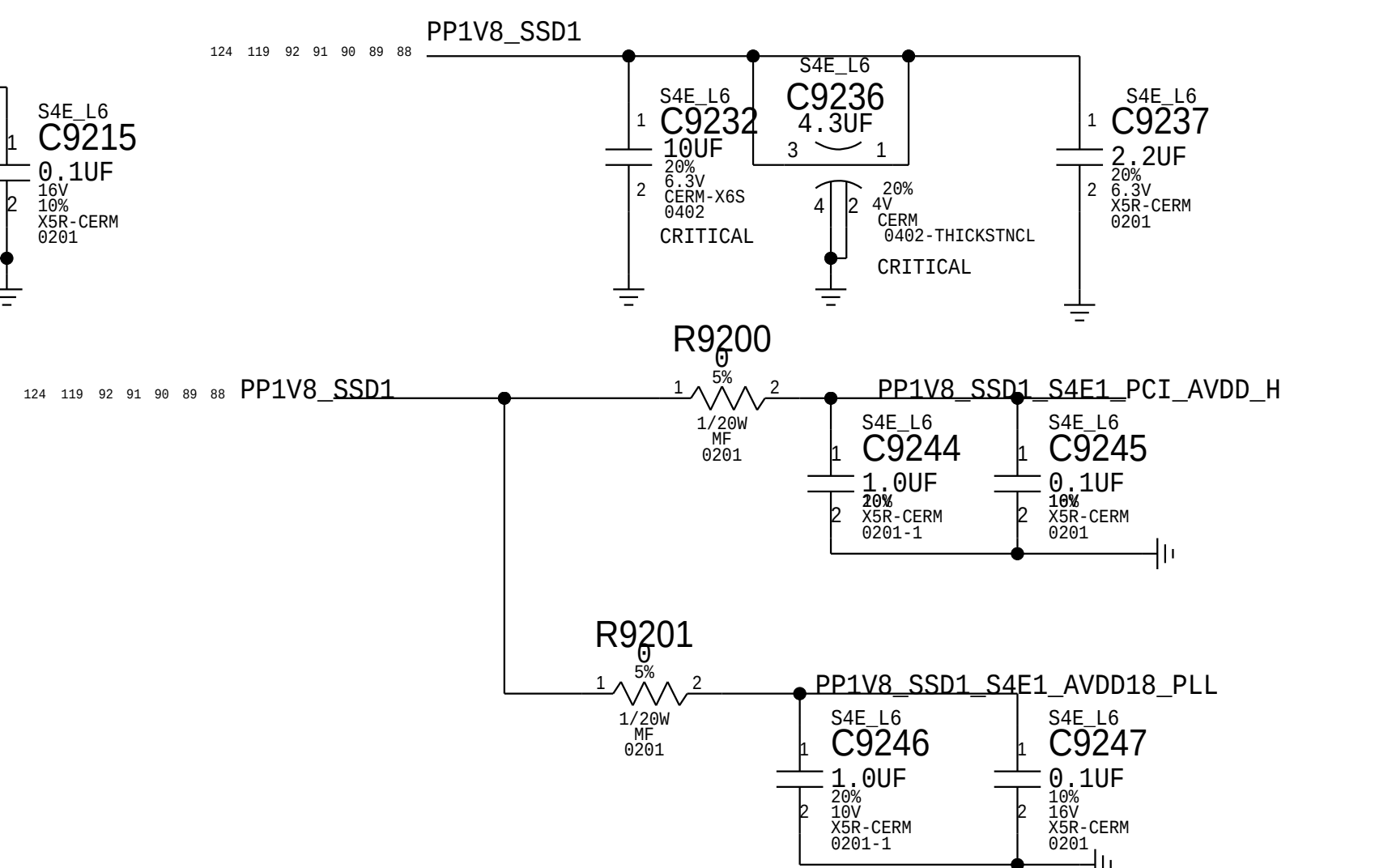
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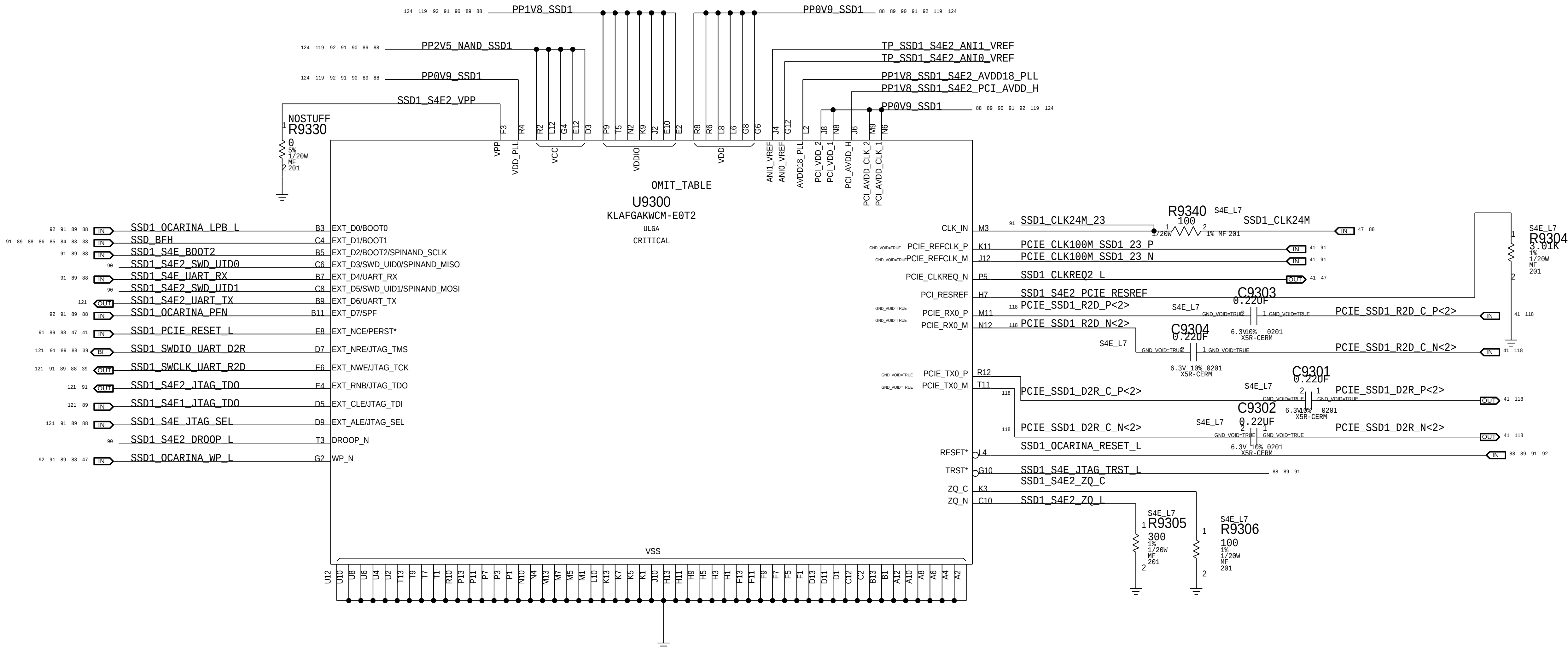
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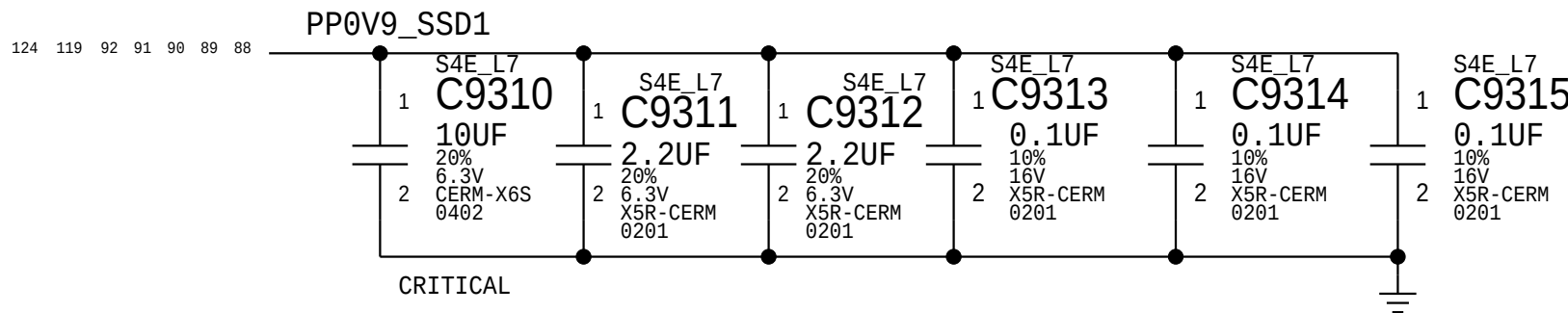
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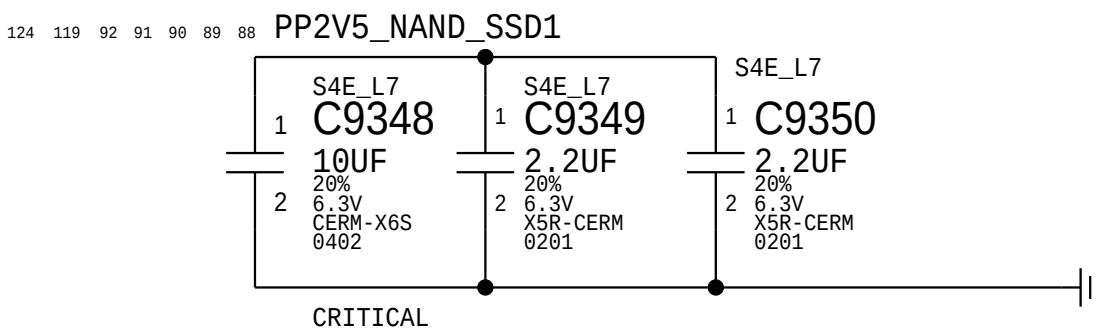
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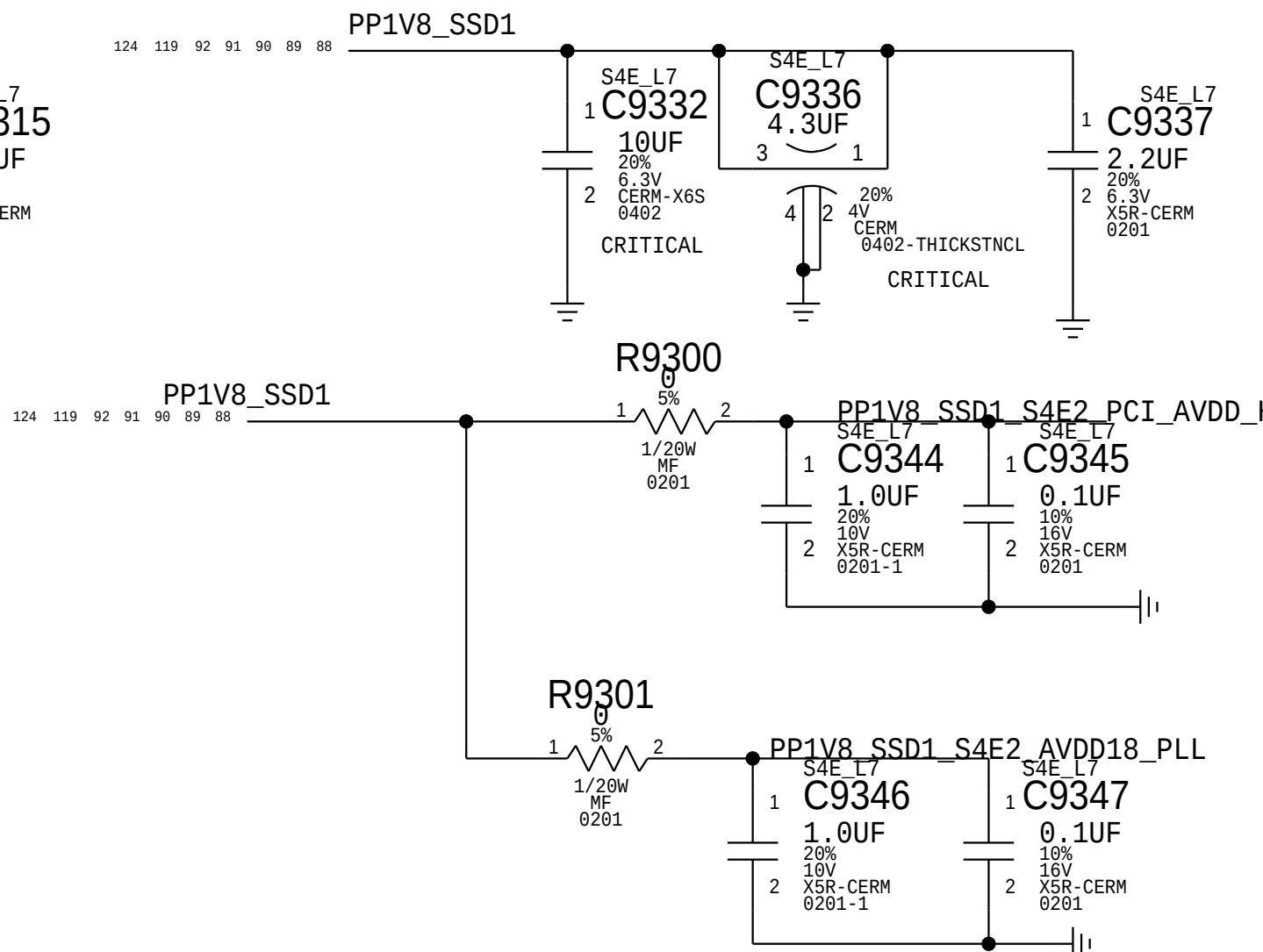
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VCC CAP



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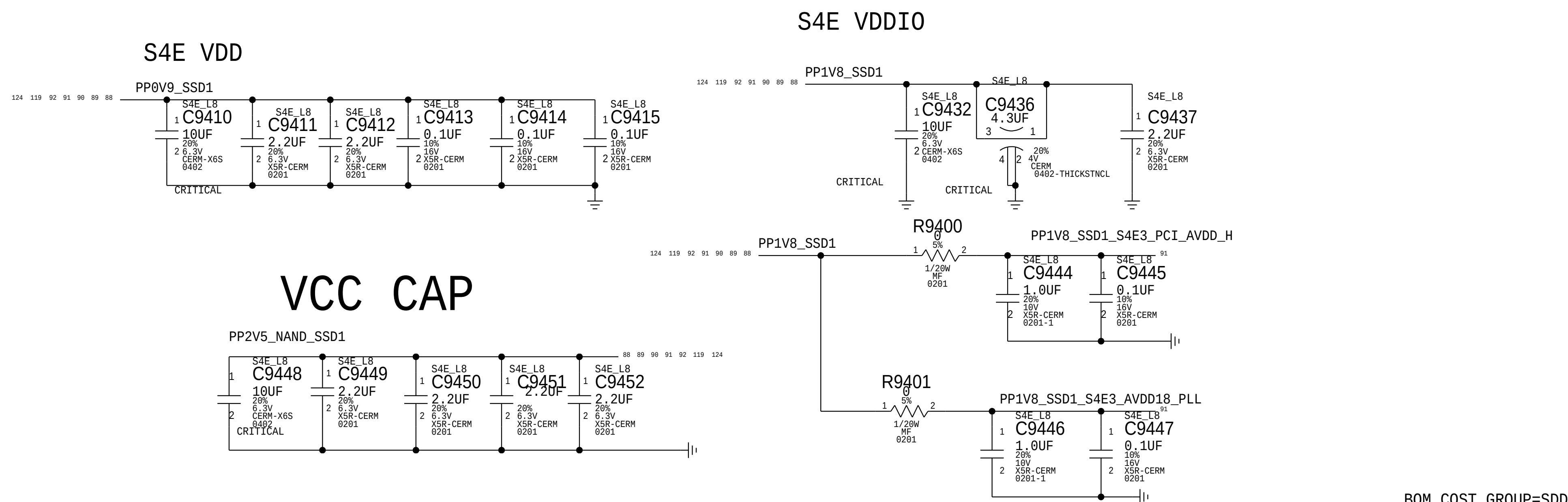
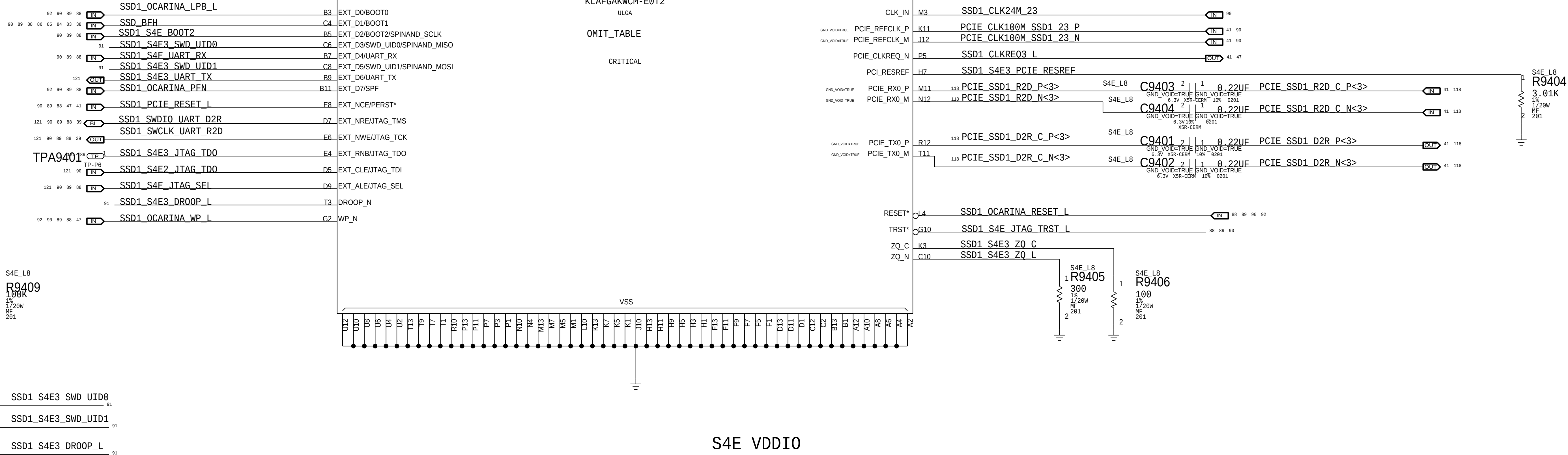


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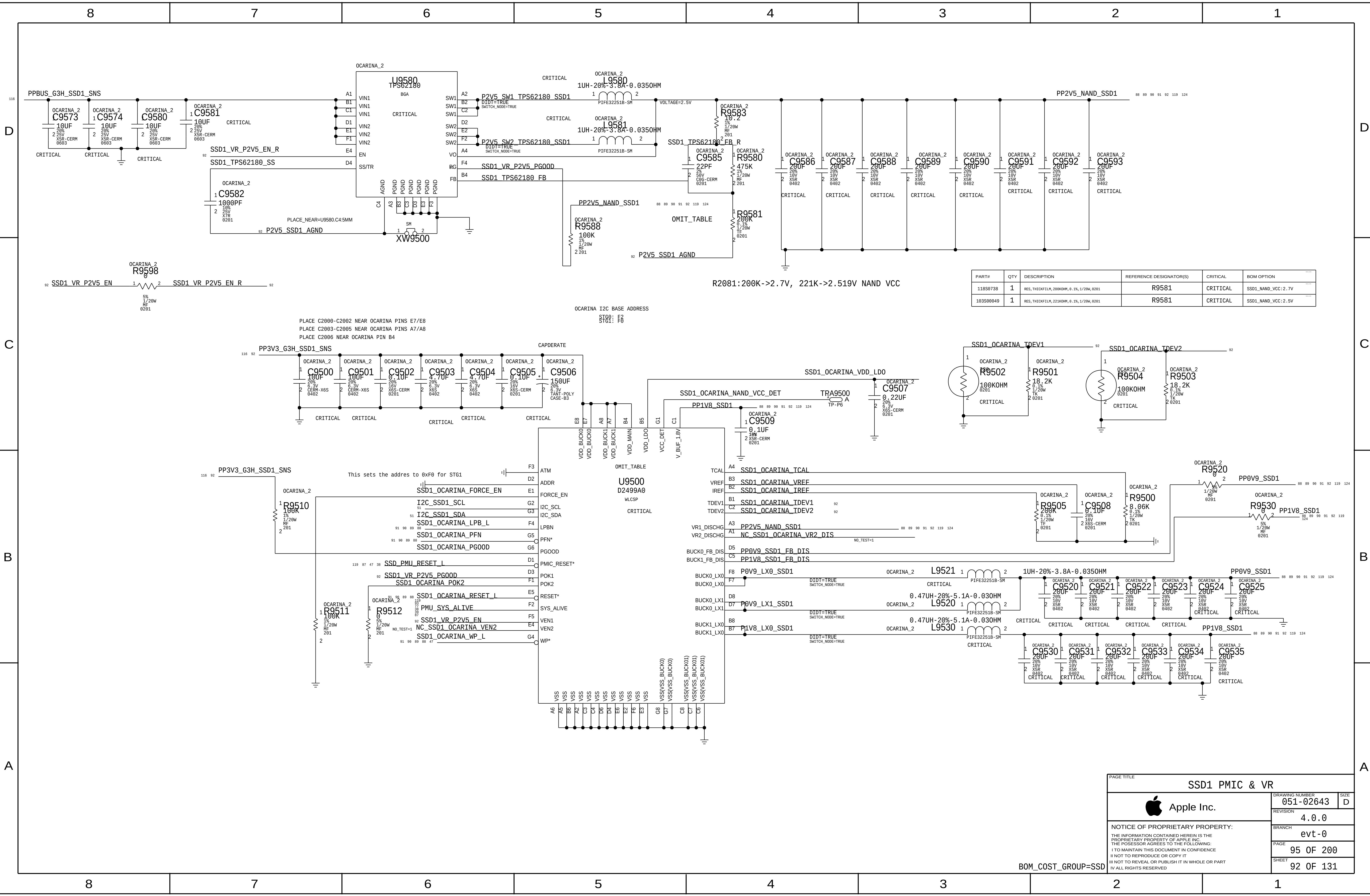
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Pinout diagram for U9400:

- VDDIO:** P9, T5, N2, K9, J2, E10
- VDD:** R8, R6, L8, L6, G8



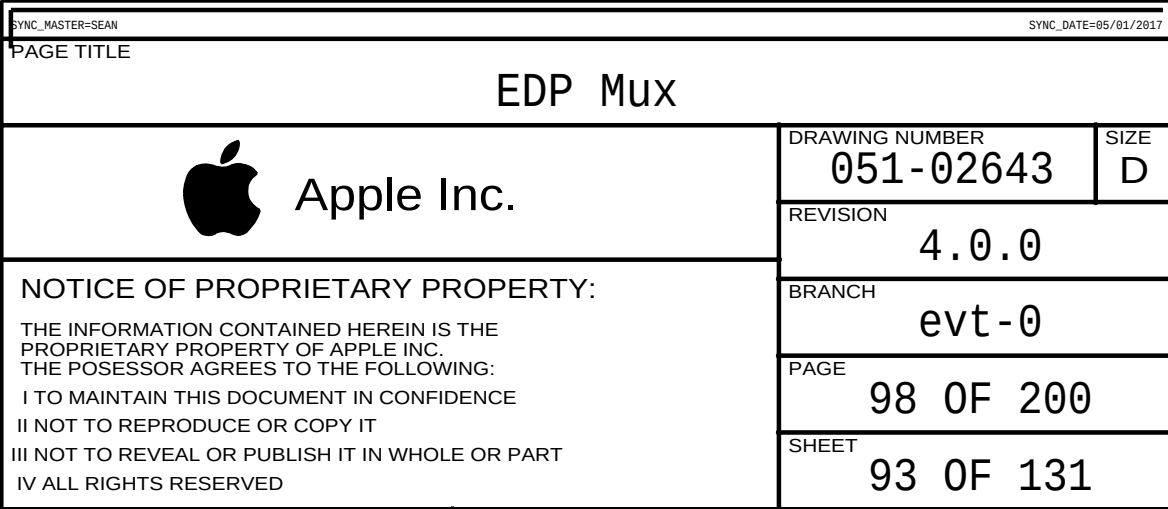
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
11850738	1	RES, THICKFILM, 200KOHM, 0.1%, 1/20W, 0201	R9581	CRITICAL	SSD1_NAND_VCC:2.7V
103500649	1	RES, THICKFILM, 221KOHM, 0.1%, 1/20W, 0201	R9581	CRITICAL	SSD1_NAND_VCC:2.5V

PAGE TITLE			
SSD1 PMIC & VR			
 Apple Inc.		DRAWING NUMBER	051-02643
		REVISION	4.0.0
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		PAGE	95 OF 200
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BOM_COST_GROUP=SSD



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	R9961		LOADRC:NO

D

C

B

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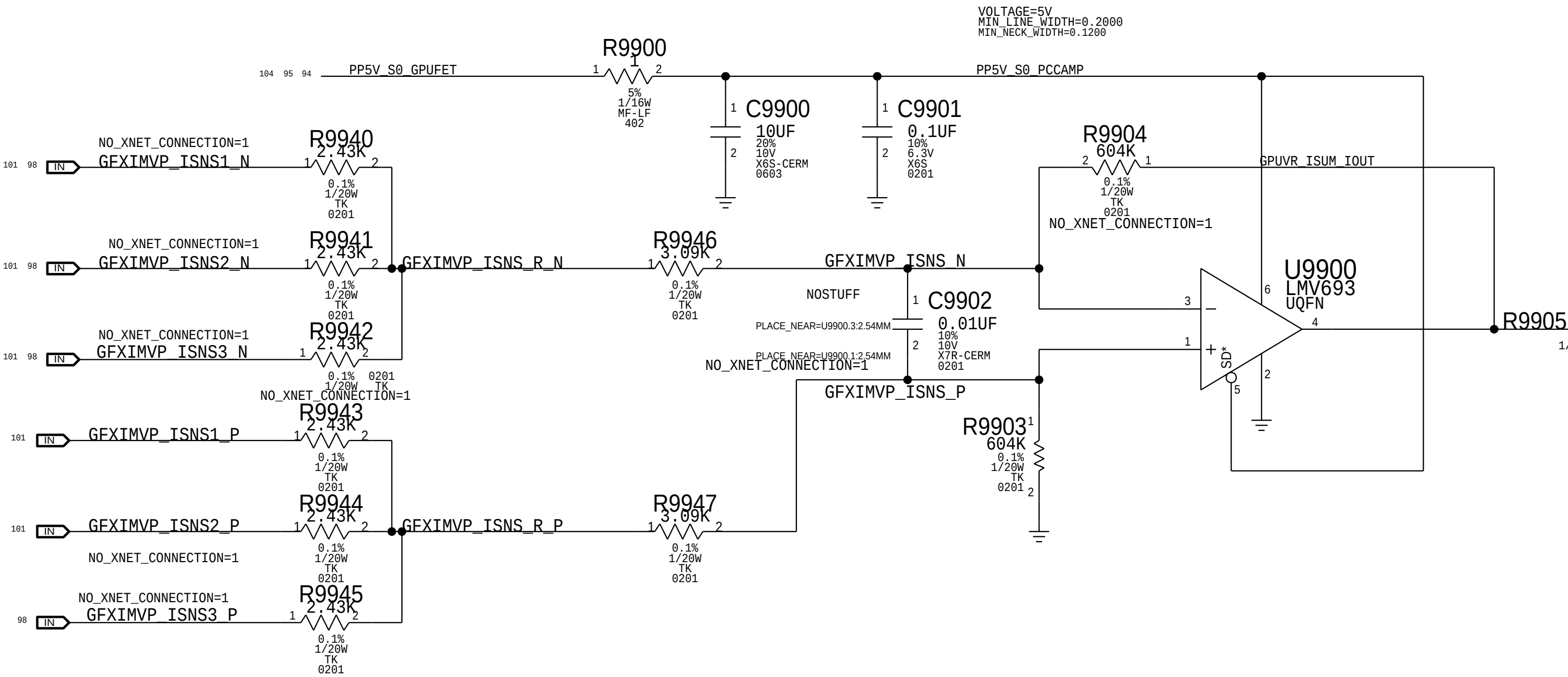
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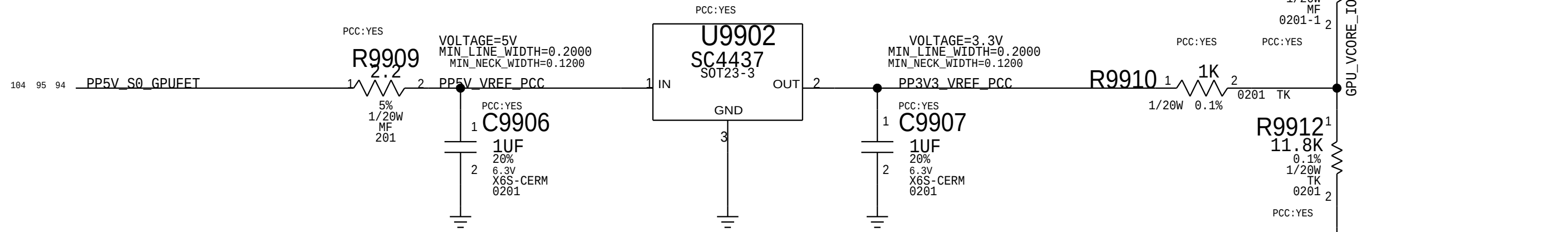
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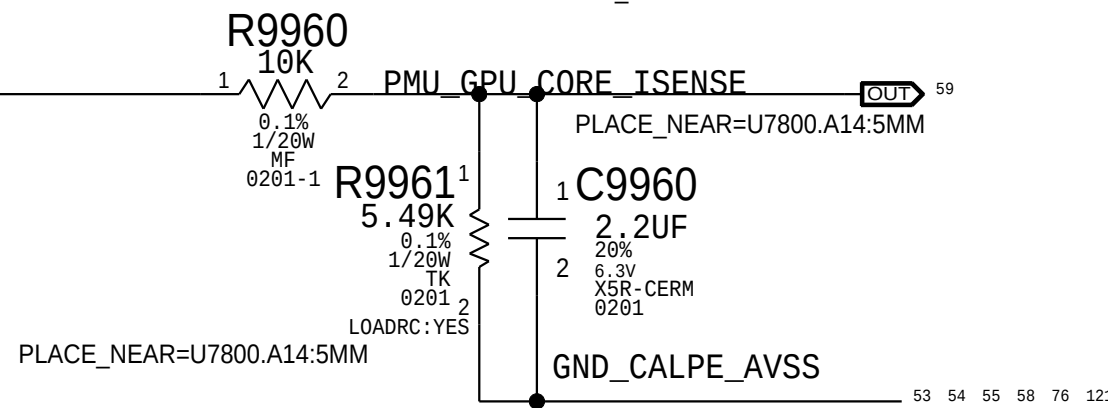
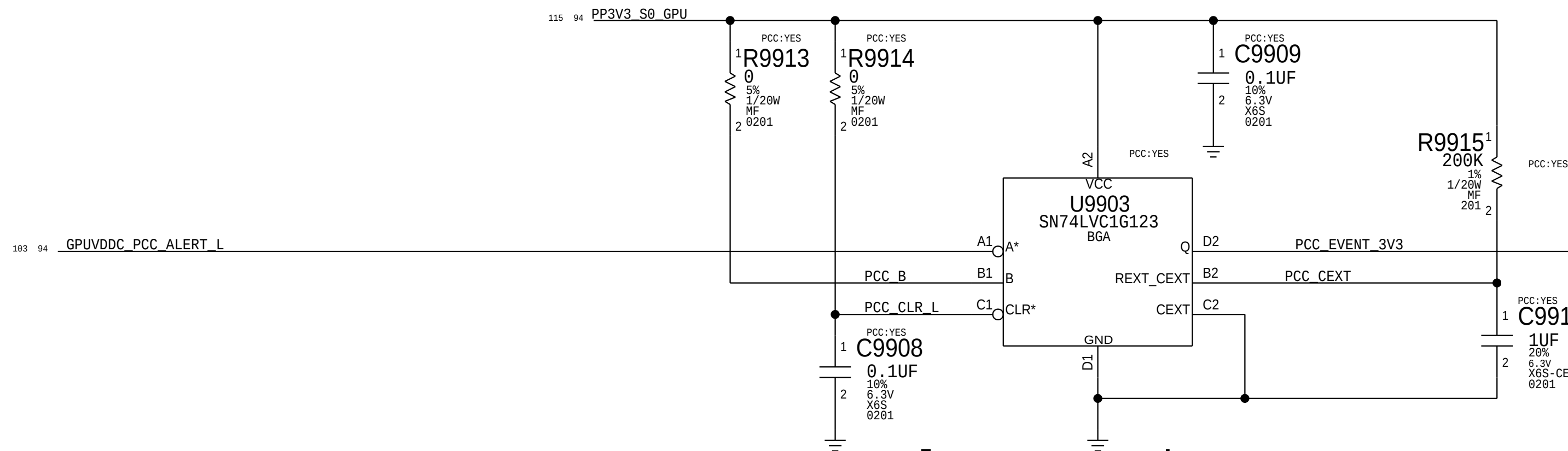
High Speed Summing AMP



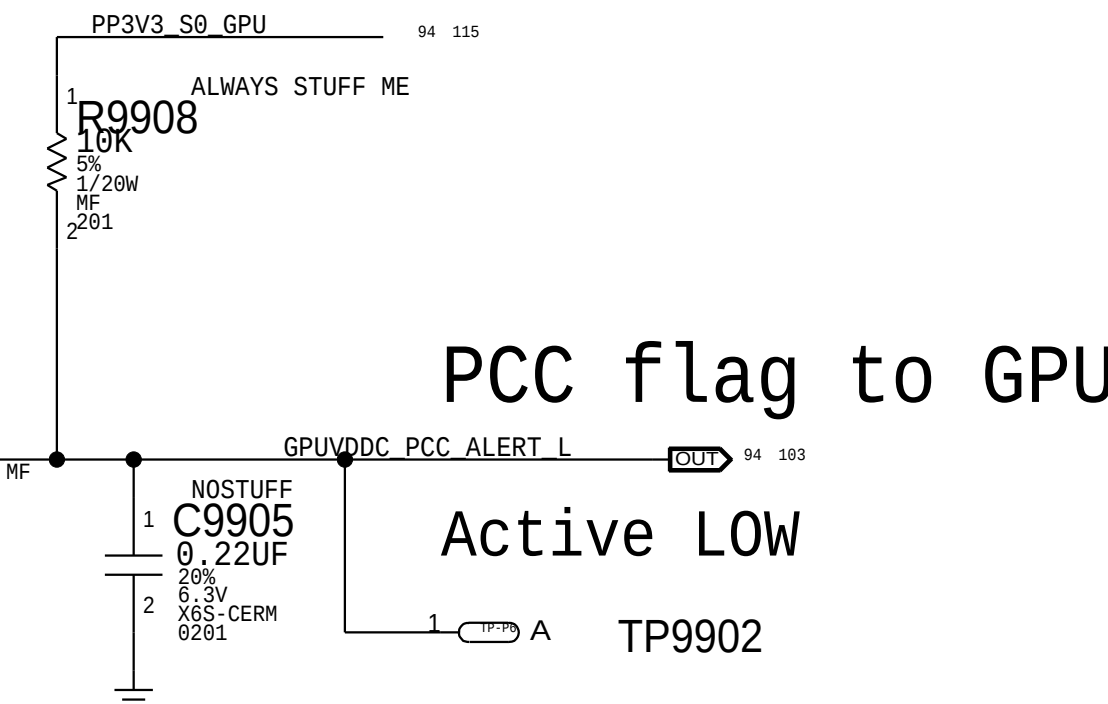
Trip Point Voltage Gen



Pulse Stretcher

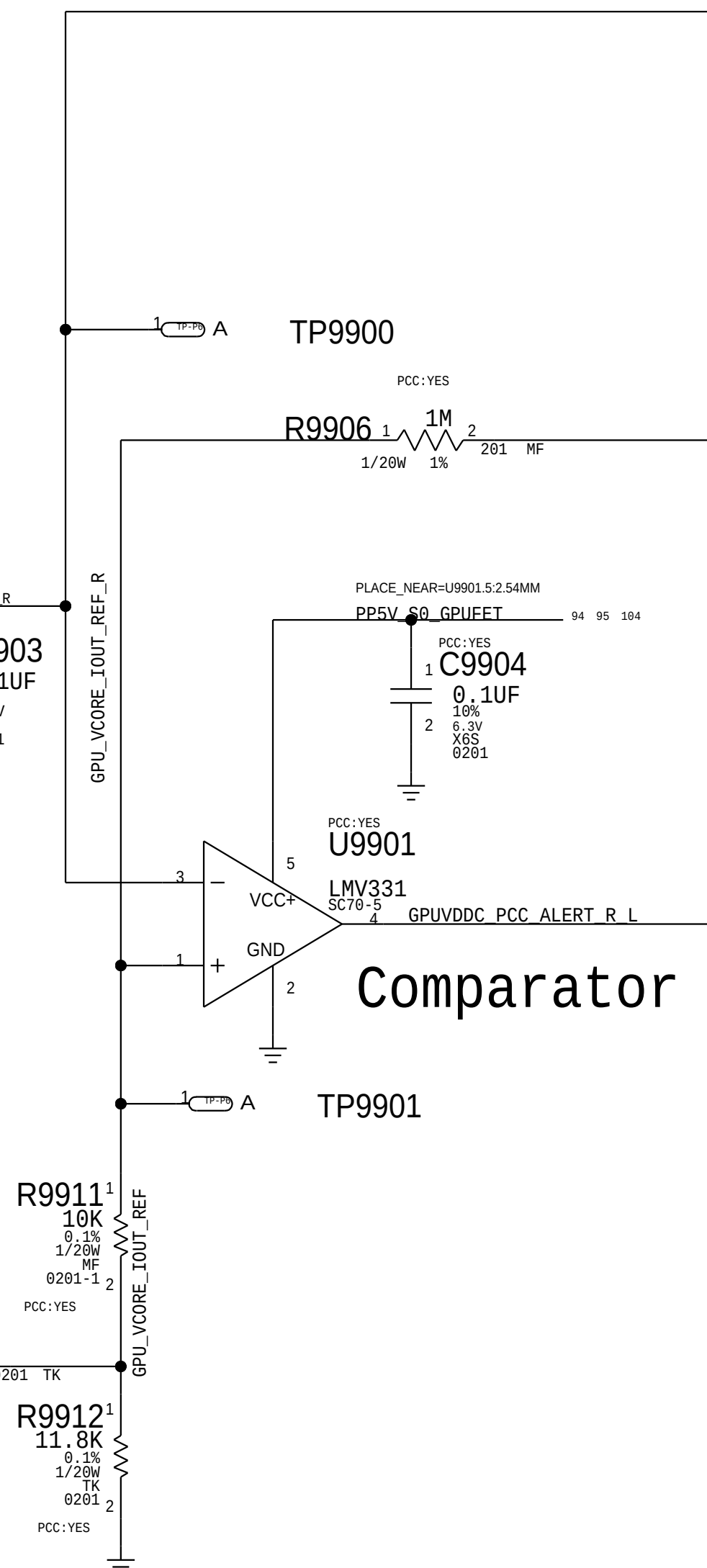


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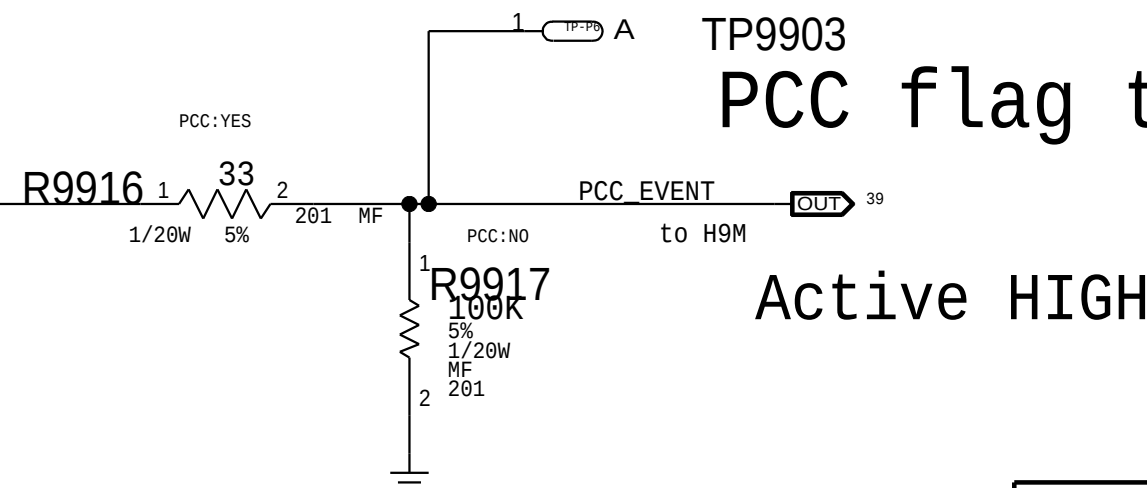


PCC flag to GPU

Comparator

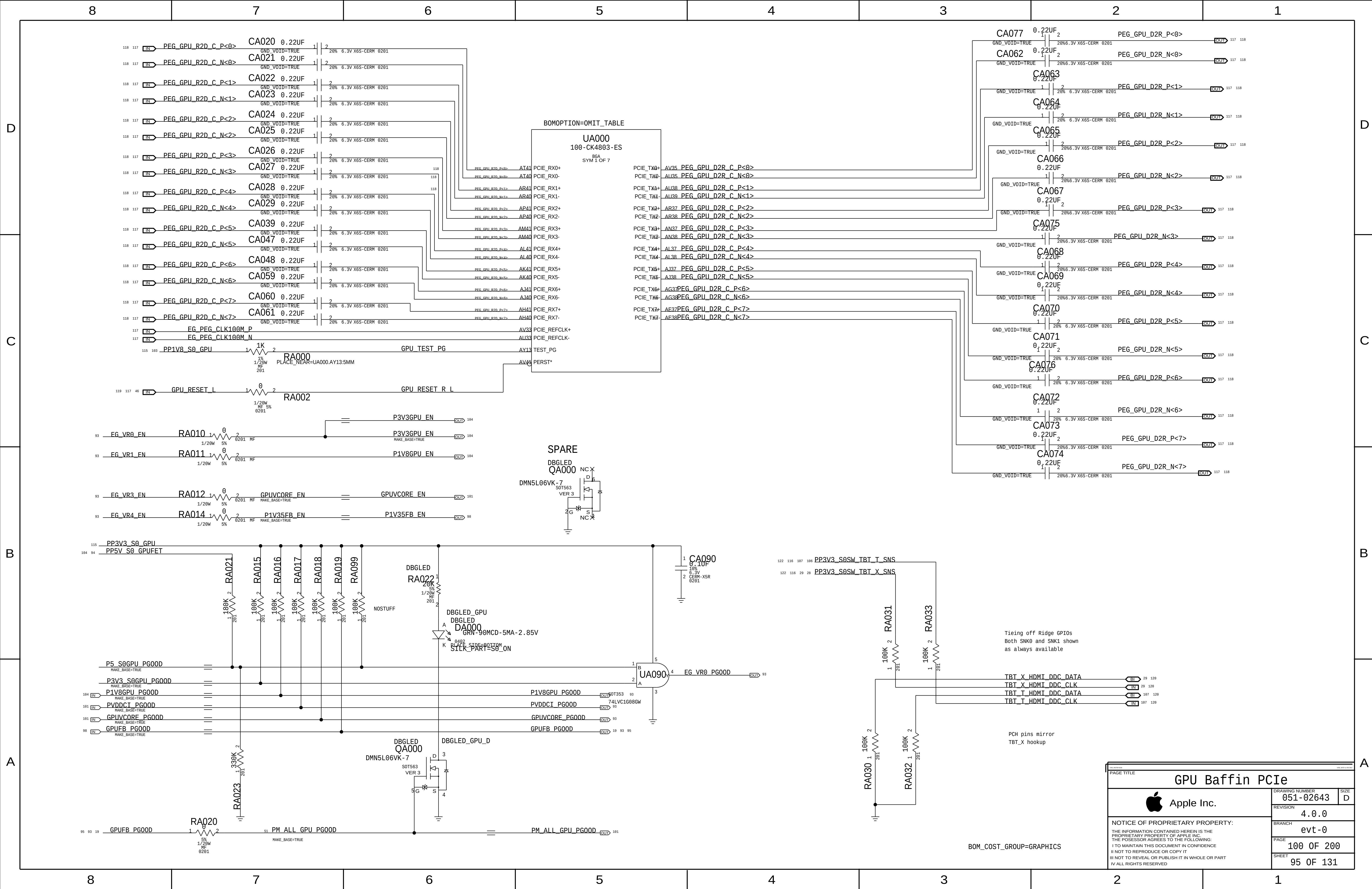


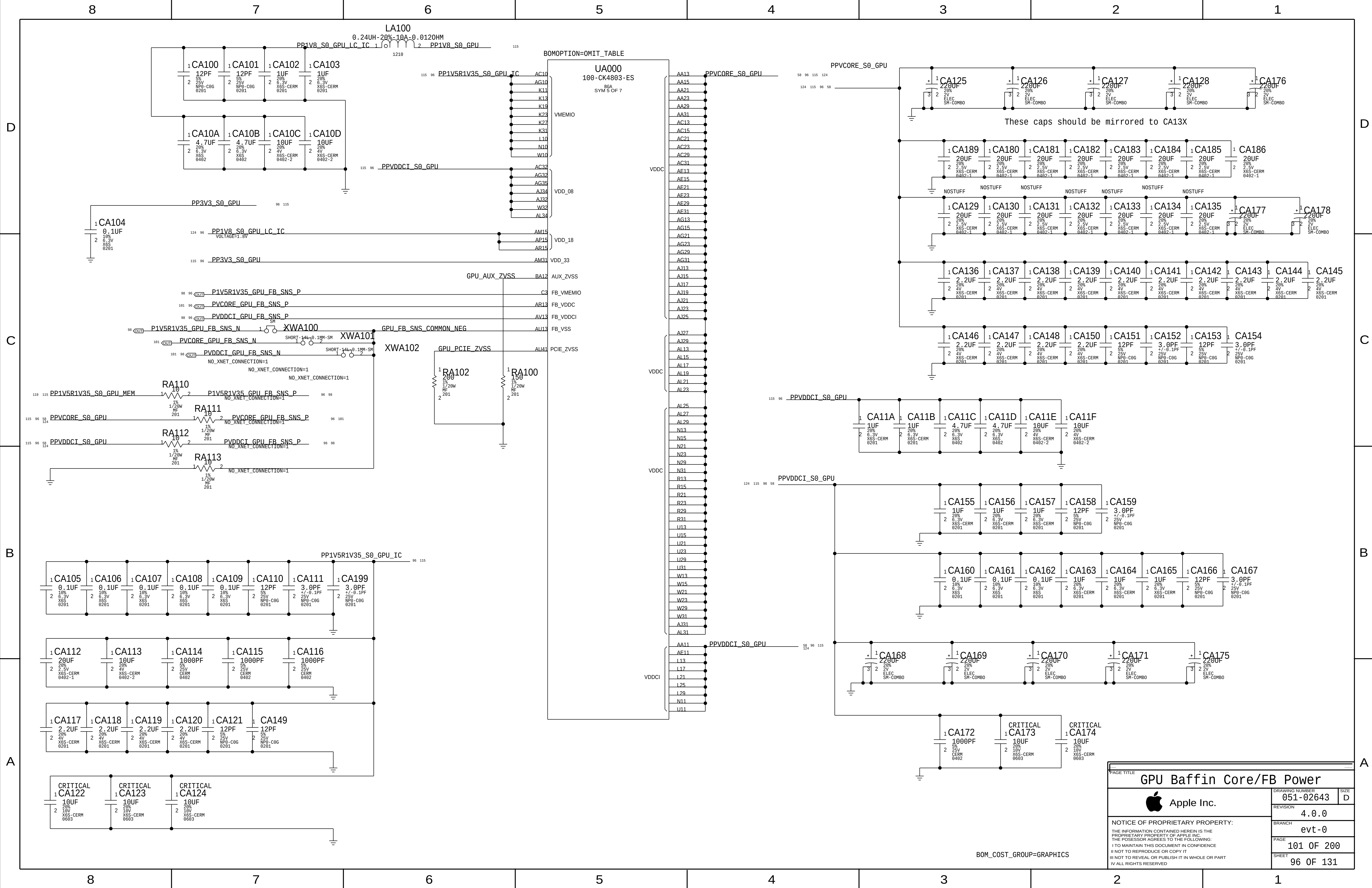
PCC flag to H9M



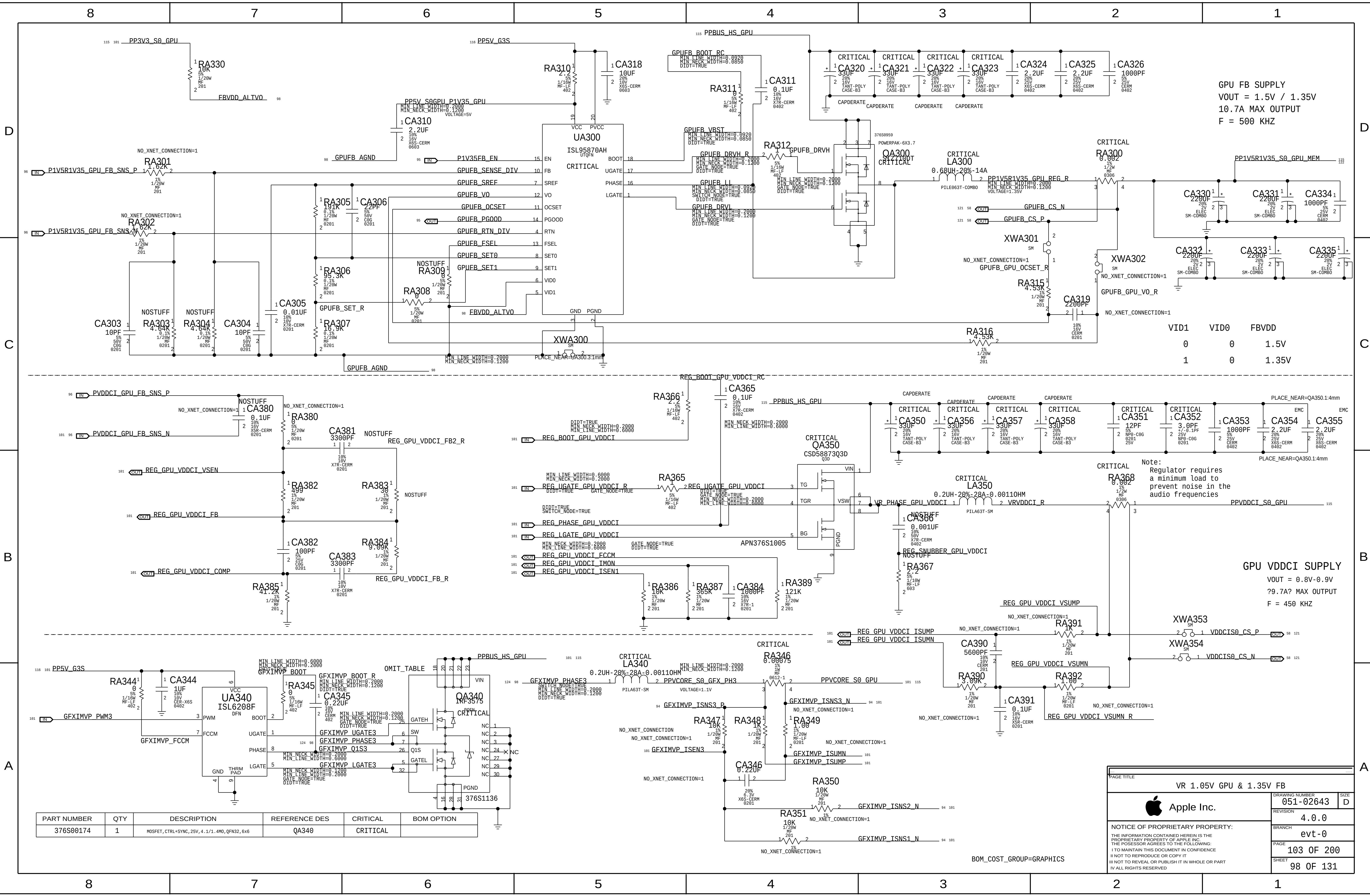
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GPU PCC		
	DRAWING NUMBER	051-02643
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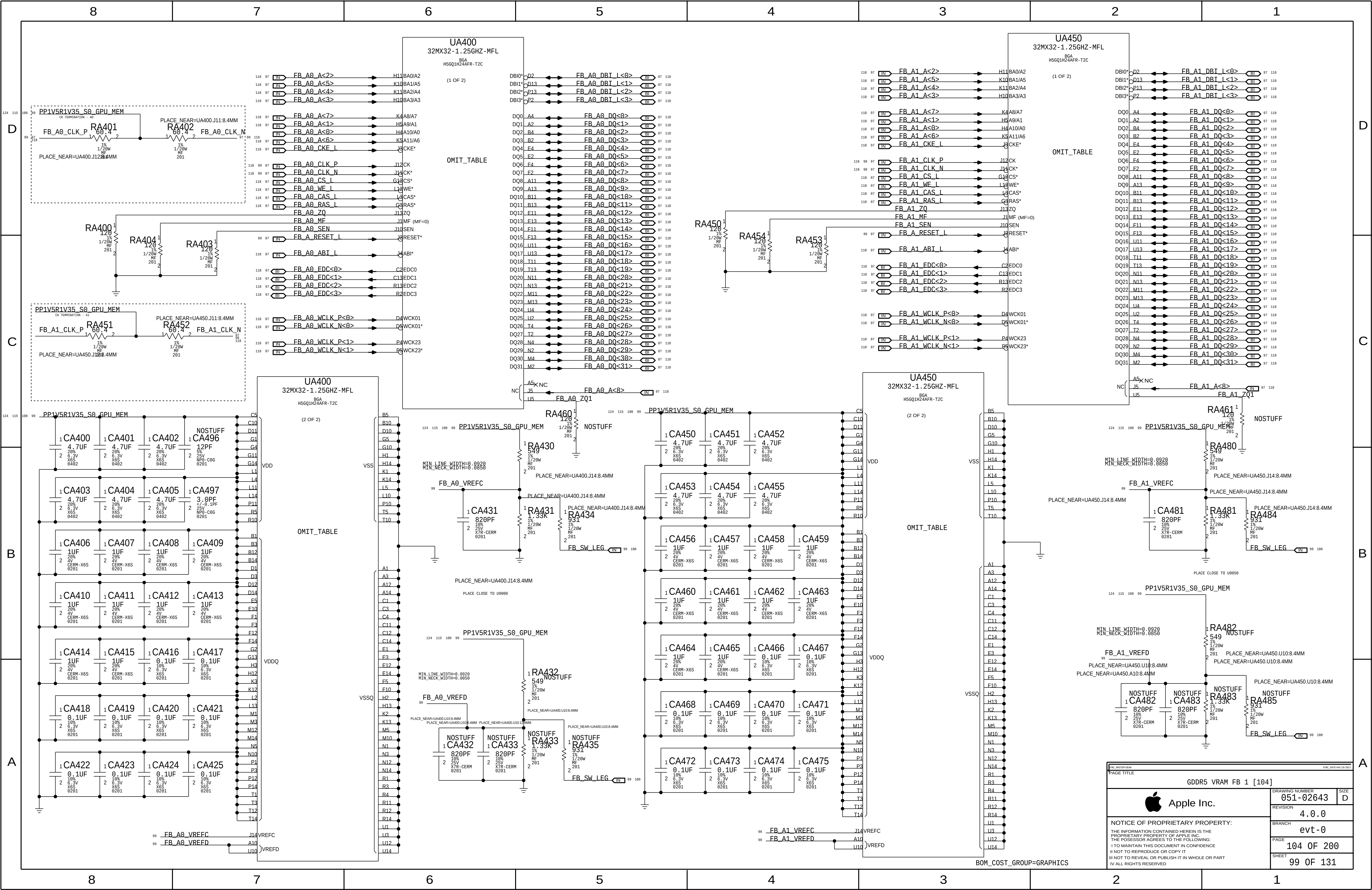




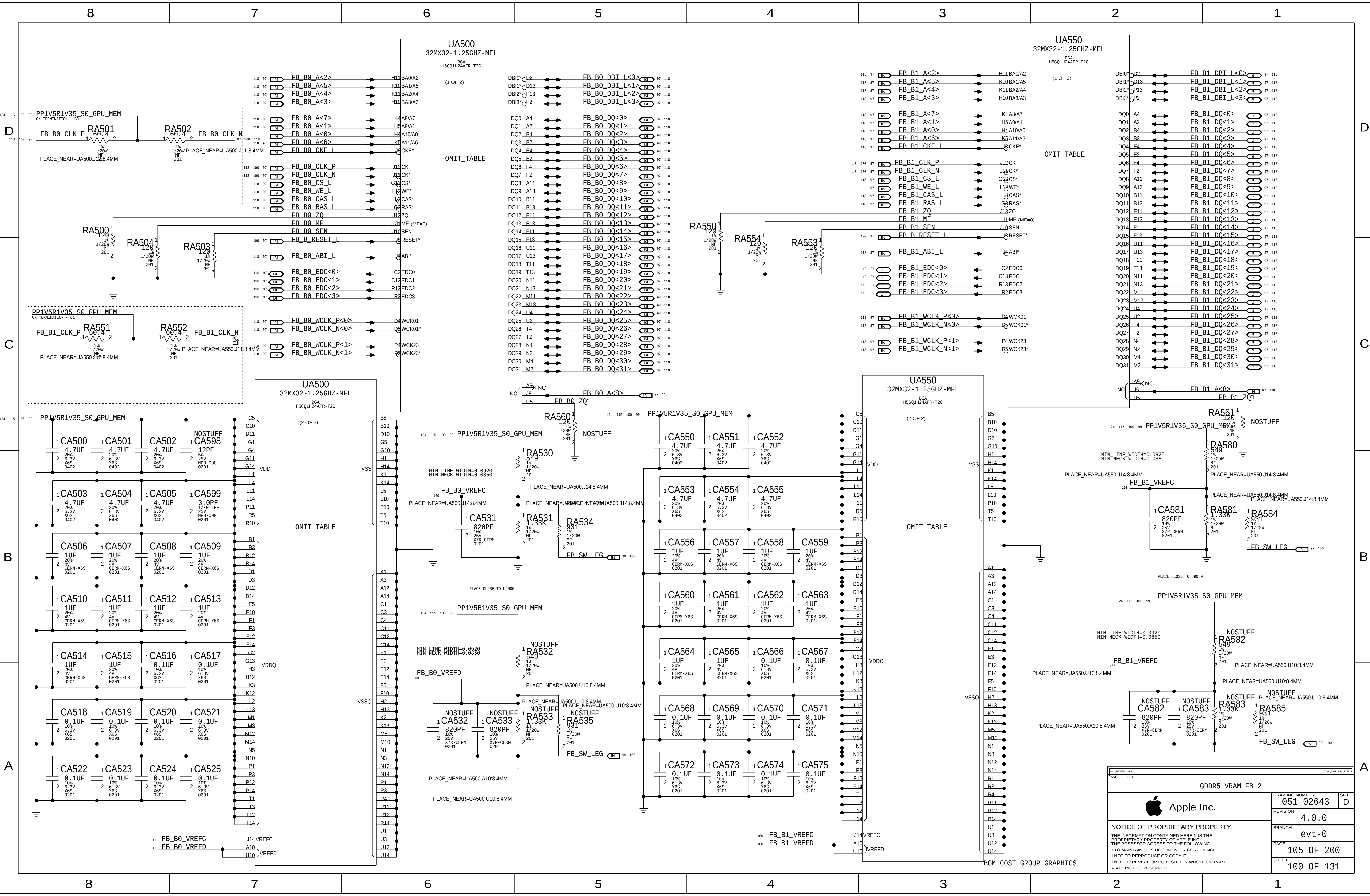


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S00174	1	MOSFET, CTRL+SYNC, 25V, 4.1/1.4MO, QFN32, 6x6	QA340	CRITICAL	

PAGE TITLE			VR 1.05V GPU & 1.35V FB	
		DRAWING NUMBER	051-02643	SIZE
		REVISION	4.0.0	D
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GDDR5 VRAM FB 1 [104]				DRAWING NUMBER			
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DOC. NUMBER/ISSUE		SYM. 3473-164-130(2007)	
PAGE TITLE			
GDDR5 VRAM FB 2			
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	051-02643		D
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	BRANCH		
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		100 OF 131	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S00174	2	MOSFET, CTRL+SYNC, 25V, 4...1/1.4M0, QFN32, 6x6	QA640, QA650	CRITICAL	

D

C

B

A

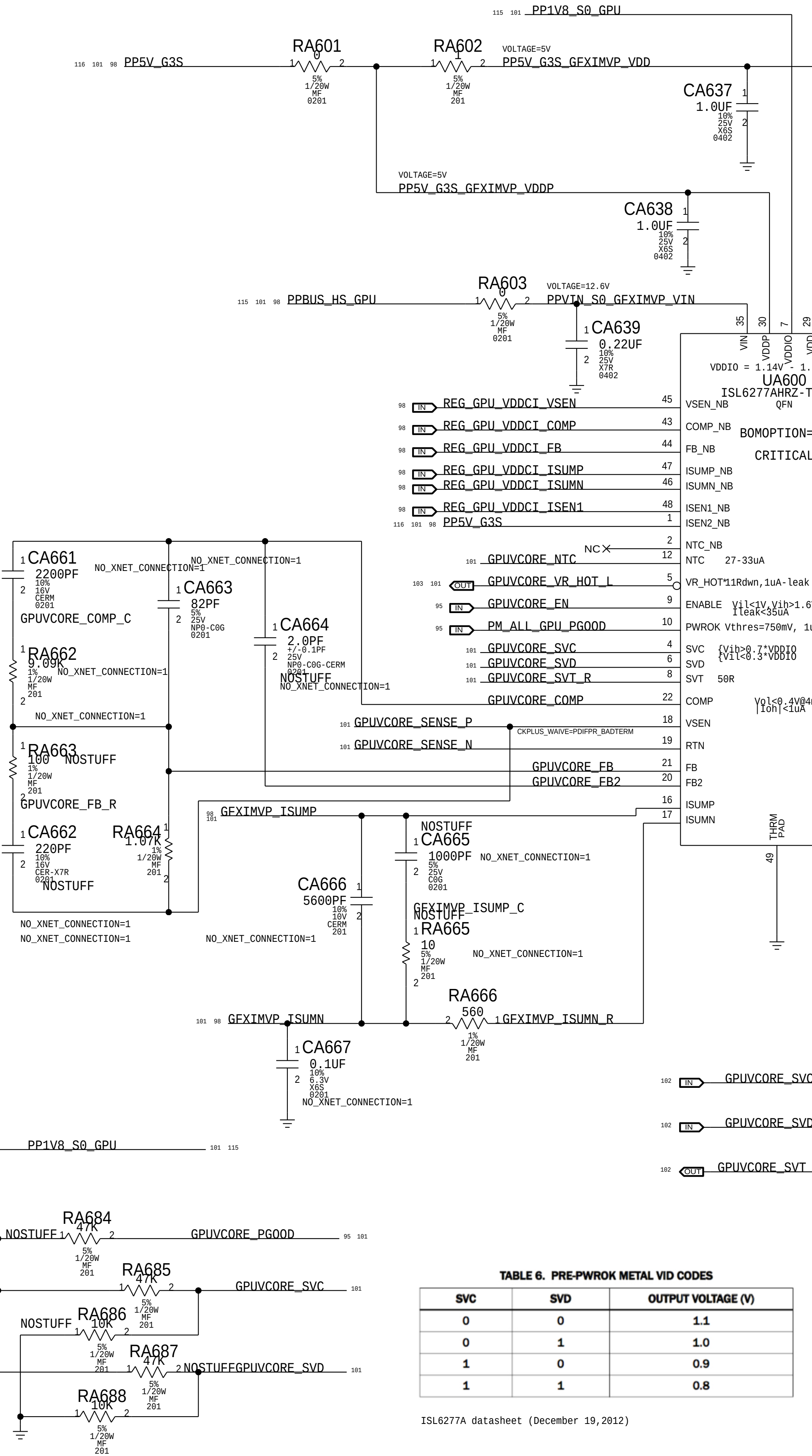
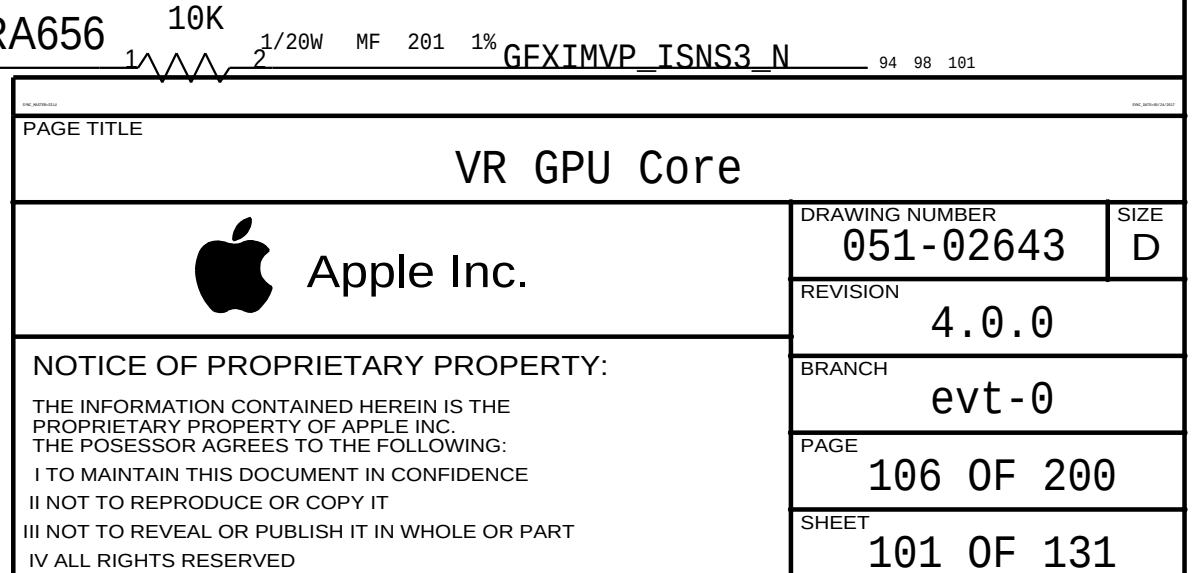
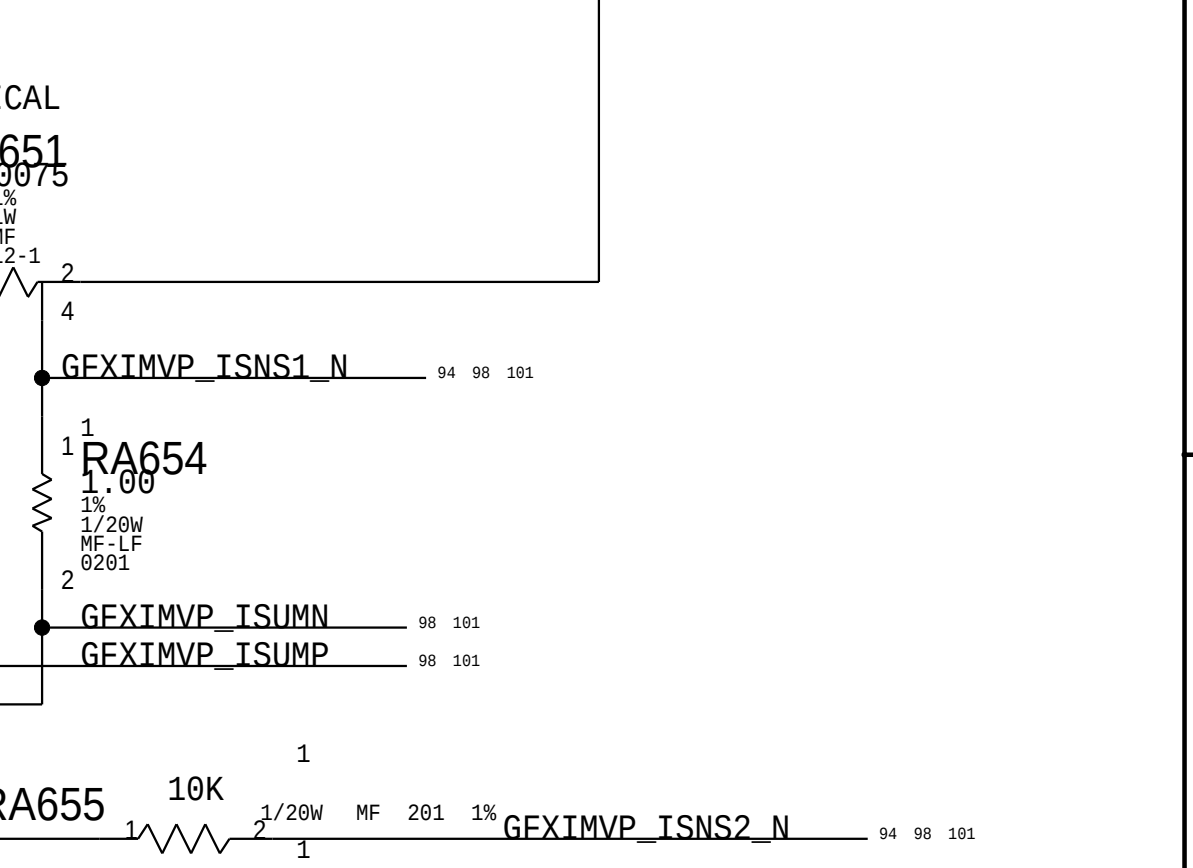
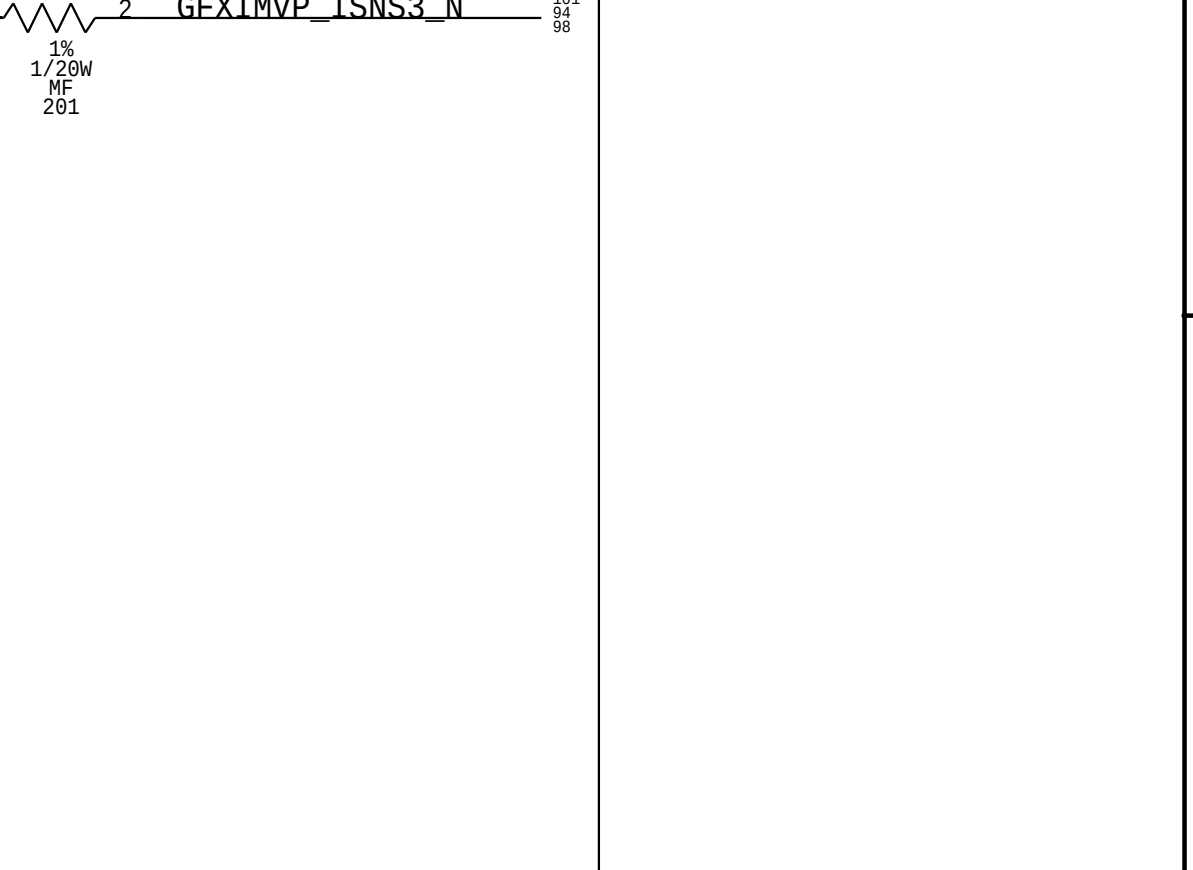
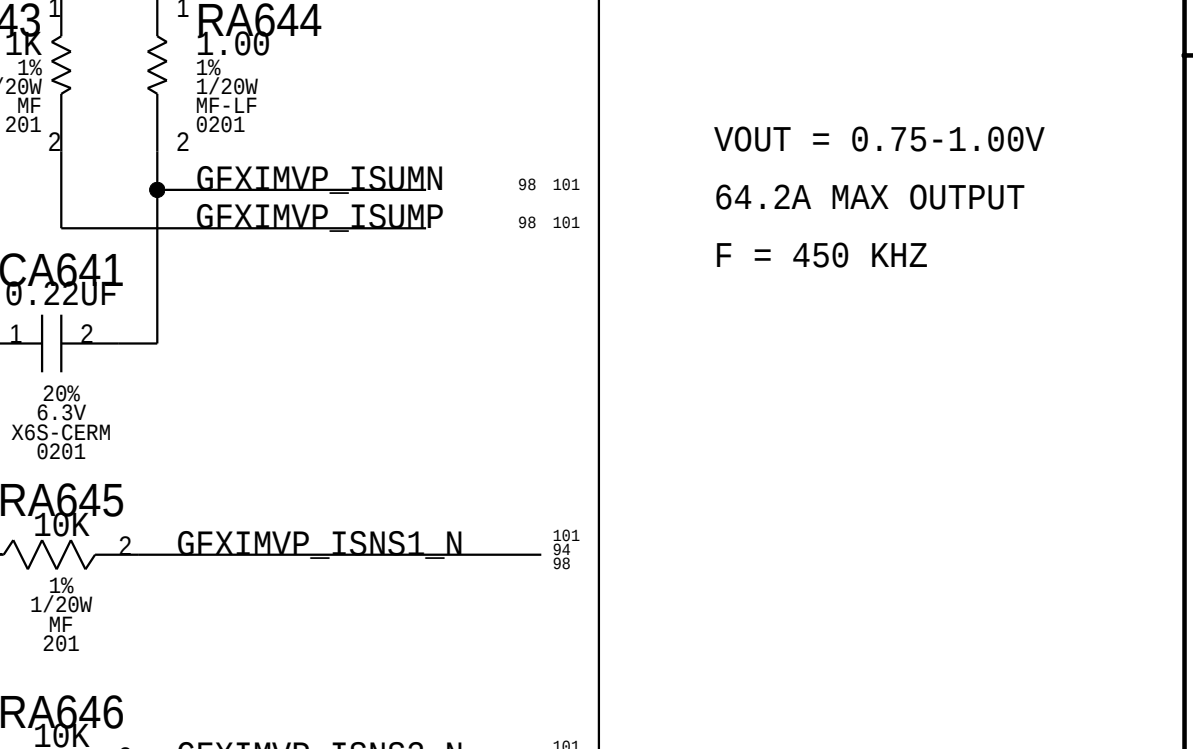
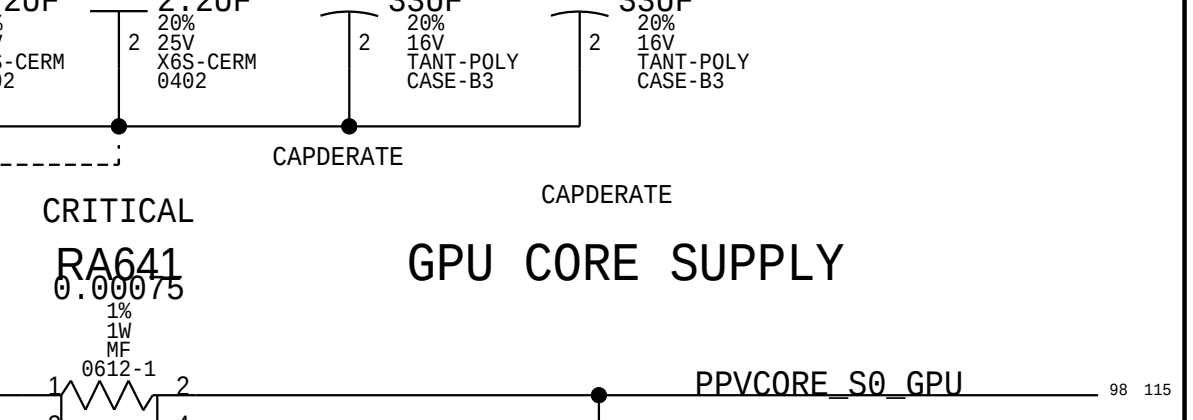
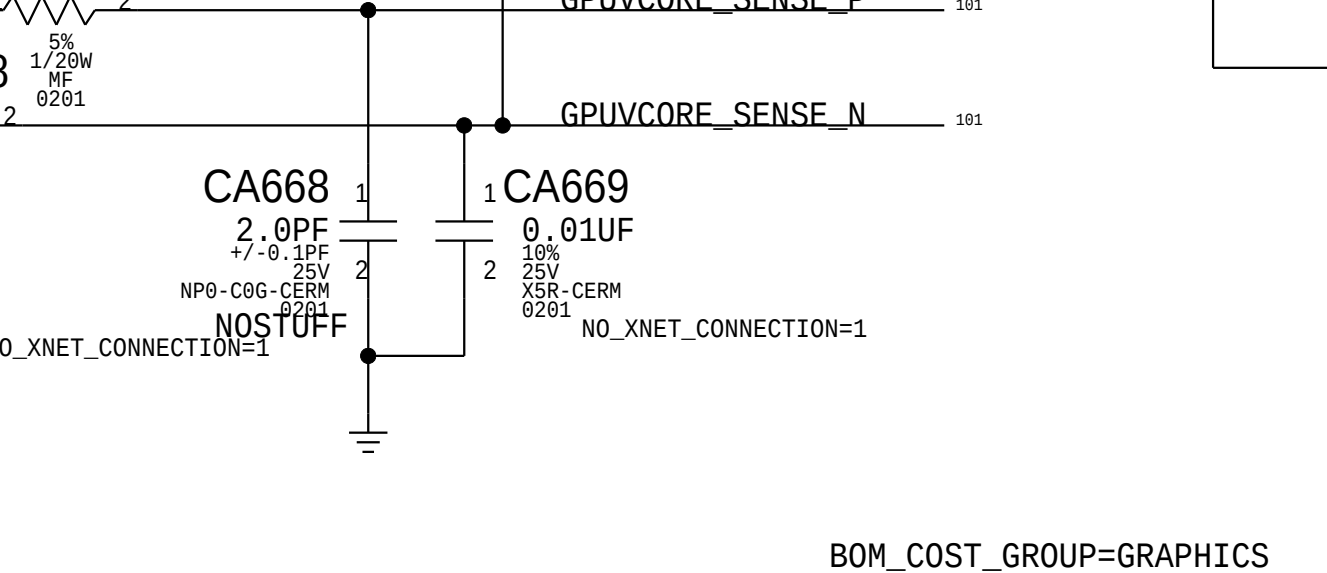
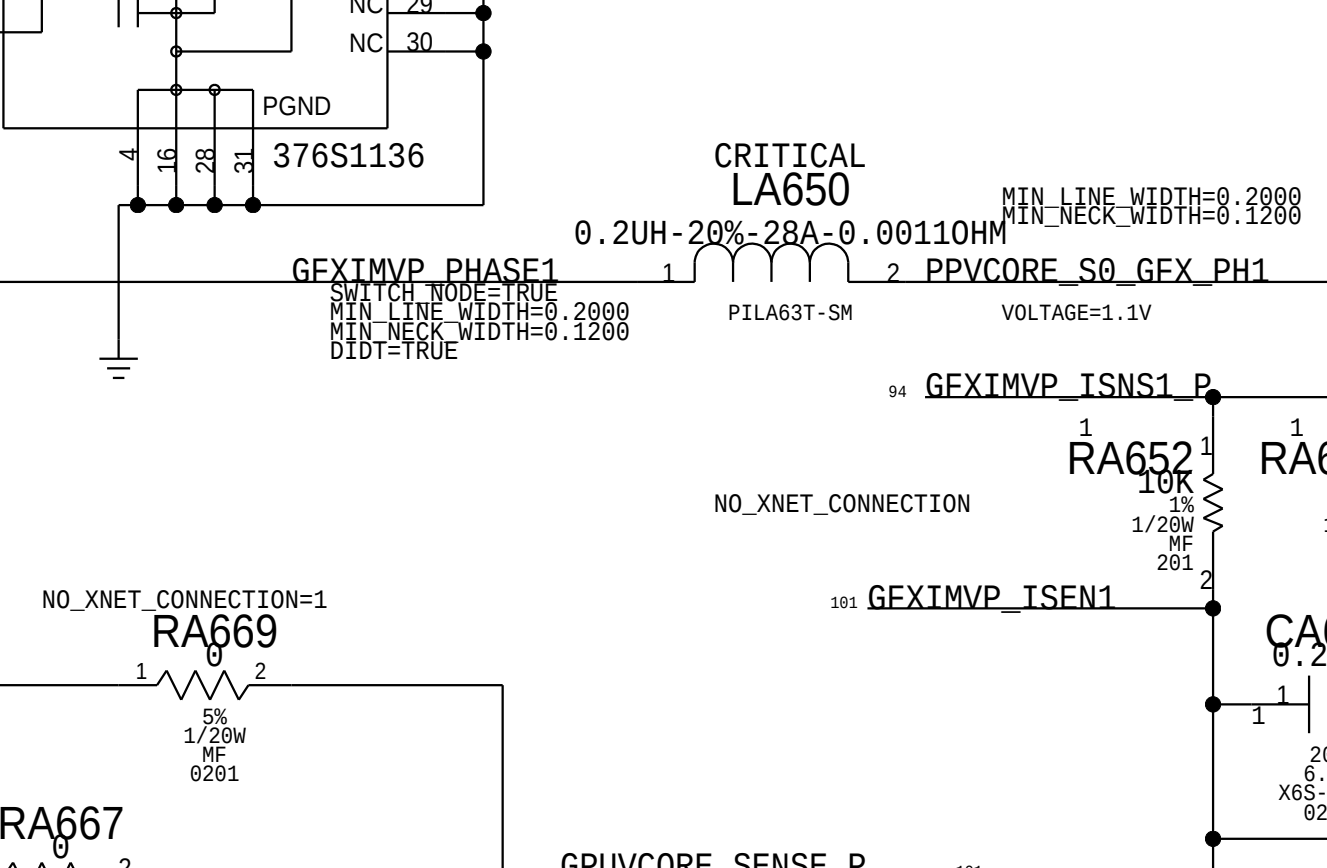
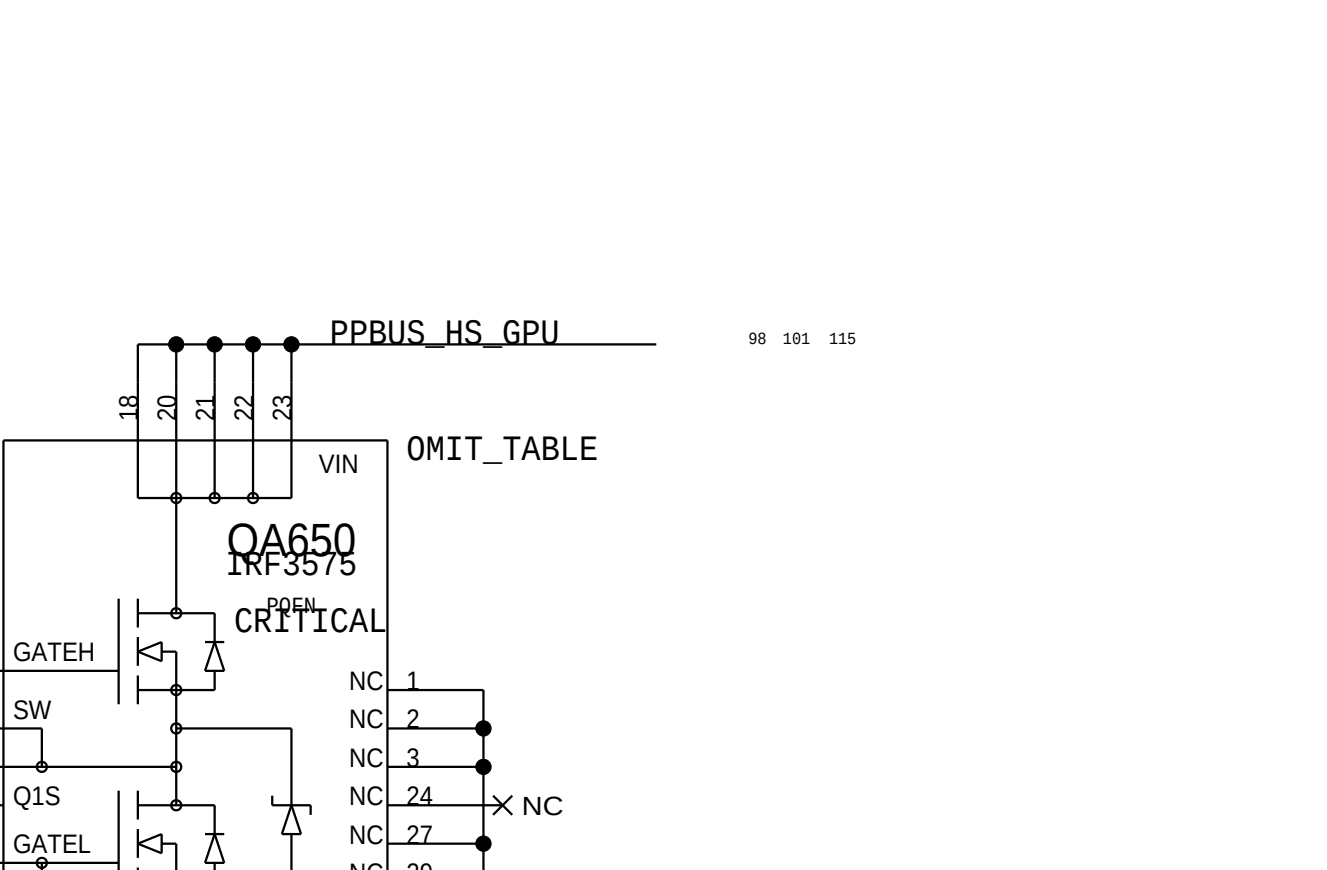
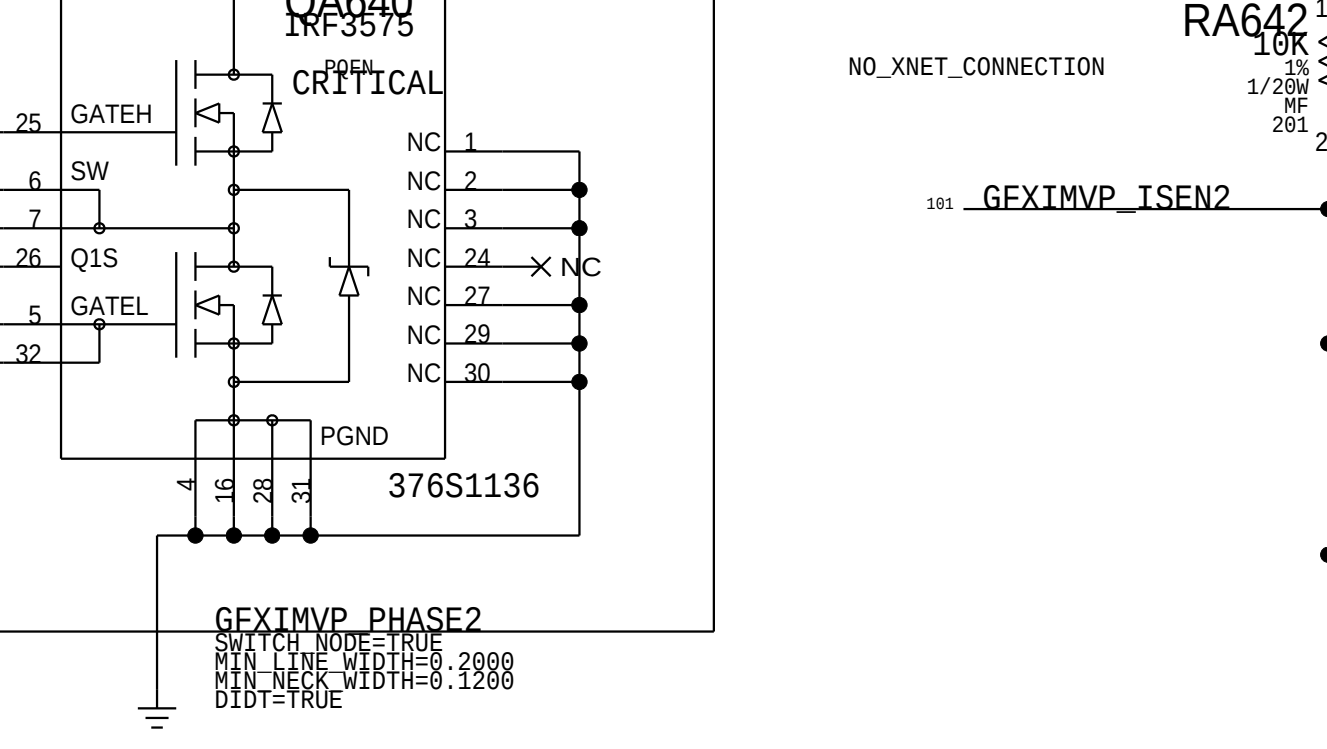
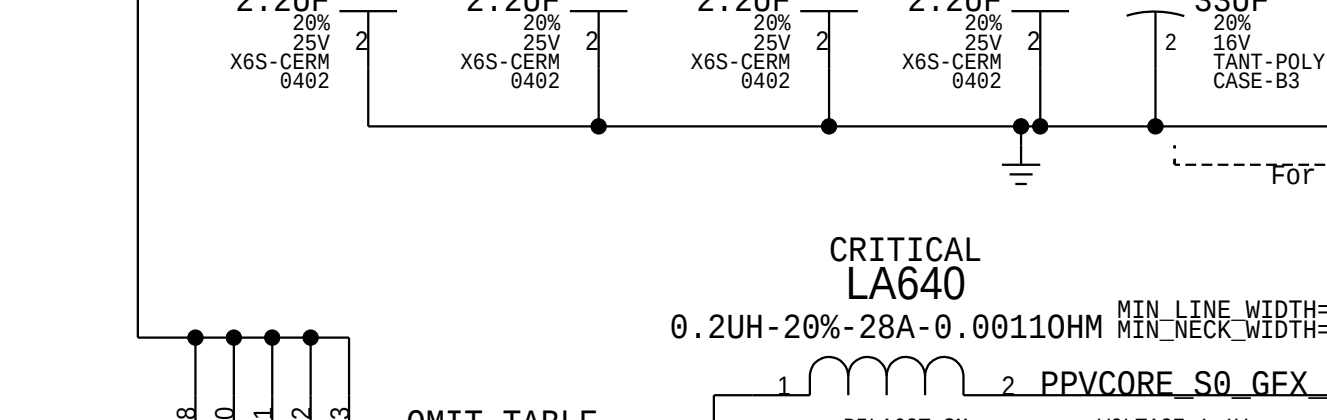
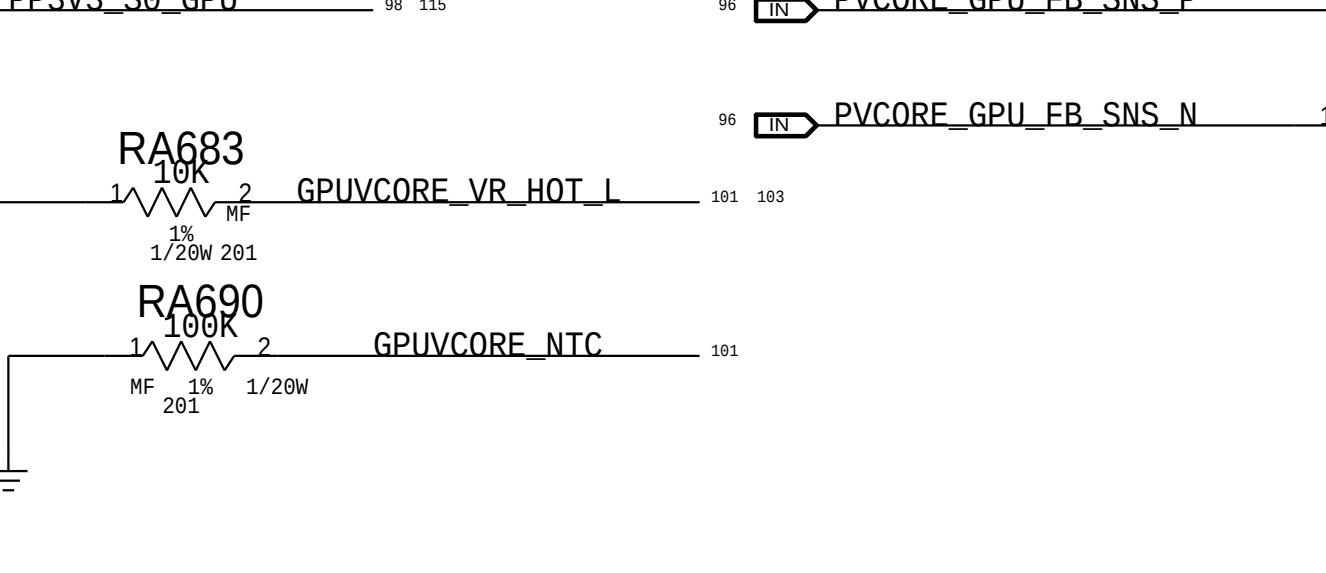
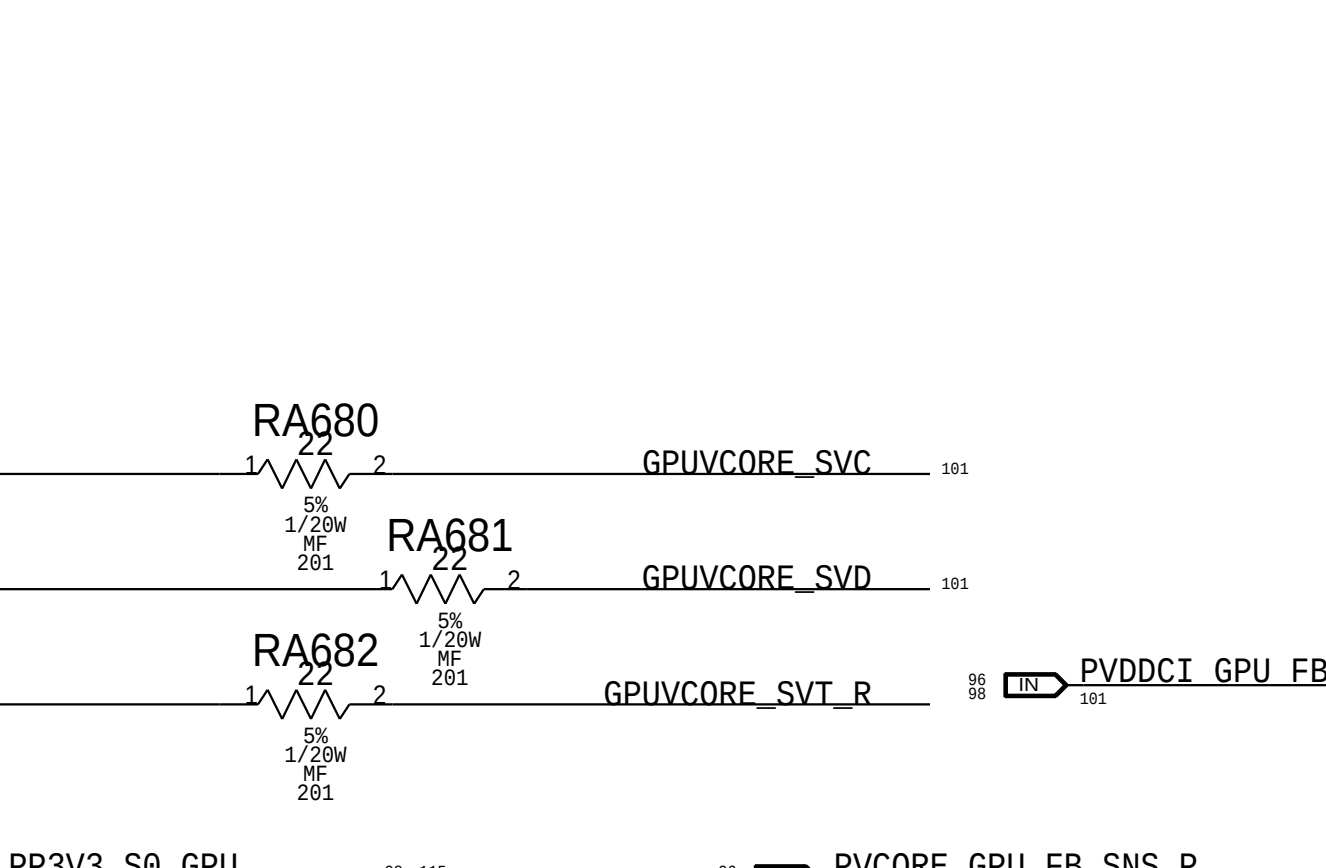
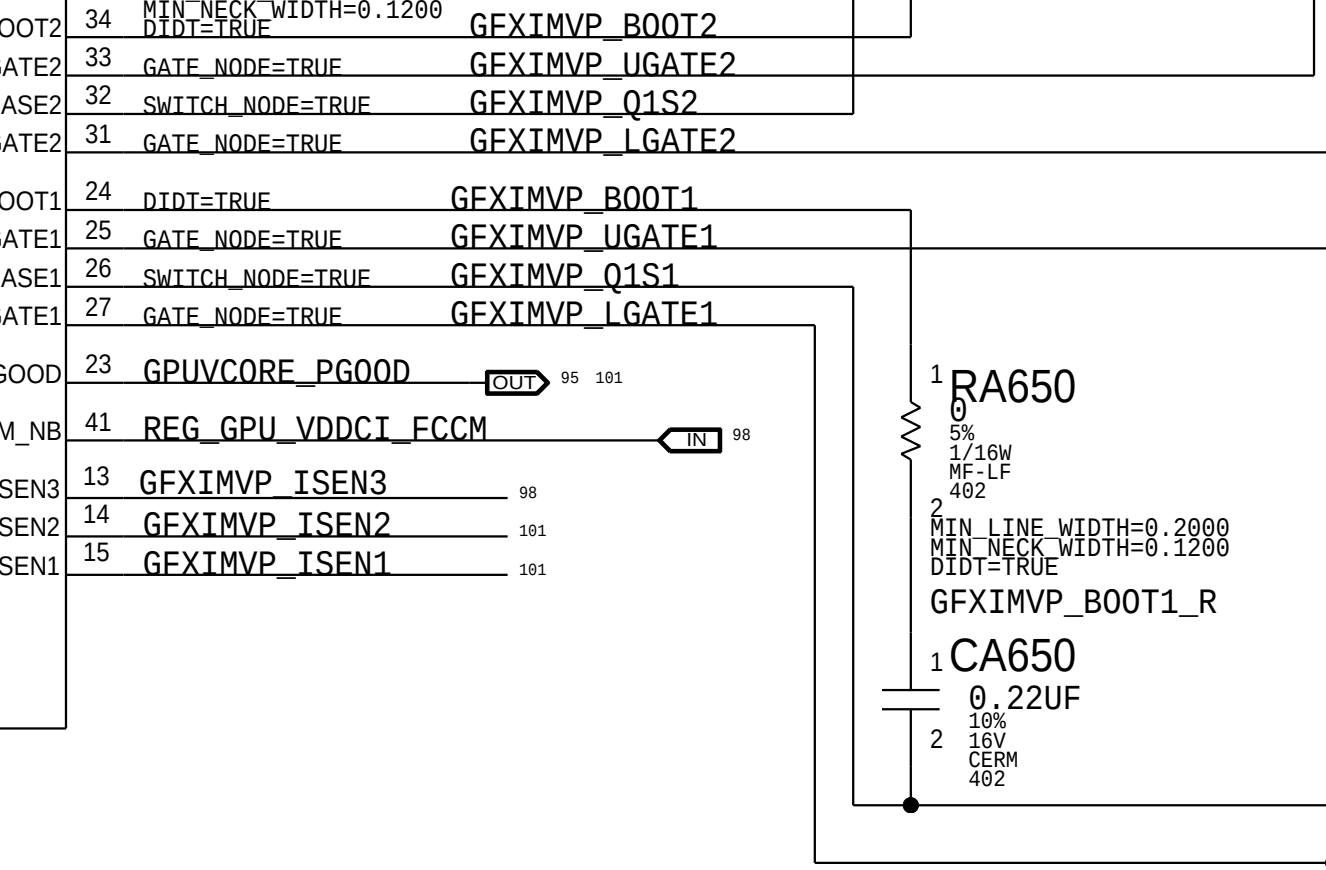
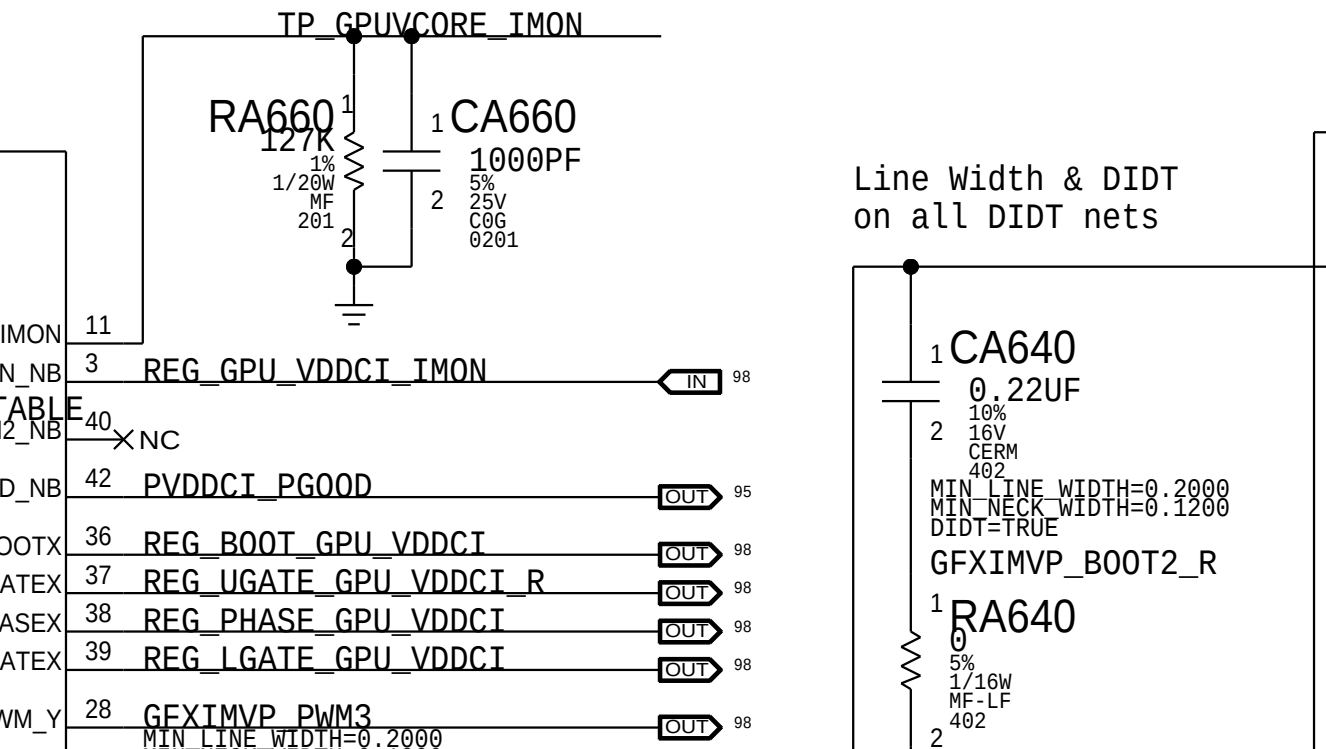
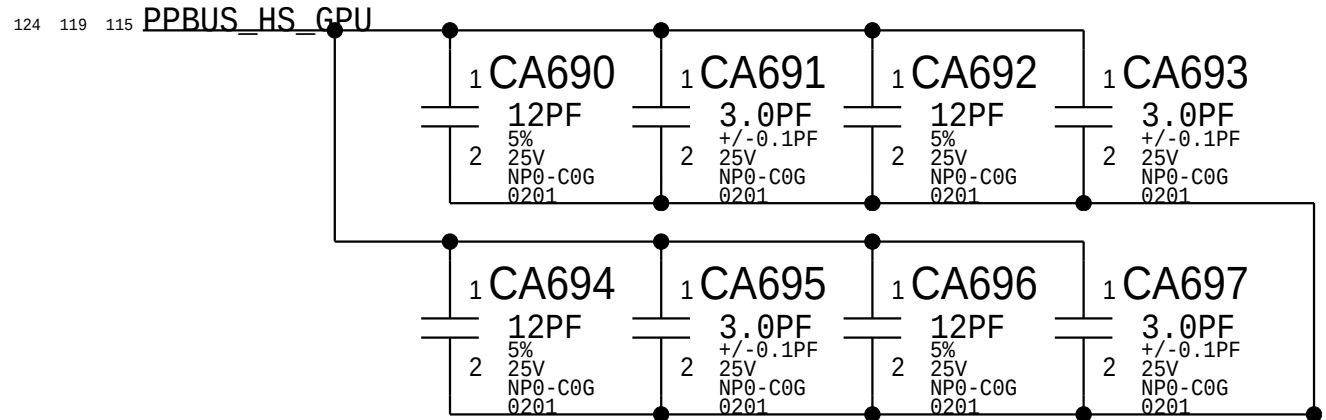


TABLE 6. PRE-WROK METAL VID CODES		
SVC	SVD	OUTPUT VOLTAGE (V)
0	0	1.1
0	1	1.0
1	0	0.9
1	1	0.8

ISL6277A datasheet (December 19, 2012)



D

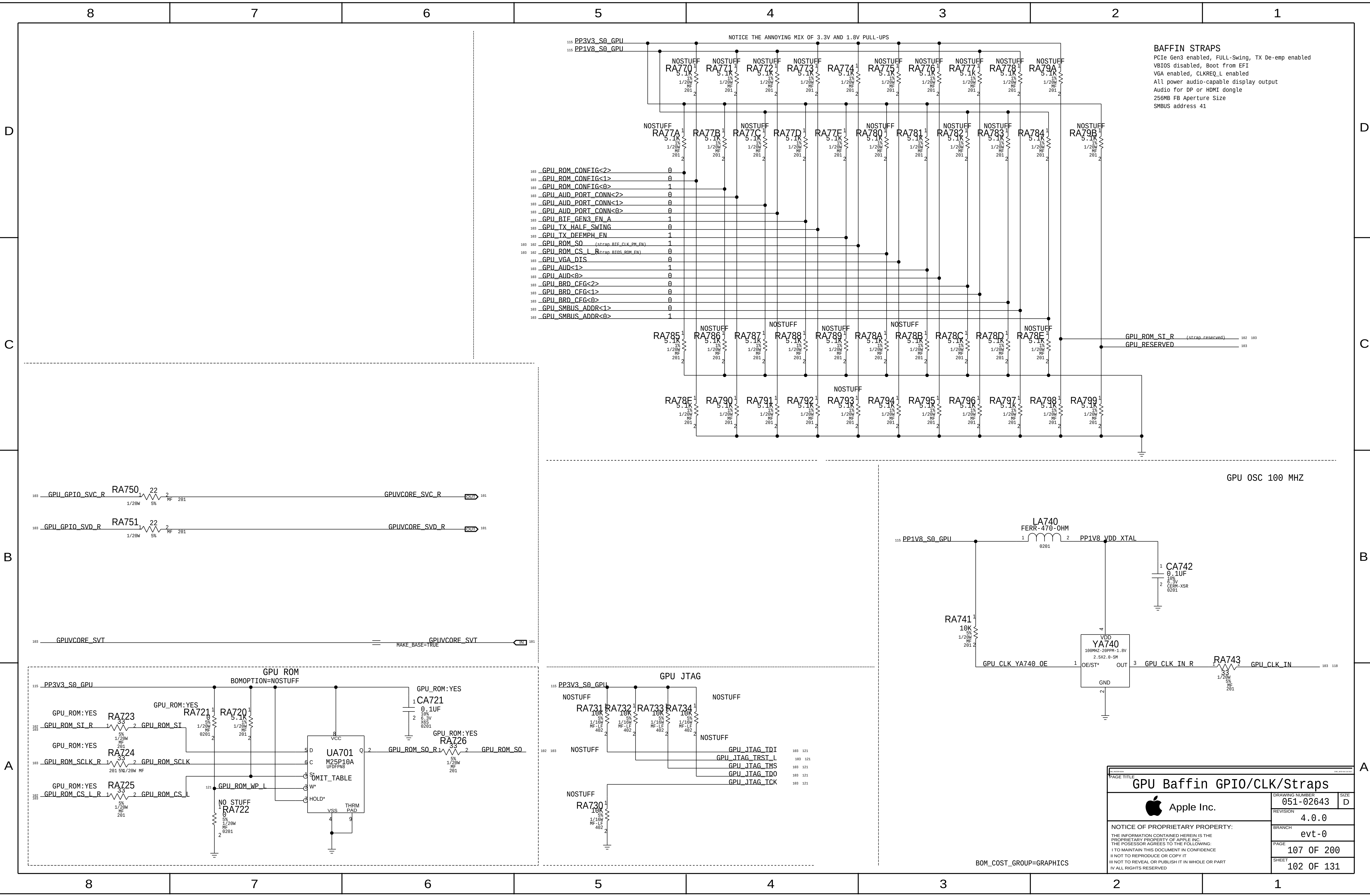
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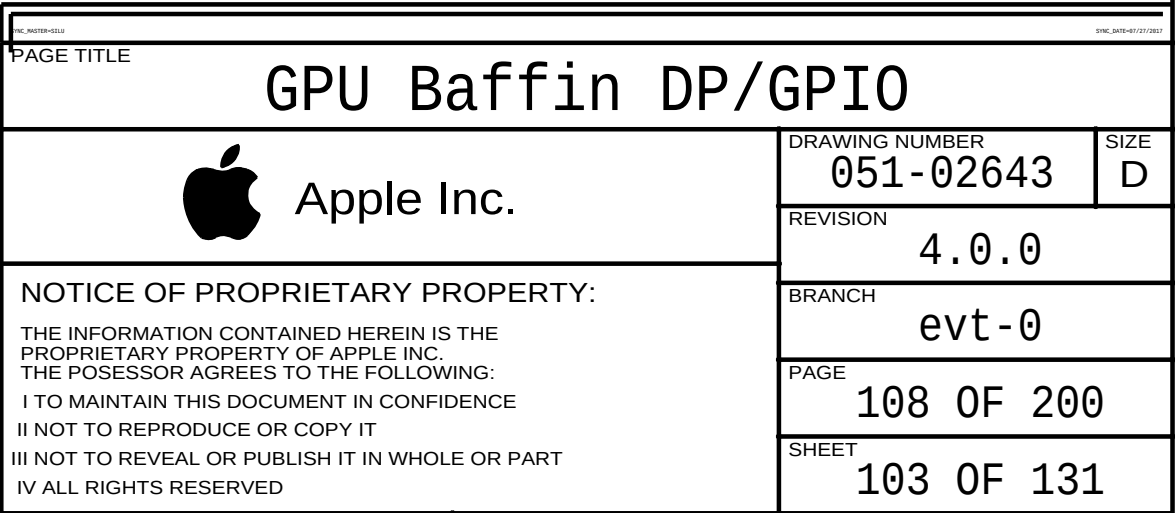
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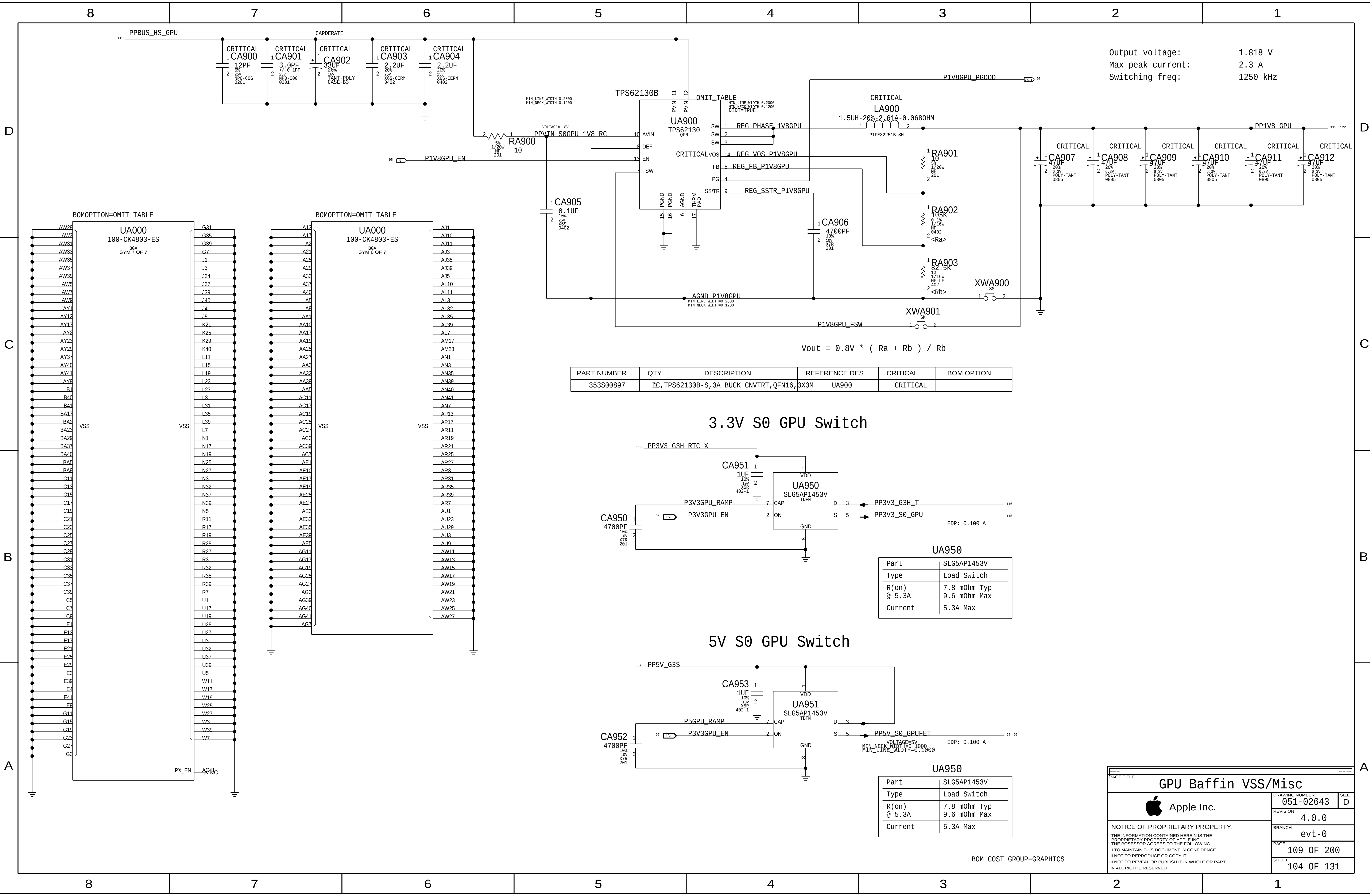
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		DRAWING NUMBER	051-02643
		REVISION	4.0.0
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GPU Baffin GPIO/CLK/Straps			
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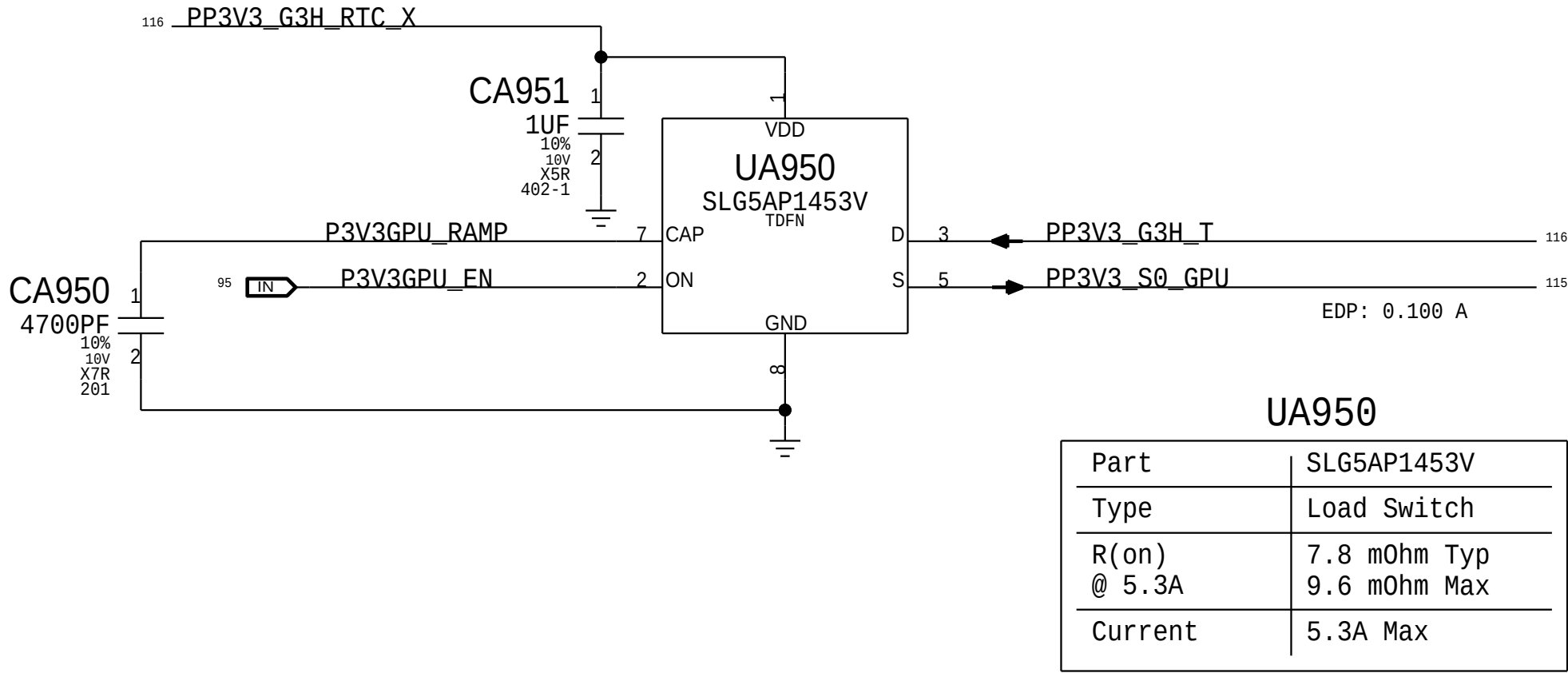


Output voltage: 1.818 V
Max peak current: 2.3 A
Switching freq: 1250 kHz

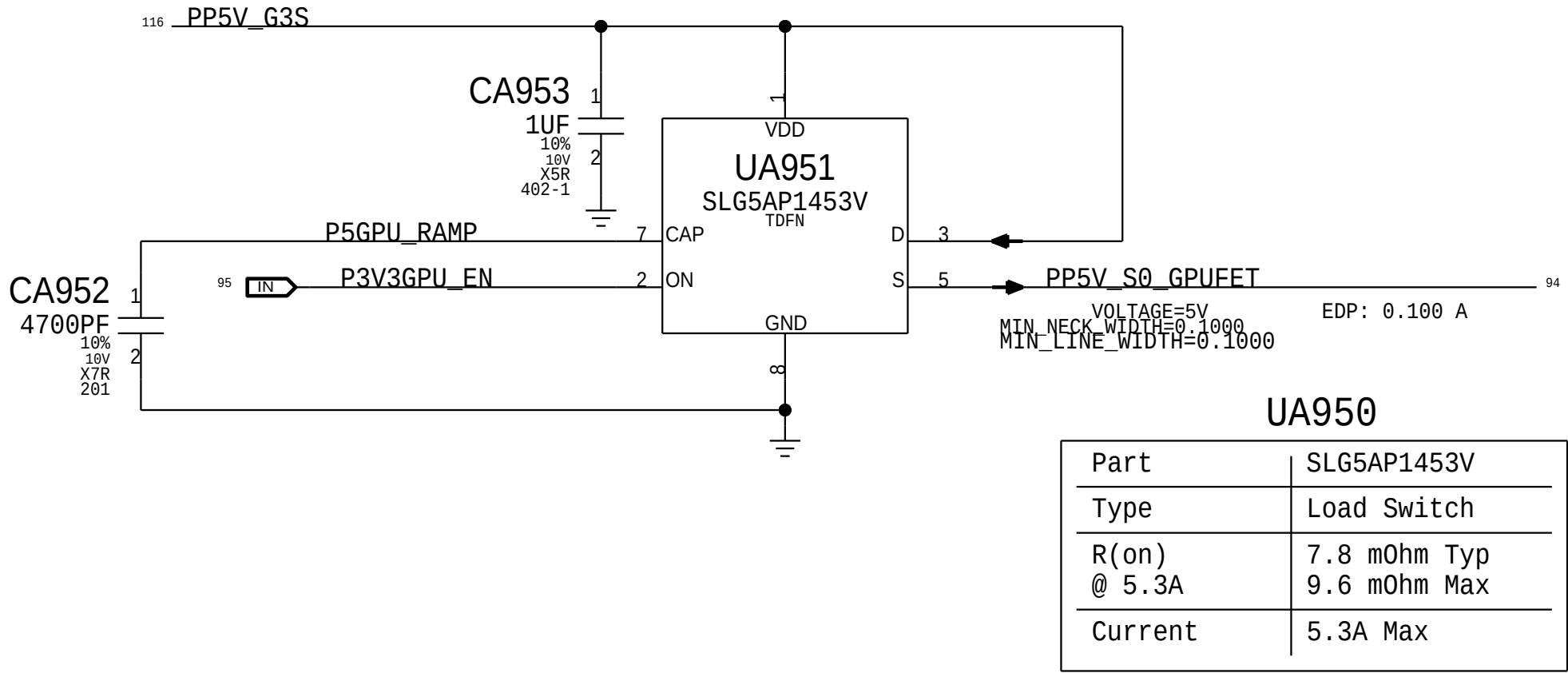
$$V_{out} = 0.8V * (R_a + R_b) / R_b$$

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00897	1C	TPS62130B-S, 3A BUCK CNVTRT, QFN16, 3X3M	UA900	CRITICAL	

3.3V S0 GPU Switch

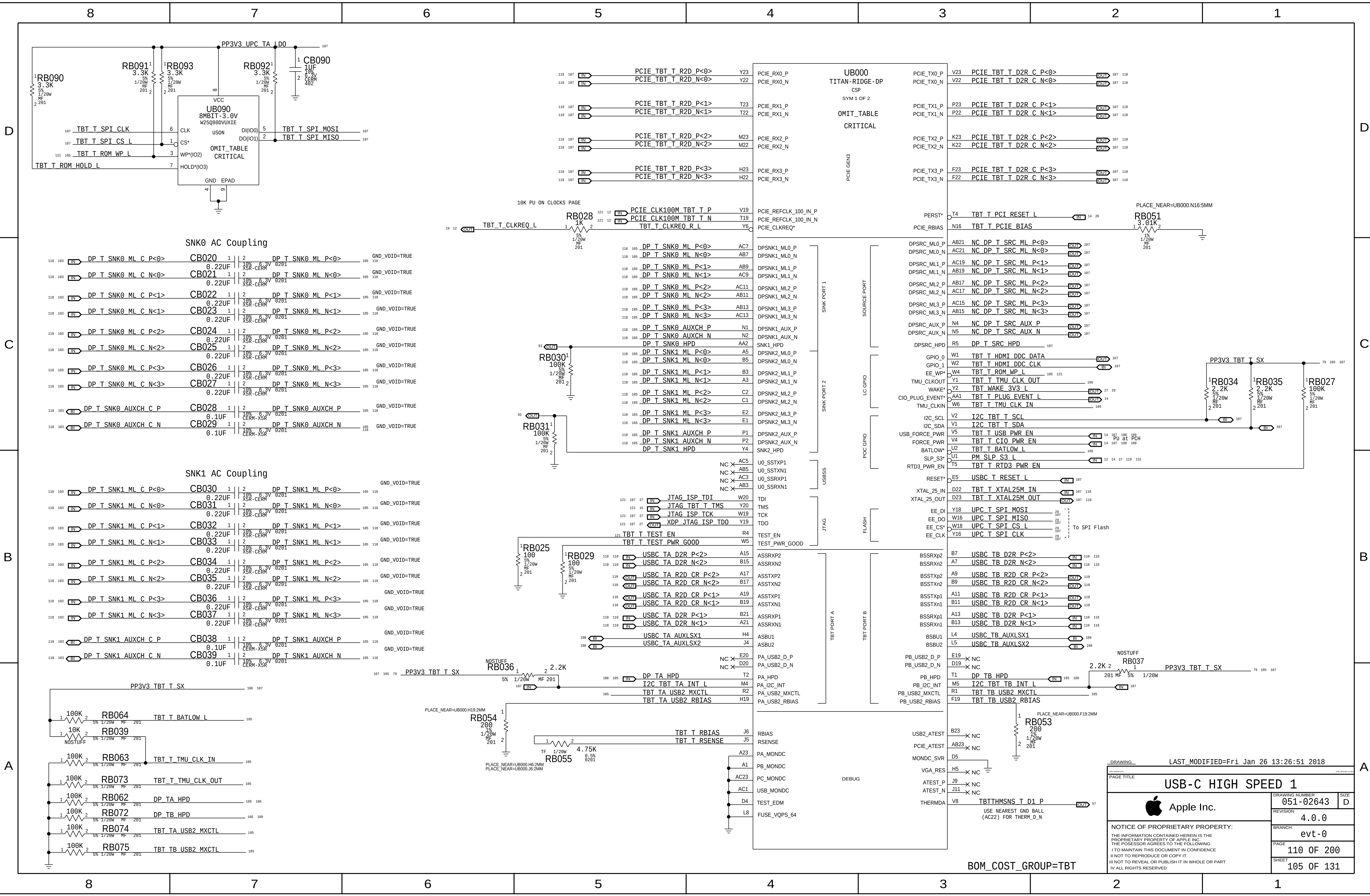


5V S0 GPU Switch



GPU Baffin VSS/Misc		
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BOM_COST_GROUP=GRAPHICS



D

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B

A

D

C

B

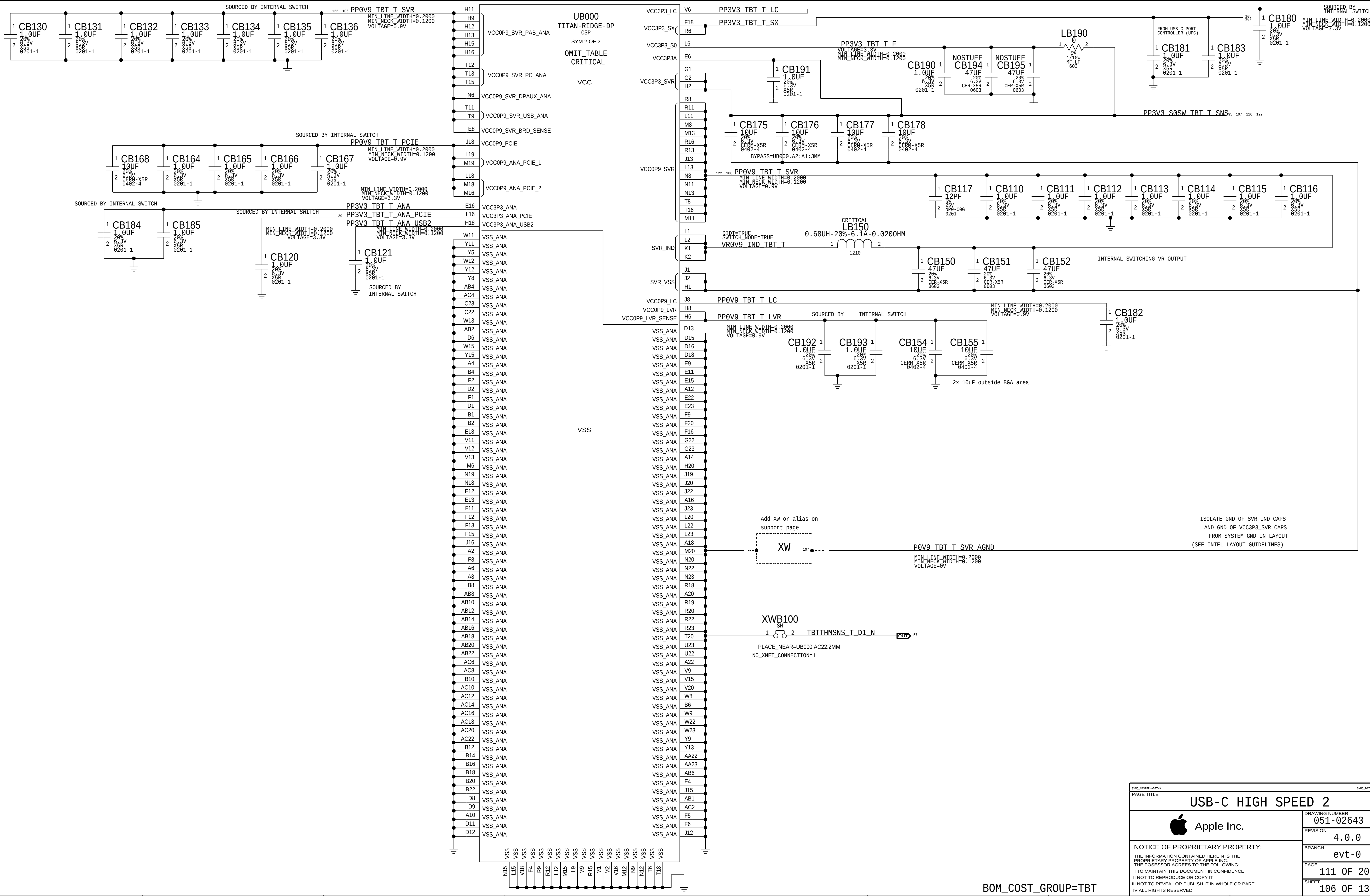
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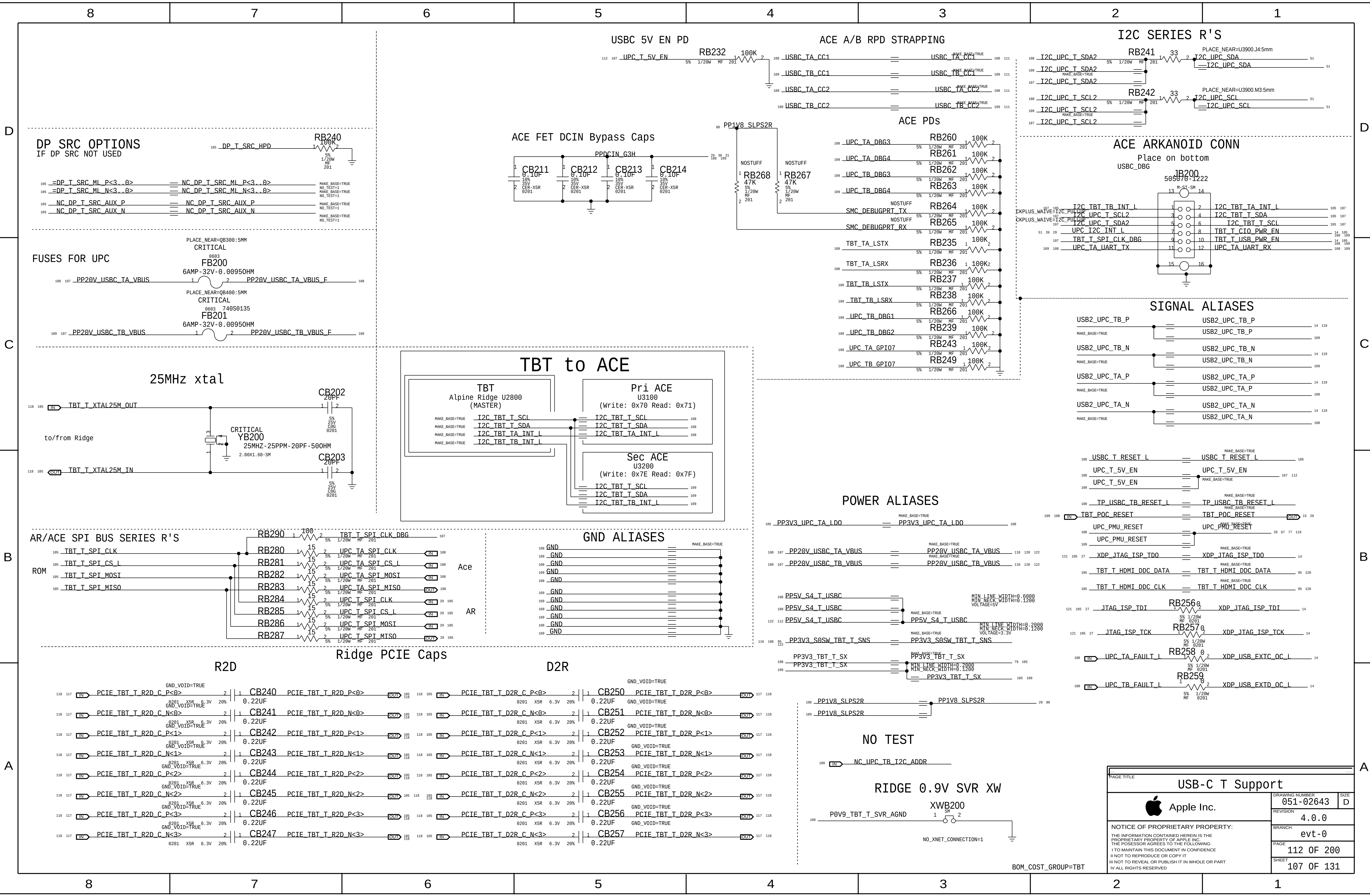
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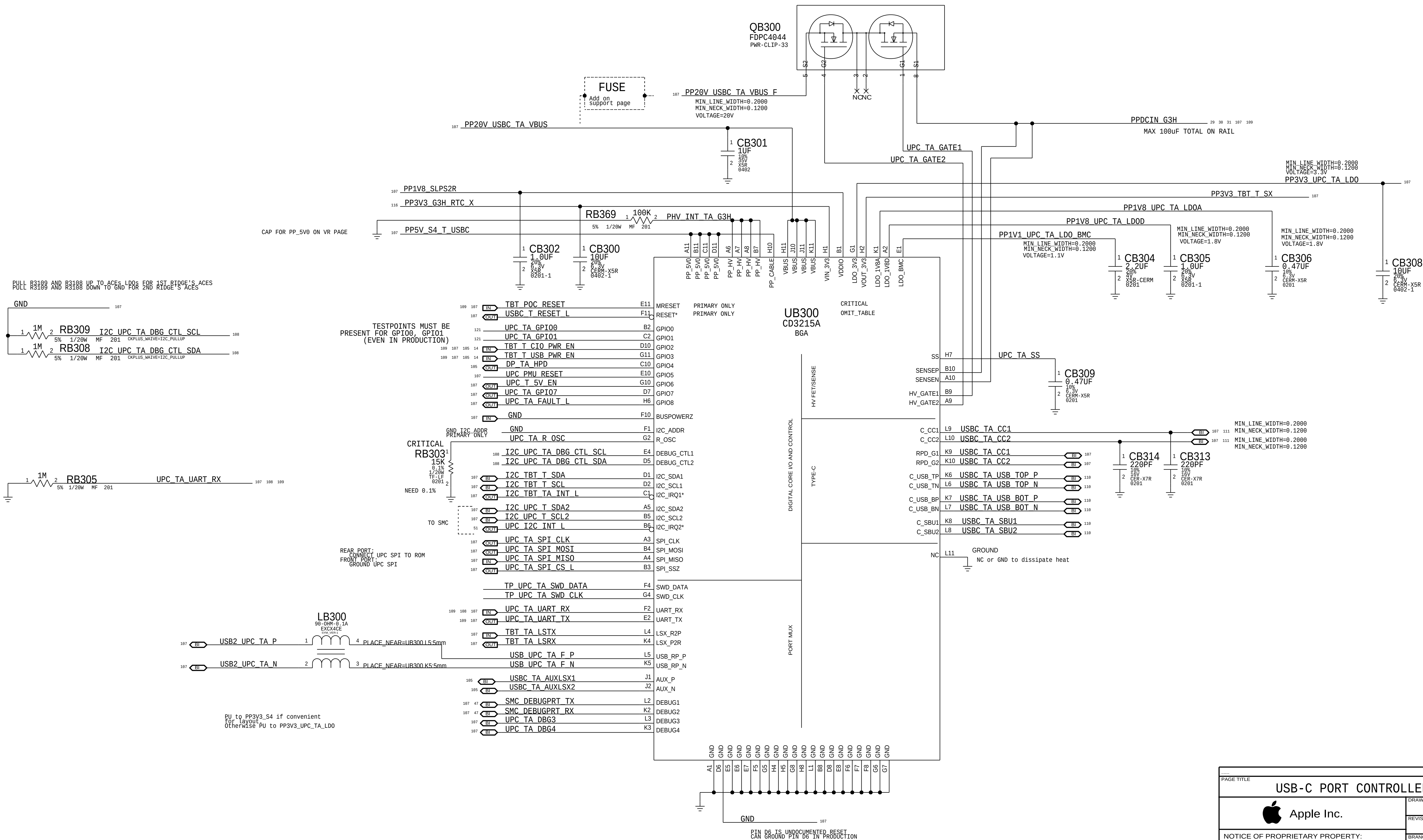
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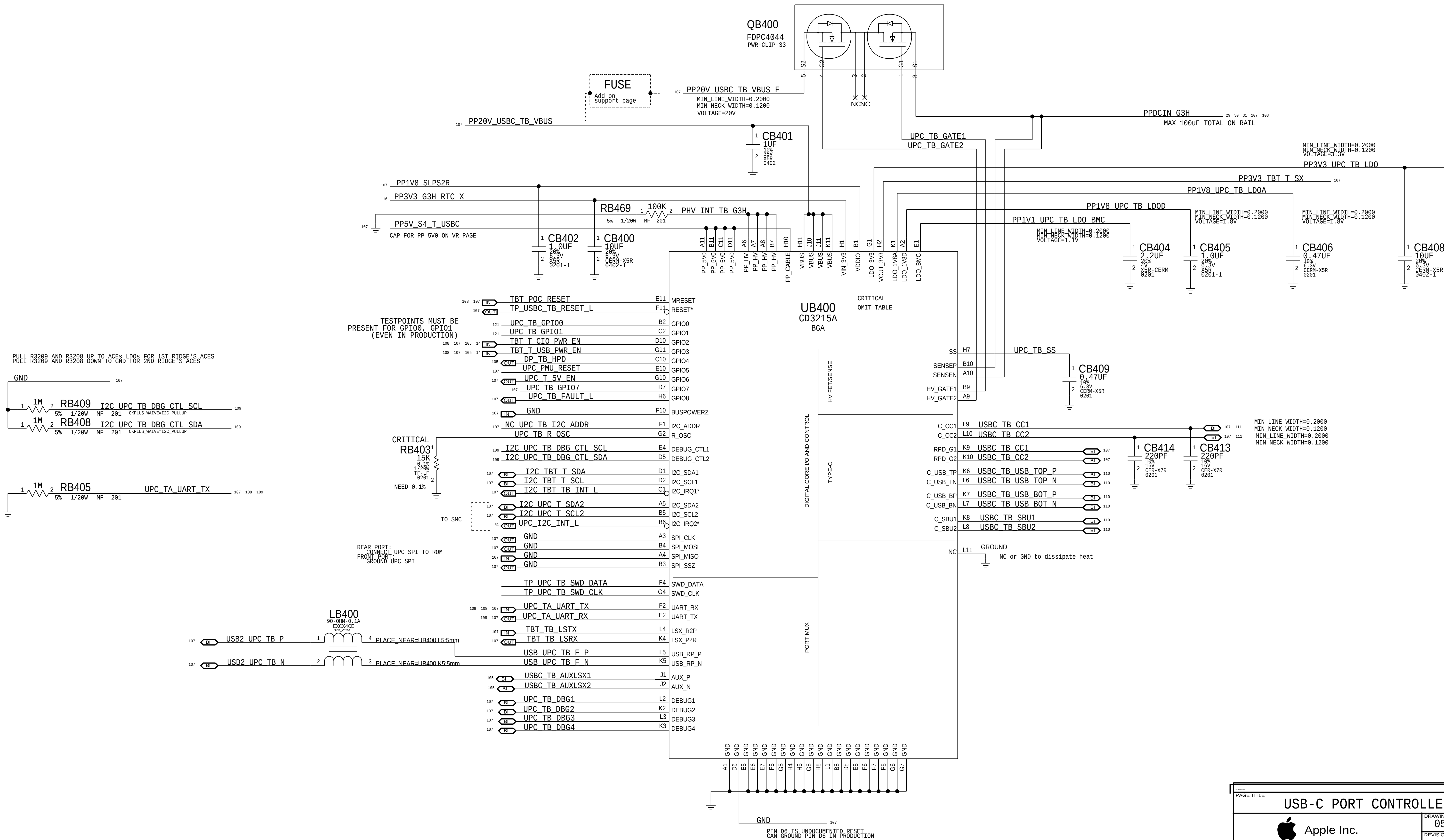
PRIMARY ACE USB-C PORT CONTROLLER (UPC)



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USB-C PORT CONTROLLER A		
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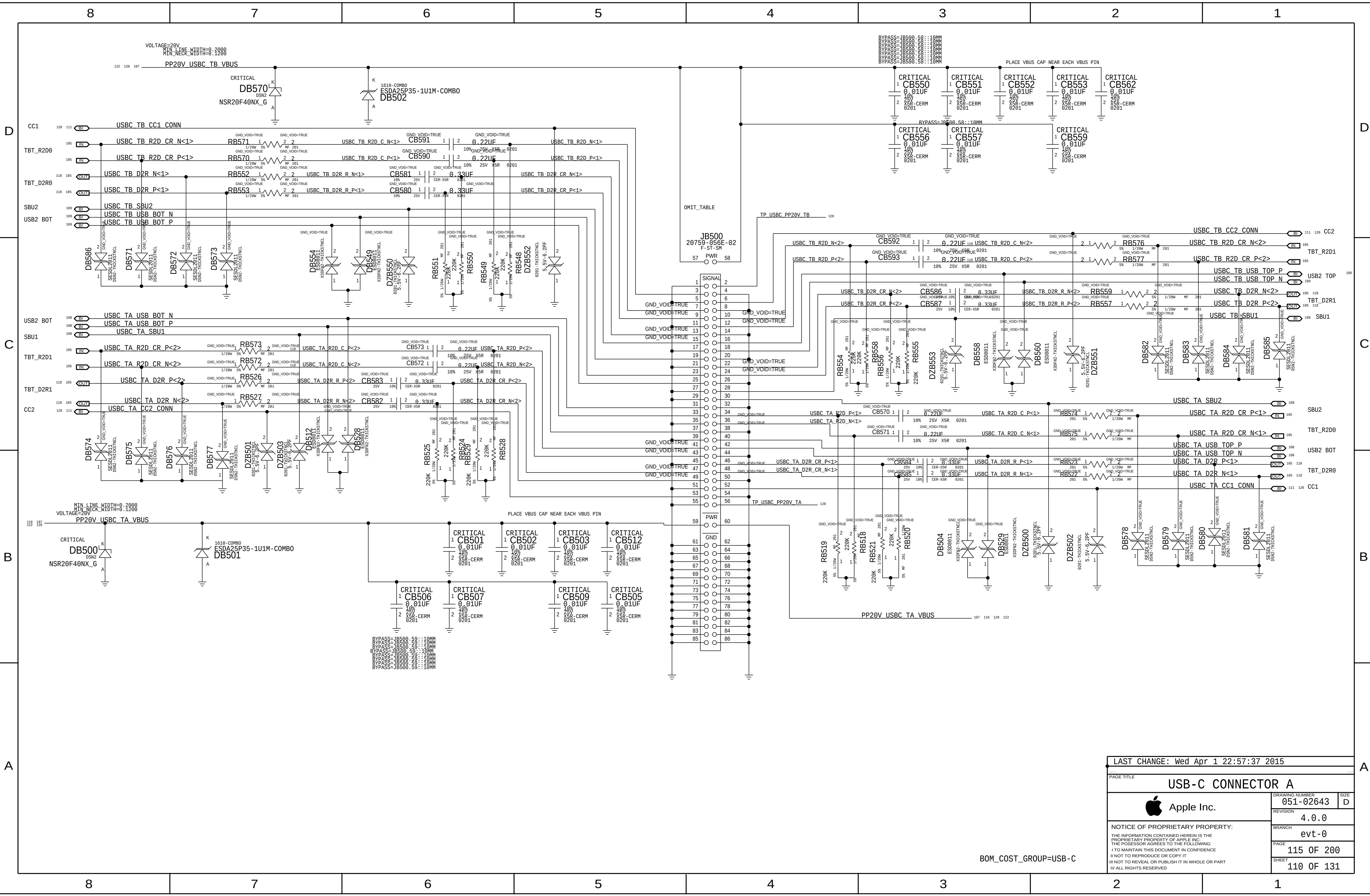
BOM_COST_GROUP=USB-C

SECONDARY ACE USB-C PORT CONTROLLER (UPC)



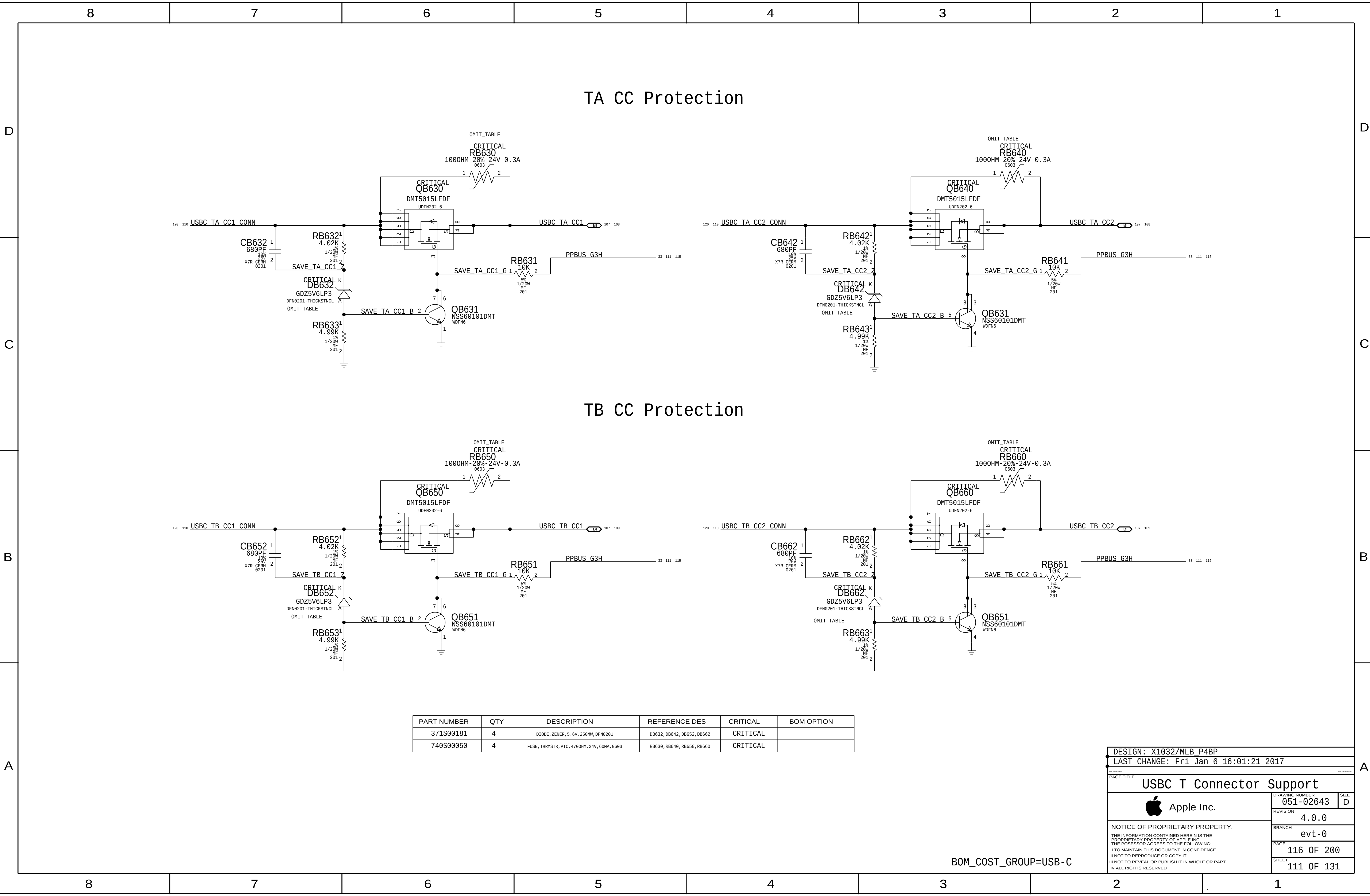
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USB-C PORT CONTROLLER B		
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BOM_COST_GROUP=USB-C



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USB-C CONNECTOR A			
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BOM_COST_GROUP=USB-C



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
371S00181	4	DIODE, ZENER, 5.6V, 250MW, DFN0201	DB632, DB642, DB652, DB662	CRITICAL	
740S00050	4	FUSE, THRMSTR, PTC, 4700HM, 24V, 60MA, 0603	RB630, RB640, RB650, RB660	CRITICAL	

DESIGN: X1032/MLB_P4BP

LAST CHANGE: Fri Jan 6 16:01:21 2017

PAGE TITLE

USB C Connector Support

 Apple Inc.

DRAWING NUMBER

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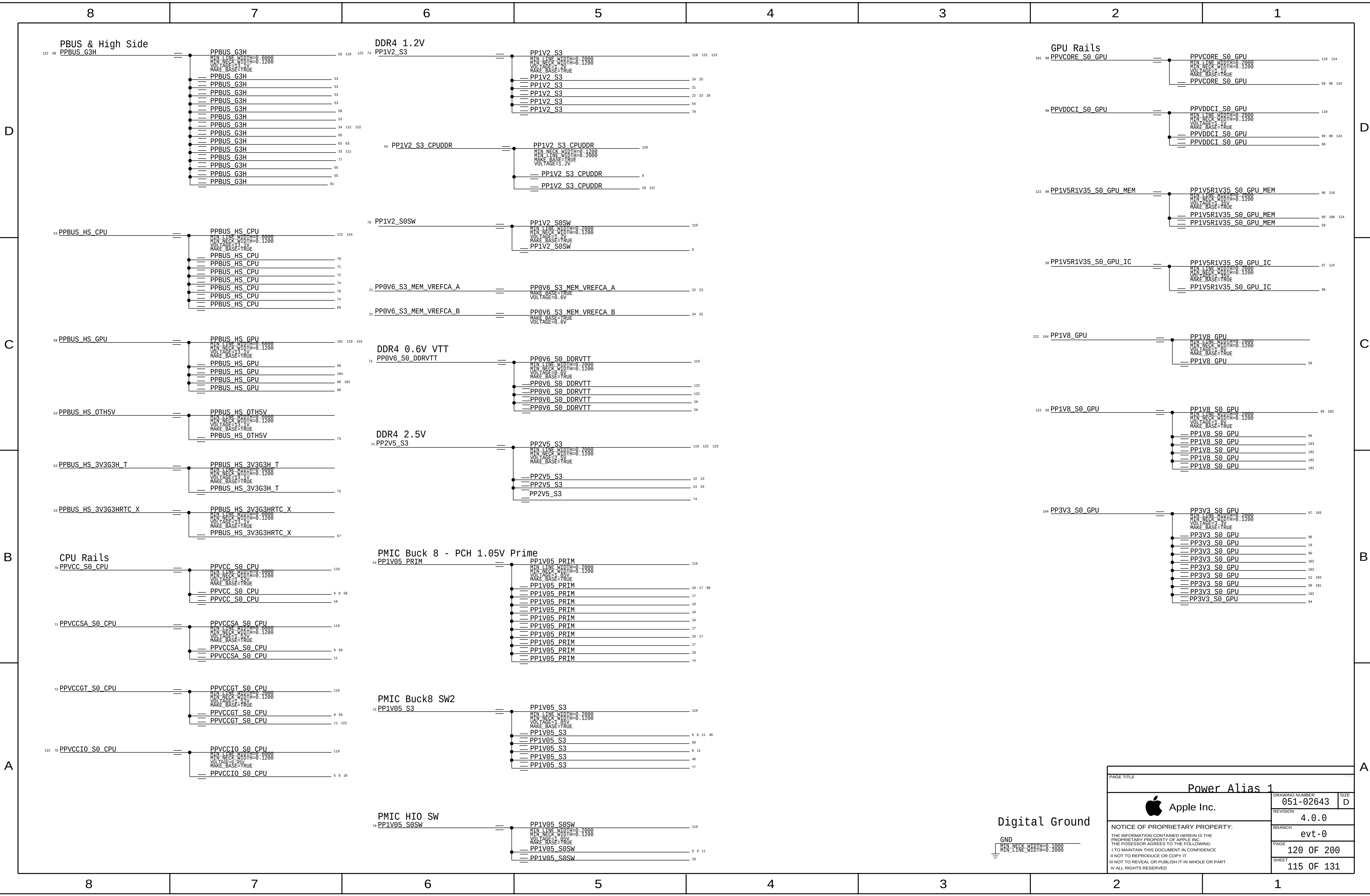
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8		7		6		5		4		3		2		1			
D																D	
C																C	
B																B	
A																A	
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PAGE TITLE			PAGE TITLE		
GDDR5 VRAM FB 4			GDDR5 VRAM FB 4		
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			PAGE		
			119 OF 200		
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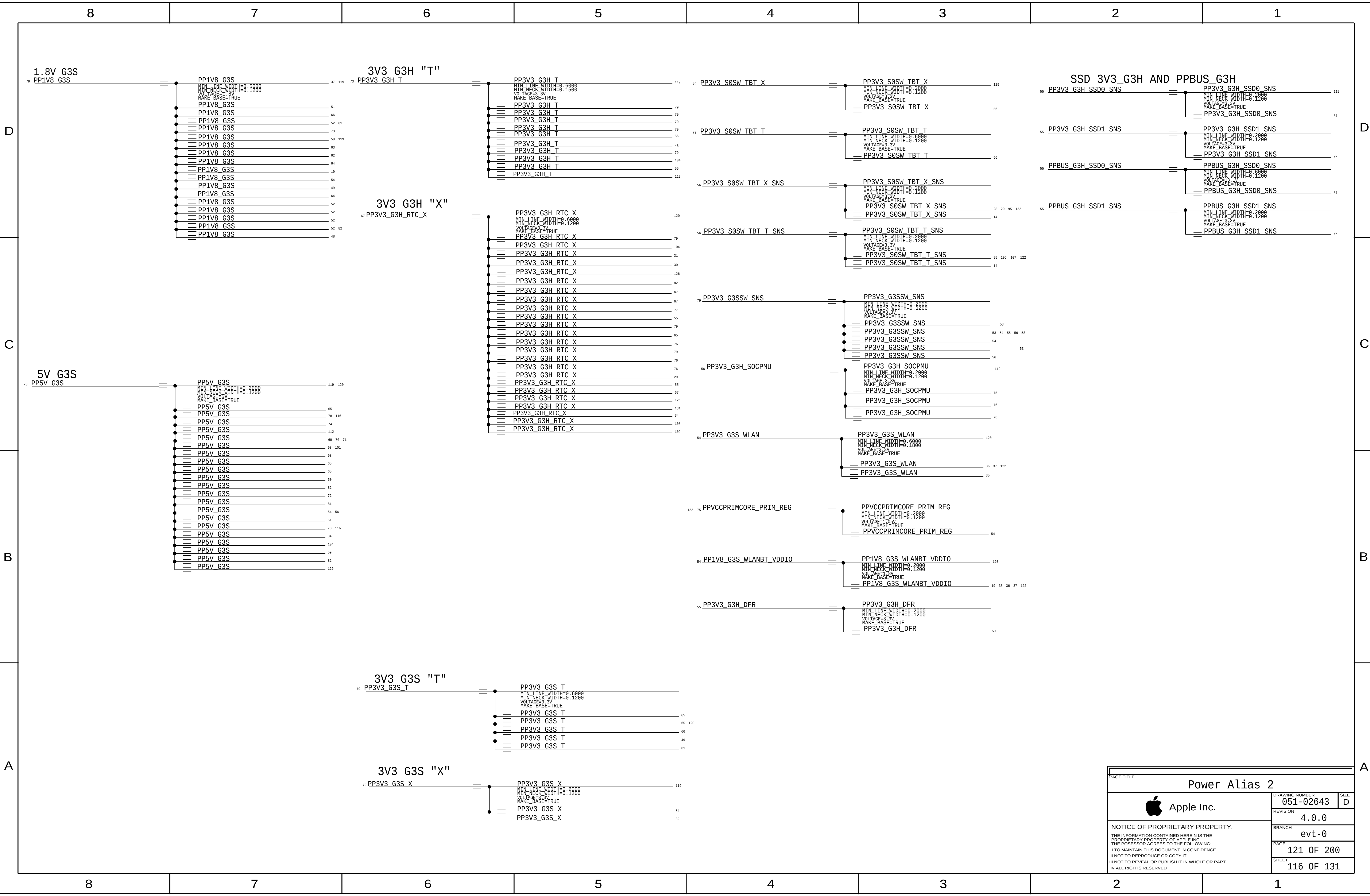
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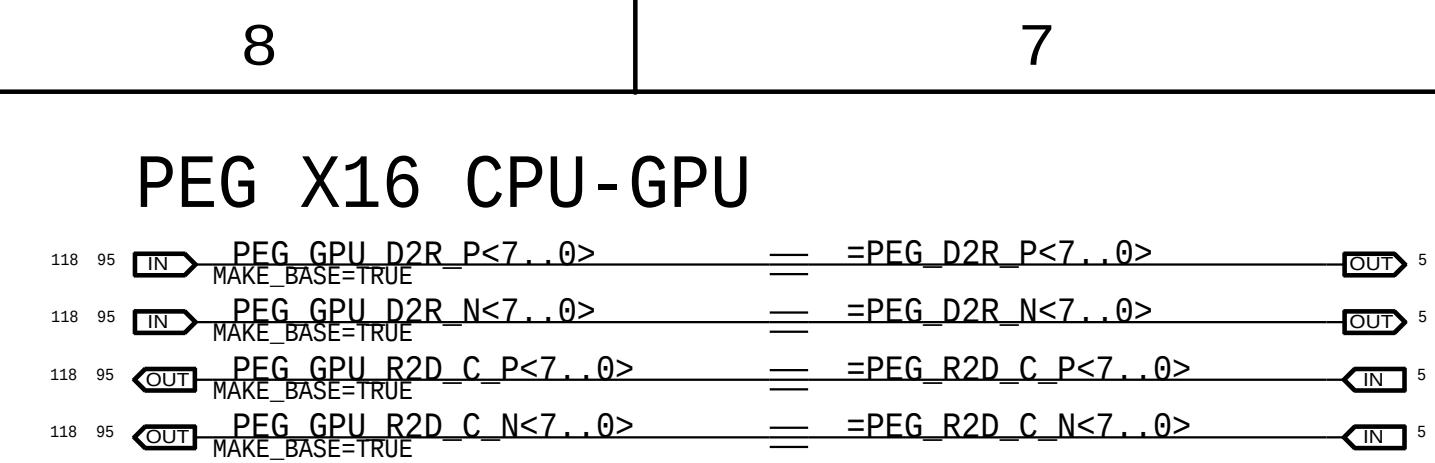


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Power Alias 1		
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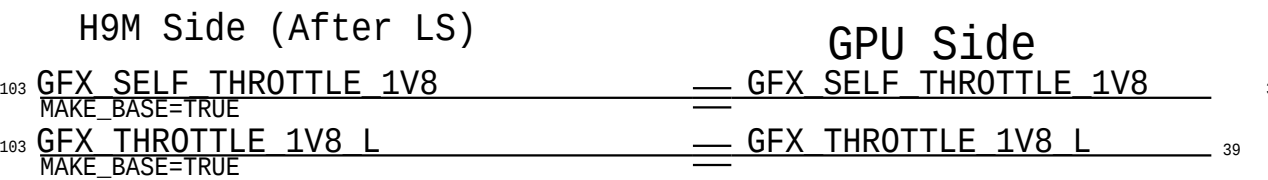
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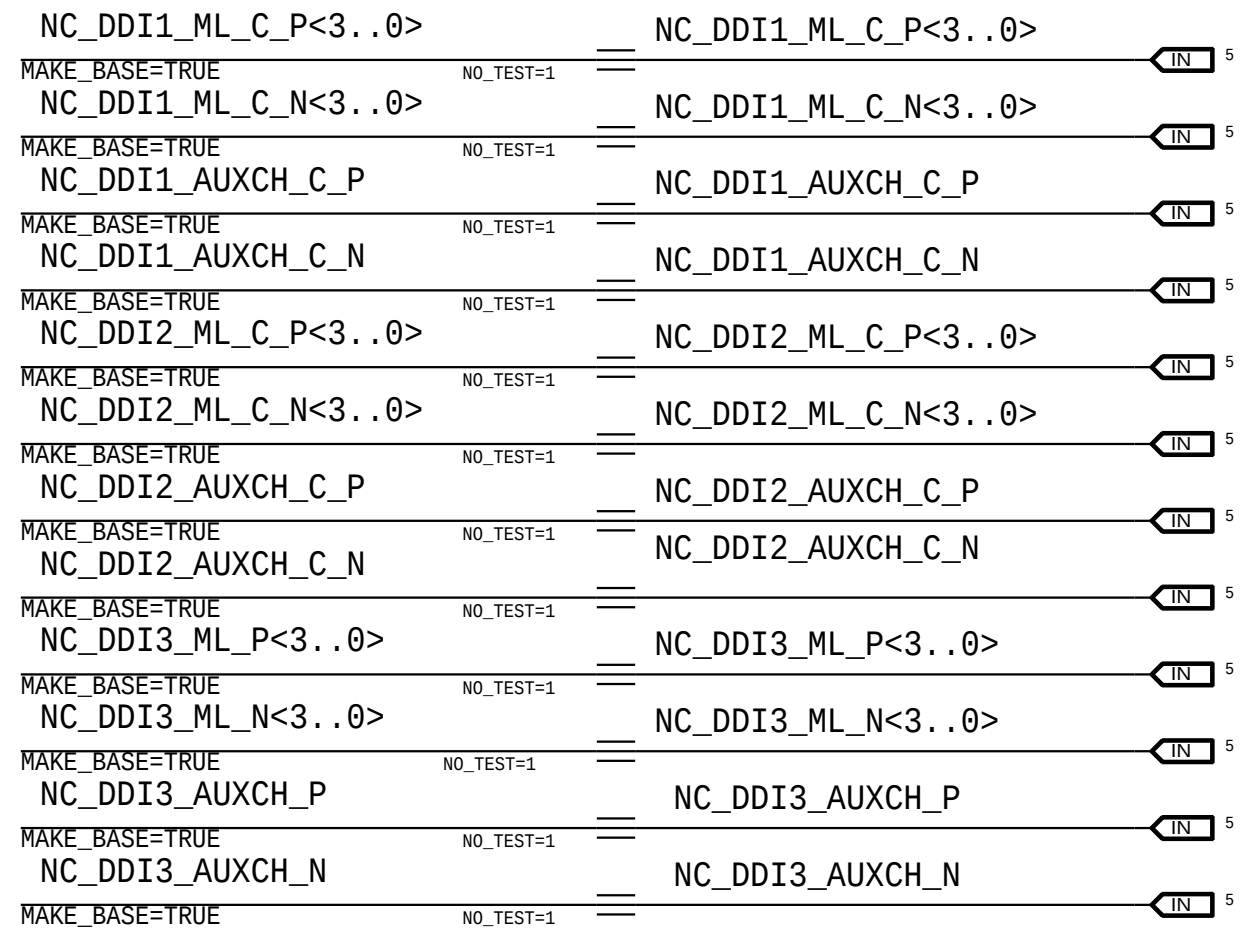




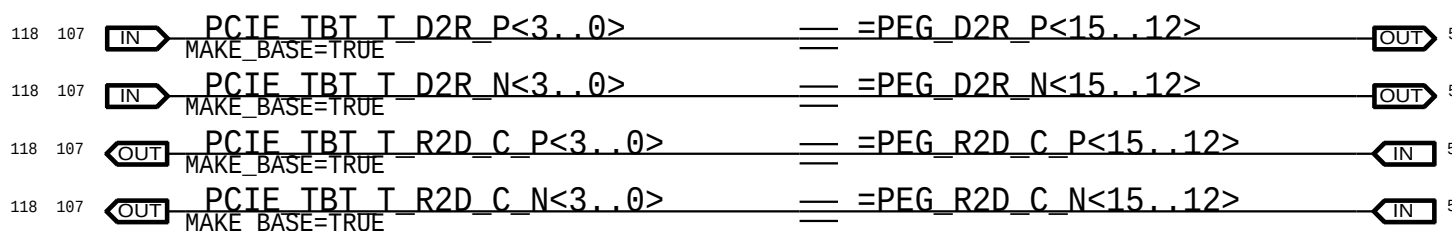
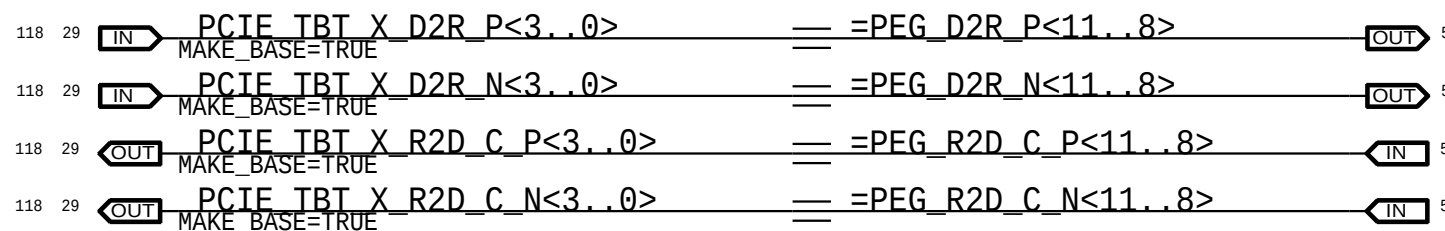
GPU Throttle



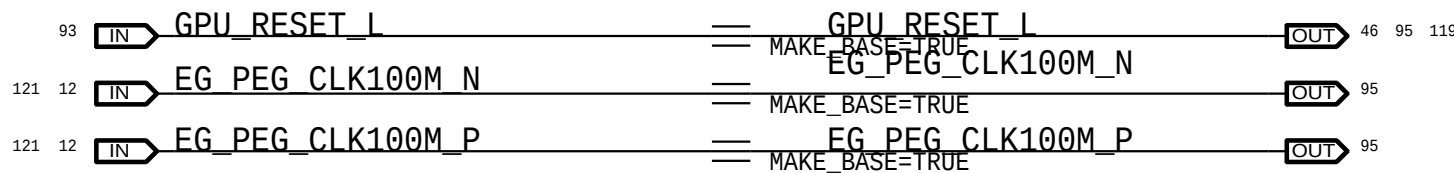
CPU Display Aliases



Thunderbolt Signals Through PEG



GPU ALIAS



ICT TESTPOINTS , High Speed NO_TEST

PEG

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117	29	IN	PCIE_TBT_X_R2D_C_N<3...0>	NO_TEST=1
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107	105	IN	PCIE_TBT_T_R2D_P<3...0>	NO_TEST=1
107	105	IN	PCIE_TBT_T_R2D_N<3...0>	NO_TEST=1
117	107	IN	PCIE_TBT_T_R2D_C_P<3...0>	NO_TEST=1
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BAFFIN FRAME BUFFER

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118	100	97	FB_B0_WE_I	NO_TEST=1
100	97	IN	FB_B0_CLK_P	NO_TEST=1
100	97	IN	FB_B1_CLK_N	NO_TEST=1
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100	97	IN	FB_B0_CAS_I	NO_TEST=1
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99	97	IN	FB_A0_ABT_I	NO_TEST=1
99	97	IN	FB_A1_ABT_I	NO_TEST=1
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99	97	IN	FB_A0_RAS_I	NO_TEST=1
99	97	IN	FB_A1_RAS_I	NO_TEST=1
99	97	IN	FB_A0_CAS_I	NO_TEST=1
99	97	IN	FB_A1_CAS_I	NO_TEST=1
99	97	IN	FB_A0_CS_I	NO_TEST=1
100	97	IN	FB_B0_CLK_N	NO_TEST=1
100	97	IN	FB_B0_DQ<31...0>	NO_TEST=1
100	97	IN	FB_B1_DQ<31...0>	NO_TEST=1
100	97	IN	FB_B0_A<8...0>	NO_TEST=1
100	97	IN	FB_B1_A<8...0>	NO_TEST=1
100	97	IN	FB_B0_WCLK_N<1...0>	NO_TEST=1
100	97	IN	FB_B0_WCLK_P<1...0>	NO_TEST=1
100	97	IN	FB_B1_WCLK_N<1...0>	NO_TEST=1
100	97	IN	FB_B1_WCLK_P<1...0>	NO_TEST=1
100	97	IN	FB_B0_EDC<3...0>	NO_TEST=1
100	97	IN	FB_B1_EDC<3...0>	NO_TEST=1
100	97	IN	FB_B0_DBI_I<3...0>	NO_TEST=1
100	97	IN	FB_B1_DBI_I<3...0>	NO_TEST=1
100	97	IN	FB_B0_ABT_I	NO_TEST=1
100	97	IN	FB_B1_ABT_I	NO_TEST=1

CPU/PCH CLK

12	6	IN	CPU_CLK24M_NSSC_CLK_N	NO_TEST=1
12	6	IN	CPU_CLK24M_NSSC_CLK_P	NO_TEST=1
12	6	IN	CPU_CLK100M_PCTBCLK_N	NO_TEST=1
12	6	IN	CPU_CLK100M_PCTBCLK_P	NO_TEST=1
12	6	IN	CPU_CLK100M_BCLK_N	NO_TEST=1
12	6	IN	CPU_CLK100M_BCLK_P	NO_TEST=1

MUX

93	5	IN	DP_INT_IG_MI_N<3...0>	NO_TEST=1
93	5	IN	DP_INT_IG_MI_P<3...0>	NO_TEST=1
93	5	IN	DP_INT_IG_AUX_N	NO_TEST=1
93	5	IN	DP_INT_IG_AUX_P	NO_TEST=1
103	93	IN	DP_INT_EG_MI_N<3...0>	NO_TEST=1
103	93	IN	DP_INT_EG_MI_P<3...0>	NO_TEST=1
103	93	IN	DP_INT_EG_AUX_N	NO_TEST=1
103	93	IN	DP_INT_EG_AUX_P	NO_TEST=1

USB-C X

32	27	IN	USBC_XB_D2R_N<2...1>	NO_TEST=1
32	27	IN	USBC_XB_D2R_P<2...1>	NO_TEST=1
32	27	IN	USBC_XA_D2R_P<2...1>	NO_TEST=1
32	27	IN	USBC_XB_R2D_C_P<2...1>	NO_TEST=1

32	27	IN	USBC_XA_D2R_N<2...1>	NO_TEST=1
32	27	IN	USBC_XB_R2D_C_N<2...1>	NO_TEST=1
32	27	IN	USBC_XA_R2D_C_P<2...1>	NO_TEST=1

32	27	IN	USBC_XA_R2D_C_N<2...1>	NO_TEST=1
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DMI

13	5	IN	DMI_S2N_P<3...0>	NO_TEST=1
13	5	IN	DMI_S2N_N<3...0>	NO_TEST=1
13	5	IN	DMI_N2S_P<3...0>	NO_TEST=1
13	5	IN	DMI_N2S_N<3...0>	NO_TEST=1

DP - CPU/ACE

103	27	IN	DP_X_SNK0_ML_C_N<3...0>	NO_TEST=1
103	27	IN	DP_X_SNK0_ML_C_P<3...0>	NO_TEST=1
27	27	IN	DP_X_SNK0_ML_N<3...0>	NO_TEST=1
27	27	IN	DP_X_SNK0_ML_P<3...0>	NO_TEST=1
103	27	IN	DP_X_SNK1_ML_C_N<3...0>	NO_TEST=1
103	27	IN	DP_X_SNK1_ML_C_P<3...0>	NO_TEST=1
27	27	IN	DP_X_SNK1_ML_N<3...0>	NO_TEST=1
27	27	IN	DP_X_SNK1_ML_P<3...0>	NO_TEST=1
103	103	IN	DP_T_SNK0_ML_C_N<3...0>	NO_TEST=1
103	103	IN	DP_T_SNK0_ML_C_P<3...0>	NO_TEST=1
103	103	IN	DP_T_SNK0_ML_N<3...0>	NO_TEST=1
103	103	IN	DP_T_SNK0_ML_P<3...0>	NO_TEST=1
103	103	IN	DP_T_SNK1_ML_C_N<3...0>	NO_TEST=1
103	103	IN	DP_T_SNK1_ML_C_P<3...0>	NO_TEST=1
103	103	IN	DP_T_SNK1_ML_N<3...0>	NO_TEST=1
103	103	IN	DP_T_SNK1_ML_P<3...0>	NO_TEST=1
103	27	IN	DP_X_SNK0_AUXCH_C_P	NO_TEST=1
103	27	IN	DP_X_SNK0_AUXCH_C_N	NO_TEST=1
103	27	IN	DP_T_SNK0_AUXCH_C_P	NO_TEST=1
103	103	IN	DP_T_SNK0_AUXCH_C_N	NO_TEST=1
27	27	IN	DP_X_SNK0_AUXCH_P	NO_TEST=1
27	27	IN	DP_X_SNK0_AUXCH_N	NO_TEST=1
27	27	IN	DP_T_SNK0_AUXCH_P	NO_TEST=1
27	27	IN	DP_T_SNK0_AUXCH_N	NO_TEST=1
103	27	IN	DP_X_SNK1_AUXCH_C_P	NO_TEST=1
103	27	IN	DP_X_SNK1_AUXCH_C_N	NO_TEST=1
103	27	IN	DP_T_SNK1_AUXCH_C_P	NO_TEST=1
103	27	IN	DP_T_SNK1_AUXCH_C_N	NO_TEST=1
27	27	IN	DP_X_SNK1_AUXCH_P	NO_TEST=1
27	27	IN	DP_X_SNK1_AUXCH_N	NO_TEST=1
103	103	IN	DP_T_SNK1_AUXCH_P	NO_TEST=1
103	103	IN	DP_T_SNK1_AUXCH_N	NO_TEST=1

CPU/EDP

103	27	IN	DP_X_SNK1_AUXCH_C_P	NO_TEST=1
103	27	IN	DP_X_SNK1_AUXCH_C_N	NO_TEST=1
103	27	IN	DP_T_SNK1_AUXCH_C_P	NO_TEST=1
103	27	IN	DP_T_SNK1_AUXCH_C_N	NO_TEST=1
27	27	IN	DP_X_SNK1_AUXCH_P	NO_TEST=1
27	27	IN	DP_X_SNK1_AUXCH_N	NO_TEST=1
103	103	IN	DP_T_SNK1_AUXCH_P	NO_TEST=1
103	103	IN	DP_T_SNK1_AUXCH_N	NO_TEST=1

H9M ESPI

39	20	12	IN	ESPI_I0<0>	NO_TEST=1	
39	20	12	IN	ESPI_I0<1>	NO_TEST=1	86
39	20	12	IN	ESPI_I0<2>	NO_TEST=1	86
39	20	12	IN	ESPI_I0<3>	NO_TEST=1	86
39	39	IN	ESPI_I0_R<0>	NO_TEST=1		
39	39	IN	ESPI_I0_R<1>	NO_TEST=1		86
39	39	IN	ESPI_I0_R<3>	NO_TEST=1		
118	39	IN		NO_TEST=1		
118	39	IN	ESPI_I0_R<3>	NO_TEST=1		

118	29	27	IN	TBT_X_XTAL25M_OUT	NO_TEST=1
29	27		IN	TBT_X_XTAL25M_IN	NO_TEST=1
107	105		IN	TBT_T_XTAL25M_OUT	NO_TEST=1

107	105	IN	TBT_T_XTAL25M_IN	NO_TEST=1
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MEM

26	23	22	7	IN	MEM_A_CLK_P<1...0>	NO_TEST=1
26	23	22	7	IN	MEM_A_CLK_N<1...0>	NO_TEST=1
125	23	22	IN	MEM_A_DQ<63...0>	NO_TEST=1	
125	25	24	IN	MEM_B_DQ<63...0>	NO_TEST=1	
125	23	22	IN	MEM_A_DQS_P<7...0>	NO_TEST=1	
125	23	22	IN	MEM_A_DQS_N<7...0>	NO_TEST=1	
125	25	24	IN	MEM_B_DQS_P<7...0>	NO_TEST=1	
125	25	24	IN	MEM_B_DQS_N<7...0>	NO_TEST=1	

26	25	24	7	IN	MEM_B_CLK_P<1...0>	NO_TEST=1
26	25	24	7	IN	MEM_B_CLK_N<1...0>	NO_TEST=1

WIFI

37	IN	50_0_ANT	NO_TEST=1
37	IN	50_1_ANT	NO_TEST=1
37	IN	50_2_ANT	NO_TEST=1
37	IN	50_0_COM	NO_TEST=1
37	IN	50_1_COM	NO_TEST=1
37	IN	50_2_COM	NO_TEST=1
37	IN	50_A_0_DIPLEXER	NO_TEST=1
37	IN	50_A_0_MATCH	NO_TEST=1
37	IN	50_G_0_DIPLEXER	NO_TEST=1
37	IN	50_G_0_MATCH	NO_TEST=1
37	IN	50_A_1_DIPLEXER	NO_TEST=1
37	IN	50_A_1_MATCH	NO_TEST=1
37	IN	50_G_1_DIPLEXER	NO_TEST=1
37	IN	50_G_1_MATCH	NO_TEST=1
37	IN	50_A_2_DIPLEXER	NO_TEST=1
37	IN	50_A_2_MATCH	NO_TEST=1
37	IN	50_G_2_DIPLEXER	NO_TEST=1
37	IN	50_G_2_MATCH	NO_TEST=1

USB-C T

110	105	IN	USBC_TA_D2R_P<2...1>	NO_TEST=1
110	105	IN	USBC_TA_D2R_N<2...1>	NO_TEST=1
110	105	IN	USBC_TA_R2D_C_P<2...1>	NO_TEST=1
110	105	IN	USBC_TA_R2D_C_N<2...1>	NO_TEST=1

110	105	IN	USBC_TB_D2R_P<2...1>	NO_TEST=1
110	105	IN	USBC_TB_D2R_N<2...1>	NO_TEST=1
110	105	IN	USBC_TB_R2D_C_P<2...1>	NO_TEST=1
110	105	IN	USBC_TB_R2D_C_N<2...1>	NO_TEST=1

H9M SSD NAND

86	85	84	83	IN	PCIE SSD0 R2D P<3...0>	NO_TEST=1	
86	85	84	83	IN	PCIE SSD0 R2D N<3...0>	NO_TEST=1	
86	85	84	83	41	IN	PCIE SSD0 D2R P<3...0>	NO_TEST=1
86	85	84	83	41	IN	PCIE SSD0 D2R N<3...0>	NO_TEST=1
86	85	84	83	IN	PCIE SSD0 R2D C P<3...0>	NO_TEST=1	
86	85	84	83	41	IN	PCIE SSD0 R2D C N<3...0>	NO_TEST=1
86	85	84	83	IN	PCIE SSD0 D2R C P<3...0>	NO_TEST=1	
86	85	84	83	41	IN	PCIE SSD0 D2R C N<3...0>	NO_TEST=1

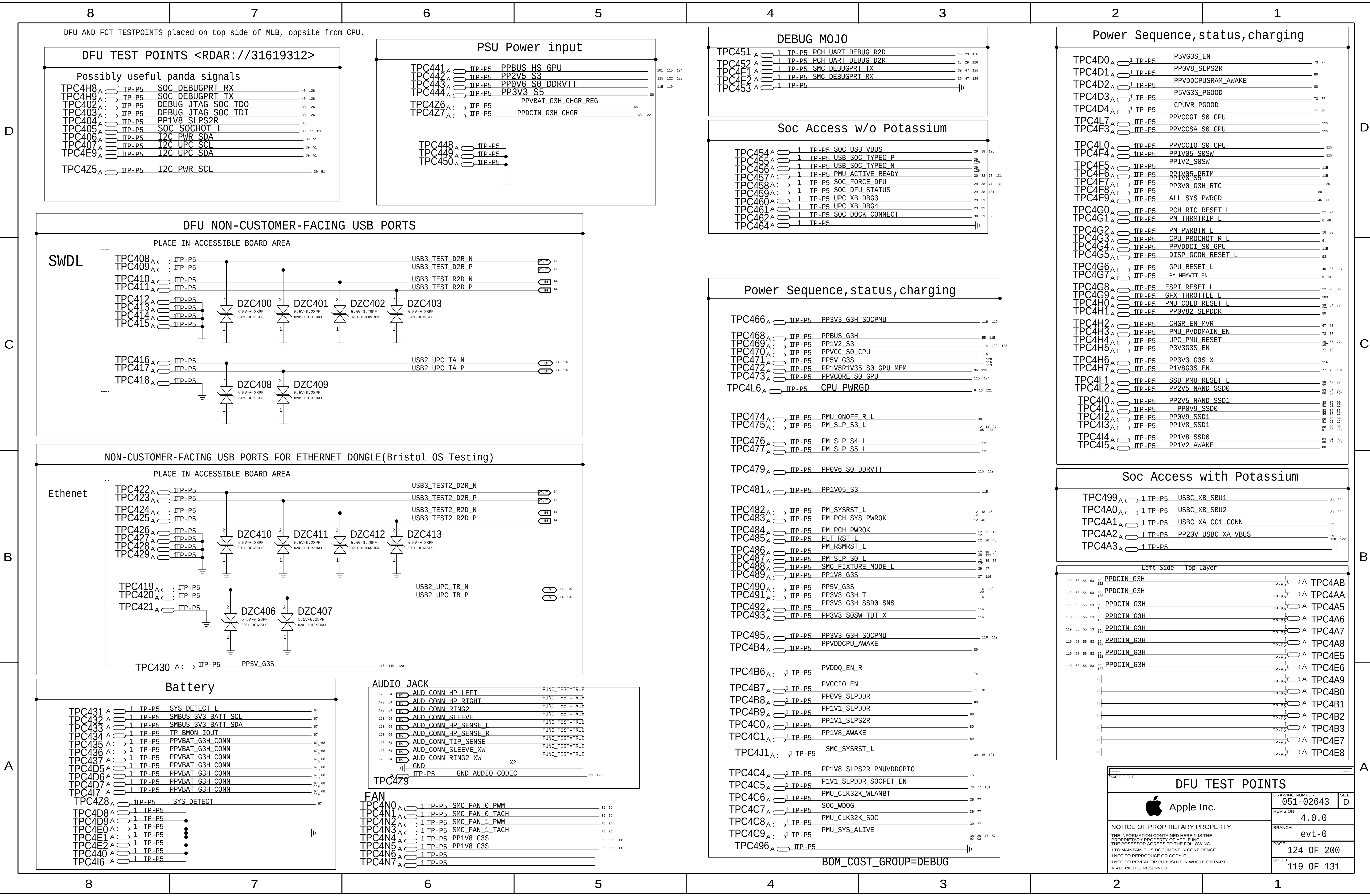
91	90	89	88	IN	PCIE_SSD1_R2D_P<3...0>	NO_TEST=1	
91	90	89	88	IN	PCIE_SSD1_R2D_N<3...0>	NO_TEST=1	
91	90	89	88	41	IN	PCIE_SSD1_D2R_P<3...0>	NO_TEST=1
91	90	89	88	41	IN	PCIE_SSD1_D2R_N<3...0>	NO_TEST=1
91	90	89	88	41	IN	PCIE_SSD1_R2D_C_P<3...0>	NO_TEST=1
91	90	89	88	41	IN	PCIE_SSD1_R2D_C_N<3...0>	NO_TEST=1
91	90	89	88	41	IN	PCIE_SSD1_D2R_C_P<3...0>	NO_TEST=1
91	90	89	88	41	IN	PCIE_SSD1_D2R_C_N<3...0>	NO_TEST=1

XTAL

118	29	27	IN	TBT_X_XTAL25M_OUT	NO_TEST=1
12			IN	PCH_CLK24M_XTALOUT	NO_TEST=1
77			IN	PMU_XTAL1	NO_TEST=1
77			IN	PMU_XTAL2	NO_TEST=1

PCH/AR

36	14	IN	PCH PCIE WLAN D2R N	NO_TEST=1
36	14	IN	PCH PCIE WLAN D2R P	NO_TEST=1
36	14	IN	PCH PCIE WLAN R2D C N	NO_TEST=1
36	14	IN	PCH PCIE WLAN R2D C P	NO_TEST=1
36		IN	PCH PCIE WLAN D2R C P	NO_TEST=1
36		IN	PCH PCIE WLAN D2R C N	NO_TEST=1
36		IN	PCH PCIE WLAN R2D P	NO_TEST=1
36		IN	PCH PCIE WLAN R2D N	NO_TEST=1



FCT TEST POINTS (TOP SIDE OF MLB)

8

7

6

5

4

3

2

1

DFR/MESA (FCT)

TPC500	A	1 TP-P5		
TPC501	A	1 TP-P5		
TPC502	A	1 TP-P5	MIPI DFR CLK CONN FILT P	59
TPC503	A	1 TP-P5	DFR DISP VSYNC	59
TPC505	A	1 TP-P5	DFR DISP TE	49 59
TPC504	A	1 TP-P5	MIPI DFR CLK CONN FILT N	59
TPC506	A	1 TP-P5		
TPC507	A	1 TP-P5	MIPI DFR DATA CONN FILT P	59
TPC524	A	1 TP-P5	MIPI DFR DATA CONN FILT N	59
TPC510	A	1 TP-P5	DFR DISP RESET L	49 59
TPC516	A	1 TP-P5	PP3V3 G3HSW DFR	59 129
TPC508	A	1 TP-P5	DFR DISP INT	39 59
TPC518	A	1 TP-P5	PP3V3 G3HSW DFR	59 129
TPC511	A	1 TP-P5		
TPC512	A	1 TP-P5		
TPC521	A	1 TP-P5	TP DFR TOUCH PANEL DETECT	59
TPC522	A	1 TP-P5	DFR LID OPEN L	59
TPC515	A	1 TP-P5		
TPC520	A	1 TP-P5		
TPC527	A	1 TP-P5	SPI DFR MISO R	47 59
TPC528	A	1 TP-P5	SPI DFR CS L	49 59
TPC533	A	1 TP-P5	DFR TOUCH INT L	39 59
TPC535	A	1 TP-P5	DFR TOUCH CLK32K RESET L	49 59
TPC534	A	1 TP-P5	I2C DFR SCL R	59 52
TPC536	A	1 TP-P5	I2C DFR SDA R	59 52
TPC538	A	1 TP-P5	DFR TOUCH RESET L	49 59
TPC530	A	1 TP-P5	SPI DFR MOSI	47 59
TPC525	A	1 TP-P5		
TPC526	A	1 TP-P5		
TPC529	A	1 TP-P5	SPI DFR CLK	47 59
TPC539	A	1 TP-P5	TP DFR TOUCH ROM WC	59
TPC540	A	1 TP-P5	PP1V8 SLPS2RSW DFR	59 52 129
TPC542	A	1 TP-P5	PP5V G3S DFR FILT	59
TPC541	A	1 TP-P5	PP1V8 SLPS2RSW DFR	59 52 129
TPC544	A	1 TP-P5	SPI MESA MISO CONN	48
TPC543	A	1 TP-P5	PP1V8 MESA CONN	48
TPC550	A	1 TP-P5	SPI MESA CLK CONN	48
TPC531	A	1 TP-P5		
TPC532	A	1 TP-P5		
TPC546	A	1 TP-P5	MESA INT CONN	48
TPC547	A	1 TP-P5	SPI MESA MOSI CONN	48
TPC548	A	1 TP-P5	MESA BOOST EN CONN	48
TPC552	A	1 TP-P5	PP16V0 MESA CONN	48
TPC537	A	1 TP-P5		
TPC554	A	1 TP-P5	PP3V0 MESA CONN	48 129
TPC554	A	1 TP-P5	PP3V0 MESA CONN	48 129
TPC555	A	1 TP-P5		
TPC545	A	1 TP-P5		
TPC553	A	1 TP-P5		

TBT/DP

TPC5L7	A	1 TP-P5	TBT X HDMI DDC DATA	29 95
TPC5L8	A	1 TP-P5	TBT X HDMI DDC CLK	29 95
TPC5L9	A	1 TP-P5	TBT T HDMI DDC DATA	95 197
TPC5N0	A	1 TP-P5	TBT T HDMI DDC CLK	95 197

FAN , Keyboard,Trackpad Test Points

TPC556	A	1 TP-P5	GND_FAN	65 129
TPC557	A	1 TP-P5	GND_FAN	65 129
TPC558	A	1 TP-P5	PP5V G3S_FAN CONN	65 129
TPC559	A	1 TP-P5	PP5V G3S_FAN CONN	65 129
TPC560	A	1 TP-P5	FAN LT TACH	59 65
TPC561	A	1 TP-P5	FAN RT TACH	59 65
TPC562	A	1 TP-P5	FAN LT PWM	59 65
TPC563	A	1 TP-P5	FAN RT PWM	59 65
TPC564	A	1 TP-P5		
TPC565	A	1 TP-P5		
TPC5Z1	A	1 TP-P5	GND_FAN	65 129
TPC5Z2	A	1 TP-P5	GND_FAN	65 129
TPC5Z6	A	1 TP-P5	I2C_TPAD3V3_SCL	65 66
TPC5Y0	A	1 TP-P5	TPAD_KBD_WAKE_L	47 65 66
TPC566	A	1 TP-P5	I2C_TPAD3V3_SDA	65 66
TPC567	A	1 TP-P5	TPAD3V3_ACTUATOR_DISABLE_L	65 66
TPC568	A	1 TP-P5	I2C_KBD_SDA	65
TPC569	A	1 TP-P5	TPAD3V3_SPI_INT_L	65 66
TPC5Z8	A	1 TP-P5	IPD_LID_OPEN	47 59 65
TPC5Z3	A	1 TP-P5	PP5V_G3S	119 119
TPC5Z4	A	1 TP-P5	PP5V_G3S	119 119
TPC5Z7	A	1 TP-P5	PMU_RSLOC_RST_R_L	65
TPC5Z5	A	1 TP-P5	PP3V3_G3H_RTC_X	116
TPC571	A	1 TP-P5		
TPC588	A	1 TP-P5	ACT_GND	65 129
TPC586	A	1 TP-P5	ACT_GND	65 129
TPC579	A	1 TP-P5	KBD_BLC_GSSOUT	65
TPC578	A	1 TP-P5	TPAD3V3_SPI_EN	65 66
TPC585	A	1 TP-P5	PP3V3_G3S_T	65 116 129
TPC584	A	1 TP-P5	PP3V3_G3S_T	65 116 129
TPC583	A	1 TP-P5	KBD_BLC_GSLAT	65
TPC582	A	1 TP-P5	PP5V_G3S_TPAD_CONN	65
TPC581	A	1 TP-P5	KBD_BLC_GSSCK	65
TPC580	A	1 TP-P5	SPI_TPAD3V3_CLK	65 66
TPC591	A	1 TP-P5	PPVIN_G3H_TPAD_FUSE	65 129
TPC590	A	1 TP-P5	PPVIN_G3H_TPAD_FUSE	65 129
TPC589	A	1 TP-P5	PPVIN_G3H_TPAD_FUSE	65 129
TPC577	A	1 TP-P5	KBD_BLC_GSSIN	65
TPC576	A	1 TP-P5	SPI_TPAD3V3_MISO	65 66
TPC575	A	1 TP-P5	KBD_BLC_XBLANK	65
TPC574	A	1 TP-P5	SPI_TPAD3V3_CS_L	65 66
TPC572	A	1 TP-P5	SPI_TPAD3V3_MOSI	65 66
TPC573	A	1 TP-P5	I2C_KBD_SCL	65
TPC587	A	1 TP-P5		
TPC5M0	A	1 TP-P5	KBD_INT_L	65

WIRELESS

TPC5H9	A	1 TP-P5	PP3V3_G3S_WLAN	116
TPC5I0	A	1 TP-P5	PP1V8_G3S_WLANBT_VDDIO	116
TPC5L0	A	1 TP-P5	PPVIN_RFLD0_WLANBT	39 122
TPC5L1	A	1 TP-P5	PP1V2_WLANBT	39 122
TPC5L2	A	1 TP-P5	PP1V5_WLANBT	39 122

HALL EFFECT

TPC5I1	A	1 TP-P5	LID_OPEN_LEFT	47 64
TPC5I2	A	1 TP-P5	LID_OPEN_RIGHT	47 64

MEMORY

TPC5I3	A	1 TP-P5	PVDDQ_PG00D	74 77
TPC5I7	A	1 TP-P5	PP1V2_S3_CPUDDR	115

AUDIO

TPC592	A	1 TP-P5	AUD_DMIC0_CLK_CONN	64
TPC593	A	1 TP-P5	AUD_DMIC0_DATA_CONN	64
TPC594	A	1 TP-P5	PP1V8_DMIC	64
TPC595	A	1 TP-P5	AUD_DMIC1_CLK_CONN	64
TPC596	A	1 TP-P5	AUD_DMIC1_DATA_CONN	64
TPC597	A	1 TP-P5		
TPC5A1	A	1 TP-P5	SPKRCONN_LW_OUTN	62 129
TPC5B1	A	1 TP-P5	SPKRCONN_LW_OUTN	62 129
TPC5B4	A	1 TP-P5	SPKRCONN_LW_OUTN	62 129
TPC5A0	A	1 TP-P5	SPKRCONN_LW_OUTP	62 129
TPC5G0	A	1 TP-P5	SPKRCONN_LW_OUTP	62 129
TPC5G1	A	1 TP-P5	SPKRCONN_LW_OUTP	62 129
TPC598	A	1 TP-P5	SPKRCONN_LT_OUTN	62 129
TPC5C2	A	1 TP-P5	SPKRCONN_LT_OUTN	62 129
TPC5M3	A	1 TP-P5	SPKRCONN_LT_OUTN	62 129
TPC599	A	1 TP-P5	SPKRCONN_LT_OUTP	62 129
TPC5M4	A	1 TP-P5	SPKRCONN_LT_OUTP	62 129
TPC5M5	A	1 TP-P5	SPKRCONN_LT_OUTP	62 129
TPC5A6	A	1 TP-P5	SPKRCONN_RW_OUTP	63 129
TPC5G6	A	1 TP-P5	SPKRCONN_RW_OUTP	63 129
TPC5M7	A	1 TP-P5	SPKRCONN_RW_OUTP	63 129
TPC5Y9	A	1 TP-P5	SPKRCONN_RW_OUTN	63 129
TPC5C8	A	1 TP-P5	SPKRCONN_RW_OUTN	63 129
TPC5M9	A	1 TP-P5	SPKRCONN_RW_OUTN	63 129
TPC5A3	A	1 TP-P5	SPKRCONN_RT_OUTN	63 129
TPC5L3	A	1 TP-P5	SPKRCONN_RT_OUTN	63 129
TPC5L4	A	1 TP-P5	SPKRCONN_RT_OUTN	63 129
TPC5A4	A	1 TP-P5	SPKRCONN_RT_OUTP	63 129
TPC5L5	A	1 TP-P5	SPKRCONN_RT_OUTP	63 129
TPC5L6	A	1 TP-P5	SPKRCONN_RT_OUTP	63 129
TPC5B8	A	1 TP-P5	AUD_CONN_RING_SENSE	64
TPC5B7	A	1 TP-P5	AUD_CONN_RING2_XW	64 119
TPC5B6	A	1 TP-P5	AUD_CONN_TIP_SENSE	64 119
TPC5B5	A	1 TP-P5	AUD_CONN_HP_SENSE_R	64 119
TPC5A7	A	1 TP-P5		
TPC5B3	A	1 TP-P5	AUD_CONN_HP_SENSE_L	64 119
TPC5A8	A	1 TP-P5	AUD_CONN_HP_LEFT	64 119
TPC5A9	A	1 TP-P5	AUD_CONN_HP_RIGHT	64 119
TPC5B0	A	1 TP-P5	AUD_CONN_RING2	64 119
TPC5B2	A	1 TP-P5	AUD_CONN_SLEEVE	64 119
TPC5K0	A	1 TP-P5		
TPC5K1	A	1 TP-P5	AUD_CONN_SLEEVE_XW	64 119
TPC5A5	A	1 TP-P5	SPKR_ID0	49 63
TPC5A2	A	1 TP-P5		

EDP , CAMERA,ALS

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TPC5C1	A	1 TP-P5		
TPC5C2	A	1 TP-P5	EDP_INT_AUX_P	62 93
TPC5C3	A	1 TP-P5	EDP_PANEL_PWR_BUF_EN	62
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TPC5C9	A	1 TP-P5	TP_LCD_IRQ_L	62
TPC5C8	A	1 TP-P5	EDP_INT_ML_P<0>	62 93
TPC5C5	A	1 TP-P5	DP_INT_HPD	62 93
TPC5C6	A	1 TP-P5	EDP_INT_ML_N<0>	62 93
TPC5C7	A	1 TP-P5	LCD_FSS	62 93
TPC5D5	A	1 TP-P5	EDP_INT_ML_P<2>	62 93
TPC5D3	A	1 TP-P5	EDP_INT_ML_N<2>	62 93
TPC5D1	A	1 TP-P5	EDP_INT_ML_N<1>	62 93
TPC5D2	A	1 TP-P5	EDP_INT_ML_P<1>	62 93
TPC5D0	A	1 TP-P5		
TPC5D7	A	1 TP-P5	I2C_BKLT_SDA	61 82
TPC5E0	A	1 TP-P5	EDP_INT_ML_P<3>	62 93
TPC5D9	A	1 TP-P5	I2C_BKLT_SCL	61 82
TPC5D8	A	1 TP-P5	EDP_INT_ML_N<3>	62 93
TPC5D4	A	1 TP-P5		
TPC5E1	A	1 TP-P5	I2C_TCON_SDA	51 82
TPC5E4	A	1 TP-P5	MIPI_FTCAM_DATA_CONN_N<0>	62
TPC5D6	A	1 TP-P5		
TPC5E3	A	1 TP-P5	I2C_TCON_SCL	51 82
TPC5E2	A	1 TP-P5	I2C_CAM_ISOL_SDA	62
TPC5F1	A	1 TP-P5	MIPI_FTCAM_CLK_CONN_P	62
TPC5F0	A	1 TP-P5	I2C_CAM_ISOL_SCL	62
TPC5E9	A	1 TP-P5	MIPI_FTCAM_CLK_CONN_N	62
TPC5E8	A	1 TP-P5	I2C_ALS_SCL	49 52
TPC5E2	A	1 TP-P5		
TPC5E6	A	1 TP-P5	MIPI_FTCAM_DATA_CONN_P<0>	62
TPC5E7	A	1 TP-P5	I2C_ALS_SDA	49 52
TPC5F7	A	1 TP-P5	PP5V_S0_ALSCAM_F	62
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TPC5F9	A	1 TP-P6	PPVIN_S0SW_LCDBKLT_R	61
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TPC5F4	A	1 TP-P5	PP3V3_S0SW_LCD	51 82 129
TPC5F6	A	1 TP-P5	PP5V_S0SW_LCD	62
TPC5F5	A	1 TP-P5	PP3V3_S0SW_LCD	51 82 129
TPC5F8	A	1 TP-P5		
TPC5M1	A	1 TP-P5	BKLT_PWM_MLB2TCON	62

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TPC5G3	A	1 TP-P5	TP_USBC_PP20V_XB	32
TPC5N3	A	1 TP-P5	TP_USBC_PP20V_XA	32
TPC5G4	A	1 TP-P5	USBC_XA_SBU1	39 32
TPC5G5	A	1 TP-P5	USBC_XA_SBU2	39 32
TPC5G7	A	1 TP-P5	USBC_XA_CC2_CONN	32 33
TPC5G9	A	1 TP-P5	PP20V_USBC_XB_VBUS	29 32 119 129 122
TPC5H0	A	1 TP-P5	PP20V_USBC_XA_VBUS	29 32 119 129 122
TPC5H1	A	1 TP-P5	TP_USBC_PP20V_TB	119
TPC5H2	A	1 TP-P5	TP_USBC_PP20V_TA	119
TPC5H3	A	1 TP-P5	PP20V_USBC_TB_VBUS	197 119 129 122
TPC5H4	A	1 TP-P5	PP20V_USBC_TA_VBUS	197 119 129 122
TPC5I8	A	1 TP-P5	PP20V_USBC_XA_VBUS	29 32 119 129 122
TPC5I9	A	1 TP-P5	PP20V_USBC_XA_VBUS	29 32 119 129 122
TPC5J1	A	1 TP-P5	PP20V_USBC_XB_VBUS	29 32 129 122
TPC5J0	A	1 TP-P5	PP20V_USBC_XA_VBUS	29 32 129 122
TPC5J2	A	1 TP-P5	PP20V_USBC_XB_VBUS	29 32 119 129 122
TPC5J3	A	1 TP-P5	PP20V_USBC_XB_VBUS	29 32 129 122
TPC5J4	A	1 TP-P5	PP20V_USBC_TA_VBUS	197 119 129 122
TPC5H5	A	1 TP-P5		
TPC5H6	A	1 TP-P5		
TPC5H7	A	1 TP-P5		
TPC5H8	A	1 TP-P5		
TPC5M2	A	1 TP-P5		
TPC5K2	A	1 TP-P5		
TPC5K3	A	1 TP-P5		
TPC5K4	A	1 TP-P5		
TPC5J5	A	1 TP-P5	PP20V_USBC_TA_VBUS	197 119 129 122
TPC5J6	A	1 TP-P5	PP20V_USBC_TA_VBUS	197 119 129 122
TPC5J7	A	1 TP-P5	PP20V_USBC_TB_VBUS	197 119 129 122
TPC5J8	A	1 TP-P5	PP20V_USBC_TB_VBUS	197 119 129 122
TPC5J9	A	1 TP-P5	PP20V_USBC_TB_VBUS	197 119 129 122
TPC5N9	A	1 TP-P5	USBC_XB_CC1_CONN	32 33
TPC5N8	A	1 TP-P5	USBC_XB_CC2_CONN	32 33
TPC5O1	A	1 TP-P5	USBC_TA_CC1_CONN	119 111
TPC5O2	A	1 TP-P5	USBC_TA_CC2_CONN	119 111
TPC5O3	A	1 TP-P5	USBC_TB_CC1_CONN	119 111
TPC5O4	A	1 TP-P5	USBC_TB_CC2_CONN	119 111

CORE OS DESBUG

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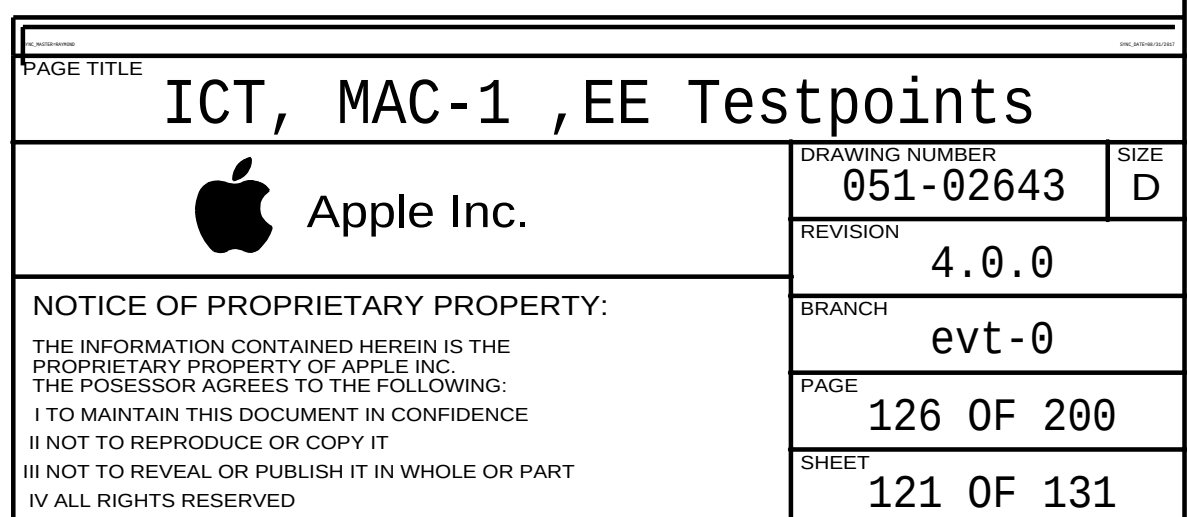
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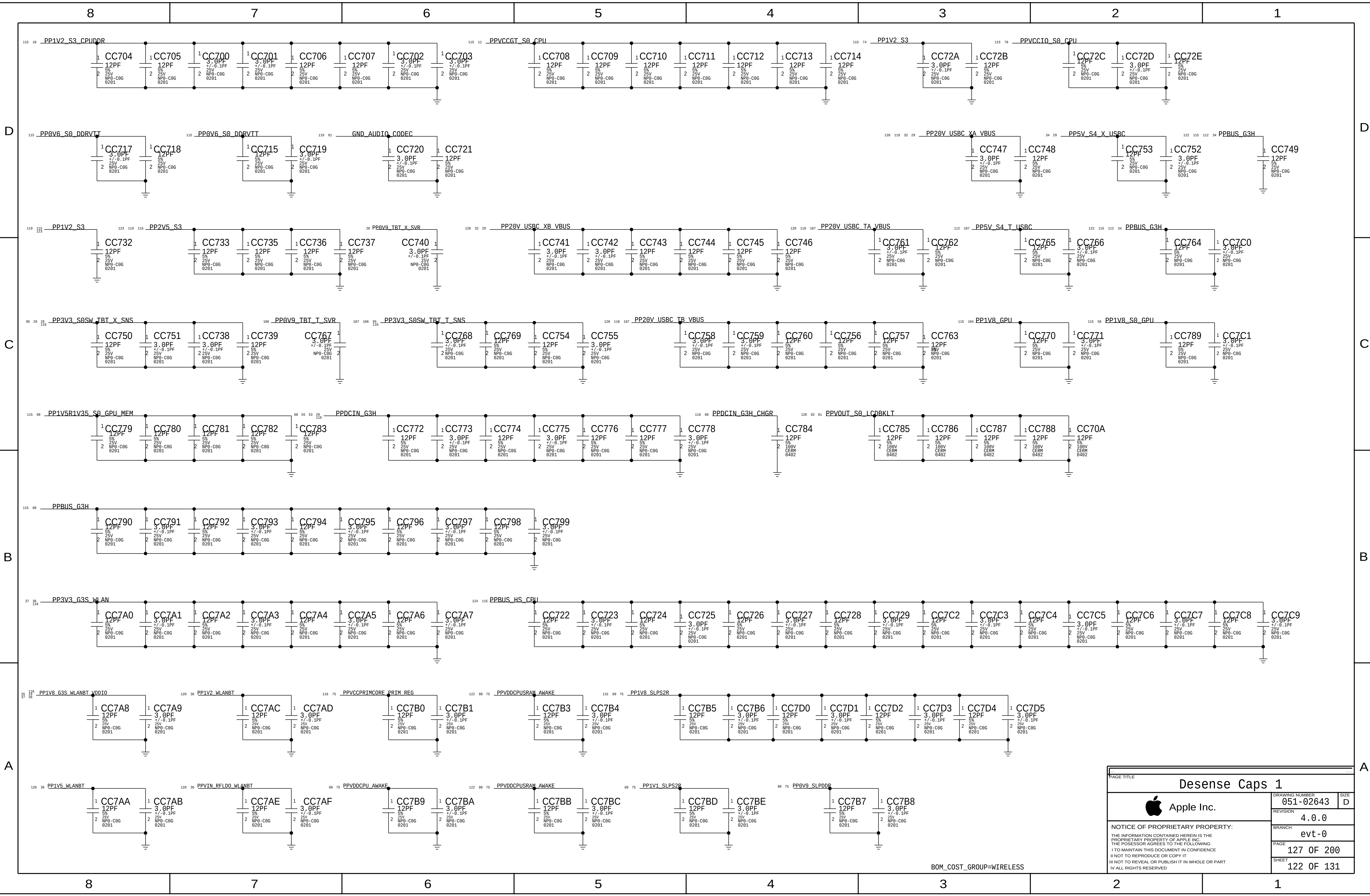
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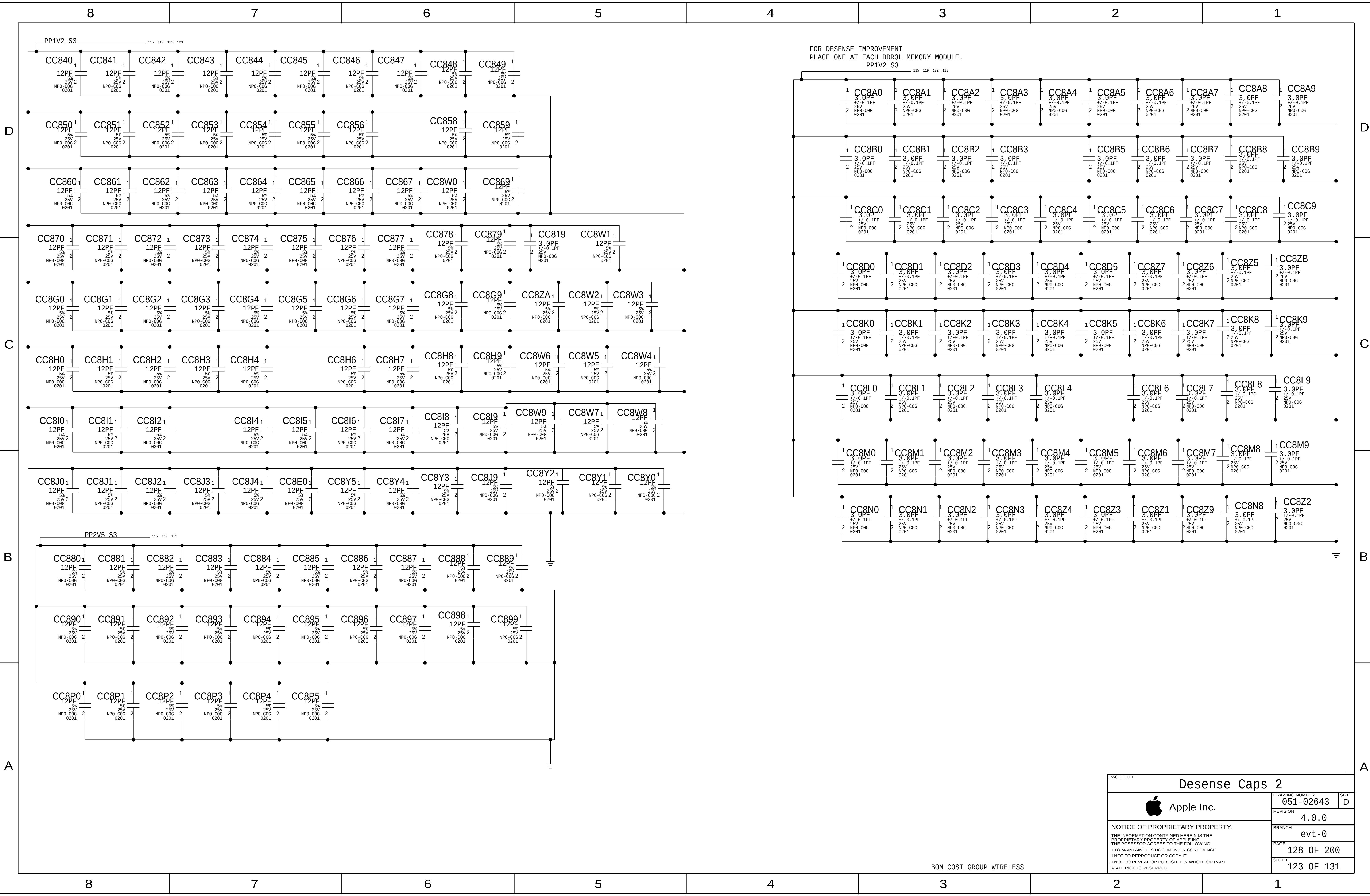
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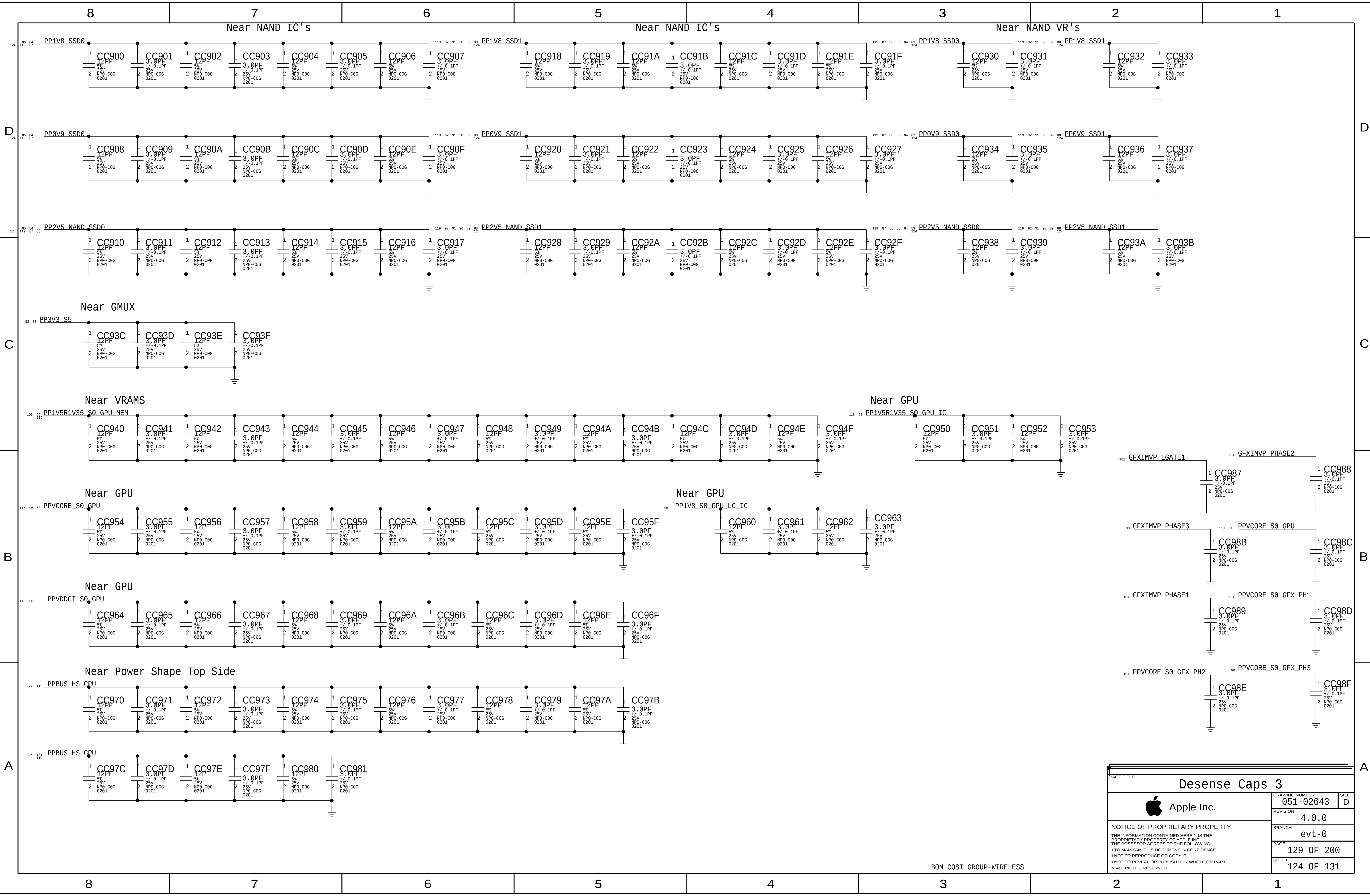
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FOR DESENSE IMPROVEMENT
PLACE ONE AT EACH DDR3L MEMORY MODULE.
PP1V2_S3

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2.6GHZ CPU, 16GB DRAM, ULX GPU, 512GB S4E		
EEEE	BOM NUMBER	BOM NAME
JP7D	639-05761	MLB, 2.6GHZ, MC-16, ULX, WD-512, X1190
JP7F	639-05762	MLB, 2.6GHZ, MC-16, ULX, TS-512, X1190
JP7G	639-05763	MLB, 2.6GHZ, HY-16, ULX, WD-512, X1190
JP7H	639-05764	MLB, 2.6GHZ, HY-16, ULX, TS-512, X1190

2.6GHZ CPU, 16GB DRAM, ULX GPU, 1TB S4E		
EEEE	BOM NUMBER	BOM NAME
JP7J	639-05765	MLB, 2.6GHZ, MC-16, ULX, WD-1TB, X1190
JP7K	639-05766	MLB, 2.6GHZ, MC-16, ULX, TS-1TB, X1190
JP7L	639-05767	MLB, 2.6GHZ, HY-16, ULX, WD-1TB, X1190
JP7M	639-05768	MLB, 2.6GHZ, HY-16, ULX, TS-1TB, X1190

2.6GHZ CPU, 16GB DRAM, ULX GPU, 2TB S4E		
EEEE	BOM NUMBER	BOM NAME
JP7N	639-05769	MLB, 2.6GHZ, MC-16, ULX, WD-2TB, X1190
JP7P	639-05770	MLB, 2.6GHZ, MC-16, ULX, TS-2TB, X1190
JP7Q	639-05771	MLB, 2.6GHZ, MC-16, ULX, SM-2TB, X1190
JP7R	639-05772	MLB, 2.6GHZ, HY-16, ULX, WD-2TB, X1190
JP7T	639-05773	MLB, 2.6GHZ, HY-16, ULX, TS-2TB, X1190
JP7V	639-05774	MLB, 2.6GHZ, HY-16, ULX, SM-2TB, X1190

2.6GHZ CPU, 16GB DRAM, ULX GPU, 4TB S4E		
EEEE	BOM NUMBER	BOM NAME
JP7W	639-05775	MLB, 2.6GHZ, MC-16, ULX, SM-4TB, X1190
JP7X	639-05776	MLB, 2.6GHZ, HY-16, ULX, SM-4TB, X1190

2.2GHZ CPU, 16GB DRAM, PRX GPU, 2TB S4E TLC

BOM NUMBER	BOM NAME	BOM OPTIONS
939-05880	MLB, 2.2GHZ, MC-16, PRX, TLC-WD-2TB, X1190	ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_TLC, MC_16G, BAFFIN_PROX, CPU_CFL:2.2
939-05879	MLB, 2.2GHZ, MC-16, PRX, TLC-TS-2TB, X1190	ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_TLC, MC_16G, BAFFIN_PROX, CPU_CFL:2.2

DC-DC BOMS

BOM NUMBER	BOM NAME	BOM OPTIONS
939-04900	MLB, CPU-SK-S, NO_GPU, X1190	ALTERNATE, BASE_BOM, DEVEL_BOM, BOM_8L, CPU_SKL:SOCKET
939-04901	MLB, NO_CPU, GPU-SK-S, X1190	ALTERNATE, BASE_BOM, DEVEL_BOM, BOM_8L, STAROUST:VDDC
939-04904	MLB, NO_CPU, GPU-SK-M, X1190	ALTERNATE, BASE_BOM, DEVEL_BOM, BOM_8L, STAROUST:VDDC1_VDD
939-04902	MLB, NO_CPU, NO_GPU, X1190	ALTERNATE, BASE_BOM, DEVEL_BOM, BOM_8L

2.6GHZ CPU, 32GB DRAM, ULX GPU, 512GB S4E		
EEEE	BOM NUMBER	BOM NAME
JP7Y	639-05777	MLB, 2.6GHZ, MC-32, ULX, WD-512, X1190
JP80	639-05778	MLB, 2.6GHZ, MC-32, ULX, TS-512, X1190
JP81	639-05779	MLB, 2.6GHZ, HY-32, ULX, WD-512, X1190
JP82	639-05780	MLB, 2.6GHZ, HY-32, ULX, TS-512, X1190

2.6GHZ CPU, 32GB DRAM, ULX GPU, 1TB S4E		
EEEE	BOM NUMBER	BOM NAME
JP83	639-05781	MLB, 2.6GHZ, MC-32, ULX, WD-1TB, X1190
JP84	639-05782	MLB, 2.6GHZ, MC-32, ULX, TS-1TB, X1190
JP85	639-05783	MLB, 2.6GHZ, HY-32, ULX, WD-1TB, X1190
JP86	639-05784	MLB, 2.6GHZ, HY-32, ULX, TS-1TB, X1190

2.6GHZ CPU, 32GB DRAM, ULX GPU, 2TB S4E		
EEEE	BOM NUMBER	BOM NAME
JP87	639-05785	MLB, 2.6GHZ, MC-32, ULX, WD-2TB, X1190
JP88	639-05786	MLB, 2.6GHZ, MC-32, ULX, TS-2TB, X1190
JP89	639-05787	MLB, 2.6GHZ, MC-32, ULX, SM-2TB, X1190
JP8C	639-05788	MLB, 2.6GHZ, HY-32, ULX, WD-2TB, X1190
JP8D	639-05789	MLB, 2.6GHZ, HY-32, ULX, TS-2TB, X1190
JP8F	639-05790	MLB, 2.6GHZ, HY-32, ULX, SM-2TB, X1190

2.6GHZ CPU, 32GB DRAM, ULX GPU, 4TB S4E		
EEEE	BOM NUMBER	BOM NAME
JP8G	639-05791	MLB, 2.6GHZ, MC-32, ULX, SM-4TB, X1190
JP8H	639-05792	MLB, 2.6GHZ, HY-32, ULX, SM-4TB, X1190

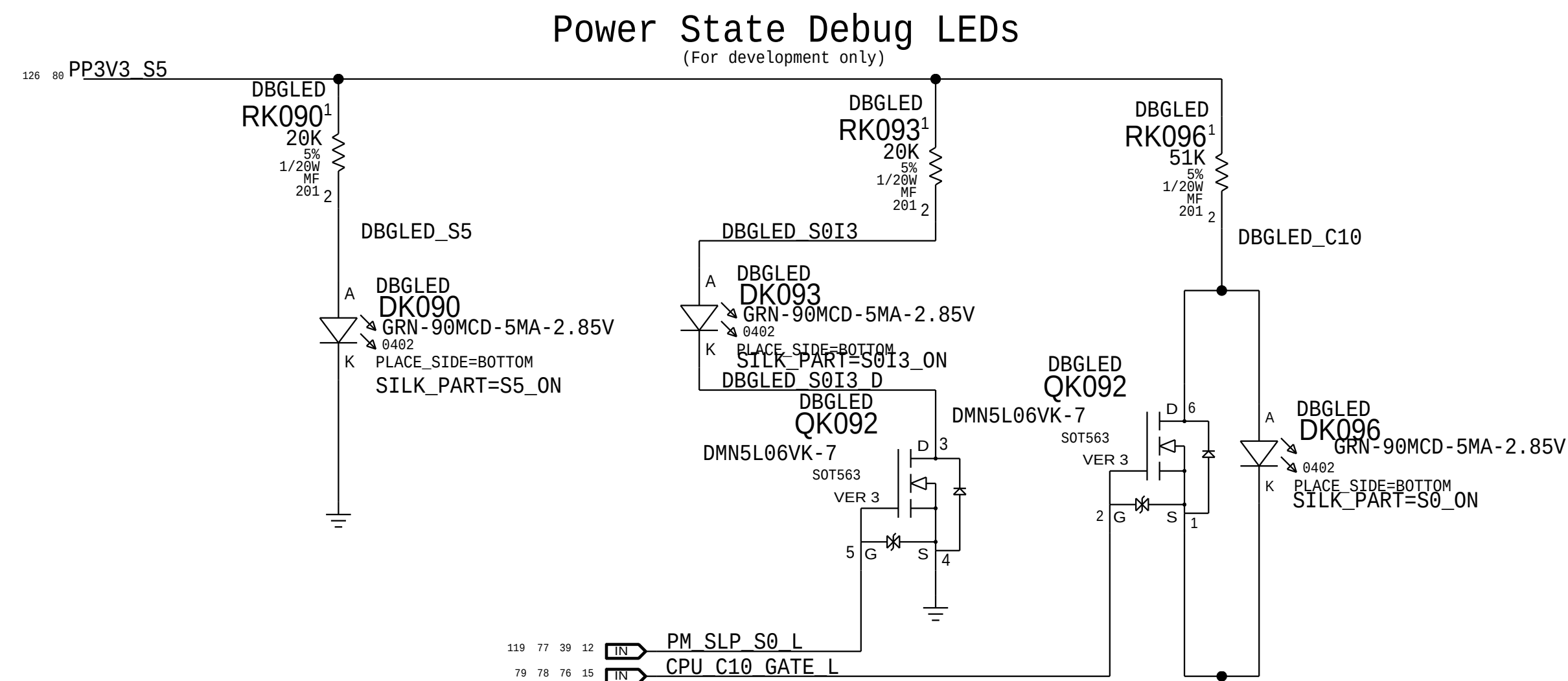
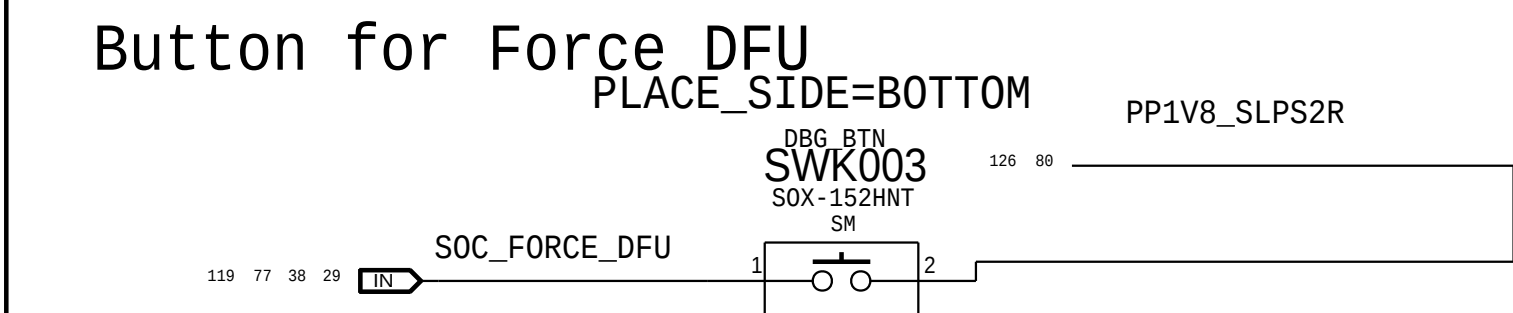
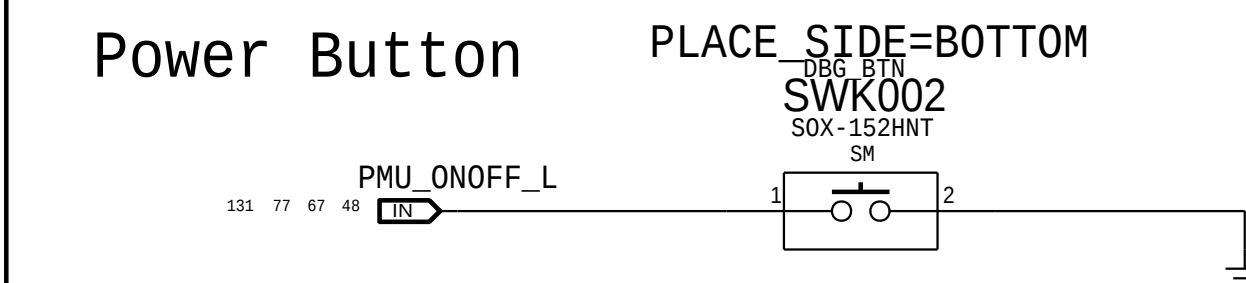
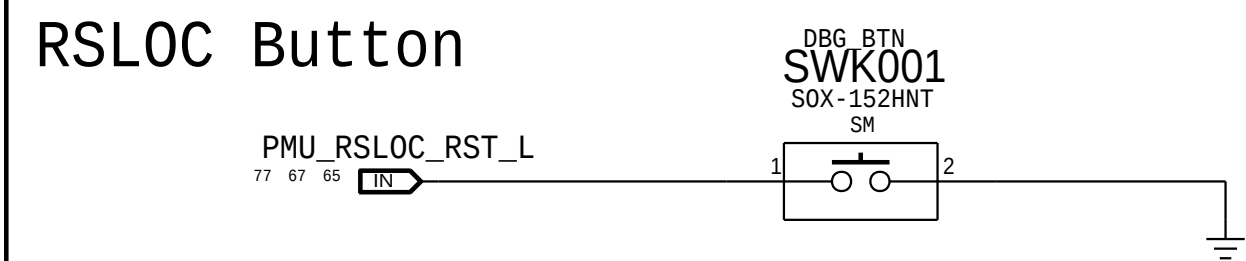
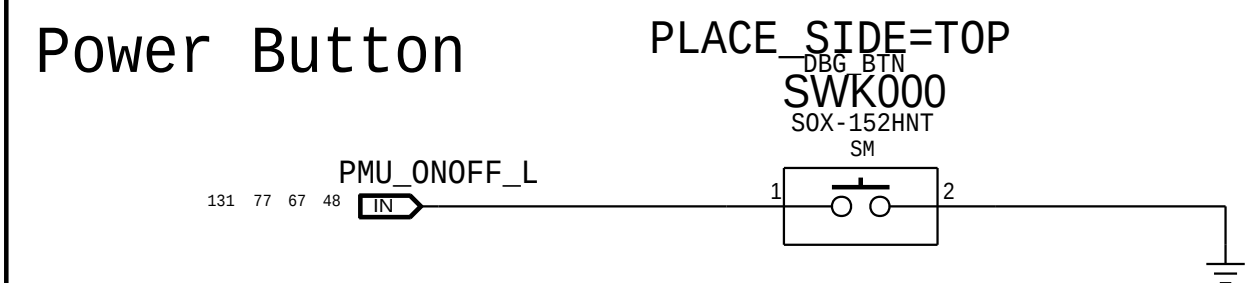
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D	EEEE	2.9GHZ CPU, 16GB DRAM, PRX GPU, 256GB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05793		MLB, 2.9GHZ, MC-16, PRX, WD-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_256_PMLC, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05794		MLB, 2.9GHZ, MC-16, PRX, TS-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_256_PMLC, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05795		MLB, 2.9GHZ, HY-16, PRX, WD-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_256_PMLC, HY_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05796		MLB, 2.9GHZ, HY-16, PRX, TS-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_256_PMLC, HY_166, BAFFIN_PROX, CPU_CFL:2.9							
		2.9GHZ CPU, 16GB DRAM, PRX GPU, 512GB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05797		MLB, 2.9GHZ, MC-16, PRX, WD-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_512_PMLC, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05798		MLB, 2.9GHZ, MC-16, PRX, TS-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_512_PMLC, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
639-05799		MLB, 2.9GHZ, HY-16, PRX, WD-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_512_PMLC, HY_166, BAFFIN_PROX, CPU_CFL:2.9									
639-05800		MLB, 2.9GHZ, HY-16, PRX, TS-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_512_PMLC, HY_166, BAFFIN_PROX, CPU_CFL:2.9									
C	EEEE	2.9GHZ CPU, 16GB DRAM, PRX GPU, 1TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05801		MLB, 2.9GHZ, MC-16, PRX, WD-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, WD_1TB_PMLC, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05802		MLB, 2.9GHZ, MC-16, PRX, TS-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, TS_1TB_PMLC, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05803		MLB, 2.9GHZ, HY-16, PRX, WD-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, WD_1TB_PMLC, HY_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05804		MLB, 2.9GHZ, HY-16, PRX, TS-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, TS_1TB_PMLC, HY_166, BAFFIN_PROX, CPU_CFL:2.9							
		2.9GHZ CPU, 16GB DRAM, PRX GPU, 2TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05805		MLB, 2.9GHZ, MC-16, PRX, WD-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_PMLC, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05806		MLB, 2.9GHZ, MC-16, PRX, TS-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_PMLC, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
639-05807		MLB, 2.9GHZ, MC-16, PRX, SM-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_2TB_3DV4, MC_166, BAFFIN_PROX, CPU_CFL:2.9									
639-05808		MLB, 2.9GHZ, HY-16, PRX, WD-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_PMLC, HY_166, BAFFIN_PROX, CPU_CFL:2.9									
639-05809		MLB, 2.9GHZ, HY-16, PRX, TS-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_PMLC, HY_166, BAFFIN_PROX, CPU_CFL:2.9									
639-05810		MLB, 2.9GHZ, HY-16, PRX, SM-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_2TB_3DV4, HY_166, BAFFIN_PROX, CPU_CFL:2.9									
B	EEEE	2.9GHZ CPU, 16GB DRAM, PRX GPU, 4TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05811		MLB, 2.9GHZ, MC-16, PRX, SM-4TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_4TB_3DV4, MC_166, BAFFIN_PROX, CPU_CFL:2.9							
		639-05812		MLB, 2.9GHZ, HY-16, PRX, SM-4TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_4TB_3DV4, HY_166, BAFFIN_PROX, CPU_CFL:2.9							
		2.9GHZ CPU, 32GB DRAM, PRX GPU, 256GB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05833		MLB, 2.9GHZ, MC-32, PRX, WD-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_256_PMLC, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05834		MLB, 2.9GHZ, MC-32, PRX, TS-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_256_PMLC, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05835		MLB, 2.9GHZ, HY-32, PRX, WD-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_256_PMLC, HY_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05836		MLB, 2.9GHZ, HY-32, PRX, TS-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_256_PMLC, HY_326, BAFFIN_PROX, CPU_CFL:2.9							
A	EEEE	2.9GHZ CPU, 32GB DRAM, PRX GPU, 512GB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05837		MLB, 2.9GHZ, MC-32, PRX, WD-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_512_PMLC, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05838		MLB, 2.9GHZ, MC-32, PRX, TS-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_512_PMLC, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05839		MLB, 2.9GHZ, HY-32, PRX, WD-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_512_PMLC, HY_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05840		MLB, 2.9GHZ, HY-32, PRX, TS-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_512_PMLC, HY_326, BAFFIN_PROX, CPU_CFL:2.9							
		2.9GHZ CPU, 32GB DRAM, PRX GPU, 1TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05841		MLB, 2.9GHZ, MC-32, PRX, WD-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, WD_1TB_PMLC, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05842		MLB, 2.9GHZ, MC-32, PRX, TS-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, TS_1TB_PMLC, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
639-05843		MLB, 2.9GHZ, HY-32, PRX, WD-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, WD_1TB_PMLC, HY_326, BAFFIN_PROX, CPU_CFL:2.9									
639-05844		MLB, 2.9GHZ, HY-32, PRX, TS-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, TS_1TB_PMLC, HY_326, BAFFIN_PROX, CPU_CFL:2.9									
	EEEE	2.9GHZ CPU, 32GB DRAM, PRX GPU, 2TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05845		MLB, 2.9GHZ, MC-32, PRX, WD-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_PMLC, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05846		MLB, 2.9GHZ, MC-32, PRX, TS-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_PMLC, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05847		MLB, 2.9GHZ, MC-32, PRX, SM-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_2TB_3DV4, MC_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05848		MLB, 2.9GHZ, HY-32, PRX, WD-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_PMLC, HY_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05849		MLB, 2.9GHZ, HY-32, PRX, TS-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_PMLC, HY_326, BAFFIN_PROX, CPU_CFL:2.9							
		639-05850		MLB, 2.9GHZ, HY-32, PRX, SM-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_2TB_3DV4, HY_326, BAFFIN_PROX, CPU_CFL:2.9							
		2.9GHZ CPU, 32GB DRAM, PRX GPU, 4TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
639-05851		MLB, 2.9GHZ, MC-32, PRX, SM-4TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_4TB_3DV4, MC_326, BAFFIN_PROX, CPU_CFL:2.9									
639-05852		MLB, 2.9GHZ, HY-32, PRX, SM-4TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_4TB_3DV4, HY_326, BAFFIN_PROX, CPU_CFL:2.9									
	EEEE	2.9GHZ CPU, 32GB DRAM, ULX GPU, 4TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05871		MLB, 2.9GHZ, MC-32, ULX, SM-4TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_4TB_3DV4, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05872		MLB, 2.9GHZ, HY-32, ULX, SM-4TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_4TB_3DV4, HY_326, BAFFIN_ULX, CPU_CFL:2.9							
		2.9GHZ CPU, 16GB DRAM, ULX GPU, 256GB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05813		MLB, 2.9GHZ, MC-16, ULX, WD-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_256_PMLC, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05814		MLB, 2.9GHZ, MC-16, ULX, TS-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_256_PMLC, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05815		MLB, 2.9GHZ, HY-16, ULX, WD-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_256_PMLC, HY_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05816		MLB, 2.9GHZ, HY-16, ULX, TS-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_256_PMLC, HY_166, BAFFIN_ULX, CPU_CFL:2.9							
	EEEE	2.9GHZ CPU, 16GB DRAM, ULX GPU, 512GB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05817		MLB, 2.9GHZ, MC-16, ULX, WD-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_512_PMLC, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05818		MLB, 2.9GHZ, MC-16, ULX, TS-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_512_PMLC, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05819		MLB, 2.9GHZ, HY-16, ULX, WD-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_512_PMLC, HY_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05820		MLB, 2.9GHZ, HY-16, ULX, TS-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_512_PMLC, HY_166, BAFFIN_ULX, CPU_CFL:2.9							
		2.9GHZ CPU, 16GB DRAM, ULX GPU, 1TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05821		MLB, 2.9GHZ, MC-16, ULX, WD-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, WD_1TB_PMLC, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05822		MLB, 2.9GHZ, MC-16, ULX, TS-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, TS_1TB_PMLC, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
639-05823		MLB, 2.9GHZ, HY-16, ULX, WD-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, WD_1TB_PMLC, HY_166, BAFFIN_ULX, CPU_CFL:2.9									
639-05824		MLB, 2.9GHZ, HY-16, ULX, TS-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, TS_1TB_PMLC, HY_166, BAFFIN_ULX, CPU_CFL:2.9									
	EEEE	2.9GHZ CPU, 16GB DRAM, ULX GPU, 2TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05825		MLB, 2.9GHZ, MC-16, ULX, WD-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_PMLC, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05826		MLB, 2.9GHZ, MC-16, ULX, TS-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_PMLC, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05827		MLB, 2.9GHZ, MC-16, ULX, SM-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_2TB_3DV4, MC_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05828		MLB, 2.9GHZ, HY-16, ULX, WD-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_PMLC, HY_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05829		MLB, 2.9GHZ, HY-16, ULX, TS-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_PMLC, HY_166, BAFFIN_ULX, CPU_CFL:2.9							
		639-05830		MLB, 2.9GHZ, HY-16, ULX, SM-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_2TB_3DV4, HY_166, BAFFIN_ULX, CPU_CFL:2.9							
		2.9GHZ CPU, 16GB DRAM, ULX GPU, 4TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
639-05831		MLB, 2.9GHZ, MC-16, ULX, SM-4TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_4TB_3DV4, MC_166, BAFFIN_ULX, CPU_CFL:2.9									
639-05832		MLB, 2.9GHZ, HY-16, ULX, SM-4TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_4TB_3DV4, HY_166, BAFFIN_ULX, CPU_CFL:2.9									
	EEEE	2.9GHZ CPU, 32GB DRAM, ULX GPU, 256GB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05853		MLB, 2.9GHZ, MC-32, ULX, WD-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_256_PMLC, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05854		MLB, 2.9GHZ, MC-32, ULX, TS-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_256_PMLC, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05855		MLB, 2.9GHZ, HY-32, ULX, WD-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_256_PMLC, HY_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05856		MLB, 2.9GHZ, HY-32, ULX, TS-256, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_256_PMLC, HY_326, BAFFIN_ULX, CPU_CFL:2.9							
		2.9GHZ CPU, 32GB DRAM, ULX GPU, 512GB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05857		MLB, 2.9GHZ, MC-32, ULX, WD-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_512_PMLC, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05858		MLB, 2.9GHZ, MC-32, ULX, TS-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_512_PMLC, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
639-05859		MLB, 2.9GHZ, HY-32, ULX, WD-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, WD_512_PMLC, HY_326, BAFFIN_ULX, CPU_CFL:2.9									
639-05860		MLB, 2.9GHZ, HY-32, ULX, TS-512, X1190				ALTERNATE, BASE_BOM, DEVEL_BOM, TS_512_PMLC, HY_326, BAFFIN_ULX, CPU_CFL:2.9									
	EEEE	2.9GHZ CPU, 32GB DRAM, ULX GPU, 1TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05861		MLB, 2.9GHZ, MC-32, ULX, WD-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, WD_1TB_PMLC, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05862		MLB, 2.9GHZ, MC-32, ULX, TS-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, TS_1TB_PMLC, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05863		MLB, 2.9GHZ, HY-32, ULX, WD-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, WD_1TB_PMLC, HY_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05864		MLB, 2.9GHZ, HY-32, ULX, TS-1TB, X1190				ALTERNATE, BASE_BOM, BOM_6L, DEVEL_BOM, TS_1TB_PMLC, HY_326, BAFFIN_ULX, CPU_CFL:2.9							
		2.9GHZ CPU, 32GB DRAM, ULX GPU, 2TB S4E													
		BOM NUMBER		BOM NAME				BOM OPTIONS							
		639-05865		MLB, 2.9GHZ, MC-32, ULX, WD-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_PMLC, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
		639-05866		MLB, 2.9GHZ, MC-32, ULX, TS-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_PMLC, MC_326, BAFFIN_ULX, CPU_CFL:2.9							
639-05867		MLB, 2.9GHZ, MC-32, ULX, SM-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_2TB_3DV4, MC_326, BAFFIN_ULX, CPU_CFL:2.9									
639-05868		MLB, 2.9GHZ, HY-32, ULX, WD-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, WD_2TB_PMLC, HY_326, BAFFIN_ULX, CPU_CFL:2.9									
639-05869		MLB, 2.9GHZ, HY-32, ULX, TS-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, TS_2TB_PMLC, HY_326, BAFFIN_ULX, CPU_CFL:2.9									
639-05870		MLB, 2.9GHZ, HY-32, ULX, SM-2TB, X1190				ALTERNATE, BASE_BOM, BOM_8L, DEVEL_BOM, SM_2TB_3DV4, HY_326, BAFFIN_ULX, CPU_CFL:2.9									
		BOM-639 2.9GHZ													
		Apple Inc.				DRAWING NUMBER		051-02643		SIZE		D			
						REVISION		4.0.0							
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<table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>197S00046</td><td>197S00036</td><td></td><td>ALL</td><td>Epson w/ TXC</td></tr><tr><td>197S00047</td><td>197S00036</td><td></td><td>ALL</td><td>Kyocera w/ TXC</td></tr><tr><td>197S00048</td><td>197S00036</td><td></td><td>ALL</td><td>Murata w/ TXC</td></tr><tr><td>376S1080</td><td>376S00820</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00726</td><td>152S000592</td><td></td><td>ALL</td><td></td></tr><tr><td>376S1053</td><td>376S00604</td><td></td><td>ALL</td><td>Diodes alt to Fairchild</td></tr><tr><td>353S000636</td><td>353S4037</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00007</td><td>155S00607</td><td></td><td>ALL</td><td></td></tr><tr><td>376S1106</td><td>376S00678</td><td></td><td>ALL</td><td>Fairchild alt to Vishay</td></tr><tr><td>371S00127</td><td>371S00182</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00769</td><td>152S000659</td><td></td><td>ALL</td><td></td></tr><tr><td>372S00016</td><td>372S000015</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00041</td><td>107S00039</td><td></td><td>ALL</td><td></td></tr><tr><td>740S00041</td><td>740S0159</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00138</td><td>311S0436</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00725</td><td>152S000590</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00041</td><td>152S00230</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00219</td><td>376S000079</td><td></td><td>ALL</td><td>TI for Fairchild</td></tr><tr><td>311S00121</td><td>311S00390</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00596</td><td>311S00593</td><td></td><td>ALL</td><td></td></tr><tr><td>353S3526</td><td>353S3528</td><td></td><td>ALL</td><td></td></tr><tr><td>353S3527</td><td>353S3528</td><td></td><td>ALL</td><td></td></tr><tr><td>197S00131</td><td>197S00116</td><td></td><td>ALL</td><td></td></tr><tr><td>197S00117</td><td>197S00116</td><td></td><td>ALL</td><td></td></tr><tr><td>197S00115</td><td>197S00116</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00398</td><td>152S00204</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00706</td><td>152S00344</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00165</td><td>311S00164</td><td></td><td>ALL</td><td></td></tr><tr><td>128S000050</td><td>128S00351</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00047</td><td>138S00073</td><td></td><td>ALL</td><td></td></tr><tr><td>353S01041</td><td>353S01042</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00605</td><td>155S000232</td><td></td><td>ALL</td><td></td></tr><tr><td>740S000051</td><td>740S000050</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00724</td><td>152S00311</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00694</td><td>155S00307</td><td></td><td>ALL</td><td></td></tr><tr><td>103S00321</td><td>103S00276</td><td></td><td>ALL</td><td></td></tr><tr><td>740S0144</td><td>740S0118</td><td></td><td>ALL</td><td></td></tr><tr><td>740S00028</td><td>740S0118</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00077</td><td>138S000035</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00093</td><td>138S000035</td><td></td><td>ALL</td><td></td></tr><tr><td>138S000056</td><td>138S1100</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00015</td><td>107S000011</td><td></td><td>ALL</td><td></td></tr><tr><td>152S000543</td><td>152S00404</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00018</td><td>155S00664</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>152S00359</td><td>152S00253</td><td></td><td>ALL</td><td>Chillisin alt to Cyntec</td></tr><tr><td>311S00118</td><td>311S0409</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00365</td><td>155S00191</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00070</td><td>107S00005</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00600</td><td>138S00604</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00046</td><td>138S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00914</td><td>155S00097</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00190</td><td>155S00097</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00117</td><td>138S00071</td><td></td><td>ALL</td><td></td></tr><tr><td>353S01346</td><td>353S01320</td><td></td><td>ALL</td><td></td></tr><tr><td>376S000201</td><td>376S1147</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00230</td><td>376S00174</td><td></td><td>ALL</td><td></td></tr><tr><td>128S000022</td><td>128S000018</td><td></td><td>ALL</td><td></td></tr><tr><td>376S000004</td><td>376S1107</td><td></td><td>ALL</td><td></td></tr></table>					PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	197S00046	197S00036		ALL	Epson w/ TXC	197S00047	197S00036		ALL	Kyocera w/ TXC	197S00048	197S00036		ALL	Murata w/ TXC	376S1080	376S00820		ALL		152S00726	152S000592		ALL		376S1053	376S00604		ALL	Diodes alt to Fairchild	353S000636	353S4037		ALL		155S00007	155S00607		ALL		376S1106	376S00678		ALL	Fairchild alt to Vishay	371S00127	371S00182		ALL		152S00769	152S000659		ALL		372S00016	372S000015		ALL		107S00041	107S00039		ALL		740S00041	740S0159		ALL		311S00138	311S0436		ALL		152S00725	152S000590		ALL		152S00041	152S00230		ALL		376S00219	376S000079		ALL	TI for 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Cyntec	311S00118	311S0409		ALL		155S00365	155S00191		ALL		107S00070	107S00005		ALL		138S00600	138S00604		ALL		138S00046	138S00011		ALL		155S00914	155S00097		ALL		155S00190	155S00097		ALL		138S00117	138S00071		ALL		353S01346	353S01320		ALL		376S000201	376S1147		ALL		376S00230	376S00174		ALL		128S000022	128S000018		ALL		376S000004	376S1107		ALL		<table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>353S000078</td><td>353S000059</td><td></td><td>ALL</td><td></td></tr><tr><td>371S000009</td><td>371S000005</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00248</td><td>107S00250</td><td></td><td>ALL</td><td></td></tr><tr><td>107S000033</td><td>107S000034</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00103</td><td>107S000050</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>311S000013</td><td>311S00500</td><td></td><td>ALL</td><td>Part w1</td></tr><tr><td>132S000064</td><td>132S00409</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00364</td><td>128S00264</td><td></td><td>ALL</td><td>Kemet w/ Panasonic</td></tr><tr><td>128S000050</td><td>128S000018</td><td></td><td>ALL</td><td>NEC w/ Rohm</td></tr><tr><td>107S00071</td><td>107S000053</td><td></td><td>ALL</td><td></td></tr><tr><td>376S000014</td><td>376S00761</td><td></td><td>ALL</td><td></td></tr><tr><td>376S000006</td><td>376S00761</td><td></td><td>ALL</td><td></td></tr><tr><td>107S000044</td><td>107S000076</td><td></td><td>ALL</td><td></td></tr><tr><td>740S000003</td><td>740S00135</td><td></td><td>ALL</td><td></td></tr><tr><td>128S000070</td><td>128S000007</td><td></td><td>ALL</td><td></td></tr><tr><td>740S000019</td><td>740S000007</td><td></td><td>ALL</td><td>Bourns w/ Polytronics</td></tr><tr><td>353S000052</td><td>353S4262</td><td></td><td>ALL</td><td>TI w/ OnSemi</td></tr><tr><td>138S00775</td><td>138S00060</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0709</td><td>138S00941</td><td></td><td>ALL</td><td>Murata w/ SS</td></tr><tr><td>152S000363</td><td>152S000040</td><td></td><td>ALL</td><td></td></tr><tr><td>152S000415</td><td>152S00140</td><td></td><td>ALL</td><td></td></tr><tr><td>155S000204</td><td>155S00731</td><td></td><td>ALL</td><td></td></tr><tr><td>311S000000</td><td>311S00273</td><td></td><td>ALL</td><td></td></tr><tr><td>128S000000</td><td>128S000007</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0730</td><td>138S1101</td><td></td><td>ALL</td><td>Samsung alt to Murata</td></tr><tr><td>107S00234</td><td>107S00235</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00156</td><td>311S00129</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00600</td><td>155S00513</td><td></td><td>ALL</td><td></td></tr><tr><td>197S00120</td><td>197S00118</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00360</td><td>152S00269</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00271</td><td>311S000000</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>311S00104</td><td>311S000091</td><td></td><td>ALL</td><td></td></tr></table>					PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	353S000078	353S000059		ALL		371S000009	371S000005		ALL		107S00248	107S00250		ALL		107S000033	107S000034		ALL		107S00103	107S000050		ALL	Cyntec w/ Yageo	311S000013	311S00500		ALL	Part w1	132S000064	132S00409		ALL		128S00364	128S00264		ALL	Kemet w/ Panasonic	128S000050	128S000018		ALL	NEC w/ Rohm	107S00071	107S000053		ALL		376S000014	376S00761		ALL		376S000006	376S00761		ALL		107S000044	107S000076		ALL		740S000003	740S00135		ALL		128S000070	128S000007		ALL		740S000019	740S000007		ALL	Bourns w/ Polytronics	353S000052	353S4262		ALL	TI w/ OnSemi	138S00775	138S00060		ALL		138S0709	138S00941		ALL	Murata w/ SS	152S000363	152S000040		ALL		152S000415	152S00140		ALL		155S000204	155S00731		ALL		311S000000	311S00273		ALL		128S000000	128S000007		ALL		138S0730	138S1101		ALL	Samsung alt to Murata	107S00234	107S00235		ALL		311S00156	311S00129		ALL		155S00600	155S00513		ALL		197S00120	197S00118		ALL		152S00360	152S00269		ALL		311S00271	311S000000		ALL	NXP w/ Diodes	311S00104	311S000091		ALL		<table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>103S00248</td><td>103S00247</td><td></td><td>ALL</td><td>8.06K 0.1% RESISTOR</td></tr><tr><td>103S00250</td><td>103S00249</td><td></td><td>ALL</td><td>10.2K 0.1% RESISTOR</td></tr><tr><td>116S000006</td><td>116S0175</td><td></td><td>ALL</td><td>00HM RESISTOR</td></tr><tr><td>128S000062</td><td>128S000067</td><td></td><td>ALL</td><td>150UF POLY CAP</td></tr><tr><td>128S000069</td><td>128S000067</td><td></td><td>ALL</td><td>150UF POLY CAP</td></tr><tr><td>128S000065</td><td>128S000067</td><td></td><td>ALL</td><td>150UF POLY CAP</td></tr><tr><td>138S000097</td><td>138S0750</td><td></td><td>ALL</td><td>4.7UF CAP</td></tr><tr><td>131S00134</td><td>131S000041</td><td></td><td>ALL</td><td>12pF Desense:31561001</td></tr><tr><td>107S00101</td><td>107S000005</td><td></td><td>ALL</td><td>5mOhm :31560708</td></tr><tr><td>107S000007</td><td>107S000029</td><td></td><td>ALL</td><td>5mOhm :31560043</td></tr><tr><td>107S000006</td><td>107S000056</td><td></td><td>ALL</td><td>3mOhm : 315600937</td></tr><tr><td>107S00100</td><td>107S000057</td><td></td><td>ALL</td><td>2mOhm : 31561003</td></tr><tr><td>107S00240</td><td>107S00255</td><td></td><td>ALL</td><td>1mOhm : 31561401</td></tr><tr><td>107S00139</td><td>107S0178</td><td></td><td>ALL</td><td>220KOhm :31561213</td></tr><tr><td>128S00296</td><td>128S0487</td><td></td><td>ALL</td><td>47uF-PT : 31561532</td></tr><tr><td>128S000012</td><td>128S0487</td><td></td><td>ALL</td><td>47uF-PT : 31561532</td></tr><tr><td>138S0706</td><td>138S0705</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>353S000032</td><td>353S000525</td><td></td><td>ALL</td><td>IMVP SA</td></tr><tr><td>353S000031</td><td>353S000519</td><td></td><td>ALL</td><td>IMVP VCORE</td></tr><tr><td>152S00702</td><td>152S000772</td><td></td><td>ALL</td><td>IMVP CORE INDUCTOR</td></tr><tr><td>138S00105</td><td>138S000037</td><td></td><td>ALL</td><td>10uF CER : 31563978</td></tr><tr><td>138S0714</td><td>138S0713</td><td></td><td>ALL</td><td>10UF : 31564912</td></tr><tr><td>138S0648</td><td>138S0703</td><td></td><td>ALL</td><td>4.7UF CER : 31564224</td></tr><tr><td>138S0715</td><td>138S0732</td><td></td><td>ALL</td><td>1UF CER : 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SS&Taiyo</td></tr><tr><td>138S0759</td><td>138S0762</td><td></td><td>ALL</td><td></td></tr><tr><td>138S000049</td><td>138S000031</td><td></td><td>ALL</td><td></td></tr><tr><td>107S000090</td><td>107S000055</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00063</td><td>138S00053</td><td></td><td>ALL</td><td></td></tr><tr><td>138S000006</td><td>138S000035</td><td></td><td>ALL</td><td></td></tr><tr><td>377S0077</td><td>377S0183</td><td></td><td>ALL</td><td></td></tr><tr><td>371S00100</td><td>371S00077</td><td></td><td>ALL</td><td>NXP alt to Diodes</td></tr><tr><td>138S00104</td><td>138S00978</td><td></td><td>ALL</td><td></td></tr><tr><td>138S000024</td><td>138S000006</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00434</td><td>152S1029</td><td></td><td>ALL</td><td></td></tr><tr><td>197S000053</td><td>197S000050</td><td></td><td>ALL</td><td>Kyocera w/ TXC</td></tr><tr><td>197S000054</td><td>197S000050</td><td></td><td>ALL</td><td>NDK w/ TXC</td></tr><tr><td>197S000055</td><td>197S000050</td><td></td><td>ALL</td><td>Murata w/ TXC</td></tr><tr><td>335S000270</td><td>335S000203</td><td></td><td>ALL</td><td>Adesto for Micronix</td></tr><tr><td>155S000034</td><td>155S00706</td><td></td><td>ALL</td><td>Taiyo w/ Murata</td></tr><tr><td>353S000079</td><td>353S00754</td><td></td><td>ALL</td><td></td></tr><tr><td>155S000203</td><td>155S00094</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00461</td><td>152S00112</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00146</td><td>376S1001</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00704</td><td>152S000413</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00199</td><td>152S000413</td><td></td><td>ALL</td><td></td></tr><tr><td>335S000066</td><td>335S00946</td><td></td><td>ALL</td><td></td></tr><tr><td>372S0186</td><td>372S0185</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00177</td><td>311S00178</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00750</td><td>353S000077</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00707</td><td>152S000609</td><td></td><td>ALL</td><td></td></tr><tr><td>376S000202</td><td>376S1128</td><td></td><td>ALL</td><td></td></tr></table>					PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	103S00248	103S00247		ALL	8.06K 0.1% RESISTOR	103S00250	103S00249		ALL	10.2K 0.1% RESISTOR	116S000006	116S0175		ALL	00HM RESISTOR	128S000062	128S000067		ALL	150UF POLY CAP	128S000069	128S000067		ALL	150UF POLY CAP	128S000065	128S000067		ALL	150UF POLY CAP	138S000097	138S0750		ALL	4.7UF CAP	131S00134	131S000041		ALL	12pF Desense:31561001	107S00101	107S000005		ALL	5mOhm :31560708	107S000007	107S000029		ALL	5mOhm :31560043	107S000006	107S000056		ALL	3mOhm : 315600937	107S00100	107S000057		ALL	2mOhm : 31561003	107S00240	107S00255		ALL	1mOhm : 31561401	107S00139	107S0178		ALL	220KOhm :31561213	128S00296	128S0487		ALL	47uF-PT : 31561532	128S000012	128S0487		ALL	47uF-PT : 31561532	138S0706	138S0705		ALL	Murata w/ Samsung	353S000032	353S000525		ALL	IMVP SA	353S000031	353S000519		ALL	IMVP VCORE	152S00702	152S000772		ALL	IMVP CORE INDUCTOR	138S00105	138S000037		ALL	10uF CER : 31563978	138S0714	138S0713		ALL	10UF : 31564912	138S0648	138S0703		ALL	4.7UF CER : 31564224	138S0715	138S0732		ALL	1UF CER : 31565015	152S000045	152S000015		ALL		138S00111	138S000036		ALL	20uF CER : 31563894	132S000012	132S0401		ALL	0.22uF : 31563715	138S000004	138S000000		ALL	47uF CER : 31564055	138S0739	138S0706		ALL	NEC w/ Vishay	138S0945	138S0706		ALL	NEC w/ Rohm	138S00641	138S0700		ALL	Murata w/ SS&Taiyo	138S0759	138S0762		ALL		138S000049	138S000031		ALL		107S000090	107S000055		ALL		138S00063	138S00053		ALL		138S000006	138S000035		ALL		377S0077	377S0183		ALL		371S00100	371S00077		ALL	NXP alt to Diodes	138S00104	138S00978		ALL		138S000024	138S000006		ALL		152S00434	152S1029		ALL		197S000053	197S000050		ALL	Kyocera w/ TXC	197S000054	197S000050		ALL	NDK w/ TXC	197S000055	197S000050		ALL	Murata w/ TXC	335S000270	335S000203		ALL	Adesto for Micronix	155S000034	155S00706		ALL	Taiyo w/ Murata	353S000079	353S00754		ALL		155S000203	155S00094		ALL		152S00461	152S00112		ALL		376S00146	376S1001		ALL		152S00704	152S000413		ALL		152S00199	152S000413		ALL		335S000066	335S00946		ALL		372S0186	372S0185		ALL		311S00177	311S00178		ALL		353S00750	353S000077		ALL		152S00707	152S000609		ALL		376S000202	376S1128		ALL		<table><tr><th colspan="5">VRAM ALT SUB-BOMS</th></tr><tr><td>605-00222</td><td>605-00221</td><td></td><td>ALL</td><td></td></tr><tr><td>605-00223</td><td>605-00221</td><td></td><td>ALL</td><td></td></tr><tr><td>605-00224</td><td>605-00221</td><td></td><td>ALL</td><td></td></tr><tr><td>605-00225</td><td>605-00221</td><td></td><td>ALL</td><td></td></tr></table>					VRAM ALT SUB-BOMS					605-00222	605-00221		ALL		605-00223	605-00221		ALL		605-00224	605-00221		ALL		605-00225	605-00221		ALL	
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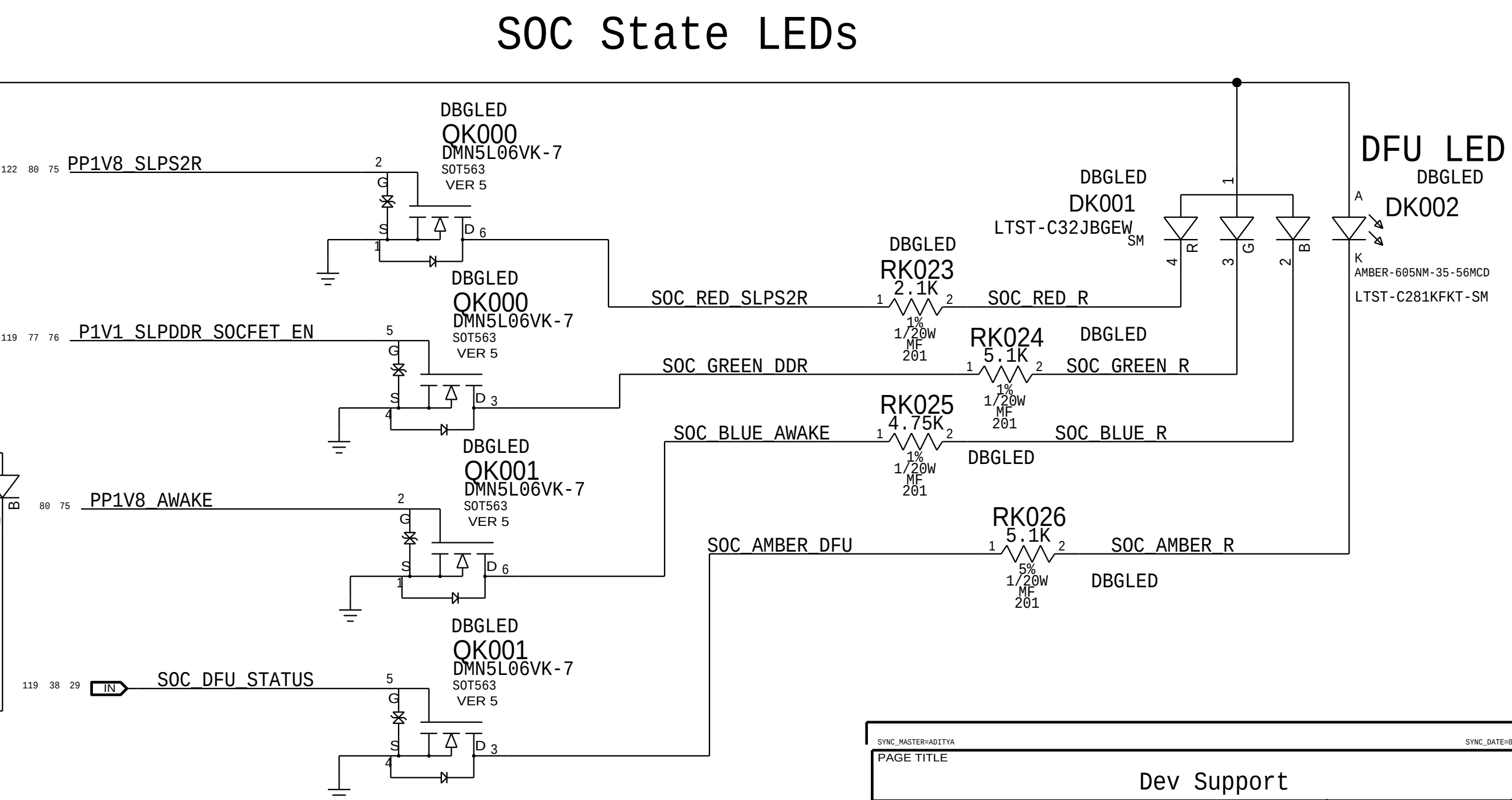
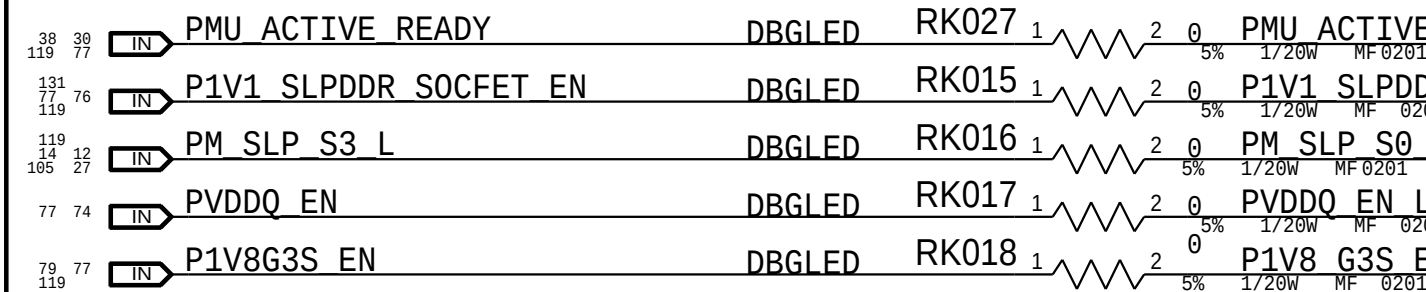
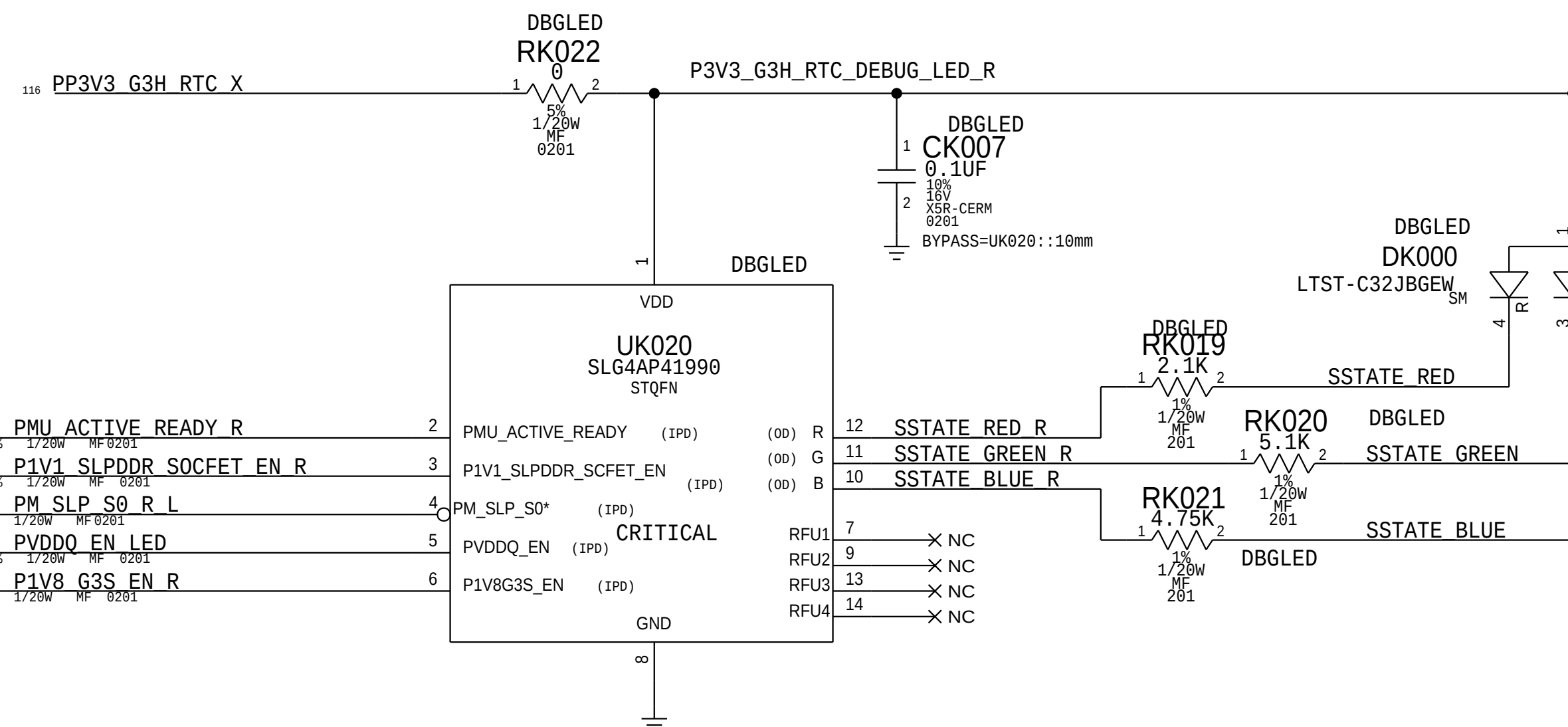
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605-00223	605-00221		ALL	
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		SHEET	130 OF 131		



System State LEDs

INPUT					OUTPUT		
PMU_ACT_RDY	SLP_SCFET_EN	SLP_S0_L	VDDQ_EN	1V8G3S_EN	R	G	B
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